Usage-Based Linguistic Informatics 1

Linguistic Informatics

State of the Art and the Future

Edited by
Yuji Kawaguchi, Susumu Zaima,
Toshihiro Takagaki, Kohji Shibano
and Mayumi Usami

John Benjamins Publishing Company

Linguistic Informatics – State of the Art and the Future

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Volume 1

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Contents

Opening Address
Setsuho IKEHATA (President, Tokyo University of Foreign Studies) 1

Center of Usage-Based Linguistic Informatics (UBLI)
Yuji KAWAGUCHI 3

I. Computer-Assisted Linguistics

One or Two Phonemes: /ø/ - /u/ in Old French, /s/ - /z/ in Dutch and Frisian
–New Solutions to an Old Problem–
Pieter van REENEN and Anke JONGKIND 9

The Lexicon-Grammar of French Verbs –A Syntactic Database–
Christian LECLÈRE 29

A Formal Analysis of Spanish Adjective Position
Masami MIYAMOTO 46

On the Language of Portuguese *Estoria do Muy Nobre Vespesiano*
–Linguistic Change and its Documental Evidence Based on the Corpus Study–
Naotoshi KUROSAWA 64

Analysing Texts in a Specific Domain with Local Grammars
–The Case of Stock Exchange Market Reports–
Takuya NAKAMURA 76

Multivariate Analysis in Dialectology –A Case Study of the Standardization
in the Environs of Paris–
Kanetaka YARIMIZU, Yuji KAWAGUCHI and Masanori ICHIKAWA 99

II. Corpus Linguistics

Corpora of Spoken Spanish Language –The Representativeness Issue–
Francisco MORENO-FERNÁNDEZ 120

Methods of "Hand-made" Corpus Linguistics - A Bilingual Database and the Programming of Analyzers -
Hiroto UEDA 145

Multilateral Interpretation of Corpus-based Semantic Analysis –The Case of the German Verb of Movement *fahren*–
Yoshiyuki MUROI 167

Tools for Creating Online Dictionaries Judeo-Spanish –A Case Study–
Antonio RUIZ TINOCO 180

III. Applied Linguistics

Socio-pragmatic Aspects of Workplace Talk
Janet HOLMES 196

What Do We Mean by "second" in Second Language Acquisition
David BLOCK 221

Integrating Applied Linguistics Research Outcome into Japanese Language Pedagogy –A Challenge in Contrastive Pragmatics–
Suzuko NISHIHARA 242

Computer Assisted Language Learning (CALL) –Moving into the Networked Future–
Mark PETERSON 248

Beyond the Novelty –Providing Meaning in CALL–
Malcolm H. FIELD 258

IV. Discourse Analysis and Language Teaching

Why Do We Need to Analyze Natural Conversation Data in Developing Conversation Teaching Materials? - Some Implications for Developing TUFS Language Modules -
Mayumi USAMI 279

An Analysis of Teaching Materials Based on New Zealand English Conversation in Natural Settings –Implications for the Development of Conversation Teaching Materials–
Takashi SUZUKI, Koji MATSUMOTO and Mayumi USAMI 295

V. TUFS Language Modules

The Creation of the TUFS Pronunciation Module
Tsutomu KIGOSHI 316

Development and Assessment of TUFS Dialogue Module –Multilingual and Functional Syllabus–
Kentaro YUKI, Kazuya ABE and Chunchen LIN 333

Concluding Remarks
Yuji KAWAGUCHI 358

Index of Proper Nouns 359

Index of Subjects 361

Opening Address

Setsuho IKEHATA *(President, Tokyo University of Foreign Studies)*

The 21st Century COE ("Center Of Excellence") Program, launched by the Ministry of Education, Sports, Culture, Science and Technology in 2002, grants subsidies to distinguished universities in our country for the establishment of a center of research and education in various fields with the highest academic standards in the world. It aims at raising the level of research in our country's universities and fostering creative academic minds, expected to become leaders of the world. Tokyo University of Foreign Studies (TUFS) submitted applications for research projects in two of the selected programs—the Humanities and Interdisciplinary/Compound/New Sphere fields. We have obtained wonderful results; both projects were selected. We are extremely pleased and encouraged by this high evaluation of the unique research projects and educational potential of our Graduate School of Area and Culture Studies.

To run the program, TUFS has outstanding experts who collaborate on education and research in a wide range of academic fields including linguistics, literature, history, philosophy, cultural anthropology, sociology, political science, and economics. Thus, we have attained an extremely consistent interdisciplinary and comprehensive approach for a single-faculty university. In an age that emphasizes the global community, it is certainly desirable for us to maximize and further develop this unique strength in both education and research.

A strong foundation in foreign languages is vital to area and culture studies. TUFS engages in education and research in over 50 languages, cultures and societies in every part of the world, which contributes to cross-cultural understanding and the development of people capable of contributing to the actualization of a harmonious global community. In addition, a double-major system that requires students to specialize in both a language and a discipline-related course of study enables TUFS to produce graduates equipped with a high degree of language competence and a deep knowledge of world cultures and societies.

Our new campus in Fuchu is proudly equipped with the state-of-the-art computing network. The most outstanding feature is the information literacy and the number of computers on campus, which ranks at the top level among liberal arts universities in our country. With such priviledged information infrastructure, TUFS endeavors to make the best use of multimedia, the inter-

net and other devices, in order to develop the most advanced language education.

The University's Usage-Based Linguistic Informatics project, selected by the 21st Century COE Program, is the concrete manifestation of our plans for the future, which I have just mentioned. The implementation team members are committed to this future vision and vigorously engaged in the project. It is my fervent desire that they will produce rewarding results. It is the intention of everyone at TUFS to combine our wisdom in a concerted effort to do our utmost to make a success of the 21st Century COE Program.

With a view to providing full support to the program, TUFS has established the "21st Century COE Program Administration Office" which is directly responsible to myself, the President. This Office is an inter-sectional organization consisted of the President, the Vice-President, the deans of each division, the Program Leader, as well as the managers of the secretariat. Its important role is to enhance the cooperation between the various sections within TUFS and to administrate the use of the space and the budget allocated for research.

In closing, let me welcome the distinguished authorities in a variety of fields from the Netherlands, France, Spain, New Zealand, and England who we have invited to be our guest speakers. Thank you so much for adjusting your busy work schedules and for coming such a long way to participate in this international conference. As President of Tokyo University of Foreign Studies, I also wish to express my deep gratitude to each of our guests from research institutions throughout Japan who have taken the time to attend the conference. I hope it will be a great success and a productive and rewarding experience for all of you.

Tokyo, December 13, 2003

Center of Usage-Based Linguistic Informatics (UBLI)

Yuji KAWAGUCHI *(COE Program Leader)*

1. Linguistic Informatics

It is widely believed that linguistic theories and computer sciences have much influenced foreign language education, while the collaboration of these three domains has not brought about new scientific results. The present program will meet such a scientific need. An overall integration of Theoretical and Applied Linguistics will be realized on the basis of Computer Sciences. We have named this new synthetic field Linguistic Informatics. When we hear this name for the first time, we may take it for a branch of natural sciences. However, since our language represents a system of information, Linguistics itself constitutes, in a broad sense, a part of Informatics. In the following lines, the limitation of space will oblige me to explain only the essence of this *21st Century COE (Center of Excellence) Program*.

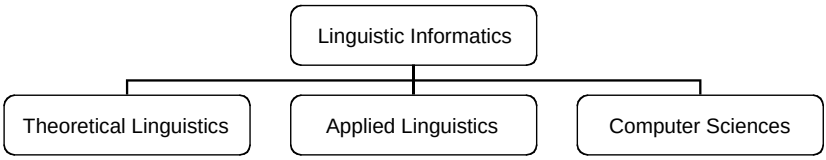
COE Program Promoters

Yuji KAWAGUCHI	French and Turkish Linguistics
Susumu ZAIMA	German Linguistics
Nobuo TOMIMORI	Romance Linguistics
Toshihiro TAKAGAKI	Spanish Linguistics
Yoichiro TSURUGA	French Linguistics
Ikuo KAMEYAMA	Russian Literature
Akira MIZUBAYASHI	French Literature, History
Hideki NOMA	Korean Linguistics
Kohji SHIBANO	Information Technology
Shigeki KAJI	Phonology
Makoto MINEGISHI	Linguistics
Mayumi USAMI	Social Psychology of Language

2. Organization and Research Projects

The present COE program is directed by the following supervisers: Susumu ZAIMA, Toshihiro TAKAGAKI, Yoichiro TSURUGA, Kohji SHIBANO, Makoto MINEGISHI, Mayumi USAMI and Yuji KAWAGUCHI .

In the academic year 2003, the following research projects are undertaken respectively in three scientific fields.



Research Projects in Academic Year 2003

THEORETICAL LINGUISTICS : Corpus Analysis, Syntax and Prosody
Responsibles: Y. KAWAGUCHI, F. KAWAMURA, T. MIYAKE, H. NAKAZAWA, I. SHOHO, K. SOHMIYA, Y. TSURUGA, T. TAKAGAKI , K. URATA

APPLIED LINGUISTICS: Discourse Analysis, Second Language Acquisition, Evaluation of TUFS Modules
Responsibles: M. NEGISHI, T. UMINO, M. USAMI, A. YOSHITOMI

COMPUTER SCIENCES: E-learning, Natural Language Processing
Responsibles: CH. LIN, H. SANO

In principle, these projects are considered as fundamental researches for the development of TUFS Language Modules, which are the very fruits of Linguistic Informatics and the significant scientific contribution of this COE.

3. TUFS Language Modules

3.1. Cohabitation of Natural Language and Machine Language

Our main objective is to innovate foreign language education by developing superior educational material and transmitting it through the Internet. At present, the following 17 languages are covered in the TUFS Language Modules.

Editors of Pronunciation and Dialogue Modules	
English	H. SAITO, A. YOSHITOMI
German	T. NARITA, A. MASAKI
French	Y. KAWAGUCHI, A. MIZUBAYASHI
Spanish	S. KAWAKAMI
Portuguese	N. KUROSAWA, CH. TAKEDA
Russian	H. NAKAZAWA
Chinese	K. HIRAI, N. MIYAKE
Korean	I. CHO, K. IKARASHI
Mongolian	Y. SAITO, R. NUKUSHINA
Indonesian	M. FURIHATA
Filipino	T. MORIGUCHI, M. YAMASHITA
Lao	R. SUZUKI
Cambodian	H. UEDA, T. OKADA
Vietnamese	Y. UNE, H. TAHARA
Arabic	R. RATCLIFFE
Turkish	M. SUGAHARA
Japanese	Y. SATO, T. UMINO

This is a large-scale project that includes more than 100 researchers and graduate students. In TUFS Language Modules, the multilingual language learning system would be one of the main characteristics. In fact, we teach more than 40 different languages at TUFS. But the novelty of TUFS Language Modules lies in another fact. For example, 17 languages are described in unicode (UTF-8), and in our system, HTML, a basic language of World Wide Web (WWW), is correlated with XML, which was first invented in 1998 and has recently begun to be applied in WWW.

This project also has educational ends for the graduate students, who undertake the role of preparing the first-hand materials for the structuring of the modules. Through this research activity, they will gain the knowledge not only of Linguistics and Applied Linguistics, but also of Computer Sciences. In this way, the program will foster new types of linguistic researchers who have full knowledge of Theoretical and Applied Linguistics and can manipulate a computer-assisted language learning system.

3.2. Modularized View of Language

With the advent of the Internet, we have become conscious of the omnipresence of information, that is, what we call ubiquity of information. On the

other hand, WWW gives us an oppurtunity to think over again how and what the information should be. On WWW, theoretically speaking, infinite ordering and combination of information are possible through their mutual linkages. In the TUFS Language Modules, we set our way of thinking free from a traditional view of language and adopt a modularized view of language. Each language unit is composed of four relatively independent modules, i.e. pronunciation, dialogue, grammar and vocabulary modules. The idea of module components allows learners and teachers to learn and teach the target language from whichever part of the modules and in whatever order.

3.3. Cross-Linguistic Syllabus

More freedom than ever will be promised to learners and teachers by these modules. However, a common measure is indispensable for the evaluation of language learning and education. In this sense, the evaluation of modules is very important for this COE program. As each module is designed independently to some extent, one may evaluate it individually. But as far as educational contents and goals are concerned, a more or less loose unity has been realized by adopting a common syllabus design for 17 different languages, so that in addition to a traditional analysis of learners' idiosyncratic characters, one can make an interesting contrastive analysis of individual or universal characteristics of second language acquisition (SLA) through 17 different languages. *Cross-linguistic syllabus* is therefore regarded as an innovation in this web-based language education system.

3.4. Linguistic Usage

The process of developing TUFS Language Modules is as follows: 1. Making language materials; 2. Implementation on WWW; and 3. Web-Based Language Education. Thus, the first step consists in making language materials appropriate for language modules. What kind of language materials must we furnish? We suppose that these language materials should be "usage-based". The key concept here is *linguistic usage*. Then, what in the world does this usage mean? The term is highly polysemous. Some researchers claim that linguistic usage will become explicit only

through quantitative analysis of an enormous corpus. Others declare that usage should be fixed in mutual speech acts between a speaker and a hearer. Moreover, some may suppose that linguistic usage is related to our cognition, for our linguistic knowledge will be accumulated through the encounter with new linguistic usages. We also find researchers who will inisist on the interaction of both linguistic and extra-linguistic aspects of linguistic usage. In short, the definition of usage is not at all unanimous among linguists. TUFS Language Modules give us an opportunity to reconsider the significance of usage for linguistic research and language education. Therefore, I believe that every researcher and graduate student involved in this program should keep their own opinion on the concept of linguistic usage.

At the end of the year 2003, the pronunciation and dialogue modules will be available in Japanese on the Internet. The development of the grammar and vocabulary modules is underway.

4. First International Conference on Linguistic Informatics

Immediately after the selection of this COE program by the Ministry of Education, Culture, Sports, Science and Technology, we began to prepare for the present conference. At the end of 2002, the outline was fixed. On December the 13th and 14th, the first International Conference on Linguistic Informatics is planned to be held at Tokyo University of Foreign Studies. The conference has three different sessions: 1.*Computer-Assisted Linguistics;* 2.*Corpus Linguistics;* and 3.*Applied Linguistics*.

It is a great honour for me to organize this international conference, because we have many guest speakers not only from other universities in Japan, but also from all over the world. We also have many graduate students, mostly PhD candidates, who give papers in this conference. As opposed to normal conferences, we prepare prepublished Proceedings before the conference. This conference covers such large scientific fields, i.e. Computer Linguistics, Philology, Dialectology, Corpus Linguistics, Discourse Pragmatics, Applied Linguistics and e-Learning, so that without assistance of prepublished papers, our audience will not be able to understand the essence of the contributions and to follow what they are discussing.

Through this Proceedings, we expect to know the state of the art of Linguistic Informatics and the problems which this new field will have to solve. We hope that this synthesis of different scientific fields is fruitful and gives us some insights into a future vision of this new science.

Finally, I'd like to express my gratitude towards my colleagues and graduate students of TUFS, and many collaborators of this COE program.

cf. Tufs Language Modules (Japanese version):
http://www.coelang.tufs.ac.jp/modules/
Tufs Language Modules (Multilingual version):
http://www.coelang.tufs.ac.jp/english/modules/
Usage-Based Linguistic Informatics:
http:// www.coelang.tufs.ac.jp/ (in Japanese)
http:// www.coelang.tufs.ac.jp/english/ (in English)

One Or Two Phonemes: /ø/ - /u/ in Old French, /s/ - /z/ in Dutch and Frisian – New Solutions to an Old Problem –

Pieter van REENEN *(Free University Amsterdam and Meertens Instituut Amsterdam)* and Anke JONGKIND *(Free University Amsterdam)*

1. Introduction

Marie de France ends her lai *Yonec* with the following couplet:

551 De la pité, de la dolur	... of the pity, of the sadness
552 Que cil suffrirent pur amur.	that they felt for love.

And in her *Fables* the following two lines of poetry form a comparable couplet:

25:3 Sa femme demeine grant dolur	His wife expresses great sadness
25:4 Sur sa tumbe e nuit e iur.	on his tomb, night and day.

The curious thing about these couplets is that the two rhyme words in them do not rhyme in Modern French, where *dolur* (in Old French also spelled *doulor, doulour, douleur*) is pronounced with /ø/ and *jur* and *amur* (in Old French also spelled *jor, jour* and *amor, amour*) are pronounced with /u/. This type of rhyme is frequent in Old French poetry. Are these bad rhymes, or has one phoneme in Old French been replaced by two phonemes in Modern French?[1]

In Modern Dutch the alveolar fricatives in *Wij willen zien/Sien* and *Wij willen geen pauze/pausen* are officially pronounced as the spelling suggests:

Wij willen [z]ien/[s]ien	We will see/want Sien (girls name).
Wij willen geen pau[z]e/pau[s]en	We do not want a break/popes

However, many speakers may pronounce either [s] or [z] in all cases: for them the words *zien* 'to see' and *Sien* 'girls name' and *pauze* 'break' and *pausen* 'popes' are simply homonyms (the *n* of *pausen* is often not pro-

[1] We thank Yves Charles Morin and especially Bettelou Los for extremely useful comments.

nounced). Spellings in Middle Dutch such as *sien/zien* suggest the same. Does Dutch always distinguish between /s/ and /z/? Or is there just one phoneme /s/ with two allophones in free variation?

This contribution will show how large quantities of data may lead to the correct linguistic analysis of pairs of sounds which seem to have undergone merger, such as /ø/- /u/ in Old French, and /s/ - /z/ in Middle Dutch, Modern Dutch and also Modern Frisian. Our data come from computerized corpora. They are classified and systematised by means of standard UNIX/LINUX tools.

2. Old French /ø/ and /u/

Modern French /u/ of *jour* does not rhyme with the /ø/ of *doleur*. These vowels were also different in Vulgar Latin. The pair /ø/ - /u/ of Modern French corresponds to Latin /o:/ (*dol*[o:]*rem*) and /u/ later /o/ (*di*[u]*rnum* later *di*[o]*rno*). *Amour*, from Latin *am*[o:]*rem*, unexpectedly has short /o/ in Old French, which became /u/, as in *jour*. It is generally assumed to be a loan from the *langue d'oc* of the south of France, imported with the poetry of the *fin amor*. Linguists have been reluctant to draw the unavoidable conclusion: *either* a distinction between two phonemes which exists in Latin and in Modern French is lacking in Old French *or* rhyme in Old French is not always perfect, i.e. not necessarily on the same phoneme, cf. DEES 1988:104. We will show that a systematic analysis of rhymes in Old French provides a more satisfactory solution: The distinction between /ø/ - /u/ has merged in part of the Old French speaking area, and the notion of perfect rhyme has to be replaced by that of whether the poet knows the difference between /ø/ - /u/, whether (s)he always respects it or not.

We examined several hundreds of rhymed texts, available on the web, cf. KUNSTMANN 2000. It concerns the corpus described in DEES et al. 1987 and REENEN & SCHØSLER 2000. From this corpus we culled couplets like those of Marie de France above and classified them as (i) rhymes on /ø/, (ii) rhymes on /u/ and (iii) mixed rhymes. The list of words on /u/ forms turned out to be rather short: *tour* 'tour', *detour* 'detour', *retour* '(I) return', *atour* 'outfit', *autour*, *entour* 'around' all derived from Latin TURNUM; *jour* 'day', *sejour* 'stay'; *tour* 'tower', *amour* 'love', and a few uncommon words like *autour* 'sparrow-hawk', and once *dor* 'almost nothing' and *aumacour* 'emir'. The list of /ø/ forms is much longer, and consists mainly of latin forms on *-orem* and *-orum*: a few examples are *seigneur* 'lord', *sereur* 'sister', *honneur* 'honour', *valeur* 'value', *empereur* 'emperor', also some verb forms such as *(je) labeur*, '(I) work', *je honeur*, '(I) honour'.

/ø/	/u/	mixed	
	cluster 1		
65	15	73	Marie de France: *Lais* (Ms. H) and *Fables* (Ms. A)
13	8	25	*Li romanz d'Athis et Prophilias*
26	2	16	*St. Modwenna*
30	0	11	*La vie du pape Saint Grégoire*, Ms. A1
	cluster 2		
29	30	0	Chrestien de Troyes: *Yvain* and *Perceval*
25	10	0	*La Genèse d'Evrat*
12	12	0	Guillaume de Lorris: *Le Roman de la Rose*
11	8	0	Jean le Marchant: *Miracles de Nostre Dame de Chartres*
	cluster 3		
37	24	25	*La Bible de Macé de la Charité*, tomes I-IV
39	14	7	Jean Renart: *Le Roman de la Rose ou de Guillaume de Dole*
7	3	4	*Fabliaux* nrs. 2 and 4 (Ms. J)
7	2	2	*Li chevaliers as devs espees*
	cluster 4		
41	15	6	*Chronique métrique attribuée à Geffroy de Paris*
24	12	2	Jean de Meun: *Le Roman de la Rose*
11	4	2	Philippe Mousket: *Chronique rimée*
10	2	1	*Leben und Wunderthaten des heiligen Martin*

Table 1. Rhyme couplets on /ø/, on /u/ and mixed, grouped into clusters. Data from Old French corpora, cf. DEES et al. 1987 and REENEN & SCHØSLER 2000. Data from Macé, Geffroy de Paris, Jean de Meun and Jean Renart have been supplemented from editions.

Some of our results are shown in table 1. The first cluster shows the number of couplets with rhymes on /ø/ and /u/ in the poetry of Marie de France: her poetry does not appear to make a distinction between the two sounds, as noted earlier. This impression is confirmed by an investigation into the probability of finding such mixed rhymes. In all, there are (65 + 15 + 73 =) 153 relevant couplets, consisting of (2 x 65 + 73 =) 203 tokens /ø/ and (2 x 15 + 73 =) 103 tokens /u/ with a total of 306 rhyme words. Consequently, the proportion of rhyme words on /ø/ is 203:306 = 0.6634, of rhyme words on /u/ is 103:306 = 0.3366. There are four types of rhyme sequences: /ø/ - /ø/, /u/ - /u/, /ø/ - /u/ and /u/ - /ø/. Under the assumption that Marie the France does not

know the difference between the two vowels, we calculated the probability of finding one of these sequences:

sequence	% x %	= chance	rhymes x chance		rhymes found
/ø/-/ø/:	0.6634 x 0.6634	= 0.4401	153 x 0.4401=	67.34	65
/u/-/u/:	0.3366 x 0.3366	= 0.1132	153 x 0.1132=	17.34	15
/ø/-/u/:	0.6634 x 0.3366	= 0.2233	153 x 0.2233=	}68.33	73
/u/-/ø/:	0.3366 x 0.6634	= 0.2233	153 x 0.2233=		

The difference between rhymes x chance (the theoretical number of rhymes to be expected) and the number of rhymes actually found shows that rhymes are slightly more mixed than expected: 73 rhymes found against (2 x 34.165 =) 68.33 calculated. When applied to all the texts of cluster 1, the test shows that the number of rhymes found in all cases comes very close to matching the number of rhymes to be expected. In other words, for these poets (it is as if) there is no difference between /ø/ and /u/.

For the second cluster headed by Chrestien de Troyes the pattern is completely different: here we see an absolute distinction between /ø/ and /u/. In other words these poets know and respect the difference between /ø/ and /u/.

At first sight the third cluster is like the first: rhyme words within couplets mix. When we calculate the number of expected rhymes and compare them to the rhymes actually found, however, we find that these poets do not mix indiscriminately. Perhaps the most striking case are the (37 + 24 + 25 =) 86 rhymes of Macé:

sequence	% x %	= chance	rhymes x chance		rhymes found
/ø/-/ø/:	0.5756 x 0.5756	= 0.3313	86 x 0.3313=	28.49	37
/u/-/u/:	0.4244 x 0.4244	= 0.1801	86 x 0.1801=	15.49	24
/ø/-/u/:	0.5756 x 0.4244	= 0.2442	86 x 0.2442=	}42.02	25
/u/-/ø/:	0.4244 x 0.5756	= 0.2442	86 x 0.2442=		

If Macé mixed his rhymes indiscriminately, we should have found considerably more than 25 mixed rhymes, i.e. some (2 x 21.01 =) 42. Statistical tests show that this considerable difference is probably not due to chance. The conclusion that these poets do know the difference between /ø/ and /u/, is unavoidable, even if they occasionally mix their rhymes.

This finding has far-reaching implications. Linguists have always assumed that the occurrence of one single mixed rhyme in the couplet of a poem is sufficient evidence for a merger. Our analysis shows that this assumption may not be right. The question to be asked is not: does a text mix rhymes or not, but: is the poet of the text aware of the distinction between /ø/

and /u/. Whether (s)he respects this distinction invariably or only occasionally is a different matter.

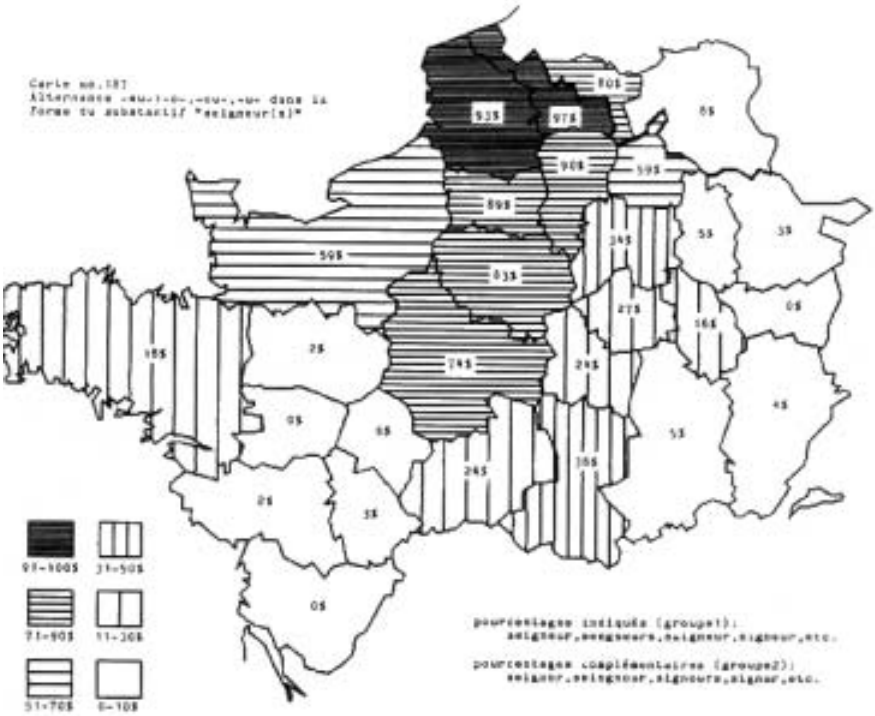
It is now clear that the fourth cluster of table 1 can be grouped with clusters 2 and 3: these poets know the distinction between /ø/ and /u/. Further analysis of this cluster shows, however, a new point: its mixed rhymes all contain as one of its members one of a small set of relatively rare words from the /ø/ series: *pascor* 'spring', *clamor* 'cry, sadness', *pastor* 'shepherd', which always rhyme with the words in the /u/ series and once the word *aumacour* 'emir' which belongs to the /u/ series but rhymes in the /ø/ series. If we accept that these words have changed category, Jean de Meun (with twice *clamor*), Saint-Martin (with once *pascor*) and Mousket (with once *clamor* and once *aumacor*) are identified as belonging in cluster 2, and Geffroy de Paris (with once *clamor* and four times *pastor*) is left with only one mixed rhyme: *valor - jor* (v. 6090) instead of six. This interpretation is reinforced by the fact that at least *clameur*, with /ø/, is extremely rare in Old French: There is only one form in our corpus against 73 others with *o*, *ou*, *u*, and *pasteur* is only slightly more frequent: it occurs 8 times with /ø/ versus 221 times with *o*, *ou*, *u*. This fits in with the observation that *patour* 'shepherd' with /u/, not /ø/, is still correct in Modern Western dialectal French. The two forms *pascor* in our corpus are not sufficient to decide to what category they belong, and the same goes for *aumacour*. If we accept this analysis, three out of four texts of cluster 4 have perfect rhymes, and belong to cluster 2. If we do not accept it, the conclusion that these poets do know the difference between /ø/ and /u/ still remains valid and their rhymes should not be grouped with cluster 1.

As long as we can assign rhymes with reasonable confidence either to cluster 1 or to clusters 2 and 3, it does not matter for linguistic analysis that they do not always rhyme. However, there are texts which cannot be assigned with reasonable confidence to any of our clusters, because their proportions of rhymes make a decision impossible. We have seen in table 1 that Jean Renart belongs to those poets who do know the difference between /ø/ and /u/ quite well, in spite of occasional mixed rhymes. There are mixed rhymes in his *Lai de l'Ombre*: out of 5, 2 are mixed (plus once /ø/, and twice /u/). This implies that statistically no reliable choice can be made whether this text has to be assigned to cluster 1 or to cluster 3. An analysis of the individual *Lais* of Marie de France gives similar, problematic results. For instance, her *Les deus Amanz* contains one rhyme on /u/ (*retour - jour*). It is evident that this does not allow us to place it in cluster 2. Such texts have to be grouped together with others of the same author in order to arrive at the correct linguistic analysis of the difference between /ø/ and /u/.

The great majority of linguists assume that rhyme in Old French is

always perfect, among them FOUCHE 1958:306-307. His view can be represented as follows:
(a) In the so-called triangle of Suchier, i.e. between Tréport (Seine-Maritime) in the west, Montargis (Loiret), Namur and Verviers in the east, there is a distinction between /ø/ and /u/. In this area Latin /o:/ > /ø/ as in *seigneur*, and /o/ > /u/ as in *jour*. Outside the triangle of Suchier (east, west and south of it, and also in Great Britain) /ø/ and /u/ have merged into /u/.
(b) Since rhyme is always perfect and mixed rhymes are found in many texts from the so-called triangle of Suchier, the vowels /ø/ and /u/ must often have merged here as well.

Our rhyme analysis shows that rhyme does not always need to be perfect, which refutes (b). The only linguist who draws the same conclusion, at least as far as /ø/ - /u/ is concerned, is NYROP 1914:210. By contrast, (a) is more or less confirmed by the maps in DEES et al. 1980 (for /ø/, cf. map A below = map 187 in DEES et al. 1980, see also maps 16, 87, 194 in DEES et al. 1980) and DEES and REENEN 1980:map 1 (this latter study also contains a more complete survey and discussion of the relevant literature). These maps, based on spellings from 3,300 charters, show that the triangle of Suchier goes further south than Fouché and Suchier claim: for instance Nièvre is in, although in this area other spellings are also well represented. Maps 13, 89, 162 in DEES et al. 1980 show that with respect to the change /o/ > /u/ as in *jour*, the area is still wider. A comparison of *seigneur* (maps 187, 188 in DEES et al. 1980) and *jour* (map 162 in DEES et al. 1980) shows that east and west of the triangle of Suchier the vowels behave more or less the same: west of the triangle (and in Great Britain, cf. DEES et al. 1987) and east of the triangle the spelling *eu*, i.e. /ø/ is virtually lacking and in this area the vowels of *jour* and *seigneur* behave more or less the same. With the exception of the region Maine-et-Loire, Mayenne/Sarthe, the tendency *jor* > *jour* is considerably stronger than the tendency *seignor* > *seignour* in becoming both *ou*, i.e. /u/. We see the same pattern, only slightly less distinct, in Wallonie, Meuse and Haute-Marne. Remarkably in Franche-Comté we see the opposite tendency: *seignor* > *seignour* considerably precedes *jor* > *jour*.



Map A. The so-called triangle of Suchier.

The provenance of literary texts is often unknown. When it is, we see a pattern that conforms to the dialectal difference established above. The language of Marie de France is Anglo-Normand or from the west of France. The language of Chrestien de Troyes is from Troyes. Macé comes from La Charité (Nièvre), Jean le Marchant: *Miracles de notre Dame de Chartres* comes from Chartres, Guillaume de Lorris and Jean de Meun from Paris or Orléanais, Geffroy de Paris from Paris, the poet of la *Genèse d'Evrat* dedicates his work to the countess of Champagne. Those who know the difference between /ø/ and /u/, whether they respect it or not, come from the enlarged triangle of Suchier. Those who do not know the difference come from the west, from the east, or from England i.e. the areas where the vowels of *seigneur* and *jour* have merged. The only exception we have come across in our corpus might be the *Lyoner Ysopet*. Although the language of this text, in which the distinction between /u/ and /ø/ is respected, locates with an extremely high probability in Franche-Comté, cf. DEES et al. 1987:531, it

seems that we have to assume that the rhymes of this text come from the so-called triangle of Suchier.

Amour with its exceptional /u/ has always been claimed to be a loan from the *langue d'oc*. FOUCHE 1958:307 Rem. IX is the only one to add that the east of Champagne may have played a part as well. The above shows that influence from the west, for instance the poetry of Marie de France, cannot be excluded either. Other words, such as *clamor*, *pastor* (as observed, *patour* is still existing in the west), *pascor* and *aumacor* may have had deviant geographical distributions. They may exhibit specific patterns of lexical diffusion.

Our rhyme analysis has shown that the key to the solution of the problem is to ask whether the poet must have known the difference between rhymes on /ø/ and /u/, not whether a poem always contains perfect rhymes or not. For those who accept this solution it is easy to see that the distinction between /ø/ and /u/ must have been known in a large, central area of the *langue d'oïl*, without being always respected in the rhymes. Here the opposition from Latin has continued in Old French and this is what we find in Modern French.

The case of /ø/ - /u/ is not an isolated one: We have found the same situation with rhymes such as *gent* and *tant* in Macé. Here, too, the poet must have been aware of the distinction, although he mixes rhymes, cf. DEES 1988 and REENEN 1989.

3. Dutch and Frisian /s/ and /z/

In Standard Dutch /s/ and /z/ may form oppositions both word-initially as in [s]*oep* 'soup', [s]*et* 'set' versus [z]*et* 'move', [z]*oon* 'son' and between vowels as in *bla*[z]*en* 'to blow' and *me*[s]*en* 'knives'. Word-finally the voice opposition is neutralized, as in all plosives and fricatives. In Frisian /s/ and /z/ oppose, or seem to oppose, only between vowels. Both in Dutch and in Frisian /s/ is found after short vowel, /z/ after long vowel: *bl*[a:z]*en* and *m*[ɛs]*en*. The opposition between /s/ and /z/ may not have existed in Old Dutch, since it did not exist in Germanic. In Modern Dutch it is weak, the number of minimal pairs being limited, and many speakers do not respect the opposition. We will examine to what extent the difference between /s/ and /z/ is real in Dutch, Flemish and Frisian dialects and in the dialects of 14th-century Middle Dutch and, if so, under which conditions. The term Frisian refers to Modern Frisian dialects only, since we have no data from older periods. Flemish refers to the area of Belgium where Dutch is spoken. Dutch is a cover term for both Flemish and Dutch. To avoid ambiguity we will use the terms Dutch, Flemish-Dutch and Dutch-Dutch.

3.1 Modern Dutch

We use data from the database Phonological and Morphological properties of Dutch and Frisian Dialects, cf. www.meertens.knaw.nl/projecten/mand/ and GOEMAN & TAELDEMAN 1996, also available on cd-rom, cf. BERG 2003. It consists of 613 dialects transcribed in a SAMPA-like computer keyboard version of narrow IPA. Each dialect is characterised by a letter number combination developed by the Dutch linguist Kloeke, which also has been used for dialect classification in Japanese, cf. KAWAGUCHI & INOUE 2002:803. Each dialect description consists of the same list of 1,876 items, mainly words, but also nominal groups and a few short clauses. All items in all dialects are presented in the same order. Items have been elicited one after the other. Informants were typically older, male, conservative speakers. Present-day Urban Dutch is underrepresented, with, for instance, the dialects of Amsterdam and The Hague lacking in the database. A strong point of this database is that it allows systematic and exhaustive comparisons across dialects. A weak point is that the speech samples have been collected in a way which may not always have been spontaneous and natural. More about the characteristics of the corpus can be found in GOEMAN 1999. Nouns and verbs occur either utterance initially or following article or pronoun: *de zoon* 'the son' versus *zoon* 'son', *ziet* 'sees', preceded by *hij* 'he', depending on whether the informant has pronounced the article, or the pronoun, or not.

The method of retrieving the data is easy, at least conceptually, allowing the collection of a series of relevant items by means of standard UNIX/LINUX tools. For our investigation we were interested in data starting with [s] or [z], or a sound in between (half voiceless [z] or half voiced [s] or a devoiced [z], i.e. a lenis voiceless sound). We assigned 100% to [s], 0% to [z], 50% to half voiced [s] or [z], and 75% to devoiced [z]. The result is a table, listing the geographical area as a Kloeke-number with a percentage. This table, containing 613 different areas, was subsequently used as input to a map program developed by E. Wattel, cf. WATTEL & REENEN 1996, resulting in maps such as 1 and 2. Maps 1 (*hij ziet* 'he sees') en 2 (*zeven* 'seven') give an idea of the distribution of /z/ when preceded by a vowel and utterance initially. The overall result of this investigation can be summarised as follows:

1. In initial position, Frisian always has [s] both for /s/ and /z/; Flemish-Dutch has [z] for /z/ and [s] for /s/, and Dutch-Dutch, the area in between Frisia and Flanders, has both [s] and [z] for /z/*ien* 'to see', and only /s/ for /s/*oep* 'soup'. A further distinction in environments for Dutch-Dutch is, however, relevant. Utterance initially, i.e. following silence, Dutch speakers have more [s] than after a voiced sound, almost as if a voiceless sound precedes. For Flemish speakers there is hardly any difference between /z/ occurring utter-

ance initially and /z/ preceded by a voiced sound: both are usually realised as [z]. After a voiceless consonant (for instance *a*/fz/*agen* 'to saw off', and especially *i*/k z/*al* 'I will, shall' and *i*/k z/*ag* 'I saw'), virtually all speakers of all dialects pronounce /z/ as [s].
2. Between vowels, virtually all dialects have [s] after a short vowel and [z] after a long vowel, diphthongs behaving usually as long vowels.

A few words behave exceptionally: /s/ in *sikkel* 'sickle' and *sap* 'juice' is pronounced [z] in Flanders, just as /s/ in *sabel* 'sabre' in the northern tip of Noord-Holland. *Kousen* 'stockings', *mossel* 'mussel', *Brussel* 'Brussels', *flessen* 'bottles', *vissen* 'fishes', *missen* 'to miss', *wassen* 'to wash' and *tussen* 'between', all with regular [s], often have [z] in Groningen, in decreasing order, a regional case of lexical diffusion.

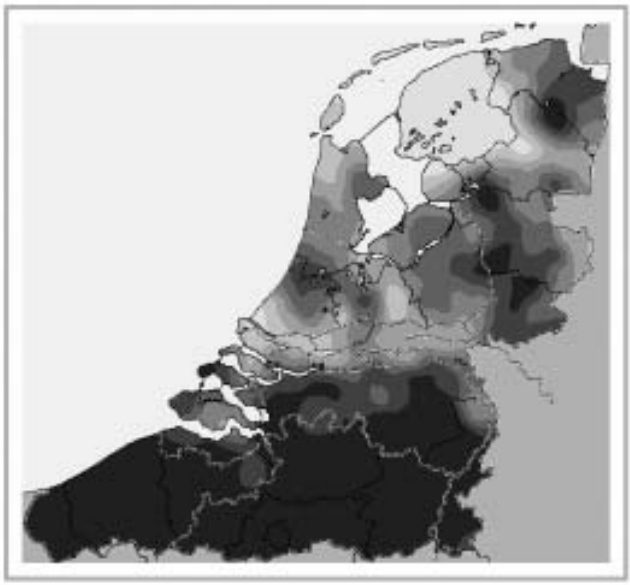
Do we have to distinguish two phonemes? In Flemish-Dutch definitely yes: the two phonemes /s/ and /z/ are well distinguished both word-initially and between vowels. The few exceptions, [z]*ap* and [z]*ikkel*, have simply changed category. The opposition is solid. It is only after a voiceless consonant that we almost invariably find [s] instead of [z]. VAN DE VELDE 1996 reports, however, that in Present-day Urban Flemish the distinction between /s/ and /z/ is tending to become slightly weaker.

In Dutch-Dutch the opposition is less solid. Not all speakers distinguish between /s/*ien* 'Sien' and /z/*ien* 'to see', and of those who do, some speakers do not always make the distinction. However, although there is much hesitation, the opposition is solid enough to allow the conclusion that /s/*abel* 'sabre' in the northern tip of Noord-Holland has exceptionally gone over to /z/*abel*. Between vowels, we find the same opposition as in Flemish-Dutch, but less well respected. Most people distinguish between *fl*/ɛs/*en* 'bottles' which after short vowel has always [s] and *bl*/a:z/*en* 'to blow' which after long vowel has [z] and sometimes [s]. A few exceptions to this rule are the short vowels in the loans *pu*/z/*el* 'puzzle' (from English) and *ma*/z/*el* '(good) luck' (from Jiddish), forming near-minimal pairs with *zu*/s/*en* 'sisters' and *pa*/s/*en* 'to fit'.

In the Present-day Urban Dutch from, for instance, Utrecht and Amsterdam speakers tend to go one step further than the speakers in our database and may pronounce both [s] and [z] in /z/*ien* and /s/*oep*. Z's, *s*'s and *c* as in *Sesam* (*street*), *zeezout* 'sea salt', *cent* 'cent' and *zend* 'sent' may be pronounced with either [s] or [z]. These speakers often do not seem to hear the difference between [s] and [z], much like Japanese speakers do not always hear the difference between /r/ and /l/. In our data we see the same in the dialects of the Rijnmond, the Betuwe and Utrecht Zuid, and Twente, where [s] is pronounced, see also maps 1 and 2. A further observation to be made is that lexical items with official /z/ differ in their degree of accepting [s]. There

seems to be an aspect of lexical diffusion here. Between vowels, Present-day Urban Dutch may have [s] also after a diphthong, as in *wij willen geen pau*[z]*e*/*pau*[s]*e(n)* 'we do not want a break/popes', see above. In the expression *doceren is doseren* 'to teach is to dose', two loan words, officially the first with [s], the second with [z], they do not distinguish these sounds, the expression consequently being ambiguous for them. For many of such speakers one phoneme /s/ would suffice, and there are few minimal pairs anyway. VAN DE VELDE 1996 shows that for speakers of Present-day Urban Dutch the distinction between /s/ and /z/ is becoming weaker. Remarkably, these speakers replace often [s] by [z] in more formal speech, showing that they are aware of the difference between the sounds, without having actually acquired them on the phonemic level. During the war in Yugoslavia, a lady newscaster used to speak of the [z]*ervische* [z]*ector* 'Serbian sector', words which in Standard Dutch officially have /s/ as in English.

Map 1. Distribution of [z] (dark) and [s] in Modern Dutch *hij ziet* 'he sees'. Production: E. Wattel cf. Wattel & van Reenen 1994.



Map 2. Distribution of [z] (dark) and [s] following silence in Modern Dutch *zeven* 'seven'. Production: E. Wattel cf. Wattel & van Reenen 1994.

In our Frisian data there are no minimal, or near minimal pairs, and the Frisian dialects in our database do not justify the distinction of /s/ and /z/. This would justify the conclusion that /s/ and /z/ represent one phoneme only. This conclusion is confirmed by the description of Frisian in VISSER 1997, in which no minimal, or near minimal pairs are found either. Therefore, it is remarkable that this study distinguishes the phonemes /s/ and /z/ since it does not provide any evidence for the distinction.

However, FOKKEMA 1971:122 mentions the existence of two, and not more than two, minimal pairs between vowels: *gêstje* [gɛ:sj@] 'to ferment' and *gêrzje* [gɛ:zj@] 'to become overgrown with grass', [bûs@] 'pocket' and [bûz@]- 'water devil'. Here the linguist has to make a choice: he may either consider the two cases as sufficient proof for the existence of the two phonemes, or he may consider them as exceptions, and conclude that there is only one phoneme /s/ in Frisian, which realizes as [z] after long vowel, with two exceptions, and as [s] elsewhere.

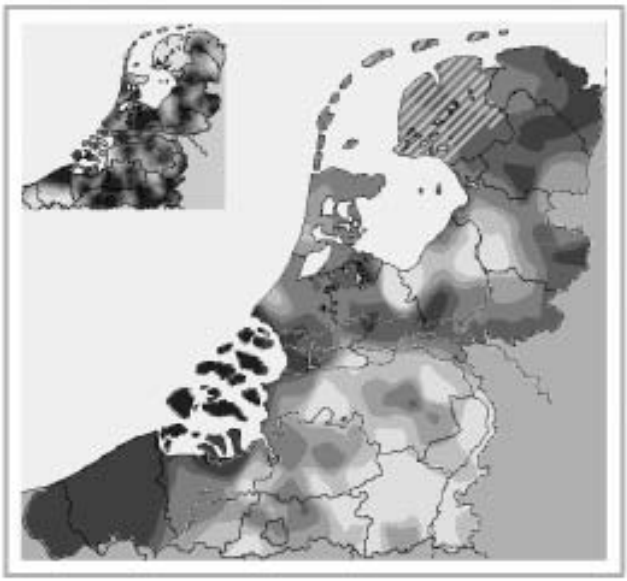
We conclude that /s/ and /z/ are well distinguished in Flemish-Dutch, whereas /s/ would suffice in Frisian. In the area between Flanders and Frisia there is much variation: for some dialects and/or speakers there are two phonemes to be distinguished, for others, especially in Present-day Urban Dutch, just one.

3.2 Middle Dutch

Our data for Middle Dutch come from the charter collection described in REENEN & MULDER 2003: more than 3,000 original charters, all from the 14th century. The provenance of all charters has been established on non-linguistic grounds. Consequently, it may be assumed that they represent the language of the place or the region of provenance. Although the Flemish part could do with the addition of more material, the corpus can be said to be representative for the entire Middle Dutch area. Although the charters come from the entire Dutch speaking area, they do not show the regular distribution of the Modern Dutch data, since in some areas no charters have survived, whereas in others not all have been collected. There are no charters available from Frisia. The charters have been transcribed, lemmatized and coded. Forms to be analysed are selected from these charters by means of UNIX/LINUX programs.

Since the data are lemmatized, relevant words can be culled from the corpus, and reduced to tables. The tables may form the input to the map program developped by E.Wattel, as mentioned above, see maps 3 and 4. We have selected a number of words from 2,773 charters in the corpus which in Modern Dutch are spelled with *z* or *s*, cf. table 2.

Map 3. The spellings *z* (dark) and *s* in Middle Dutch *zien* 'to see'. The small map shows the frequency distribution of the data: The darker the area, the better it is documented. Production: E. Wattel cf. Wattel & van Reenen 1994.



Map 4. The spellings *z* (dark) and *s* in Middle Dutch *duizend* 'thousand'. The small map shows the frequency distribution of the data: The darker the area, the better it is documented. Production: E. Wattel cf. Wattel & van Reenen 1994.

z	s	total	%z	item
30	7	37	81.0	Someren (place-name)
74	21	95	77.9	Zeger (boy's name)
272	249	521	52.2	zeven ‘seven’
2547	2366	4913	51.8	(be)zegel(en) ‘(to) seal’
162	165	327	49.5	zeker(heid) ‘certain(ty)’
360	537	897	40.1	zien ‘to see’
374	650	1024	36.5	zaak ‘case, thing’
242	459	701	34.5	zes ‘six’
37	80	117	31.6	zaterdag ‘Saturday’
1945	4590	6535	29.8	zoon ‘son’
1341	3869	5210	25.7	zullen ‘to shall, will’
123	374	492	25.0	zeggen ‘to say’
130	390	520	25.0	zelf ‘self’
164	771	935	17.5	zonder ‘without’
32	158	190	16.8	zondag ‘Sunday’
415	2325	2741	15.1	zijn ‘to be’
881	5072	5993	14.7	zijn ‘his’
443	3441	3884	11.4	zo ‘so, thus’ (adverb)
251	2809	3060	8.2	sint ‘saint’
5	381	386	1.3	som ‘sum’
2	368	370	0.5	Simon ‘Simon’
0	18	18	0.0	simpel ‘simple’

0	32	32	0.0	solemniteit ‘solemnity’
0	59	59	0.0	saluut ‘salute’

Table 2. Absolute and relative frequencies of word-initial *z* or *s* in a series of words from 14th-century Middle Dutch charters.

A first point to be observed is that the loanwords from Latin in table 2 have more *s* than native words. *Sint* ‘saint’ with 8.2% *z* may be a borderline case, on its way to becoming a native word. It has *z* almost as often as native *zo*. *(Be)zegel(en)* ‘(to) seal’ has *z* in 51.8% of the words. It also comes from Latin, but is no longer felt to be a loan. *Som* ‘sum’ with only 1.3%, *Simon* with 0.5%, *simpel*, *solemniteit* and *saluut*, never spelled with *z*, apparently have kept their Latin [s]-pronunciation. Modern Dutch spelling usually reflects this difference: All words with more than 10% *z* in Middle Dutch are spelled with *z* in Modern Dutch, all words with less than 10% *z* in Middle Dutch have *s* in Modern Dutch. The only exception is the placename *Someren*. Since the absolute number of forms is low and all the forms of this placename come from the place itself or the area around it, the relatively high number of *z*'s may not be representative. Perhaps the same holds with respect to the proper name *Zeger*. Relatively unstressed function words like *zonder*, *zijn*, *zo* also have relatively low percentages of *z*. This suggests that initial *z* is especially popular in words with a relatively prominent place in speech.

Regular as these results may be, words vary considerably with respect to the frequency with which they are spelled with *z*. *Zeven* has *z* relatively often, *zeggen* relatively seldom. Geographically speaking, the south-west (West Flanders) and Amsterdam area have usually the highest *z* score. There is, however, considerable variation per region as regards individual words: *zeven* and *zullen* often have *z* in Deventer, unlike other words. Groningen also varies per word: *zien*, *zoon*, *zaterdag*, *zondag*, *zijn* (verb) often have *z*; *zes*, *zullen*, *zonder*, *zo*, *zeggen* do not. We have seen already that *Someren* from the east of Noord-Brabant has the highest *z* score of all words examined. Other words in this area behave completely differently.

z	s	total	%z	item
38	213	251	15.1	Elisabeth (proper name)
85	713	798	10.7	lezen ‘to read’
38	388	426	8.9	Gijsebert (proper name)
482	6848	7330	6.6	deze ‘this’
111	1714	1825	6.1	duizend ‘thousand’

Table 3. Absolute and relative frequencies of the spellings *z* and *s* between vowels in a series of words from 14th-century Middle Dutch. Forms with *ss* (such as 4x *ss* in *Elisabeth*, 1x *ss* in *lezen*, and 468 x *ss* in *deze*) have not been included.

Between vowels the percentages of *z* in *Elisabeth*, *lezen*, *Gijsebert*, *deze* and *duizend* are lower than word-initially. In the latter position we usually have *s*, but we also find *z*, usually in the north, and in Holland/Utrecht, hardly or not at all in Flanders. Although geographically the patterns are far from uniform, the main findings are that *z* is a coastal feature, some words having more *z* than others. The geographical distributions of *z* and *s* on maps 3 and 4: *zien* and *duizend* are rather typical.

A question often asked is whether spellings from medieval documents can be used for phonological research. Several linguists in the past and even today still believe that medieval spelling variation and spelling conventions do not represent a phonetic record. FRANCK 1910:74, for instance, observes that *s* preceding a vowel in Middle Dutch is pronounced as [z], although it was spelled both *s* and *z*. Franck is apparently of the opinion that scribes could not spell very well, when he observes about *z*: "Diese Schreibung ... wird selten konsequent, sondern in willkürlichem Wechsel mit *s* angewandt." In more recent publications on older Dutch, we usually find an echo of this view. Franck and his colleagues apparently do not accept these spellings as evidence for the existence of /s/ and /z/. Our results above show, however, that although there is much variation the spelling distribution of *s* or *z* is far from random, just as we have seen with respect to the patterns of the maps of the modern data. In both cases we can conclude: (i) many words can be pronounced with both [s] and [z], and (ii) some areas have more [z] than others.

A second question to be asked is whether the letters *s* and *z* represent a phonetic record in the same sense as the narrow IPA-transcription of Modern Dutch dialects. For instance, if a word is written with *z*, will this *z* be influenced by a preceding voiceless sound and become *s*? Do we find a tendency to replace the letter *z* by *s* in *ik zie* ‘I see’ as opposed to *wij zien* ‘we see’? To answer this question we have examined the forms of the verb *zullen* ‘to will’

and the noun *zoon* ‘son’. The result is represented in table 4. Table 4 shows that there is no influence at all of a preceding voiceless sound in the case of *ik zal*. In the case of *zoon* there may be some marginal influence, but the effect is not significant. We conclude that scribes write the underlying form, i.e. phonologically, and that we will not need to take into account the nature of a preceding sound.

	z	s	total		z	s	total
zullen, zal, etc.	317	1006	1323	zoon, etc.	1092	506	1598
sullen, sal, etc.	856	2954	3810	soon, etc.	3433	1423	4856
	1173	3960	5133		4525	1929	6454
$X^2 = 1.2425$		$p = .30$		$X^2 = 3.1972$		$.05 < p < .10$	

Table 4. *Zullen* and *zoon* preceded by voiceless and voiced sounds.

A final question concerns the influence of unstressed prefixes such as *ge-* and *be-* in the past participle. Is there a tendency to spell more *z* in *bezegeld*, where the *z* occurs between vowels, than in *zegel*, where it occurs word-initially? The results of this investigation was that the forms with the prefix behave virtually the same as the forms without.

We conclude that the spellings *s* and *z* are used to reflect [s] and [z]. Yet they are not directly comparable to the modern phonetic transcriptions since they are not influenced by a preceding sound: *s* and *z* can be interpreted as the phonemes /s/ and /z/.

What can we conclude about the question whether Middle Dutch has one phoneme /s/ or two phonemes /s/ and /z/? The word-initial difference between loan words and native words shows that we have two phonemes. Native words tend to be pronounced with [z], especially in the west of Flanders and Holland, whereas loan words are pronounced with [s]. The south-east is less affected by the apparently new distribution of phonemic values than the coastal area. The tendency to write *z* is considerably weaker between vowels than word-initially. It shows up in the west and in the north, hardly ever in Flanders. Again, the south-east has in general /s/. We can certainly not conclude that the distribution of *z* and *s* is arbitrary, as claimed by Franck. It looks as if we are witnessing the development of a phonological opposition which did not exist in older Germanic.

3.3 Middle Dutch and Modern Dutch

We have shown that, although regional differences between dialects in the 14th century are usually not the same as in modern times, there is much

variation between [s] and [z] in the two periods examined, as is evident from a comparison of the Modern Dutch map 1 *hij ziet* with Middle Dutch map 3 *zien*. In Middle Dutch /z/ is mainly found in the coastal area, in Modern Dutch /z/ is especially strong in Flanders. Distribution patterns of /s/ and /z/ are also quite different word-initially and between vowels, as a comparison of all four maps shows.

In the 14th century we see the beginnings of a phonemic split: /z/ is in the process of being introduced: words like *zoon*, *zien*, *zegel* are systematically spelled with either *s* or *z*, especially in the western areas, whereas *som*, *simpel*, *Simon* are virtually always spelled with *s*. Especially the south-east appears to have kept /s/ in the 14th century. Whether the use of [s] and [z] had any social connotations, as in Modern Dutch, we do not know.

There is no doubt that Flemish distinguishes /s/ and /z/. In Dutch there is much hesitation, Present-day Urban Dutch showing that the distinction between /s/ and /z/ is in the process of merging. It could be that /z/ was introduced slowly and hesitantly in Middle Dutch, achieved its widest distribution in Modern Dutch dialects and is disappearing again in Present-day Urban Dutch. It is unclear whether the distinction between /s/ and /z/ may have peaked not in Modern Dutch, but in the centuries between the 14th and the 20th.

In Modern Frisian there is much to recommend the claim that there is only one phoneme /s/, as in Older Germanic. Although data from the 14th century are lacking for Frisian, the situation in medieval times is unlikely to have been different.

Finally, our analysis shows that phonemic oppositions can be weak during many centuries without either disappearing or becoming well established.

4. Conclusion

Several conclusions can be drawn from this study on problematic phoneme distinctions.

1. The opposition /ø/- /u/ in Old French has survived from Latin into Modern French, and was always well established in the central French speaking area. The opposition /s/-/z/ in Dutch was introduced some time before the 14th century, becoming well established only in Flanders. It probably never reached Frisian.
2. Regional differences are considerable, both in French and in Dutch: There are areas in which the phonemic opposition is solid and areas where it is almost non-existent.
3. Analysis of the Old French data shows that the problem in Old French is a pseudo-problem as a consequence of a generally accepted but invalid

assumption about how to interpret rhymes. Analysis of the Middle Dutch data shows that there are systematic spelling patterns that have long remained undetected, so that no connection was made between the variation between /s/ and /z/ in both Middle Dutch and Modern Dutch.
4. The analyses could not have been carried out without the availability of large corpora and UNIX/LINUX computer tools to search them. This study has demonstrated how these modern tools may detect patterns which have always been overlooked.

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The Lexicon-Grammar of French Verbs – A Syntactic Database –

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Introduction

The LADL (Laboratoire d'Automatique Documentaire et Linguistique)[2], headed by Maurice GROSS from 1968 to 2002, aimed to classify all grammatical word classes in French according to their syntactic properties, and the distributional constraints that could characterize the sentences in which they occur. At the outset, it was essentially a linguistic approach, with no intention to build a tool for computational applications. But the way in which the description was formalized allowed us to incorporate the data within a general system capable of tagging very large corpora, analyzing texts and producing a syntactic description of sentences.

Our electronic dictionary provides information about the grammatical category (part-of-speech of each item), its possible inflected forms, and, in the case of verbs, a code indicating which syntactic class(es) it belongs to (COURTOIS 1997). For example, an entry like the following:

afficher V6 + 6, 35R, 38LD

indicates that the verb *afficher* has a V6 type of conjugation (i.e. together with the associated inflected forms), and that it belongs to syntactic classes 6, 35R and 38LD.

I shall briefly describe how the classification of verbs has been organized, and what kind of information it contains.

1. Syntactic description

1.1 General problem

The Lexicon-Grammar is organized into a series of tables, each of them grouping items which share at least one main construction. This basic construction is considered as the "defining property" of the item. For example,

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[1] I would like to thank Antoinette Renouf for her help. The translations provided are as close as possible to the French examples, rendering some of them rather unnatural.

[2] The LADL belongs to the CNRS (French National Research Center). It is now part of the Institut Garpard Monge at the University of Marne-la-Vallée (http://infolingu.univ-mlv.fr).

the verb *comparer* [compare] has the construction: [3]

N_0 *V* N_1 *avec* N_2

where the preposition *avec* [with] can alternate with *et* [and]:

(1) *John a comparé Jane (avec + et) sa mère* [4]
John compared Jane (to + and) his mother

This construction has been considered characteristic of this verb, for a number of reasons, the main one being that it contains all the "arguments" that the meaning of the verb implies, the second one being that other verbs have the same characteristics -- like *marier* [marry], for example:

(2) *Le prêtre a marié John (avec + et) Jane*
The priest married John (to + and) Jane

This group of verbs constitutes a "natural class" of 129 "symmetrical" transitive verbs which are classified in the same "table" (Table 36S, see Figure 1 below).

1.2 Properties

Constructions (1) and (2) are obviously not the only ones for these verbs. For example, we can have a construction $[N_1$ *et* $N_2]_{No}$ *se V*:

(3) *John et Jane se marient*
John and Jane get married (Lit. John and Jane marry each other)

where the two complements of (2) are in subject position (the verb is in pronominal form in this case). In each table of this type, various properties are encoded (in column) to indicate what other constructions are possible (Figure 1).

On the other hand, a verb like *permuter* [switch], which is in the same class because we can have sentences like:

John a permuté la bouteille (avec + et) le verre
John switched the bottle (with + and) the glass

---

[3] N_0 is always the subject and N_1, N_2, etc. the complements, prepositional or not.

[4] "+", in parenthesis, means that there is a choice.

doesn't accept a pronominal construction of type (3), $[N_1\ et\ N_2]_{No}$ *se V*:

(4) **La bouteille et le verre se sont permutés*
The bottle and the glass switched each other

N0						N1										Prép		N2										
N1 V avec N2	N1 se V avec N2	N1 et N2 se V	N2 V N1	**Table 36S**	Pfx nég/source	N1 =: Nhum	N1 =: Npc	N0 lui V N1pc	N0 V N1 Loc N1pc	N0 V N0pc	N1 =: -Nhum	N1 =: le fait Qu P	N1 =: V-n	Ppv =: le	N0 V N1	Prép =: à	Prép =: de	N2 =: Nhum	N2 =: N-hum	N2 =: le fait Qu P	N2 =: V-n	Ppv =: lui	Ppv =: y	Ppv =: en	N0 est Vpp (Prép N1)	N1 et N2 font un V-n	V-n entre N1 et N2
-	+	+	-	maquer	+	+	-	-	-	-	-	-	-	+	+	+	-	+	-	-	-	-	-	-	+	-	-
-	+	+	+	**marier**	+	+	-	-	-	-	-	-	+	+	+	+	-	+	-	-	+	-	-	-	+	+	+
-	+	+	-	marier	+	-	-	-	-	-	+	-	-	+	+	+	-	-	+	-	-	-	-	-	-	-	+
-	+	+	-	mélanger	-	+	+	+	-	+	+	+	-	+	-	+	-	+	+	+	-	+	-	-	+	+	-
-	+	+	-	mêler	+	+	+	-	-	-	+	+	-	+	-	+	-	+	+	+	-	+	-	-	+	-	-
-	+	+	-	mésallier	-	+	-	-	-	-	-	-	-	+	+	+	-	+	-	-	-	-	-	-	+	+	+
-	+	+	-	mixer	-	-	-	-	-	-	+	-	-	+	-	-	-	-	+	-	-	-	-	-	+	-	-
-	+	+	-	opposer	-	+	+	-	-	-	+	+	-	+	-	+	-	+	+	+	-	+	-	-	+	+	-
+	-	-	-	**permuter**	-	+	+	+	-	-	+	+	-	+	+	-	-	+	+	+	-	-	-	-	+	+	-
-	+	+	-	rabibocher	-	+	-	-	-	-	-	-	-	+	-	-	-	+	-	-	-	-	-	-	+	-	-

Figure 1: Extract of Table 36S

Instead, we would say (structure $[N_1\ et\ N_2]_{No}\ V$):

(5) *La bouteille et le verre ont permuté*
The bottle and the glass switched (are switched)

which construction is not possible for *marier* [marry]:

**John et Jane ont marié*
John and Jane married

Not all the verbs accepting sentences of type (5) are classified the same way. See for example:

(6) *John et Jane flirtent*
John and Jane flirt

The structure is the same as (5):

$N_0\ V$ (with N_0 = *Na et Nb*)

but there is no transitive construction as in (2) which (6) can relate to:

**Quelqu'un a flirté John (avec + et) Jane*
Somebody flirted John (with + and) Jane

So *flirter* cannot be in class 36S. On the other hand, (6) can be associated to (7):

(7) *John flirte avec Jane*
John flirts with Jane

Constructions (6) and (7) define Table 35 S (134 intransitive "symmetrical" verbs, see Figure 2).

N0 =: Num									Prép					N1										
N0 =: Nhum	N0 =: N-hum	N0 =: Nnr	N0 =: Vinf W	N0 =: V-n	**Table 35S**	N0 V	N0 est V-ant	N0 est Vpp	Prép =: dans	Prép =: à	Prép =: contre	Prép =: avec	Prép =: de + d'avec	N1 =: Nhum	N1 =: N-hum	N1 =: le fait Qu P	Ppv =: lui + y	Ppv =: en	Nhum V sur ce point	N0 est V-ant Prép N1	il y a V-n entre N0 et N1	N0 est en V-n avec N1	il V N0 W
+	-	-	-	-	fauter	+	-	-	-	-	-	+	-	+	-	-	-	-	-	-	-	-	-	+
+	-	-	-	-	ferrailler	+	-	-	-	-	+	+	-	+	+	-	-	-	-	-	-	-	-	+
+	-	-	-	-	**flirter**	+	-	-	-	-	-	+	-	+	+	-	-	-	-	-	+	-	-	+
+	-	-	-	-	forniquer	+	-	-	-	-	-	+	-	+	-	-	-	-	-	-	+	-	-	+
+	-	-	-	+	fraterniser	-	-	-	-	-	-	+	-	+	-	-	-	-	+	-	+	-	-	+
+	-	-	-	-	frayer	-	-	-	-	-	-	+	-	+	-	-	-	-	-	-	+	-	-	+

Figure 2: Extract from Table 35S

1.3 "Defining" properties

As I said, the primary use of each given verb is defined by a main construction which defines all the verbs which have the same behaviour at the first level. The properties that are involved in the definition of tables are of three types: syntactic, distributional and semantic (LECLÈRE 2002).

As far as simple verbs are concerned, we distinguish 60 different classes of verbs (i.e. 60 tables) (M. GROSS 1975, J-P. BOONS, A. GUILLET & C. LECLÈRE 1976a, 1976b, A. GUILLET & C. LECLÈRE 1992). We have first taken into account the formal structure of the sentences in which each verb

can occur. There are six of them:

N_0 *V*
N_0 *V* N_1
N_0 *V Prép* N_1
N_0 *V* N_1 *Prép* N_2
N_0 *V Prép* N_1 *Prép* N_2
N_0 *V* N_1 *Prép* N_2 *Prép* N_3

It's important to say here that, in all these analyses, the adverbial phrases are not taken into account, because they are not considered as characteristic of the verb. Everyone will agree that the adverbial phrases of place or time in:

John a flirté avec Jane (dans le jardin + ce matin)
John flirted with Jane (in the garden + this morning)

are not arguments that characterize the verb *flirt*. Nevertheless, at least some types of locative complements, considered as adverbial in traditional grammars, have been retained in the description of several classes of verbs. See:

John a mis sa voiture dans le jardin
John put his car in the garden
John a enlevé sa voiture du jardin
John removed his car from the garden

Although they can take the same form as for other verbs, these locative complements do not have the same syntactic role when they are used with verbs like *mettre* [put] or *enlever* [remove]. There are dozens of verbs like these, for which these complements have to be considered as crucial arguments and not as adverbials.

Each of the *N* positions in the sentence structures above can be occupied by a noun or a sentence (noted *Qu P* [That S]). For example, the structure N_0 *V* N_1 corresponds to three constructions:

(1)	N_0 *V* N_1	Tables 32
(2)	*Qu P* *V* N_1	Table 4
(3)	N_0 *V* *Qu P*	Table 6

which can be illustrated respectively by the following three examples:

(1) *John a abimé le livre* [John damaged the book]
(2) *Que Jane vienne amuse John* [That Jane comes amuses John]
(3) *John pense que Jane est folle* [John thinks that Jane is crazy]

The presence of a *Qu P* [that S] complement in the construction is one determining factor in the choice of the class which the verb belongs to and thus of the table in which it appears. The verb *confier* [confide, entrust], for example, as in:

(4) *Paul confie son problème à Marie*
Paul entrusts his problem to Mary
(5) *Paul confie à Marie qu'il doit partir*
Paul confides to Mary that he must go

will be classified as a sentence complement verb (Table 9) because of (5). Sentence (4) is considered as being derived from (5) (*That he must go is his problem*), and inventoried as such in Table 9. In contrast, the sentence:

(6) *Luc confie sa valise à Max*
Luc entrusts his suitcase to Max

cannot be derived from a sentence complement, and so it appears in a table for constructions with nominal complements (Table 36DT in this case). It's interesting to note that, in many cases, the uses we distinguish between have different translations, but not always in the same constructions, as here for *confide* and *entrust*.

Such a purely formal classification, though useful at a first level, appears to be too coarse. To obtain more homogeneous classes of verbs, we need to associate the syntactic definitions with distributional properties; that is to say: specify what kind of preposition is possible, which features are attached to the different possible nouns in subject and complement positions, and so on. For example, *obéir* [obey] and *changer* [change] have the same construction N_0 *V Prép* N_1, but different prepositions, corresponding to two different tables:

N_0 *V à* N_1 Table 33 *John obéit à Jane* [John obeys Jane]
N_0 *V de* N_1 Table 35R *John a changé de voiture* [John changed his car]

Other properties can be used to separate the uses of verbs more precisely, so that the final classes we obtain appear to be more or less homogeneous

(when they are, we speak of "natural classes"). For example, the feature "obligatorily plural" attached to the direct complement of a few verbs (147 verbs) of structure N_0 *V* N_1 leads us to put in the same table (32PL) verbs of which the meaning is roughly "gather things or people": *centraliser* [centralize], *collectionner* [collect], *rallier* [rally], *rassembler* [gather], etc.

Several properties can of course be combined to define a class. This is the case in Table 4, for example, where one can find a class of "psychological verbs" (*amuser* [amuse], *étonner* [surprise], *effrayer* [frighten], etc.

Structure: N_0 *V* N_1
Properties: N_0 =: *Qu P*
N_1 =: *NhumObl* (*N* "human" only)
Que John vienne (amuse + surprend + effraye) Jane
That John comes (amuses + surprises + frightens) Jane

Note that the two properties do not have the same status: the direct object is obligatorily a human, but the subject can be a noun as well as a completive (*That John comes amuses Jane, John amuses Jane*).

One requirement in the selection of such properties is that they can be formally defined, from a linguistic point of view. Many of the features that we have chosen as properties are easily recognized because they are formally marked (like "obligatorily plural", which is generally marked by "s" or "x" in French). For others, it is necessary to use classification tests. For instance, a noun has the property *Nhum* (human) when it answers the question "*qui* ?" [who ?]: *Who is amused by John's coming* ?

In certain cases, only semantic properties can be used. The condition in this case (as in others, in fact), is that the intuition is "reproducible", whoever the native speaker is:

> "Consensus among specialists is reached through experiments, but facts and experiments must be *reproductible*." (M. Gross 2002:58)

For example, among those verbs with the construction N_0 *V* N_1 , we have defined a sub-class, on the basis that the verb means "transformer en *V-n*" [transform into *V-n*][5]. One can find, in this table, 131 verbs like *caraméliser* [caramelize], *gazéifier* [gasify] or *pronominaliser* [pronominalize]:

John a caramélisé ce morceau de sucre

[5] *V-n* stands for any noun which is morphologically related to the verb (*caramel / caramelize*).

John caramelized this piece of sugar = transformed it into caramel
On peut pronominaliser ce complément
One can pronominalize this complement = transform it into a pronoun

1.4 General processes of classification

To summarize, one can imagine a giant "super-table" which could take the form of Figure 3. The lines correspond to verbal entries (about 15,000 in French for simple verbs), and the different properties are in columns (about 300 of them have been tested). This super-table does not actually exist, but it represents what our work of classification has involved over several years. Theoretically, it represents 4,500,000 types of sentence. In fact, not all of them are studied for a given verb: to take a simple example, for an intransitive verb, it's clearly unnecessary to test all the properties of direct objects. Moreover, certain properties have been selected because of their relevance to a particular class of verbs but hold no significance within other classes. Of course all the defining constructions have to be tested for each verb, before the table it belongs to can be decided.

	Defining properties			Other properties			
	>>>>>>>>>Priority>>>>>>>>>						
	P1 (defining Table 1)	P2 (defining Table 2)	P3 (defining Table 3)	P4	P5	P6	P7
Verb 1	+	-	+				
Verb 2	-	-	+				
Verb 3	-	+	-				
Verb 4	+	+	-				

etc.

Figure 3: Theoretical general table

Let us consider only these defining properties[6]. They cannot be chosen so that they define separate classes of verbs. A verb generally has more than one defining property. To take a simple example: a verb often 'accepts' that one or

[6] It should be noticed that I often use "property" and "construction" (or "sentence") interchangeably. The reason is that each property corresponds to a sentence. We consider that every feature has to be studied in context, the sentence being the minimal significant unit.

the other of its complements are deleted (*Paul mange un sandwich/ Paul mange* [Paul eats a sandwich/Paul eats]). We consider here that this is the same verb *manger* (there are many other verbs *manger*, with other meanings); so we do not create two entries, one in a table defined by N_0 *V* N_1 and the other in a table defined by N_0 *V*. Instead, we give priority to the longer construction, because it is the one containing more information about the arguments of the verb. The fact that N_0 *V*, which can be a defining property for other verbs, is possible, will be regarded here as a simple property, and encoded in a column in the table defined by N_0 *V* N_1 (Table 38L0 in this case)[7]. In the schematic case of Figure 3, property 1 (P1) has been given priority over P2, which has priority over P3. So the verb V1 will be classified within Table 1, in which property 3 will be noted in column, as P2 will be noted for V4, which is classified within Table 1 as well. On the other hand, V2, which has P3 but no other property, will be classified within Table 3. The consequence is that, if somebody is interested in a given property and wants to know the list of all the verbs which have it, s/he may have to look in different places. For example, the verbs which accept P3 are:

- all the verbs of Table 3 by definition (V2 here)
- all the verbs for which P3 is encoded "+" in other tables (V1 here, in Table 1).

1.5 Splitting entries

While it appears that the verb *flirter* of (6), in §1.2, is the same as the one of (7), this is not always the case. A morphological verb has as many entries as it has uses that have been judged to be distinct. The distinction between two entries for the same verb, based on intuition at the beginning, has to be underpinned by appropriate properties. That becomes obvious when the different meanings correspond to different constructions. Take for example the verb *réaliser*: among its several meanings, it is easy to distinguish between one which can take a completive as direct object and another for which this is impossible:

John a réalisé que Jane était partie (Table 6)
John realized that Jane was gone (had gone)
John a réalisé une œuvre d'art (Table 32A)
John realized / created a masterpiece

[7] This would not be the same for the verb *boire* [drink]: the sub-structure *John boit* [John drinks] having a special meaning ("John is an alcoholic"), it deserves an entry in a table defined by the structure N_0 *V*.

But sometimes, two meanings (or more) can correspond to the same primary construction. In this case, we create two entries (or more) in the same table. Other properties encoded in this table allow us to justify the distinction. Look, for example, at the verb *communiquer* [communicate]. It has two entries in Table 35S (the same table of intransitive symmetrical verbs as *flirter* [flirt] above) (see Figure 4)):

La chambre communique avec la cuisine / La chambre et la cuisine communiquent
The bedroom communicates with the kitchen / The bedroom and the kitchen communicate

John communique avec Jane / John et Jane communiquent (par e-mail)
John communicates with Jane / John and Jane communicate (by e-mail)

N0 =: Num									Prép					N1										
N0 =: Nhum	N0 =: N-hum	N0 =: Nnr	N0 =: Vinf W	N0 =: V-n	**Table 35S**	N0 V	N0 est V-ant	N0 est Vpp	Prép =: dans	Prép =: à	Prép =: contre	Prép =: avec	Prép =: de + d'avec	N1 =: Nhum	N1 =: N-hum	N1 =: le fait Qu P	Ppv =: lui + y	Ppv =: en	Nhum V sur ce point	N0 est V-ant Prép N1	il y a V-n entre N0 et N1	N0 est en V-n avec N1	il V N0 W
-	+	-	-	-	communiquer	-	+	-	-	-	-	+	-	-	+	-	-	-	-	+	+	+	+	+
+	-	-	-	-	communiquer	+	-	-	-	-	-	+	-	+	-	-	-	-	+	-	+	+	+	+

Figure 4: Entries of *communiquer* [communicate] in Table 35S

Apart from the feature "human", attached to both subject and object in one case, and impermissible in the other one:

* *John communique avec la cuisine*
John communicates with the kitchen
* *La chambre communique avec Jane*
The bedroom communicates with Jane[8]

[8] These sentences are possible if there is metonymy (*room / people in the room*). This is a question I cannot discuss here, but it is obvious that the processes consisting in asking systematic questions about such features as "human" and "non human" about every argument of the verbs is a fruitful way to investigate a lot of problems of this type and provides good examples to illustrate them.

two other properties, N_0 *est V-ant* and N_0 *est V-ant Prép* N_1 confirm the difference:

La chambre est communicante (avec la cuisine)
The bedroom is communicating with the kitchen
* *John est communicant (avec Jane)*
John is communicating with Jane

At the present stage of classification, about 5,000 morphological French verbs yield about 15,000 different entries in 60 tables; that is to say, an average of 3 entries per verb; but of course a lot of them have only one entry, and some polysemic verbs may yield as many as 30 entries.

In conclusion, each entry for a verb in a table supposes that:

- the verb can be used in the defining construction of the table;
- it cannot be used with the same meaning in any more complex defining construction with higher priority;
- the construction in question is not a derived sentence. If this is the case, it is the source sentence that must be considered.

2. Support verbs and compound verbs

So far, I have considered only simple verbs. While carrying out our classification, we found that the description of the sentence predicate frequently required us to take into account not only the verb itself, but a combination of the verb and one or more nouns.

2.1 Support verbs

Let us consider the following examples:

(1) *John projette de partir*
John plans to leave
(2) *John [a le projet] de partir*
Lit. John [has the plan] to leave

It is clear that in (2), the predicative role is taken by the noun *projet* [plan] and not by the verb *avoir* [have]. It is the noun that decides the distribution of subjects and complements, in the same way as does the simple verb *projeter* [plan] in (1). The verb *avoir* [have] is only what we call a "verbe support" [support verb] (*Vsup*) of the predicative noun (*Npréd*). Such combinations [*Vsup Npréd*] are very numerous. Some of them correspond to a verb as in (1)-(2), but, in most cases, there is no equivalent simple verb (at least in

French). See for example:

John [fait un signe] à Jane
John [makes a sign] to Jane
John [donne un rendez-vous] à Jane
John [gives a rendez vous] to Jane

Hundreds of such combinations have been itemized and have entries in special tables (see for example J. GIRY SCHNEIDER 1978, G. GROSS 1989 and R. VIVÈS 1983), organized in the same way as the tables of simple verbs (except that the entries are nouns).

2.2 Compound verbs and "frozen" sequences

An other case where it is necessary to consider compound predicates is where a verb is associated with one or more nouns, so that it is impossible to deduce the meaning of the expression from the meaning of the words of which it is composed. See the following sentences:

(3) *John [brûle les planches]*
John gives a spirited performance (Lit. John burns the boards)
(4) *[Le rideau est tombé] sur cette affaire*
The curtain came down on this affair

The simple verbs *brûler* [burn] and *tomber* [fall] do exist as entries in tables of simple verbs (Table 32C and 35L respectively). These tables, of course, can only describe the proper meaning of (3) and (4). But we have here specialised uses of these verbs: in (3), nothing is really burnt, and in (4), there is no curtain. The sentences are not comprehensible if you only know the meaning of *brûler* [burn], *planches* [boards], *rideau* [curtain] and *tomber* [fall]. The only way to describe such idiomatic cases is to take $V\ N_1$ [*brûler les planches*] and $N_0\ V$ [*le rideau tombe*] as complex units. We then create entries in tables of "frozen expressions". Other constraints can be observed in these complex units, in particular syntactic ones, as for example in:

John garde / perd son sang-froid
John keeps / loses his head (Lit. keeps / loses his cold blood)

where the determiner of *sang-froid* can only be a possessive (co-referential with the subject).

There are thousands of so called "frozen" combinations of this type, which do not obey the normal rules of simple verbs and deserve special treatment.

The electronic dictionary of LADL also contains other compound words, such as compound nouns like *perte de temps* [loss of time] or adverbs like *à toute vitesse* [in a hurry].

2.3 Example of the processes in the classification of a verb

We show here (Figure 5), with the verb *afficher*, an example of the way we have created entries for a verb and put them into appropriate tables, according to its syntactic and distributional properties. The sentences I give here are only examples of some of the sentences one can find in the tables (*C* = constraint noun, *Loc* = locative preposition and *V-n* = noun morphologically linked to the verb).

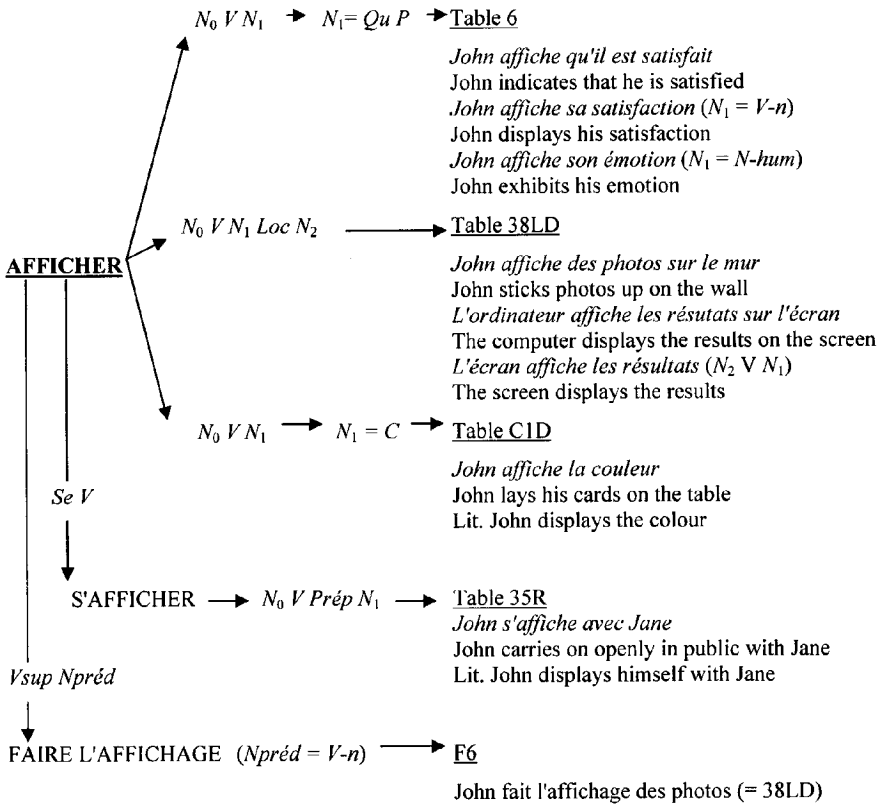


Figure 5: Part of the classification of the verb *afficher*

2.4 Results

Our **electronic dictionaries** make up what we call the 'DELA' system[9]. It contains:

- a dictionary of about 90,000 simple words (DELAS);
- a dictionary of corresponding phonetic forms (DELAP);
- a dictionary of more than 100,000 compound words (DELAC).

The inflected forms of simple words are automatically generated to produce the 'DELAF' dictionary.

In our **Lexicon-Grammar**, as far as verbs are concerned, we have entries in tables for:

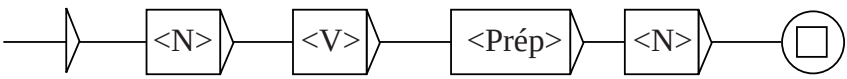
- about 15,000 "free" constructions with simple verbs;
- about 25,000 "frozen" constructions with compound verbs;
- about 50,000 constructions with support verbs and predicative nouns.

As I said, with each verbal entry in our dictionary is associated the code(s) of the table(s) in which it is classified. This allows us to associate each verbal entry with all the main types of sentence in which it is likely to appear in texts.

3. Local grammars and graphs

The third part of our system consists of a series of "local grammars" which are formalized as FST (finite state automata). They have been created to describe sets of sentences which are used in a specific domain: expressions of dates, of temperature, stock exchange market reports (see T. NAKAMURA in this volume). I shall not describe these automata here. The interesting point here is that the Lexicon-Grammar, or at least part of it, can be converted into such graphs, and so applied to texts (see E. ROCHE 1999, S. PAUMIER 2001).

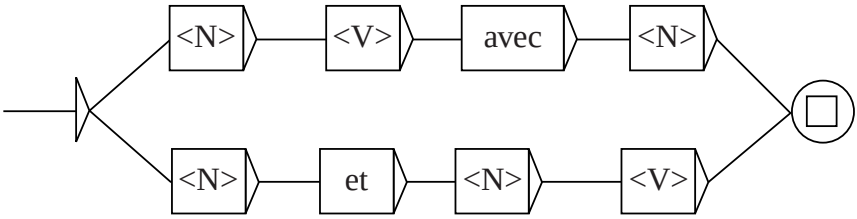
Schematically, a simple intransitive sentence can have the form:



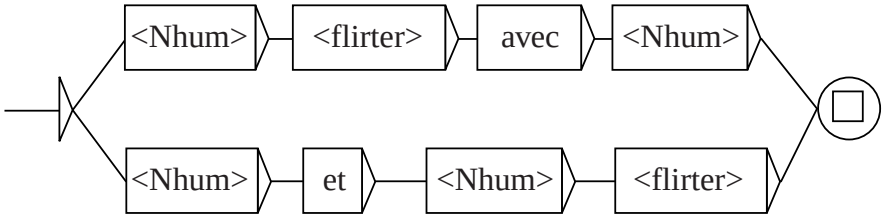
(<N>, <V> and <Prép> stand for any noun, verb or preposition respectively).

[9] DELA stands for 'Dictionnaire Electronique du LADL' [Electronic Dictionary of LADL].

The defining property of Table 35S, for example, corresponds to a more precise graph:



The properties encoded in Table 35S, corresponding to different constructions, can be converted into as many paths in the graph as there are "+" signs in the line of a given entry (the paths corresponding to "-" are of course eliminated). The verb *flirter* [flirt], for example, has the properties N_0 = *Nhum* and N_1 = *Nhum*. It can be associated with the following graph:



(<flirter> stands for all the inflected forms of the verb)

In theory, all the possible sentences described in the tables can be represented by graphs of this type. So, with each verb of the dictionary, or, more precisely, with each pair [V + code of table], we can associate a complex graph representing all the sentences we have retained as characteristic of the corresponding use of this verb. These graphs can be applied to a tagged corpus, but of course a lot of problems have not yet been solved:

- in practice, many properties (semantic, for example) cannot be exploited computationally;
- many derived constructions (imperative, for example) are not represented in tables;
- adverbial phrases, as well as various kinds of sequence which can be inserted at several places in sentences, are not taken into account (some of them have already been studied; see, for example, FAIRON 2000)
- etc.

Conclusion

The systematic description of verbs (and other items) in syntactic tables is valuable, from a linguistic point of vew, in raising many questions which have never been examined. The final result constitutes a very large formalized database which is an invaluable set of information for researchers.

As for the computational applications, interesting results have already been obtained: dictionaries and various types of graph have already been incorporated into platforms like INTEX (M. SILBERZTEIN 1993, 1994) and UNITEX (S. PAUMIER 2002) for tagging and parsing very large corpora. The computational application of all the information contained in the lexicon-grammar raises some problems which are now being studied: it opens a lot of interesting avenues of research in the automatic treatment of texts, information retrieval, and even automatic translation (many other languages like English (M. SALKOFF 1995), Italian, Spanish, Korean are being described according to the same theoretical principles as for French).

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A Formal Analysis of Spanish Adjective Position[1]

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1. Introduction

In Spanish there are various words which agree with nouns in number and, if necessary, in gender, adding a lexical meaning to them: *mi* 'my', *mío* 'mine', *este* 'this', *doscientos* 'two hundred',... *claro* 'clear', *difícil* 'difficult', *fuerte* 'strong', *importante* 'important',... Among these are words in the closed class like possessives, demonstratives, quantifiers, etc., and they generally have their fixed position in relation to the noun:

(1) a. mi libro
'my book'
b. *libro mi
c. una amiga mía
'a girlfriend of mine'
d. *mía amiga

On the other hand, there are words in the open class like *claro* 'clear' and *difícil* 'difficult' which generally have much more flexible position:

(2) a. una clara idea
'a clear idea'
b. una idea clara
'a clear idea'
c. la difícil situación'
'the difficult situation'
d. la situación difícil
'the difficult situation'

In this paper, we call the words in the latter group adjectives, excluding the former (possessives, demonstratives, quantifiers, etc.) and we will consider some rules of the position of adjectives in noun phrases.

A great proportion of conventional explanations about Spanish adjective position has been from a syntactic, semantic, and/or pragmatic point of view[2]. For example, adjectives are classified semantically and/or syntacti-

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[1] This paper is, in principle, an English version based on the data of Miyamoto(1997a). I wish to thank Junichi Murata for his comments and suggestions on this paper.

cally, and their position is explained. Classifying adjectives are post-posed like (3a), and most qualitative adjectives can take both postposition as (3c) and pre-posed position as (3d):

(3) a. la amiga madrileña
'the girlfriend of Madrid'
b. *la madrileña amiga
c. la amiga joven
'the young girlfriend'
d. la joven amiga
'the young girlfriend'

Although nonrestrictive adjectives take pre-posed position (4a) and post-position (4b), the adjectives of restrictive use are post-posed (4c). Moreover, there are some adjectives which have different meanings with their position like *pobre*. It is said that a pre-posed adjective (5a) expresses a subjective and figurative meaning, and a post-posed one (5b) expresses an objective and original meaning:

(4) a. su pequeña buhardilla
'his small attic'
b. sus ojos pequeños
'his small eyes'
c. un país pequeño
'a small country'

(5) a. una pobre mujer
'a poor woman (= unfortunate)'
b. una mujer pobre
'a poor woman (= not rich)'

Apart from such a syntactic, semantic, and/or pragmatic point of view, there is a formal analysis which pays attention to the length[3] of the noun and the adjective, although it is in the extreme minority. One of the pioneers in this respect is Salvá(1830:12.4.2), who describes that when the noun has one syllable and the adjective has three or more syllables, the adjective follows the noun (6a), even if the adjective shows essential character of the noun. And, he adds that, however, when a definite article is attached and the adjec-

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[2] For example, Bosque (1993), Bosque (1996), Bosque and Picallo (1996), Demonte (1982), Demonte (1999) are recent and very interesting studies from this viewpoint.

[3] As a unit which measures the length of a word, the number of characters, the number of phonemes, the number of morphemes, etc. can be considered in addition to the number of syllables. Refer to Grotjahn & Altmann (1993) in this respect.

tive has three or less syllables, it can also be pre-posed (6b)[4]. Later, Fernández Ramírez (1951: 84) selects from fragments of 13 works the constructions of *con + un(a) + {NA / AN}* and *con + {NA / AN}* in a literary style which describe people's talk, voice, act or gesture, and he indicates that a long constituent is clearly placed back in the construction of *con + un(a) + {NA / AN}* and that the tendency to post-pose a long constituent can be found also in the more literary and affected construction of *con + {NA / AN}*[5]. In fact, according to Miyamoto (1995: 66), (7a) and (7b) are more natural in Spanish than (7a') and (7b'):

(6) a. el sol resplandesciente
'the gleaming sun'
b. la dorada luz del sol
'the golden light of the sun'

(7) a. con una ternura infinita
'with infinite tenderness'
a' con una infinita ternura
'with infinite tenderness'
b. con una sonrisa inocente
'with an innocent smile'
b' con una inocente sonrisa
'with an innocent smile'

In this paper, we will try to clarify some rules of the adjective position from a formal viewpoint, i.e., the number of syllables of the adjective and/or the noun[6].

2. Procedure for creating the text for analysis

The KWIC data text which contains "adjective + noun" phrases and "noun + adjective" phrases for analysis was created in the following procedures:

(8) a. We make an adjective list and a noun list by taking out adjectives and nouns from the dictionaries of Shogakukan(1990), Hakusuisha(1990), Kenkyusha (1993), Academia (1995), Vox (1992), and Arco/Libros (1994)[7].
b. We make an adjective data list and a noun data list in which

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[4] It must not be disregarded that this is a phrase of "noun + of + (article) + noun".

[5] A slight reference to the length of an adjective and its position is also found in Szadziuk (1994: 83), De Bruyne & Pountain (1995: 106) and Demonte (1999: 201).

[6] In Miyamoto (1997a) the accent position of the adjective and/or the noun is also considered as an element determining adjectives position.

the data of "(part-of-speech sign: number of syllables: accent position from the end of a word)" is attached to each word.[8]

The adjective data list is as follows:

abacial (a:3:1)
abaciales (a:4:2)
... omitted...
zuro (a:2:2)
zuros (a:2:2)
zura (a:2:2)
zuras (a:2:2)

c. We make a file of 5,000 lines taken out at random from the file of *ABC Cultural 1991-1995*[9], which is our object text for analysis.
d. We make a "text with data" by attaching the data of the adjective data list and the noun data list to the text of 5,000 lines.
e. We extract[10] the portion of "adjective + noun" combinations or "noun + adjective" combinations from the "text with data" with five words each on its left and right sides to make a "partial text" in KWIC form.
f. We check the "partial text" of KWIC form manually, and complete the "KWIC data text" containing "adjective + noun" or "noun + adjective" found in the text of 5,000 lines.

3. Analysis of the KWIC data text and its results

The following analysis was performed on the KWIC data text created in the above procedure. First, the number of syllables of each component of "adjective + noun" is compared with that of "noun + adjective". Next, its total frequency (token frequency) is counted, and then the minimum, maximum, and average values of the number of syllables of adjectives and nouns

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[7] Each list including a female form and a plural form is made, based on the list of 9,547 adjectives and that of 24,165 nouns. In the case of the adjectives, quantifiers, possessives, demonstratives, indefinite words and negative words are deleted.

[8] I have written all processing scripts used in this paper by AWK or Perl. Refer to Miyamoto (1997b: 337-339), for example, about a syllabication script of Spanish words, and to a retrieval script *y30104* of Appendix used in (8e). See also Aho, Kernighan and Weinberger (1988), Wall, Christiansen and Schwartz (1996), etc. about AWK and Perl.

[9] The *ABC Cultural* is a collection of cultural columns of *ABC*, one of the most important dailies in Spain. The number of the text lines of *ABC Cultural* is 284,170.

[10] See in Appendix the retrieval script *y30104* by AWK used for this processing.

are calculated:

Table 1

	short + long	same	long + short	Total
noun + adjective	2,336 52.14%	1,227 27.39%	917 20.47%	4,480 62.92%

number of syllables	minimum	average	maximum
Noun	1	3.03	7
Adjective	1	3.57	8

	short + long	same	long + short	Total
adjective + noun	1,129 42.77%	805 30.50%	706 26.74%	2,640 37.08%

number of syllables	minimum	average	maximum
Adjective	1	2.86	6
Noun	1	3.16	8

noun + adjective adjective + noun	short + long	same	long + short	Total
Subtotal	3,465 48.67%	2,032 28.54%	1,623 22.79%	7,120

In order to compare with Table 1, whose target language is newspaper Spanish, the following Table 2 for the spoken Spanish in Madrid is mentioned here from Miyamoto (1993:37):

Table 2

	short + long	same	long + short	Total
noun + adjective	844 52.5%	406 25.3%	357 22.2%	1,607 78.9%

number of syllables	minimum	average	maximum
Noun	...	2.82	...
Adjective	...	3.40	...

	short + long	same	long + short	Total
adjective + noun	246 57.3%	104 24.2%	79 18.4%	429 21.2%

number of syllables	minimum	average	maximum
Adjective	...	2.33	...
Noun	...	2.98	...

noun + adjective adjective + noun	short + long	Same	long + short	Total
Subtotal	1,090 53.5%	510 25.0%	436 21.4%	2,036

By comparing Table 1 with Table 2, the following points seem clear:
In the Spanish noun phrases which consist of a noun and an adjective,

(9) a. The adjective is overwhelmingly post-posed.
b. The word order of "short + long" components is most frequently used[11].

On the other hand, the following differences are found between spoken Spanish and written Spanish:

(10)a. The percentage of pre-posed adjectives is higher in written Spanish than in spoken Spanish.
b. In the case of "adjective + noun", the percentage of "short + long" is higher in spoken Spanish.
c. Pre-posed adjectives are remarkably shorter than post-posed adjectives in spoken Spanish.

Next, we will calculate the total frequency (token frequency) of pre-posed position (a+n) and of posposition (n+a) for each combination of the number of syllables of "adjective, noun", and also its pre-posed percentage of "(a+n)/(a+n)+(n+a)":

[11] The ratio of "short + long" to "(short + long) + (long + short)" is high in spoken Spanish and written Spanish, and they are 71.0% and 68.1% respectively.

Table 3

a[12]	n[13]	a + n	n + a	(a+n)/(a+n)+(n+a) %	a	n	a +n	n + a	(a+n)/(a+n)+(n+a)%
1	1	0	0	-	5	1	1	3	25
1	2	43	2	95.56	5	2	39	121	24.38
1	3	86	2	97.73	5	3	61	210	22.51
1	4	35	1	97.22	5	4	34	111	23.45
1	5	10	0	100	5	5	17	31	35.42
1	6	0	0	-	5	6	4	8	33.33
1	7	0	0	-	5	7	1	1	50
1	8	0	0	-	5	8	1	0	100
2	1	12	7	63.16	6	1	0	1	0
2	2	292	253	53.58	6	2	3	33	8.33
2	3	349	203	63.22	6	3	11	60	15.49
2	4	212	115	64.83	6	4	3	25	10.71
2	5	67	36	65.05	6	5	4	13	23.53
2	6	6	4	60	6	6	0	4	0
2	7	1	1	50	6	7	0	0	-
2	8	0	0	-	6	8	0	0	-
3	1	7	18	28	7	1	0	2	0
3	2	199	525	27.49	7	2	0	1	0
3	3	346	564	38.02	7	3	0	7	0
3	4	187	273	40.65	7	4	0	6	0
3	5	52	102	33.77	7	5	0	3	0
3	6	11	13	45.83	7	6	0	0	-
3	7	0	3	0	7	7	0	0	-
3	8	0	0	-	7	8	0	0	-
4	1	1	15	6.25	8	1	0	0	-
4	2	114	526	17.81	8	2	0	0	-
4	3	217	650	25.03	8	3	0	1	0
4	4	152	373	28.95	8	4	0	1	0
4	5	44	122	26.51	8	5	0	0	-
4	6	15	27	35.71	8	6	0	0	-
4	7	1	1	50	8	7	0	0	-
4	8	1	0	100	8	8	0	0	-

For example, the real data of the combination of two-syllable adjectives and one-syllable nouns is as follows, and the pre-posed adjectives account for 63.16% in the level of total frequency (token frequency) because there are 12 pre-posed and 7 post-posed cases:

la memoria(n:3:2), la [propia(a:2:2)] [voz(n:1:1)] es su ma'scara(n:3:3), y
A todo ese [amplio(a:2:2)] [haz(n:1:1)] de lecturas(n:3:2) corres
(Madrid(n:2:1), 1930) el [alto(a:2:2)] [sol(n:1:1)] de su jornada(n:3:2) marc
y establece una(n:2:2) [sutil(a:2:2)] [red(n:1:1)] de correspondencias(n:5:2)
ese mandari'n(a:3:1) de [buena(a:2:2)] [fe(n:1:1) de las letras(n:2:2) ameri

[12] a: adjective
[13] n: noun

or(n:2:1) que estalla a [plena(a:2:2)] [luz(n:1:1)] del di'a(n:2:2) en la ser
tte>, cantada una(n:2:2) [sola(a:2:2)] [vez(n:1:1),] en el teatro(n:3:2) de l
o' el di'a(n:2:2) 8 del [mismo(a:2:2)] [mes(n:1:1)] con una(n:2:2) sonata(n:3

... omitted ...

ados, sin pensar que los [pies(n:1:1)] [claros(a:2:2)] de las Mari'as pudiese

los colores(n:3:2) es la [luz(n:1:1)] [solar(a:2:1)] que envuelve todas las

Pero lo dicen en [voz(n:1:1)] [baja(a:2:2)(n:2:2)] porque saben que
:3:1) de alzar una(n:2:2) [voz(n:1:1)] [propia(a:2:2),] inconfundible(a:5:2)

Table 4, made from Table 3, shows the number of syllables of an adjective in the horizontal line, the number of syllables of a noun in the vertical column, and the pre-posed adjective percentage in the intersection slots:

Table 4

Noun									
8 syll.	---	---	---	100	100	---	---	---	
7	---	50	0	50	50	---	---	---	
6	---	60	45.83	35.71	33.33	0	---	---	
5	100	65.05	33.77	26.51	35.42	23.53	0	---	
4	97.22	64.83	40.65	28.95	23.45	10.71	0	0	
3	97.73	63.22	38.02	25.03	22.51	15.49	0	0	
2	95.56	53.58	27.49	17.81	24.38	8.33	0	--	
1	---	63.16	28	6.25	25	0	0	--	
	1	2	3	4	5	6	7	8 syll.	Adjective

(--- shows that there is no combination corresponding to it.)

Table 4 specifies some interesting facts about the position of Spanish adjectives in noun phrases. If we follow the line of three-syllable nouns from the left to the right, we will notice the pre-posed adjective percentage falling gradually like 97.73%, 63.22%, 38.02%, 25.03%, 22.51%, 15.49% and 0%. Thus, it turns out that pre-posed adjective percentage is falling in inverse proportion to the number of syllables of the adjectives, clearly irrespective of the number of syllables of a noun. On the other hand, if each column of an adjective is followed upwards from the bottom, it can be said that the pre-posed adjective percentage becomes slightly higher as the number of syllables of a noun increases, but the correlation of the pre-posed adjective percentage and the number of syllables of a noun is not so explicit as in the case of the num-

ber of syllables of an adjective. That is, at the level of the total frequency (token frequency), the following facts are pointed out:

(11)a. The position of adjectives is determined by the number of syllables of adjectives rather than by the number of syllables of nouns.

b. As adjectives have fewer syllables, its pre-posed percentage becomes higher.

c. Adjectives of two or less syllables are pre-posed more often than post-posed, and adjectives of three or more syllables are distinctly post-posed[14].

Now, when all adjectives are reduced to a basic form, i.e., a masculine singular form, we have Table 5 below, which represents the percentage of pre-posed adjectives with a total frequency (token frequency) of ten or more times. The adjectives marked with * are plural forms, and are included in Table 5, because they have one more syllable than their singular forms:

Table 5

Adjective of ten or more frequencies in its basic form							
Adjective	pre-posed	post-posed	pre-posed percentage	Adjective	pre-posed	post-posed	pre-posed percentage
cierto	57	0	100	dicho	11	0	100
enorme	14	0	100	espléndido	16	0	100
inmenso	15	0	100	numeroso	15	0	100
grande	215	2	99.08	(gran)	(149)		
vario	34	1	97.14	pequeño	36	2	94.74
verdadero	32	2	94.12	bueno	46	3	93.88
(buen)	(14)			amplio	14	1	93.33
doble	11	1	91.67	pleno	11	1[22]	91.67
largo	29	3	90.62	auténtico	9	1	90
complejo	9	1	90	precioso	9	1	90
máximo	24	3	88.89	solo	15	2	88.24
diverso	21	3	87.50	mismo	122	18	87.14
último	108	16	87.10	bello	13	2	86.67
reciente	13	2	86.67	malo	12	2	85.71
(mal)	(6)			propio	89	15	85.58
medio	18	4	81.82	viejo	18	4	81.82
presente	13	3	81.25	excelente	12	3	80
nuevo	103	26	79.84	único	27	8[23]	77.14

[14] This is clear from the fact that the boundary of 50% of pre-posed position is between two and three syllables of an adjective.

simple	10	3	76.92	determinado	16	5	76.19
magnífico	9	3	75	múltiple	9	3	75
interesante	11	4	73.33	singular	11	4	73.33
distinto	24	9	72.73	sucesivo	8	3	72.73
alto	18	7	72	breve	14	6	70
notable	8	4	66.67	terrible	10	5	66.67
profundo	11	6	64.71	extraño	8	5	61.54
joven	11	7	61.11	puro	12	8	60
importante	17	12	58.62	extraordinario	7	5	58.33
antiguo	17	14	54.84	difícil	6	5	54.55
fuerte	6	5	54.55	diferente	13	11	54.17
posible	15	14	51.72	especial	10	10	50
principal	5	5	50	claro	7	9	43.75
siguiente	7	9	43.75	oscuro	5	7	41.67
absoluto	8	12	40	supremo	4	6	40
negro	5	8	38.46	actual	16	27	37.21
corto	5	9	35.71	total	5	9	35.71
pasado	11	20	35.48	perfecto	4	8	33.33
necesario	4	9	30.77	anterior	8	23	25.81
mágico	5	21	19.23	entero	2	9	18.18
común	5	23	17.86	personal	5	24	17.24
moderno	6	30	16.67	tradicional	2	14	12.50
original	2	15	11.76	clásico	2	16	11.11
universal	2	18	10	definitivo	1	10	9.09
fantástico	1	10	9.09	monumental	1	10	9.09
simbólico	1	10	9.09	anteriores*	1	11	8.33
populares*	1[16]	11	8.33	concreto	1	12	7.69
expresivo	1[17]	14	6.67	lírico	1	16	5.88
real	1	17	5.56	crítico	1[20]	18	5.26
poético	2	47	4.08	final	1[21]	27	3.57
popular	1[18]	30	3.23	histórico[19]	1	43	2.27
español[15]	1	81	1.22	abierto	0	12	0
abstracto	0	11	0	ajeno	0	11	0
alemán	0	19	0	amoroso	0	19	0
artístico	0	31	0	biográfico	0	12	0
central	0	13	0	científico	0	13	0
civil	0	21	0	contemporáneo	0	33	0
creador	0	17	0	cubano	0	10	0
cultural	0	27	0	ecomómico	0	18	0
escrito	0	13	0	españoles*	0	18	0
estético	0	22	0	europeo	0	14	0
familiar	0	14	0	francés	0	23	0
humano	0	44	0	individual	0	12	0
inglés	0	11	0	italiano	0	16	0

literario	0	53	0	madrileño	0	10	0
masónico	0	16	0	moral	0	19	0
musical	0	63	0	nacional	0	12	0
narrativo	0	22	0	natal	0	10	0
natural	0	12	0	norteamericano	0	10	0
pictórico	0	11	0	plástico	0	10	0
político	0	53	0	privado	0	10	0
religioso	0	28	0	social	0	29	0
urbano	0	10	0	vienés	0	10	0
vital	0	11	0				

Table 6 below, made from Table 5, shows the comparison between the adjectives of two or less syllables exceeding 50% of pre-posed position and the adjectives of three or more syllables of the same kind:

Table 6[24]

	number of adjectives exceeding 50%	total number	its percentage
Adj. of two or less syllables	23	40	57.50
Adj. of three or more syllables	33	98	33.67

This table will support our argument in (11c):

(11c) Adjectives of two or less syllables are pre-posed a little more often than post-posed, and adjectives of three or more syllables are distinctly post-posed.

From Table 5 we can confirm that there are some (groups of) adjectives which have a strong tendency to be pre-posed or post-posed, for example, the

[15] The only example of pre-posed *español* is: *La caja de sorpresas que es nuestra [española] [guitarra], fue adivinada por el músico creador en toda su mágica potencia expresiva*. 'The jack-in-the-box which is our Spanish guitar, was explored by the creative musician in all his magic expressive power.' This is a typical literary style as De Bruyne and Pountain(1995:106) have pointed out.

[16] las más *populares* óperas italianas

[17] *expresivas* muestras de un temperamento

[18] la *popular* presentadora de televisión

[19] el *histórico* jefe de la estación de (...)

[20] la *crítica* situación del Liceo

[21] la *final* trilogía

[22] Una pintura *plena* y delicada, (...)

[23] For example, un tema *único.*

[24] To make Table 6 the data of the adjectives marked with * in Table 5 are excluded.

so-called "classifying adjectives" represented by *español* 'Spanish', *musical* 'musical', etc. are generally post-posed, while the adjectives which lose its endings in front of a noun like *grande* 'big, great', *bueno* 'good', etc. are pre-posed in many cases.

Finally, we use our bigger KLM *Corpus*[25] to find every noun phrase of the 15 adjectives which are located between 60% and 40% of pre-posing percentage included in Table 5. The following Table 7 represents the average number of syllables[26], type frequency[27], and total frequency (token frequency) of the nouns which appear in each combination of "adjective + noun" and "noun + adjective":

[25] The *KLM Corpus* elaborated by Miyamoto consists of three types of Spanish texts: (1) journalistic ones, which are a portion of one of the representative dailies in Spain: *El Mundo*, 1995, (2) literary ones, which are a collection of about 20 novels of the contemporary Spain, and (3) (semi-)spoken ones, which are composed, on one hand, of texts of the conversations recorded in Spain like *El habla de la Ciudad de Madrid*, CSIC, 1981, on the other hand, of a collection of about 30 dramas of the present-day Spain. Each group of the texts has the quantity of 12 megabytes.

[26] This "average number of syllables" is based not on the total frequency (token frequency) but on the type frequency of nouns.

[27] The type frequency of pre-posed *difíciles*, for example, is 6 and the total frecuency is 8 as indicated in Table 7, because the nouns modified by pre-posed *difíciles* are: *relaciones* 2 times, *tiempos* 2, *cuestiones* 1, *elecciones* 1, *momentos* 1, and *vicisitudes* 1.

Table 7

Data of nouns combined with the indicated adjective						
	Adjective + Noun			Noun + Adjective		
Adjective	average num. of syllables	type frequency	total freq.	average num. of syllables	type frequency	total freq.
puro	3.26	170	246	3.15	72	108
importante	3.46	180	223	2.90	164	239
extraordinario	3.26	39	42	2.89	98	160
antiguo	3.17	281	414	2.69	197	273
difícil	3.08	50	73	2.78	69	121
difíciles*	3.50	6	8	2.90	20	61
fuerte	3.28	206	351	2.64	85	148
diferente	3.29	140	180	2.90	116	153
posible	3.47	278	365	3.26	114	149
especial	3.42	74	107	3.06	250	433
especiales*	4.25	4	4	3.18	68	117
principal	3.43	248	431	2.97	115	187
principales*[28]	3.53	123	224	3.00	24	32
claro	3.45	110	134	2.91	131	213
siguiente	3.00	95	126	2.70	54	494
oscuro	3.11	84	87	2.67	161	272
absoluto	3.68	79	102	3.15	104	242
supremo	2.85	13	14	2.55	29	143

Table 7 shows clearly that the post-posed nouns are longer than the pre-posed ones in every adjective. In other words, we can say that there is a clear tendency that adjectives precede relatively long nouns and follow relatively short nouns.

4. Conclusion

As mentioned above, although some facts have become clear about the position of Spanish adjectives in noun phrases by analysis from the formal viewpoint of the number of syllables, especially we want to make the following five points:

a. Adjectives are overwhelmingly post-posed.
b. The word order of "short + long" constituents is valid.
c. The adjectives is more pre-posed in written Spanish than in spoken Spanish.
d. The position of an adjective is determined by the number of syllables of an adjective rather than by the number of syllables of a noun. For example, if an adjective is shorter, it is more pre-posed. Adjectives of

[28] The unexpected high pre-posed percentage of *principales*, in spite of four syllables, may be explained from the constructions as follows: *los [principales] barrios artísticos; los [principales] problemas de la filosofía*. The former noun has another adjective post-posed, and the latter forms the "noun + of + (article) + noun" construction referred in (6b)

two or less syllables are pre-posed more often than post-posed, and adjectives of three or more syllables are distinctly post-posed.

e. Adjectives precede relatively long nouns and follow relatively short nouns.

Data

1. Object texts for analysis

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Appendix: y30104

```
BEGIN {
for(i=1; ARGV[i] ~ /^[0-9{\/@]/; i++)
        {key[++nkey] = ARGV[i]; ARGV[i] = ""        }
        for(h=2; h<=nkey-1; h=h+2)
            {if(key[h] ~ "/")
              {if(key[h] ~ "@/")
                {keyword[++nkeyword] = substr(key[h] , 3, length(key[h])-3)
                yes_no[++k] = "yes"
```

```
                if(key[h] ~ "[\^\$]")
                    {sub("\\^", "", key[h])
                    sub("\\$", "", key[h])
                    principal_keyword = substr(key[h] , 3,  length(key[h])-3)
                    }
                }
              if(key[h] !~ "@/")
                {keyword[++nkeyword] = substr(key[h] , 2, length(key[h])-2)
                yes_no[++k] = "yes"
                }
              }
            if(key[h] ~ "{")
                {keyword[++nkeyword] = substr(key[h] , 2, length(key[h])-2)
                yes_no[++k] = "no"
                }
            }
        for(g=1; g<=nkey; g=g+2)
            distan[++ndistan] = key[g]
        }
$0 ~ principal_keyword        {f = 0
for(j=1; j<=NF; j++)
        {f = 0
        if(yes_no[1] =="yes")
           if($j ~ keyword[1])
            {x[2] = j
            for(k=2; k<=nkeyword; k++){
            split(distan[k], s, "/")
            minidistan[k]=s[1]
            maxdistan[k]=s[2]
                for(m=x[k]+1+minidistan[k];m<=(x[k]+1+maxdis-
                tan[k]);m++){
                    if(yes_no[k] =="yes"){
                        if($m ~ keyword[k])
                        {f++
                        x[k+1] = m
                        break
                        }
                    }
                if(yes_no[k] =="no"){
                    if($m !~ keyword[k])
                        {f++
```

```
                    x[k+1] = m
                    break
                    }
                }
              }
            }
        if(nkeyword == f+1)
            {if((x[2]-distan[1]) <= 1)
                {printf("%s: ", $1)
                    for(p=1; p<=x[2]-1; p++)
                        printf("%s ", $p)
                    printf("[%s] ", $x[2])
                }
            if((x[2]-distan[1]) > 1)
                {printf("%s: ", $1)
                    for(p=x[2]-distan[1]; p<=x[2]-1; p++)
                        printf("%s ", $p)
                    printf("[%s] ", $x[2])
                }
            for(q=2; q<=nkeyword; q++)
                {for(p=x[q]+1; p<=x[q+1]-1; p++)
                    printf("%s ", $p)
                printf("[%s] ", $x[q+1])
                }
            for(r=x[nkeyword+1]+1;r<=x[nkeyword+1]+distan[ndis-
            tan];r++)
                printf("%s ", $r)
            printf("\n")
            }
        }
    if(yes_no[1] =="no")
      if($j !~ keyword[1])
        {x[2] = j
        for(k=2; k<=nkeyword; k++){
        split(distan[k], s, "/")
        minidistan[k]=s[1]
        maxdistan[k]=s[2]
            for(m=x[k]+1+minidistan[k];m<=(x[k]+1+maxdis-
            tan[k]);m++){
                if(yes_no[k] =="yes"){
                 if($m ~ keyword[k])
```

```
                    {f++
                    x[k+1] = m
                    break
                    }
                }
              if(yes_no[k] =="no"){
               if($m !~ keyword[k])
                    {f++
                    x[k+1] = m
                    break
                    }
                }
              }
            }
        if(nkeyword == f+1)
            {if((x[2]-distan[1]) <= 1)
                {printf("%s: ", $1)
                    for(p=1; p<=x[2]-1; p++)
                        printf("%s ", $p)
                    printf("[%s] ", $x[2])
                }
            if((x[2]-distan[1]) > 1)
                {printf("%s: ", $1)
                    for(p=x[2]-distan[1]; p<=x[2]-1; p++)
                        printf("%s ", $p)
                    printf("[%s] ", $x[2])
                }
            for(q=2; q<=nkeyword; q++)
                {for(p=x[q]+1; p<=x[q+1]-1; p++)
                    printf("%s ", $p)
                printf("[%s] ", $x[q+1])
                }
            for(r=x[nkeyword+1]+1;r<=x[nkeyword+1]+distan[ndis-
            tan];r++)
                printf("%s ", $r)
            printf("\n")
            }
        }
    }
}
```

On the Language of Portuguese *Estoria do Muy Nobre Vespesiano* – Linguistic Change and its Documental Evidence Based on the Corpus Study –

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1. *Estoria do Muy Nobre Vespesiano* is a prose text written in medieval Portuguese, which is known from only one copy of the book printed at Lisbon in 1496. This incunabulum is now owned by the Lisbon National Library, *Biblioteca Nacional de Lisboa* with a shelf mark, *incunábulo 571* and, as for the contemporary edition, we have one critical edition prepared by David Hook and Penny Newman in 1983, in addition to a facsimile edition, also published in the late eighties (Anselmo 1981). The text is a medieval fiction, which, using some real historical personalities, relates the cure of the Imperator Vespasian from leprosy by St. Veronica's sudarium, (his name in the text spelled as *Vespesiano* five times and as *Vespesiann* one time, the latter would probably be a mere printing error), the sacking of Jerusalem and the punishment of Pilatus. Also, in the course of the narrative, this tale tells of the release of Joseph of Arimathaea (in the text called *Josep Abaramatia*), and for this reason this text sometimes is referred to as related to the cycle of the Holy Grail stories.

According to the study of David Hook, the Portuguese text is supposed to be a direct translation of the Spanish text (Hook, 1983), which in turn, with other versions of the Iberian Peninsula, ultimately originated from the French medieval prose text *La Vengeance de Notre-Seigneur.* Alvin E. Ford reports that at least 45 manuscripts known today for the prose versions of this story are in Old and Middle French (Ford 1984, p.1), but which family or which branch of these continental testimonies could have closer relationship with the peninsular traditions is still unknown.

For the date of the translation from Spanish to Portuguese, Leite de Vasconcelos (1858-1941), a leading authority of Portuguese Philology in the first half of the twentieth century, presumed it to be in the first quarter of the fifteenth century for the existence of some archaisms demonstrating of that period (Vasconcelos 1922, p.95). On the contrary, one of the contemporary experts on this material, David Hook, regards the problem of dating as an open question in the introduction to his edition. From his study, we now

know that the Portuguese 1496 text would have an antecedent written in Portuguese, and that this and the Spanish 1492 one, (the latter is a printed book), have one intermediate stage of transmission as their common source. For this reason, this intermediary text, presumably in the manuscript form, must have existed before 1492. We will return to this problem of dating with respect to some linguistic archaisms shown in the text.

2. Portuguese, in the course of its evolution, lost its intervocalic *d* in the Second Person Plural verbal endings, like its Spanish counterpart did. This loss of *d* is supposed to have occurred in the first quarter of the fifteenth century. The Present Indicative and the Imperfect Indicative of the verbs of the first conjugation have the following paradigms before and after the loss of the intervocalic *d*. We exemplify with the verb *falar*, "to speak."

	Present Indicative		Imperfect Indicative	
	Old Portuguese[1]	Modern Portuguese	Old Portuguese	Modern Portuguese
1st sg.	falo	falo	falava	falava
2nd sg.	falas	falas	falavas	falavas
3rd sg.	fala	fala	falava	falava
1st pl.	falamos	falamos	falávamos	falávamos
2nd pl.	**falades**	**falais**	**falávades**	**faláveis**
3rd pl.	falam	falam	falavam	falavam

(The accented vowels are underlined.)

According to the prosodic pattern of the form, the development of the endings can be systematized as follows[2]:

I. Old Portuguese paroxytonic forms (Modern Portuguese oxytonic)

1) OP *-ades* (< Latin -ātĭs) >
-aes (dissyllabic to monosyllabic) > MP *-ais*
2) OP -edes (< Latin -ētĭs and -ĭtĭs)> *-ees* > MP *-eis*
3) OP *-ides* (< Latin -ītĭs) > *-ies* > MP *-is*
4) OP *sodes* (< Latin *sŭtĭs) > *soes* > MP sois
5) OP *-ade* (< Latin -āte) > -ae > MP *-ai*
6) OP *-ede* (< Latin -ēte and -ĭte)> -ee > MP *-ei*

[1] Here we divide roughly the history of Portuguese language into two successive stages: old Portuguese and modern Portuguese. The boundary of these two divisions is supposed to be in the first half of the sixteenth century.

[2] This scheme is based on the description of Williams, 1962, pp.188-206. We made it with slight simplifications, which are not relevant to the discussion.

7) OP *-ide*	(< Latin -īte)	>	-ie	>	MP *-i*
II. Old Portuguese proparoxytonic forms (Modern Portuguese paroxytonic)					
8) OP *-ávades*	(< Latin -ābātĭs)	>	*-ávaes*	>	MP *-áveis*
9) OP *-íades*	(< Latin -(i)ābātĭs)	>	*-íaes*	>	MP *-íeis*
10) OP *-árades*	(< Latin -ārātĭs)	>	*-áraes*	>	MP *-áreis*
11) OP *-êrades*	(< Latin *-ērātĭs)	>	*-êraes*	>	MP *-êreis*
12) OP *-érades*	(< Latin *-ĕrātĭs)	>	*-éraes*	>	MP *-éreis*
13) OP *-írades*	(< Latin *-irātĭs)	>	*-íraes*	>	MP *-íreis*
14) OP *’-ssedes*	(< Latin -ssetĭs)	>	*-ssees*	>	MP *’-sseis*

The changes are divided into two groups according to whether a vowel preceding the vanishing *d* is accented or not, that is to say, if the verbal word form in question was paroxytonic or proparoxytonic before this change took place in Old Portuguese.

Group I includes the forms of the Present Indicative (1,2,3,4), Present Subjunctive (1,2) and Imperative (5,6,7). The endings became single tonic syllables in Modern Portuguese.

In contrast to these, the verbal tenses belonging to the group II are Imperfect Indicative (8,9), Pluperfect Indicative (10,11,12,13) and Imperfect Subjunctive (14). Here the syncope of *d* happened in the posttonic position and, though the final outcome is unaccented *-eis* in all the cases, there were still two types of endings at the starting point, *-ades* and *-edes*. With regard to the change of the unaccented *-ades* > *-eis*, Edwin B. Williams mentions the lack, in the historical documentation, of chronologically intermediate forms like **-aes* or **-ais* in that posttonic position, but he does not refer to the *-edes* forms of the Imperfect Subjunctive. He only makes the distinction of tonic or atonic types of ending. (Williams 1962, pp.170-176)

The Future Indicative and the Conditional forms in Portuguese are, as in other Romance languages, etymologically formed from the infinitive and the finite verb forms of an auxiliary verb, *haver*. So are the endings of the Future tense originating from the Present Indicative (2) of *haver* and the Conditional from the Imperfect Indicative (9) of the same verb.

We provide here some examples of the change with the entire verbal form for 1) to 14). The infinitives are given in round brackets.

1) *falades* > *falais* (*falar* "to speak") ; *digades* > *digais* (*dizer* "to say")
2) *fazedes* > *fazeis* (*fazer* "to do") ; *faledes* > *faleis* (*falar*)
3) *abrides* > *abris* (*abrir* "to open")
4) *sodes* > *sois* (*ser* "to be", unique example)
5) *falade* > *falai* (*falar*)

6) *fazede* > *fazei* (*fazer*)
7) *abride* > *abri* (*abrir*)
8) *falávades* > *faláveis* (*falar*)
9) *fazíades* > *fazíeis* (*fazer*)
10) *falárades* > *faláreis* (*falar*)
11) *vendêrades* > *vendêreis* (*vender* "to sell")
12) *fezérades* > *fizéreis* (*fazer*)[3]
13) *abrírades* > *abríreis* (*abrir*)
14) *falássedes* > *falásseis* (*falar*), *vendêssedes* > *vendêsseis* (*vender*), *fezéssedes* > *fizésseis* (*fazer*)[3] ; *abríssedes* > *abrísseis* (*abrir*), *fôssedes* > *fôsseis* (*ser*)

On the other hand, for the elimination of *d* we have two exceptions. First, in the Future Subjunctive and in the Personal Infinitive the *d*, from the very outset of the language, was no longer in the intervocalic position because of the fall of the posttonic vowel. So it survived like in the example below (the verb *fazer*):

Future Subjunctive: Lat. **fēcĕrĭtĭs* > OPtg. *fezerdes* > MPtg. *fizerdes*
Personal Infinitive : Lat. *facĕrētĭs* (?)[4] > **fazeredes*[5] > OPtg.&MPtg. *fazerdes*

It is interesting that these two tenses are very characteristics of Portuguese Language: the Personal Infinitive is, in the Iberian Peninsula, only found in Portuguese and Galician, a romance dialect closely related to Portuguese, and, as for the Future Subjunctive, in contrast to the Spanish corresponding tense, Contemporary Portuguese still uses it vigorously.

The second exception is, several contemporarily monosyllabic infinitives (*crer* "to believe", *ler* "to read", *ter* "to have", *ver* "to see", *ir* "to go", *rir* "to laugh", *vir* "to come", *pôr* "to put") have their corresponding Second Person Plural forms in the Present Indicative and in the Imperative with *d* in the end-

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[3] The change of radical vowel *fez-* to *fiz-* is an analogical levelling of the verbal conjugation. It has nothing to do with our discussion here.

[4] This Latin form is of the Imperfect Subjunctive, but the Latin antecedent of Portuguese Personal Infinitive is still controversial.

[5] A form without syncope like **fazeredes* is, as for as we know, still unknown, but unsyncopated forms for the First Person Plural like *fazeremos* are sporadically known. Azevedo Maia reports the existence of such forms in the notarial documents in Old Portuguese (Maia 1986, p.757). On the other hand, it is supposed from the Latin origin that such unsyncopated forms like **fezeredes* for the Future Subjunctive must have preceded Old Portuguese *fezerdes*.

ings. In addition to these, *vades*, the form of Present Subjunctive of the verb *ir*, and *sede*, the Second Person Plural form of the Imperative of *ser* "to be" also have its *d*.

	Latin	Old Portuguese	Modern Portuguese
crer	**credētĭs*	*creedes*	*credes*
	**credēte*	*creede*	*crede*
ler	**lĕgētĭs*	*leedes*	*ledes*
	** lĕgēte*	*leede*	*lede*
ter	*tenētĭs*	*tẽedes*	*tendes*
	tenēte	*tẽede*	*tende*
ver	*vĭdētĭs*	*veedes*	*vedes*
	vĭdēte	*veede*	*vede*
ir	*ītĭs*	*ides*	*ides*
	īte	*ide*	*ide*
	vadātĭs	*vaades*	*vades*
rir	**rīdītĭs*	*riides*	*rides*
	**rīdīte*	*riide*	*ride*
vir	*venītĭs*	*vĩides*	*vindes*
	venīte	*vĩide*	*vinde*
pôr	**pōnētĭs*	*põedes*	*pondes*
	**pōnēte*	*põede*	*ponde*
ser	*sĕdēte*	*seede*	*sede*

The retention of *d* in these forms is generally explained, for some verbs (*ter, vir, por*), by the newly created nasal environment and, for the others, by a possible confusion with Second Person Singular form of the respective verb or by some prosodic limitation possibly having existed in Old Portuguese.

In the case of the Preterit tense, it does not have *d* in the ending from the beginning. Its Latin ancestor *-stĭs* did not yield the results with *d* in Portuguese, for the Latin *t* was not in the intervocalic position and consequently it does not become voiced: the Portuguese endings are *-astes, -estes, -istes* in both weak and strong Preterits, and so they have nothing to do with our case.

So we can summarize this state of affairs in Portuguese with regard to the retained or syncopated *d* in the Second Person Plural endings.

A. the d retained : Future Subjunctive and Personal Infinitive of all verbs. Some sporadic cases.

B. the fall of *d* : According to whether a vowel preceding the *d* is

accented or not, the phonetic shapes of the endings develop as follows.

1) Accented: *-ade(s)* > *-ai(s)*, *-ede(s)* > *-ei(s)*, *-ide(s)* > *-i(s)*, *sodes* > *sois*
2) Unaccented: *-ades* > *-eis*, *-edes* > *-eis*

This loss of intervocalic *d* in particular verbal endings, clearly is not normal sound change as is meant by classical historical linguistics. Its scope was limited to certain verbal morphology. So the change is, in a sense, morphologically motivated, but conditioned by its phonetic environment. Since the *d* in the endings of the other two tenses and in some specific verbs continues uninterruptedly, there must have constantly been the possibility of analogical restoration, or of stylistic expansion in the case of literary texts.

As for the period during which this *d* disappeared, Edwin B. Williams carried out a pilot research in his historical grammar of Portuguese (Williams 1962, p.172). He concluded there that *d* seems to be fallen "in the sixteen years between 1418 and 1434 (ibid.)," though he himself was sufficiently cautious about the oversimplification of this sort of question. Another information given by him is that King Edward of Portugal (Dom Duarte, 1391-1438, the King 1433-), in his philosophical essay, *Leal Conselheiro* "Loyal Counselor," "retained the *d* only in passages quoted from older texts (Williams 1962, p.173)" and otherwise "used forms without d (ibid.)." In that time this work of the King was supposed to be written between 1428 and 1438. Recent study shows that the final redaction of the work was performed between 1435 and 1438 (Castro 1998, p.XVI).

3. In our study, a complete scrutiny was carried out for the Second Person Plural forms of verbal conjugation in the corpus text, *Estoria do Muy Nobre Vespesiano*. The text was transcribed from the facsimile edition into the electronic form and, when necessary, the edition of Hook and Newman was referred to but, as far as possible, the original graphic notations in incunabulum were respected. In the edition of Hook and Newman, in addition to some corrections based on the Spanish source, the graphic notation of nasals was unified. We maintained the original graphics and some correctable texts, as far as a grammatical interpretation could be possible in Portuguese.

The text was processed through a concordance program. We used the Simple Concordance Program Ver.4.07. The Verbal forms in question are the following:

A) Forms in which the *d* is retained still in Modern Portuguese (27 instances)[6]:

1 deixardes	1 quiserdes	2 creede	1 veedes
1 entẽderdes	5 quiserdes	1 tẽdes	1 vede
2 fezerdes	1 poderdes	1 tẽde	1 veede
1 mandardes	1 verdes	1 vades	2 vinde
1 fordes	1 virdes	2 vedes	

B) Forms in which the d was lost in Modern Portuguese:

1) Accented

α) Retained (**<u>8 instances</u>**):

1 bautizedes	2 aparelhade	1 emxalçade	1 rogade
		1 exalçade	2 sabede

β) fallen (**<u>143 instances</u>**):

6 ajaes	1 defendaes	2 façaes	1 queyraes
1 alegrai	1 demãdaes	1 guardae	1 saibaes
1 anojaes	1 demãday	1 guarday	3 sejaes
1 assaaes	1 desmajaees	2 mãdae	1 temaees
1 bautizae	1 deyxay	1 mãdaees	1 temaes
1 comaaes	2 digaaes	1 mandae	1 tenhaes
1 cõuertaees	1 digaes	1 mãtenhas*[7]	5 tomae
1 cũpraaes	1 escuytaae	1 p*re*gae	1 tomay
1 dae	1 ẽuiae	1 q*ue*eyraes	1 tornae
2 day	1 façaaes	1 queyraaes	

[6] Numeral indicates the number of occurrences in our corpus; the letters expanded from abbreviations are indicated in italics.

[7] The form with asterisk indicates a verbal form in which the vowel of the ending is notated by one graphic vowel letter. That is to say, in the case of *foses*, above, this and *fosseis* are the same Second Person Plural form of Imperfect Subjunctive of Verb *ser*, "to be". They are oxytone, so in the former case the distinction with the Second Person Singular form of the same tense is made by the position of the stress: *fosés* (Plural) ~ *fósses* (Singular). (The graphic tradition of Old Portuguese did not distinguish -*s*- and -*ss*- usually in spite of its phonemic opposition between /s/ and /z/.)

2 acharees	1 dees	7 fazee	8 sabee
1 aparelhees	1 desauiees	1 filhees	4 sabees
1 auees	2 deuees	1 leyxees	2 saberees
1 avees	1 d*e*uees	2 parees	5 serees
2 auee	1 deues*	1 podeeis	1 series
1 avee	2 dizee	7 podees	3 soees
1 avõdarees	2 dizees	1 pod*e*es	2 soes
1 bautizarees	2 enuies *	1 poderees	2 tomees
1 bautizareis	1 ereys	1 podes *	1 tornees
3 bautizees	1 escaparees	1 querees	2 vereis
1 busquees	3 ẽuiees	1 queres *	
1 comee	1 ẽuies *	1 rogueis	

2) Unaccented

α) Retained (0 instances):

β) fallen (8 instances):

2 fosseis	1 teuesseis	1 uyseis
1 foses *	1 tomasseis	2 tornasseis

In our text, other than above examples, there occurred the following 12 Second Plural verbal forms, although they are not directly relevant to our analysis.

1 acusastes	1 estabeleçestes	1 partistes	2 tomastes
1 entregastes	1 mandastes	3 p*ro*metestes	2 ouuestes

As for the Second Person Singular forms in the text, we have the following 34 examples:

1 abras	1 crerias	1 segurares	1 desprezaste
1 acharas	1 es	1 seras	2 enuiaste
1 aguardasses	1 mercaras	1 soyas	1 ẽuiaste
1 duuides	3 teẽs	1 temas	2 liuraste
1 fazes	1 quiseres	1 vees	3 deste
1 fosses	1 quiseres	1 achaste	1 acolheste
1 rregesses	1 sairas	1 arrenegaste	1 viste

We have 193 Second Person Plural forms and 34 Second Person Singular forms in the text.

4. As for the development of the endings, if we can postulate intermediate stages after the loss of *d*, a first step would be two vowels in hiatus, and then comes the stage in which two successive vowels became diphthongized. Otherwise, the endings in question were replaced by some analogical process without chronologically successive phonetic modifications.

Chronological possible successive stages would be as follows:

1) Accented: ***-ade(s) > -a·e(s) > -a·i(s) > -ai(s)***
-ede(s) > -e·e(s) > -e·i(s) > -ei(s)
-ide(s) > -i·e(s) > -i·i(s) > -i(s)[8]
sodes > so·es > so·is> sois

2) Unaccented: ***-ades > -a·es > -a·is > -ais > -eis***
-edes > -e·es > -e·is > -eis

Out of these, the four evolutions have the corresponding documentation in the text, which we systematize as follows:

-ade- (5) ~ *-aae-* (8) ~ *-aee-* (4) ~ *-ae-* (34) ~ *-ai(ay)-* (7) ~ *-a-* (1)
-ede- (3) ~ *-eei-* (1) ~ *-ee-* (71) ~ *-ei(ey)-* (5) ~ *-e-* (7)[9]
sodes (0) ~ *soees* (3) ~ *soes* (2)
-edes (0) ~ *-eis* (7) ~ *-es* (1)

On the other hand, we have the following variants of the grammatically identical forms:

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[8] In Portuguese canonically a sequence of two identical vowels develops into one single vowel:
v.g. a + a ➞ [a], e + e ➞ [ɛ], o + o ➞ [ɔ], i + i ➞ [i].

[9] Here *i* and *y* are meaningless graphical variations. The figures in round brackets indicate the occurrence in the corpus.

1 bautizedes ~ 3 bautizees	
2 sabede ~ 8 sabee	2 mãdae ~ 1 mandae
1 bautizarees ~ 1 bautizareis	1 podeeis ~ 7 podees
1 dae ~ 2 day	~ 1 pod*e*es ~ 1 podes *
2 deuees ~ 1 d*e*uees	1 querees ~ 1 queres
~ 1 deues*	1 queyraaes ~ 1 queyraes
2 digaaes ~ 1 digaes	3 soees ~ 2 soes
3 ẽuiees ~ 2 enuies	1 temaees ~ 1 temaes
~ 1 ẽuies*	5 tomae ~ 1 tomay
1 façaaes ~ 2 façaes	2 fosseis ~ 1 foses
1 guardae ~ 1 guarday-os	

What can we conclude from all of this? The forms mentioned above with intervocalic *d* (8 occurrences, *aparelhade* "Prepare!", *bautizedes* "Baptize!", *emxalçade* "Heighten!", *exalçade, rogade* "Pray!", *sabede* "Know!") could be something like stereotyped expressions. Edwin B. Williams reports that "stereotyped expressions such as *comprades e façades comprir, sabede*[10]...found commonly as late as the end of the fifteenth century (op.cit. p.172)." So we think that these cases can be considered as sporadic phenomena which do not reflect the state of the language of that time, that is to say, the time of the redaction of the Portuguese text. The translation of the text must have been performed after the general loss of the intervocalic *d* in endings.

There is another point of question about whether the endings were dissyllabic or monosyllabic. Forms like *-aae-* (8), *-aee-*, *-eei-* and *soees* (3) seem to support the dissyllabic interpretation of the ending, but, considering that in Old Portuguese the duplicated vowel notation was normally used to indicate the tonic vowel and that in *-aee-*, and *soees* the *e* was never stressed historically, we can not take them for the proof of the dissyllabic character of those endings. But clearly we can not deny the existence of the dissyllabic morphological variants in the text. However, the very existence of monophthongized ending forms like *deues* (1), *enuies* (2), *ẽuies* (1), *mãtenhas* (1), *podes* (1), *queres* (1), *foses* (1) seems to give strong support for the monosyllabic interpretation. Or we could conclude that the process of the contraction of the hiatus was rather progressing. Concerning the endings, *-eis* and *-es*, we can explain their difference by the lack of the assimilation process in the stage in which two vowels were formed with hiatus status:

[10] Approximately the meanings are as follows: *comprades e façades comprir* "Carry out and make (them) fulfil, too!", *sabede* "Know!"

-e·es > *-e·is* (assimilation)> *-eis* (diphthongization)
or *-e·es* > *-es* (contraction of hiatus)

Or we can presume some influence of southern Portuguese dialects which normally simplify the diphthong *ei* to the monophthong *e*. Anyway, it can be said that the chronological stratification is reflected in our verbal morphology.

5. When Leite de Vasconcelos discussed the dating of the text (op. cit., p.95), he mentioned the alternation of the syncopated forms with the unsyncopated forms, citing *rogade, bautizedes* as the example of forms with *d*. It was presumed that the redaction of the Portuguese text must have been before the disappearance of *d*, that is to say, in the first quarter of the fifteenth century. But from our examination it became clear that this kind of alternation in the text is of a sporadic character and probably can be explained by some stylistic factors. So the redaction of Portuguese text would be after that time, and, in the end, this conclusion will match better the textual history of this text as demonstrated by David Hook.

(Here I would like to express my gratitude to Prof. Robert R. Ratcliffe who reviewed the final redaction of this text.)

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Analysing Texts in a Specific Domain with Local Grammars – The Case of Stock Exchange Market Reports[1] –

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Introduction

A series of brief journalistic columns, written and published regularly to report facts in a specific domain, give a reader an impression of being clichéd. This impression is characterisable from a linguistic point of view and exploitable from an IT point of view.

Taking as my corpus financial columns published in a French daily newspaper, I shall demonstrate in this article that expressions of variation in value of stocks exchanged in a market represent essential and therefore recurrent information in this corpus. These expressions are reduced formally to one sentence schema, corresponding to diverse formal variants. In the second part of this article, I shall describe this variation in the form of sentences, corresponding to a specialized lexicon-grammar.

In the last part of this article, I shall show a way of analysing the electronic corpus automatically with local grammars. The preliminary linguistic description proves to be a help in the construction of local grammars. About 22 % of the corpus is concerned by analysis and it is automatically analysed without ambiguity.

1. Corpus

The corpus which will be analysed is made up of an accumulation of the columns entitled *Valeurs France* which used to be published daily in the French newspaper *Le Monde*[2]. Each day, we found between three and five (or six, occasionally) reports in this column, each of which dealt with a variation in value of specific stocks. I collected these columns electronically from the web version of *Le Monde* from January 2001 to August 2001. This collection of columns amounts to 146 columns, containing approximately 560 reports. The entire document represents 175 kilo bytes of electronic text and 28,983

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[1] I wish to express my gratitude to Antoinette Renouf, Christian Leclère and Eric Laporte for their generous help.

[2] This column no longer exists in *Le Monde*. This is due to an editorial change which took place recently.

running words.

2. Linguistic analysis of the corpus

2.1. Types of sentence in the corpus

Several types of sentence are observed to occur very regularly and they can be described syntactically without problem. In this study, I shall take only one example of these types of sentence to show the advantages of a preliminary linguistic analysis of a corpus for running a syntactic parsing on it successfully.

The syntactic description I shall apply to the corpus is based on a lexicon-grammar[3], the principle of which is 1) to describe, for each predicative element given, the nature and the number of its objects, so as to construct a simple sentence, and 2) to verify for each simple sentence constructed a certain number of syntactic, distributional and transformational properties. The results of the description take the form of a matrix of lexicon-grammar, each row of which represents one particular lexical element, i.e. an entry word, and each column one particular syntactic, distributional or transformational property. In each intersection of a row and a column, if the sentence endorses the property, we put a "+" sign; if it doesn't endorse the property, we put a "-" sign. (see Annex 1.)

2.1.1. Sentences describing a variation in value

The type of sentence which I shall analyse, that appears most frequently in the corpus, contains stock exchange market information which reports a variation in value of particular stocks, exchanged in the Paris' stock exchange market (la Bourse de Paris). Here are some examples:

(1) *ABC <u>gagnait</u> 2,84 %, vendredi 24 août dans les premières transactions,* à 18,10 euros

(2) *L'action ABC <u>s'appréciait</u> de 0,94 %, vendredi dans les premiers échanges,* à 37,48 euros

(3) *Le titre d'ABC <u>(a fait un bond + a bondi)</u> de 3,29 %, à 16,78 euros, mercredi matin*[4]

(4) *L'action du spécialiste de la carte à puce ABC <u>restait stable</u> dans les premières transactions, lundi 20 août, à 3,09 euros*

These sentences express the most important information found in daily stock exchange market reports in this newspaper. They are so organized as to take as their pivotal element several dozens of specific predicative elements

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[3] See for more details: M. GROSS (1975, 1981), Ch. LECLERE (2002, 2003), T. NAKAMURA (2002, 2003)

[4] The brackets are for the paradigm and the "+" sign shows a disjunction.

(verbal, nominal and adjectival: the sequences underlined in the examples (1)-(4)) and their objects. They present a variety of shapes but this formal variation corresponds to a stable unit of meaning, i.e. the expressions of numerical change. This meaning of numerical change is made syntactically explicit by means of two numerical objects, one being a relative numeral, the other an absolute numeral (see § 2.2.2 - § 2.2.3. below).

The subjects of these sentences are relatively easy to describe because of their regularity in form and syntax. The class of subjects is made up of sequences of noun phrases, which are mainly juxtaposed in an appositive manner (cf. (2)-(4)), around such head nouns as *action*, *titre* or *valeur*. The proper nouns of companies the transfer of whose stocks is reported in the corpus have been listed and they can function syntactically as the subject of predicative elements of variation, on the basis of metonymy (cf. (1) and see § 2.2.5.).

The two numerical objects have a strong relation with verbs of variation, in spite of their adverbial nature. In particular, the relative numerical object is essential to define a certain number of verbs I call here "verbs of variation". It functions more as an argument to verbs of variation than as the absolute numerical object.

Other clearly adverbial elements in a sentence, like adverbial phrases of time and location, are also considered to be a typical constituent of this type of sentence. Adverbs of location tend to disappear from a surface structure, because the location is always the same. In contrast, adverbial expressions of time are always present in a sentence, information on time being crucial in the report as an integral part of the observation. (see § 2.2.4. below)

The same predicative element can take several formal variants. Some verbs which are classified as verbs of variation go through a transformation of nominalisation with support verbs and the resulting expressions are included in the same syntactic paradigm as they are (see § 2.2.1.2.-5. below). The example (3) above gives one such example of nominalisation.

The linearisation of these predicates and their objects is quite regular. Because of the nature of the corpus, which is a factual report, we find no interrogative, negative or imperative variants of the sentences under analysis. Only the declarative form of the sentence is observed.

I shall mention very briefly the semantic regularity of the predicates. It is possible to group the predicates of variation into three categories: one contains predicates of upward movement, one predicates of downward movement and the last one contains predicates of stability.

The types of sentence I have described briefly in this section are hereafter represented by a simple sentence schema, which is a sequence of a few symbols, as seen below:

A V D% DN AT AL

The arbitrary symbol *A* represents a class of noun phrases functioning as the subject of predicates of variation, and *V* the predicates of variation. The symbols *D*% and *DN* designate the relative numerical object and the absolute numerical object respectively, *AT* and *AL* meaning adverbial phrases of time and of location respectively.

I shall give a detailed description of each constituent below.

2.2 Individual study of each constituent

2.2.1. The predicates of variation

The predicates of variation symbolized by *V* in the sentence schema are primarily the verbs like *gagner*, *s'apprécier*, *chuter*, *etc*. and an adjective "actualised[5]" in a sentence with a support verb like *rester stable*. The examples (1)-(4) given above contain these predicates.

Although I call these predicates "predicates of variation", it is not always the case that any one of them intrinsically has the meaning of "variation of quantity". Compared with verbs like *augmenter* or *baisser*, in which the meanings of variation of quantity seem to be inherent, the verbs like *glisser* or *chuter* are here used metaphorically. What confers the meaning of variation of quantity to these predicates is their combination with specific types of subjects and objects.

In the corpus under analysis, these predicates take as subject the noun phrases designating the stocks, or the proper noun of a company whose stocks are exchanged, or the head nouns which are to do with an account, but the second most recurrent types of sentence, describing the commercial results of a company, are not treated in this study.

These predicates are always followed by a relative numerical object and there is a high possibility that they are followed by an absolute numerical object.

2.2.1.1. The syntactic nature of the predicates

The verbal predicates of variation can be transitive or intransitive, with regard to a relative numerical object. The transitive verbs are ones like *gagner*, *céder*, *perdre*, *etc*. They are followed by it directly. They are sub-classified as *Vvt*, in the *V* category. This syntactic property figures in the column *NVDnum*% within the matrix of the lexicon-grammar I provide in the annex 1.

The intransitive verbs are verbs like *baisser*, *chuter*, *monter*, *etc*. They are followed by a relative numerical object indirectly with the aid of the preposi-

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[5] See 2.2.1.2. for the explanation. -*n* attached to *V* means a suffix of nominalisation.

tion *de*. They are grouped within the sub-category *Vvi*, in the larger class of *V*. This property figures in the *NVDeDnum*% column.

Only one example of an adjectival predicate with a support verb is found in the corpus. The expression is *rester stable*.

2.2.1.2. Nominalisation of the predicates

Between the sentences below:

Max adore Léa = N V N

Max a de l'adoration pour Léa = N Vsup Dét V-n Prép N

Max est en adoration devant Léa = N Vsup Prép V-n Prép N [6]

it is possible to observe a close synonymy, in which the first of the three sentences has a verbal predicate (*adore*), and the next two sentences are structured around a predicative noun (*adoration*), which is morphologically related to the verbal predicate in the first sentence. This is considered a nominalisation relation between the sentences. Verbs like *avoir* in the second sentence and the verb *être* and the preposition *en* in the last sentence are considered to be *support verbs*[7] whose role is to supply sentences with information about time, person and number, which the noun predicate alone is not able to give. This procedure of attribution of morphological frame is called "actualisation".

The three sentences observed above form a class of equivalence, related one to the other by a transformation of nominalisation. To put this another way, all three sentences are to be considered as formal variants of the same unit of meaning, i.e. the relation of a predicate to its arguments which could be rendered explicit by notation like; *adorer (Max, Léa)*.

2.2.1.3. N0 VsupEn en Vvn de Dnum %

Certain verbs of variation examined here have nominalised forms, with or without a suffix. This property figures lexically in the matrix within the *Vvn* column, for each case concerned. The nominalised predicates form a 'nominal' sentence with support verbs, which is equivalent to a corresponding 'verbal' sentence. The support verbs followed by the preposition *en* are symbolized as *VsupEn*.

- VsupEn = être (en)

The semantically neutral support verb which applies most frequently to the verbs of variation is *être (en)*. For each verbal predicate, the possibility of

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[6] The symbol *Vsup* represents support verbs, *Dét* determiners, *Prép* preposition, *N* noun phrases, and *Vvn* are nouns morphologically related to the verbs of variation.

[7] See for details M. GROSS (1981), J. GIRY-SCHNEIDER (1978), J. LABELLE (1974) and A. MEUNIER (1981).

creating a construction with this support verb is marked within the column *être (en)*, dependent on the column *N VsupEn en Vvn de D*% within the matrix of lexicon-grammar given in the annex. Here are some examples of nominalisation:

(5) *L'action ABC (augmente + baisse + progresse +...) de 0,03 %, à 12 euros*

(6) = (13) *L'action ABC est en (augmentation + baisse + progression +...) de 0,03 %, à 12 euros*

- *VsupEn* = other than *être en*

Besides the support verb *être (en)*, there are several verbs whose normal syntactic behaviour in everyday language does not need the object *en Vvn*, but which, in this particular corpus, need it and are obligatorily followed by it, so as to make an informationally sufficient sentence:

(7) *L'action ABC (s'inscrivait + pointait +...) en (hausse + baisse +...) de 0,03 %...*

(8) ** ? L'action ABC (s'inscrivait + pointait +...)*

Sentence (8) is insufficient syntactically and semantically. But, in the corpus, these verbs often appear accompanied by the sequence *en Vvn De Dnum* %, as is seen in example (7). I shall consider these verbs as a sort of stylistic variant of the support verb *être (en)*[8]. In the matrix, under the column *VsupEn en Vvn de Dnum* % we find some pseudo support verbs functioning in parallel with *être (en)*.

- *Vtps*

Within the column *VsupEn en Vn de Dnum* %, we find the *Vtps* column. The verbs classified under this sub-category are verbs like *ouvrir*, *débuter*, *finir*, *etc*., which accept as their direct objects the head nouns dealing with the stock exchange activities like *séance*, *échanges*, *etc*. These noun phrases are in this context both totally omissible and reconstitutable.

(9) *L'action ABC <u>a (ouvert + débuté + clôturé + fini) (E + la séance + les échanges)</u> en (progression +baisse +...) de 0,9 %* [9]

The underlined sequences above, constituted by *Vtps* and these nouns, take an aspect-time meaning, as do the adverbs of time. This sentence is synonymous with the one given below:

(10) *L'action ABC était en (progression + baisse) de 0,9 % <u>à LE (ouverture + début + clôture + fin) de LE (séance + échanges)</u>*[10]

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[8] This analysis is, of course, only applicable to this specific corpus.

[9] *E* signifies a zero element in the paradigm.

[10] *LE* represents all the variants of the definite article.

The underlined parts in these two sentences (*Vtps Dét N* and *Prép Dét Vtps-n de Dét N*[11]) could be considered equivalent and there exist two types of interpretation for this equivalence.

In one way, it is possible to consider that a transformation of re-structuration was applied to (9) to engender (10), but the phenomena is not so clear-cut. The time adverb can appear in (9), although this semantic notion is thought to be carried by the *Vtps Dét N* sequence.

(11) *ABC a ouvert en progression de 0,04 %, à 42 euros, à LE (ouverture + début) de la Bourse de Paris*

Another comment on these sentences is that the verbal tense of sentences of the type (9) is always punctual (*passé composé*) while the tense of support verbs in sentences like (10) is always static (*imparfait*).

Another interpretation is to see the *Vsup Dét N* sequences underlined in the example (9) as support verb sequences, which support a true predicate *en Vvn de Dnum* %. This type of sentence without the information of variation is unacceptable in this corpus, due to the lack of specific information. Here is an example:

(12) *? *L'action ABC a (ouvert + débuté + clôturé + fini) (E + la séance + les échanges)*

This argumentation is sufficiently strong to opt for the last analysis, which is to recognise these *Vtps* verbs and their objects as support verb sequence.

2.2.1.4. N0 Vsupt Dét Vvn de Dnum %

Transitive support verbs[12] like *enregistrer*, *afficher* and *connaître* also occur frequently. They are symbolized by *Vsupt*.

(13) *La valeur ABC (gagnait + progressait de + baissait de + ...) 0,04 %, à 12 euros*

(14) *= La valeur ABC (affiche + connaît + enregistre +...) un(e) (gain + progression + baisse) de 0,04 %, à 12 euros*

2.2.1.5. N0 VsupA à Dét Vvn

In this corpus, we find several sentences whose role is to indicate an approximate direction of the variation in value of stocks. These are sentences like the ones below:

(15) *L'action ABC (était (E + réservée + orientée) + reprenait +...) à la (hausse + baisse)*

As was the case with the *Vtps* verbs, sentences like (15) without the con-

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[11] Here *Vtps-n* means a nominalized *Vtps* with a suffix (-*n*) (which can be zero).

[12] The verbs here examined do not present all the properties of a transitive verb.

stituent *à LE Vvn* would have no sense. I consider these verbs as support verbs and symbolize them by *VsupA*. They are lexically enumerated in the rubric *N0 VsupA à Dét Vvn*.

2.2.2. Relative numerical object

In this corpus, the movement of a value, whether upward or downward, is calculated from two distinct numeral pieces of data. With regard to stocks, these two are 1) their value at the end of the exchange the day before the report and 2) the one at the moment of observation in the morning after the Bourse opens[13].

The difference between the two values is represented by objects of percentage and they can appear directly or indirectly, according to the syntactic nature of the verbs, within a sentence:

(16) *L'action ABC (monte + augmente + chute + ...) <u>de 0,03 %</u>, à 32 euros*

(17) *L'action ABC a enregistré une (chute + hausse +...) <u>de 0,03 %</u>, à 32 euros*

(18) *L'action ABC (gagne + perd + prend +...) <u>0,03 %</u>, à 32 euros*

Examples (16) and (17) show this object (the underlined sequences) in an indirect syntactic position, with the indirect verbs of variation for (16) and the nominal predicates with support verb for example (17). Example (18) is constructed with a transitive verb, with a direct relative numeral.

These relative numerical objects are symbolised by *D*%.

2.2.2.1. Relative numerical indirect object

The relative numerical indirect object takes the form below and is represented by *de Dnum* %.

de Dnum % = de Dnum (% + pour cent)

The adverbial nature of this object seems to be clear, in the light of the following discourse:

(19) *- De combien l'action ABC a (augmenté + chuté + baissé) ?*
- De 0,03 %

The object in question is introduced by a composite adverbial interrogative pronoun *de combien*.

The fact that this object can be replaced by a morphologically simple adverb of quantity would be another reason to consider this type of object as an adverbial type. In fact, the co-occurrence of an adverb and this type of object within a sentence gives an impression of redundancy, the measure

[13] This is a peculiarity of *Le Monde* which appears at about noon each day, so in this newspaper we can read the information on a stock exchange market in the morning.

object furnishing the precision on quantity:

(20) *L'action ABC a (beaucoup + peu) (augmenté + chuté + baissé), à 32 euros*

(21) *L'action ABC a beaucoup augmenté, de 2,1 %, à 32 euros*
L'action ABC a peu baissé, de 0,01 %, à 32 euros

Within the class of intransitive verbs classified as verbs of variation, some do not take this object as their obligatory object in everyday language. This is the case of verbs like *s'envoler, s'effondrer, etc.* These verbs are used here metaphorically and in this use the verbs obligatorily take a relative numerical object.

2.2.2.2. Relative numerical direct object

The syntactic nature of the direct object of measure, which I represent simply by *Dnum* %, is not as clear as it seems to be. Its pronominalisation by direct pronouns is not entirely satisfactory, as the following examples show:

(22) ? 0,03 %, l'action ABC l'a (gagné + perdu + pris +...), à 32 euros*

(23) - Qu'est-ce que l'action ABC a (gagné + perdu + pris+...), à 32 euros ?
*- * 0,03 %*

The morpho-syntactic nature of this object is challenged if sentences like (24) enter the paradigm:

(24) *L'action ABC (abandonne + perd + cède +...) <u>0,03 % de sa valeur</u>, à 32 euros*
= *L'action ABC (abandonne + perd + cède +...) <u>0,03 %</u>, à 32 euros*

The two sentences in example (24) are strictly synonymous and they suggest that the sequence *Dnum* % is in fact not a noun phrase but a sort of nominal determiner of fraction, which is followed by a sequence of definite noun phrase with preposition *de Dét N*, like *un quart* or *un tiers* in the example below:

(25) *Max a mangé un (quart + tiers) de ce gâteau*
= *Max en a mangé un (quart + tiers)*

This interpretation still does not explain the phenomenon entirely, because the object *Poss valeur*[14] does not follow the verbs with a meaning of positive movement:

(26) ** ? ABC a (gagné + pris + regagné +...) 0,3 % de sa valeur*

The distribution of the noun phrase *Poss valeur* in sentences (24) and (26) being complementary, it is possible to maintain the interpretation of the

[14] *Poss* here represents the genitive adjectival determiner.

complement *Dnum* % as a nominal determiner.

2.2.3. Absolute numerical object

The absolute numerical object, which I represent here by the symbol *DN*, behaves here like a purely adverbial object. This interpretation is supported by the syntactic mobility that this object presents within a sentence. In the sentence (27) below, the syntactic positions the absolute numerical object *à 30 euros* can occupy are indicated by the bracketed exclamation marks:

(27) <!> *ABC* <!> *a (gagné* <!> *+ chuté de) 0,3 %, à 30 euros*

The absolute numerical object *à Dnum N* which appears within the sentences of variation in this corpus indicates the numerical value of a stock at the moment of observation by a journalist.

2.2.3.1. Possible transformational analysis of the absolute numerical object

With regard to this syntactic constituent, there is a possible alternative analysis to considering it as a simple adverb. In the following sentence:

(28) *ABC a (gagné + chuté de) 0,3 %, pour (atteindre + s'établir à) 32 euros*
= *ABC a (gagné + chuté de) 0,3 %, en (atteignant + s'établissant à) 32 euros*

The adverbial constituents like *pour (atteindre + s'établir à) 32 euros* and the gerund *en (atteignant + s'établissant à) 32 euros* have exactly the same meaning as the absolute object *à 32 euros*; that is to say, the sentences (27) and (28) are exactly synonymous. With this distribution, it would be possible to explain the presence of the absolute object in example (27) by a deletion of the sequences *pour (atteindre + s'établir à)* and *en (atteignant + s'établissant à)* in example (28).

This solution poses a problem, however. It does not explain why the constituents in question which are supposed to be the origin of the absolute numerical object do not have the mobility of the latter one. The fact is that objects in the form of *pour...* cannot be moved from final position to anywhere within the sentence. The gerundive, which, normally, can be moved quite freely as an adverb, does not have the mobility of an adverb in this corpus.

If it is allowed that example (28) is the complex sentence, it is difficult to restore the original discourse. The coordinated sentences make a slightly bizarre impression, they are not exactly synonymous with the sentence type (28):

(29) *La valeur ABC a (gagné + perdu) 0,4 % et elle (s'établissait à + atteignait) 32 euros*

This clumsiness seems to come from the conjunction *et (and)*, but the

inherent logic dominating the predicates and their objects in examples (27) and (28) must be expressed by a coordinated sentences like (29).

The fact that the transformational solution to introduce the absolute numerical object in the sentence (27) does not work well leads to the adoption of the local transformational solution. In this frame of analysis, I consider to be equivalent the sequences below:

(30) *pour (atteindre + s'établir à + coter) Dnum Nunité = en (atteiganant + s'établissant à + cotant) Dnum Nunité = à Dnum Nunité*

2.2.4. Adverbial phrases of time and location

2.2.4.1. Locative adverbs

The locative adverbs, which I symbolise by *AL*, present several formal variants but they correspond, in this corpus, all to one location, i.e. la Bourse de Paris. These variants are due to a truncation:

(31) *à la Bourse de Paris, à Paris, en Bourse, à la Bourse*

They all have adverbial mobility.

2.2.4.2. Adverbial phrases of time

It has to be observed that in this corpus, the time adverbs, which I symbolise by *AT*, when they appear with the sentences of variation, are to do with the notion of date, because they mark the punctual time of observation. The frequently occurring punctual time adverbs are organized around nouns of stock exchange market activities like *transactions, échanges* or *cotation* with the precision of time made by certain nouns whose meanings have inherently something to do with the notion of aspect, like *début, milieu*. Here are some examples:

(32) *(dans LE première(s) + au début de + au cours de) LE (transactions + échanges + cotation)*

Besides these specialized adverbial phrases of time, it is also possible to observe adverbs of time used in everyday language:

(33a) *dans la matinée du mardi 10 octobre = le mardi 10 dans la matinée*

(33b) *le mardi 10 = le mardi*

The examples (40) and (41a-b) can be combined:

(34a) *(dans LE première(s) + au début de + au cours de) LE (transactions + échanges + cotation) (du mardi 10 dans la matinée + de la matinée du mardi 10 octobre)*

(34b) *(dans la matinée du mardi 10 octobre + le mardi 10 octobre), (dans LE première(s) + au début de + au cours de) LE (transactions + échanges + cotation)*

2.2.5. Subjects

The subjects of predicates of variation can take the form of long sequences of noun phrases, as are seen below:

(35) *(L'action + le titre + la valeur) ((de + E) Npr + de N Npr + E)*[15]

The syntactic particularity of sequence (35) is the apposition of the proper noun to head nouns like *action* or *titre*, and that, with or without the appositive preposition *de*:

(36) *(l'action + le titre + la valeur) de ABC = (l'action + le titre + la valeur) ABC*

the sequences in the example (36) can ultimately be reduced to the following sequences:

(37) = (36) *((l'action + le titre + la valeur) + ABC)*

There is a group of nouns which describes the activity of a company and which serves to designate a company. Here are some examples: *constructeur d'automobiles, fabricant de téléphones portables, distributeur de produits cosmétiques, etc*. The head nouns like *constructeur, fabricant, distributeur* are nouns derived from verbs which describe the activities of companies. Example (38) shows a class of equivalence of the sentences describing the activity of a company:

(38) *ABC construit des automobiles = ABC est un constructeur d'automobiles*
ABC distribue des produits cosmétiques = ABC est un distributeur de produits cosmétiques

These nouns occupy the same paradigm as nouns designating the social or juridical status of a company like *société, groupe, enterprise, etc*. These nouns enter a classifying construction:

(39) *ABC est (une société (E + privée) (E + à responsabilité limitée) + un groupe)*

In example (35), they are to be found in the *N* position just before the proper noun. Here are some examples of the complete paradigm for subjects of predicates of variation:

(40) *(l'action + le titre + la valeur) de l'équipementier de télécommunication ABC*
= *l'équipementier de télécommunication ABC*
= *ABC*
= *l'équipementier de télécommunication*
= *(l'action + le titre + la valeur)*

[15] *Npr* here means proper nouns.

2.3. Conclusion of the linguistic analysis

With the aid of syntactic transformations, it was possible to group a variety of forms into one class, on the grounds of unity of meaning. This is true of the sentence level, where the notion of variation of value serves to describe a multitude of forms of sentences within one category, schematised as follows:

A V D% DN AT AL

and this is equally true of each constituent level. To take an example, a class of predicates has a paradigm like this:

V = (Vvi + Vvt +VsupEn en Vvn + VsupA à Vvn + Vsupt Dét Vvn)

In the following part of this article, I shall show that the type of transformational analysis and lexicon-grammar applied to the corpus is very easy to represent in the form of local grammars. The result of the automatic analysis corresponds to the linguistic analysis as follows.

3. Automatic syntactic parsing of the corpus

The syntactic description of the most frequent types of sentence in the corpus here examined is represented in the form of a matrix of lexicon-grammar. This result is exploitable with ease in the domain of automatic analysis of electronic texts.

3.1. Local grammars

The local grammars are graphs which have one initial state and one final state, linked together by several paths which contain boxes. The boxes contain words and if a automatic parsing starts from the initial state and arrives at the end state, matching one to one a word of text and a word in a box at each point on a path, the sequence of words found on that path is said to be recognised by the local grammar. Here I give some examples of local grammar:

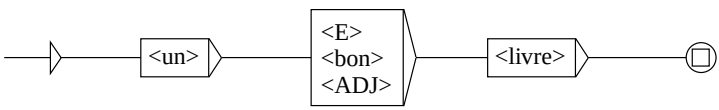

Figure 1: un bon livre

Figure 2: un livre (ADJ + E)

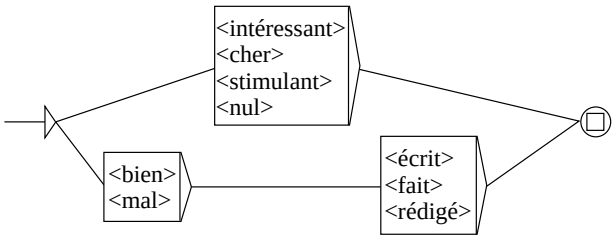

Figure 3: The ADJ graph

In these three graphs, the leftmost arrow represents the initial state, and the rightmost square in a circle represents the final state. The application of the first of the three local grammars given above recognises the sequences of words like: *un bon livre, un livre, un gros livre, etc*. In the graph, a word within the brackets represents a canonical form, if the word varies morphologically. The word *livre* within the brackets, for example, recognises in the texts both *livre* and *livres*. The upper-case letter *E* signifies a zero element, so that the graph of Figure 1. could recognise in the text the sequence *un livre*. The name of a grammatical category within brackets can match all the words that appear in the context that correspond to this category. So, the name of the grammatical category adjective represented by *ADJ* in brackets in the local grammar of the Figure 1. recognises an adjective like *gros* which appears in the appropriate context.

A local grammar can contain another local grammar within it, and the grey box within the local grammar of the Figure 2. is a call for an embedded graph. The embedded one here is the local grammar called *ADJ*, which is a graph with two paths appearing in Figure 3. The local grammar of Figure 2. recognises the sequences such as *un livre intéressant, un livre, un livre bien écrit, un livre nul, etc*.

3.2. Electronic dictionaries

In the previous section, it was stated that the local grammar of Figure 1. recognises whatever word in the text is tagged as an adjective. This procedure presupposes the application of electronic dictionaries to the words of the texts.

The INTEX system (M. SILBERTZTEIN (1993)) and the UNITEX system (S. PAUMIER (2003)) are platforms which permit an application of electronic dictionaries and local grammars to a text. The text analysed here is processed by these systems.

3.3. Organization of local grammars

It has been seen that the types of sentence described in section 2 are represented by simple syntactic schema, composed of a series of constituents clearly distinguished. This representation is quite easily transportable into a local grammar description. The graph I have created to parse this corpus is given below:

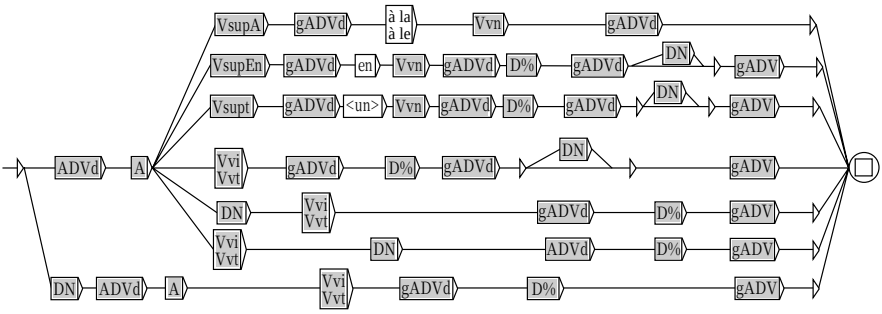

Figure 4: The local grammar for the sentences of variation

This graph is divided into two major parts. The three paths from the topmost path recognise sentences with support verbs, i.e. nominal sentences in the corpus. The last four paths in the graph recognise verbal sentences.

The grey boxes in the local grammar above correspond roughly to an embedded graph corresponding in turn to a syntactic constituent set for the sentences of variation. To take one example, the downmost path recognises in the corpus sentences like the ones below:

(41a) *(A 30 euros,)*$_{DN}$ *(l'action ABC)*$_{A}$ *(gagnait)*$_{Vvt}$ *(1,9 %)*$_{D\%}$ *(à l'ouverture de la Bourse de Paris, le lundi)*$_{gADV}$

(41b) *(A 30 euros,)*$_{DN}$ *(l'action ABC)*$_{A}$ *(,à l'ouverture de la Bourse de Paris, le lundi,)*$_{gADVd}$ *(gagnait)*$_{Vvt}$ *(1,9 %)*$_{D\%}$

(41c) *(A 30 euros,)*$_{DN}$ *(,à l'ouverture de la Bourse de Paris, le lundi,)*$_{gADVd}$ *(l'action ABC)*$_{A}$ *(gagnait)*$_{Vvt}$ *(1,9 %)*$_{D\%}$

In examples (41a-c), each part of the sentence recognised by a particular local grammar is bracketed and tagged by its name. For example, all three examples must start by recognising an absolute numerical, so the constituent *à 30 euros* is enclosed in brackets *DN*. The shape of the *DN* graph is given below:

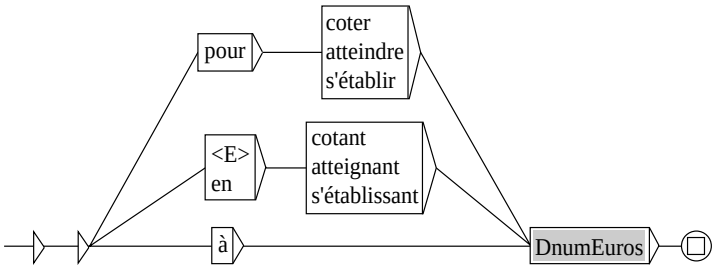

Figure 5: The graph *DN*

In the graph of Figure 5, the variant relations I have described (see §2.2.3.1.) for three forms of the absolute numerical objects are reflected. In this sentence initial position, as I have said, only a simple adverbial phrase *à Dnum* % can be observed. In the grey box *DnumEuros* in the local grammar above, the embedded graph describes a sequence of numeral determiners and the proper nouns of units of money.

The subject of the sentences is recognised by the following local grammar *A*. Local grammar *A* is designed for sequences of several noun phrases designating stock. There are several ways to designate stock, as has been mentioned (see § 2.2.5.). Here is an *A* graph:

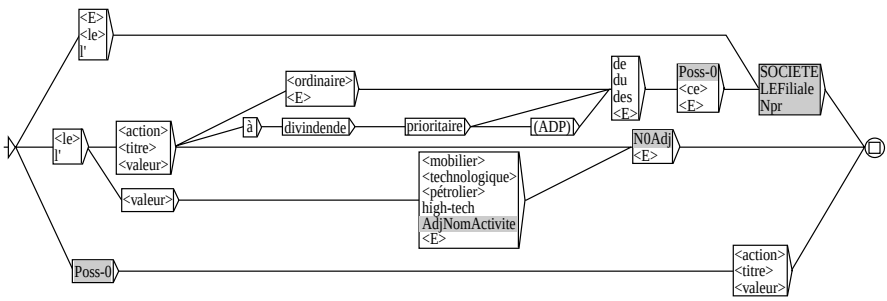

Figure 6: The local grammar recognising A

The local grammar of Figure 6. recognises 1) expressions whose head nouns are *action*, *titre* or *valeur*, followed by the proper noun of a company or a description of activity of a company (followed or not by a proper noun) and 2) metonymic expressions where noun phrases designating a company replace the head nouns:

(42a) *le titre du groupe de services de télécommunications d'entreprises Equant*

(42b) *l'action de l'éditeur de logiciels de jeux*

(42c) *le groupe de services de télécommunications d'entreprises Equant*
(42d) *l'éditeur de logiciels de jeux*
(42e) *Equant*
(42f) *EDF*

Examples (42a) and (42b) are the longest sequences which contain the head nouns *action* and *titre*, followed by noun phrases designating a company (*groupe de services de télécommunications* for (42a) and *éditeur de logiciels de jeux* for (42b)), with or without an appositive proper noun of a company (*Equant* for (42a), zero for (42b)). The central path of the graph corresponds to them and the embedded graph *SOCIETE* recognises a range of expressions for companies, including a proper noun. Examples (42c-d) can replace in this syntactic position the sequences (42a-b) and these equivalences are shown by the topmost path in the graph, which, after the determiner part of the graph, passes directly to the embedded graph *SOCIETE*. Example (42e) is equivalent to (42a-c). This is the most reduced form of (42a), a proper noun of a company functioning as a longer sequence.

Proper nouns like (42e-f) of company names are classed in a specialized *electronic dictionary* for proper nouns. This permits the tagging of these nouns in the corpus as *N+propre*.

In Figure 4., the predicative part *V* does not appear as such, but the subcategories of it appear individually. *Vvt*, *Vvi* group lexical local grammars of individual verbs, but support verb expressions are divided into two parts; on the one hand local grammars for support verbs like *VsupEn*, *VsupA* and *Vsupt*, and on the other, prepositional or non-prepositional predicative parts. *Vvn* groups the lexical predicative elements.

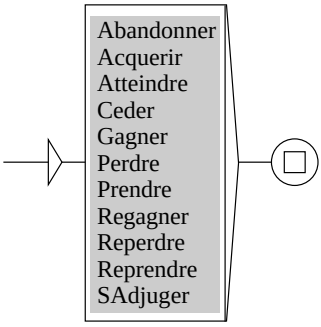

Figure 7: Vvt

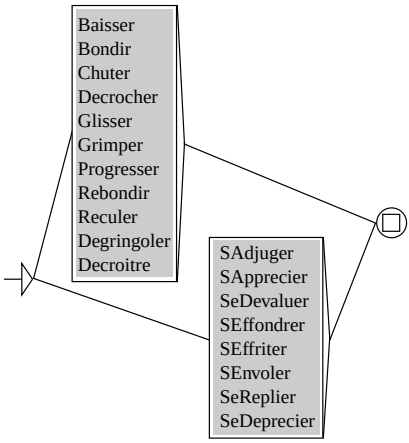

Figure 8: Vvi

The *D*% part is recognised by a following *D*% graph:

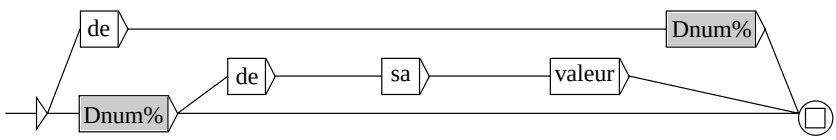

Figure 9: D%

In this local grammar, the two paths each represent the relative numeral object. The topmost path is for the indirect object and the one below is for the direct object.

Adverbial expressions of time and location, which I symbolized by *AT* and *AL* respectively, are here grouped under local grammar *ADV*, which takes the form of the following local grammar:

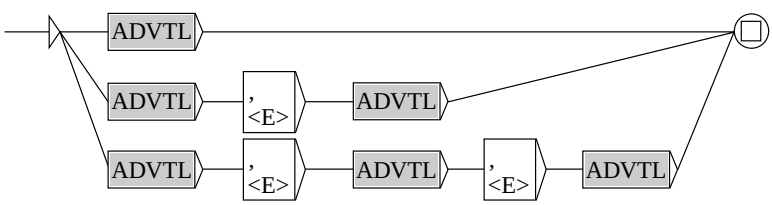

Figure 10: ADV

This local grammar shows a structure of juxtaposition for adverbial expressions. It admits a repetition of adverbial units up to a maximum of

three times. So, in examples like (43a-b):

(43a) *(à Paris)*$_{AL}$, *(le 6 mai dans la matinée)*$_{AT}$, *(lors des premières transactions)*$_{AT}$

(43b) *(lors des premières transactions)*$_{AT}$, *(à Paris)*$_{AL}$, *(le 6 mai dans la matinée)*$_{AT}$

it is possible to observe an adverbial expression three times. The bracketed part of the sequence shows a unit of recognition by means of a local grammar.

3.4. Results of application of local grammars

The application to the corpus of the main graph exemplified in Figure 4 yielded a recognition score of 22 % of the total text of corpus. The number of embedded graphs in the main graph amounts to approximately 250. Here is a partial example of a concordance:

(3) L'action Schneider Electric s'appréciait de 0,28 %, à 72,6 euros, mardi matin.(8) Le groupe a relevé de 18,1 % à
(3) L'action Sonr s'échangeait en nette hausse, de 1,48 % à 47,95 euros, à le début des cotations, mercredi 4 avril.
(3) L'action Sema gagnait 0,11 %, à 8,92 euros, vendredi matin.(3) Schlumberger Ltd. a annoncé avoir reçu le feu vert
(3) L'action Société générale gagnait 0,24 %, lundi matin, à 62,8 euros.(3) Selon l'édition de lundi du magazine all
(3) L'action Société générale progressait de 0,28 %, à 71 euros.(3) La banque française a confirmé mercredi matin le
(3) L'action Sodexho Alliance perdait 1,75 %, lundi matin, à 53,3 euros.(3) L'agence Standard & Poor's maintient sou
(3) L'action Sodexho enregistrait, en début de séance, une hausse de 3,92 %, à 53,95 euros après l'annonce par le gr
(3) L'action Sopra gagnait 5,73 %, à 67,90 euros, jeudi matin.(3) De son côté, le titre Cap Gemini Ernst & Young pro
(3) L'action ST Microelectronics a cédé 3,13 %, jeudi matin, à 37,09 euros.(8) Le titre Thomson Multimédia a perdu 3
(3) L'action STMicroelectronics a débuté la séance de jeudi en forte baisse, pendant 5 % à l'ouverture, à 47,50 euro
(3) L'action Stmicroelectronics gagnait 2,77 %, jeudi 12 juillet dans les premiers échanges, à 31,40 euros.(3) Le gr
(3) L'action Suez Lyonnaise des Eaux était en hausse de 3,45 %, à 138,6 euros.(3) Le groupe de services collectifs,
(3) L'action Suez Lyonnaise gagnait 0,17 %, à 179 euros, mardi matin.(8) Le Financial Times de mardi révèle que ST
(3) L'action Suez perdait 0,24 %, à 37,38 euros, mardi 7 août à l'ouverture de la Bourse de Paris.(3) Le groupe a dé
(3) L'action Suez s'appréciait de 0,94 %, vendredi dans les premiers échanges, à 37,46 euros.(3) Jeudi, des rumeurs
(3) L'action Technip chutait de 10,33 %, à 148,5 euros, jeudi 28 juin dans les premiers échanges, et le titre Cofiex
(3) L'action TF1 gagnait 1,73 %, à 55,95 euros, mercredi 31 janvier.(8) Le groupe a enregistré un résultat net provi
(3) L'action TF1 grimpait de 5,31 %, jeudi, dans les premières transactions, à 34,11 euros.(3) Les opérateurs ont ét
(3) L'action TF1 perdait 0,62 %, à 45,79 euros, vendredi matin.(3) La société a annoncé une hausse de 1,7 % de son c
(3) L'action TF1 s'appréciait de 1,72 % mardi matin, à 41,3 euros.(8) Le titre figure dans une liste, publiée par le
(3) L'action Thales gagnait lundi matin 1,05 %, à 43,45 euros. (8) Le groupe de cosmétiques Clarins doit publier son
(3) L'action Thales s'inscrivait en hausse de 1,43 %, mardi matin, à 44,02 euros.(3) Le groupe de électronique de dé
(3) L'action Thomson Multimédia grimpait de 4,2 %, lundi matin, à 51,6 euros.(3) Le groupe a enregistré en 2000 un b
(3) L'action Thomson Multimédia s'échangeait en hausse de 5,43 % à 48,38 euros.(3) Le groupe a publié de bons résult
(3) L'action TotalFinaElf gagnait 1,14 %, à 177 euros.(8) Le groupe pétrolier a obtenu 30 % de un contrat pour le va
(3) L'action Transgène rédait 0,86 %, lundi matin, à 11,50 euros.(3) Les 63 millions de euros levés dans le cadre de
(3) L'action Trédi Environnement reculait de 2,47 %, vendredi dans les premières transactions, à 39 euros.(3) Le spé

Figure 11: Concordance

The recognised sentences of the corpus are underlined in the example above. We have seen that the local grammars are organized according to a syntactic analysis of the sentences, which are decomposed into a sequence of several syntactico-semantic categories. Instead of producing a simple parsing like Figure 11, we could have produced a text which would be tagged according to this syntactic analysis. It is easy to do this using a transducer, which gives an output when an input is accepted. With this extension of local grammars, it would be possible to use results to run automatic translation in the future.

4. Conclusion

I started this analysis of the corpus to see if the Harrisian view of dis-

course analysis[16] is also applicable to a series of texts taken as a single discourse. In effect, there is a theoretical difference between a Harrisian analysis of discourse and what has been done in this study. The point Harris made in his studies was that "a coherent discourse can be reduced to sequences of formal categories", and what is done in this study is to see what the types of sentence are which are formally characterisable and which are highly recurrent. Harris took only one discourse to reduce to a sequence. I have taken a series of texts and reduced 22 % of the totality to a simple schema and I exclude the possibility of reducing all the sentences contained in this corpus analysed to a sole sequence of categories, but do not exclude the possibility of postulating several schemata for it.

[16] See HARRIS (1952, 1963).

ANNEX 1. Matrix of a specialized lexicon-grammar

A	Prédicats de variation		Vvi	Vvt																	
					VsupEn en Vvn De Dnu						Vsup à Dét V-n						Vsupt Dét V-n De Dnu				
N0 = ACTION	Vv ou Vsup Adj	Vvn	NVDeDnum%	NVDnum%	NVDnum%DePoss0	être	s'échanger	s'inscrire	pointer	se négocier	Vtps	être réservé	être orienté	s'orienter	reprendre	repartir	pointer	faire un V-n	enregistrer UN V-n	connaître UN V-n	afficher UN V-n	accuser UN V-n
+	apprécier(s')	(appréciation)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
+	augmenter	augmentation	+	-	-	+	?	?	?	?	+	?	?	?	?	?	?	-	?	?	?	?
+	baisser	baisse	+	-	-	+	+	+	+	+	+	+	?	?	?	?	?	-	+	?	?	?
+	bondir	bond	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	?	-	?	?
+	chuter	chute	+	-	-	?	?	?	?	?	-	-	-	-	-	-	-	?	?	?	?	?
+	décrocher	(décrochage)	+	-	-	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?
+	dégringoler	(dégringolade)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	-	-	-
+	déprécier(se)	(dépréciation)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?
+	effondrer(s')	(effondrement)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?
+	effriter(s')	(effritement)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?
+	envoler(s')	(envolée)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?
+	glisser	(glissement)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	-	-	-
+	grimper	(grimpée)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
+	progresser	progression	+	-	-	+	?	?	?	?	+	?	?	?	?	?	?	-	+	?	?	?
+	rebondir	(rebond)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?	?
+	reculer	recul	+	-	-	+	?	?	?	?	?	-	-	-	-	-	-	-	+	?	+	+
+	replier(se)	repli	+	-	-	+	?	?	?	+	+	?	?	?	?	?	?	?	?	?	+	?
+	reprendre(se)	(reprise)	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?	?
+	stable(être + rester)		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
+		hausse	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	?	+	+	+	?
+	abandonner	(abandon)	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
+	adjuger(s')		-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
+	céder	(cession)	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?
+	gagner	gain	-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	+	?
+	perdre	(perte)	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?	?
+	prendre		-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
+	regagner	(regain)	-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	?	?	?	?

Note: This matrix is a lexicon-grammar tailored to my corpus. Intersections where we find an exclamation mark signify the fact that we cannot find the sentence in the corpus, whereas minus signs in columns signify agrammaticality. Words in brackets in the *Vvn* column are those not found in the corpus.

ANNEX 2. Example of *Valeurs France*

LE MONDE | 17.08.01 | 12h46

L'action Avenir télécom s'envolait de 21,58 %, vendredi 17 août dans les premiers échanges, à 2,33 euros. Le distributeur de produits et de services de téléphonie a enregistré une hausse de 30,5 % de son chiffre d'affaires annuel pour l'exercice 2000/2001, à 1 004 millions d'euros. A périmètre comparable, la progression ressort à 24 %.

Le titre Fi System bondissait de 7,48 %, à 3,45 euros, vendredi matin. L'agence Web a vu son chiffre d'affaires consolidé reculer de 13 % au deuxième trimestre, à 13,26 millions d'euros. Sur le semestre, l'activité reste en hausse de 1,5 % à 29,15 millions d'euros.

Le titre Ipsos cédait 0,58 %, vendredi, à 68,5 euros. La société d'études a annoncé un chiffre d'affaires consolidé en hausse de 64 % au premier semestre, à 217 millions d'euros. A périmètre constant, la hausse s'établit à 9,9 %.

L'action Tredi Environnement reculait de 2,47 %, vendredi dans les premières transactions, à 39 euros. Le spécialiste du traitement et de la valorisation des déchets nucléaires a annoncé un chiffre d'affaires en hausse de 5,4 % au premier semestre, à 81 millions d'euros. Ses dirigeants ont néanmoins précisé qu'ils anticipaient une croissance à deux chiffres de l'activité au cours du second semestre.

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Multivariate Analysis in Dialectology – A Case Study of the Standardization in the Environs of Paris –

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1. Introduction

The present article is a case study of a multivariate analysis applied to French dialects. We will examine here the problems of standardization of French dialects in the environs of Paris. Our dialect source comes from three volumes of *L'Atlas Linguistique et Ethnographique de l'Ile-de-France et de l'Orléanais (ALIFO)*, which was edited by Mme Marie-Rose Simoni-Aurembou and published by C.N.R.S. at 1966, 1969 and 1978. It is composed of 687 maps with 76 research points.

We know that dialect differences are qualitative in nature. But we also recognize that quantitative analysis of dialect differences has much influence on dialectology. In the history of quantitative analysis of French dialects, three different streams must be taken into account among the previous studies.

The first stream is the traditional study of dialect boundary or division. The combination of the frequency of words with their geographical distribution has been an effective method for the demarcation of dialect boundary or division. Therefore, the frequency of dialect forms and their geographical distribution are traditionally considered as a discrete phenomenon rather than a continuous one. It can be said that in traditional dialectology, the differences in frequency and distribution of dialects have been treated as qualitative differences.

The second is the study of language standardization. WOLF (1970), DAHMEN (1985) and KAWAGUCHI (1994) are all based on the "simple statistics" of the questionnaire. In order to examine the standardization process, DAHMEN 1985 introduced a quantitative method into his analysis of *L'Atlas Linguistique et Ethnographique du Centre (ALCe)*. WOLF (1977) and KAWAGUCHI (1995) are the two papers relevant to the standardization in the environs of Paris. We will review their results later.

The third stream is the statistical analysis of dialect differences. SÉGUY (1971) was the first full-scale statistical analysis in French dialectology.

More recently, the quantitative dialectology at Salzburg University deserves special attention. GOEBL (2002) obtained important results from a large-scale quantitative analysis of *L'Atlas Linguistique de la France (ALF)*. We will comment on his paper in the next section.

As far as the multivariate analysis in dialectology is concerned, we should refer to some studies in Japanese dialectology. In Japan, the national-scale linguistic atlases, such as *The Linguistic Atlas of Japan (LAJ)* and *The Grammar Atlas of Japanese Dialects (GAJ)*, have been published since the 1960s by The National Institute for Japanese Language (NIJLA). The databases of these atlases are being constantly updated and released on the Internet.

The new trend in Japanese dialectology since the 1980s is similar to that in European dialectology, i.e. an increasing use of statistical techniques. For instance, having applied the factor analysis to the standard Japanese usage for every prefecture of LAJ, INOUE and KASAI (1982) clarified the relevant factors for the historical formation of the standard Japanese. In his article on the geographical and historical constitution of Japanese, INOUE (1986) applied to the data of GAJ "the quantification method type three" developed by Chikio Hayashi (almost the same technique as the correspondence analysis; see also KAWAGUCHI and INOUE (2002): 816-829).

On the other hand, SIBATA and KUMAGAI (1985) invented their original calculation method called the "network method" intended for the examination of the similarity among research points, and tried to establish the quantitative divisions of Japanese dialects.

2. Previous Studies

2.1 GOEBL's dialectometrical analysis

We will review here some results of previous quantitative studies on French dialects. GOEBL (2002) is the first large-scale dialectometrical analysis of ALF. Having calculated the dialect similarity between adjoining points, he showed that the regions, such as Walloon, Limousine, and Franco-provençal, are not so distant from their contiguous dialect areas. As these regions are considered as independent dialect regions in the traditional dialectology, GOEBL's findings will arouse a controversy against the traditional view, GOEBL (2002): 17-18 and Carte 1, p. 40. In his article, setting up an imaginary standard point, GOEBL calculated the relative similarity of each point, and plotted it on the map, ibid., Carte 2, p. 41. He illustrated how the standard French spread in all directions from that imaginary standard point. The process of standardization reconstructed by GOEBL deserves attention. GOEBL also applied the cluster analysis to the similarity matrix for every point. In his analysis, the most effective methods of clustering were the com-

plete linkage method and the Ward method. The dendrogram showed clearly the diachronic divergence from Latin to French, ibid., p. 32.

We regret, however, that in GOEBL's article, the procedure in measuring the similarity between research points, i.e., "the procedure of taxation" in GOEBL's terminology, needs more explicit explanations. The procedure is described too briefly in his paper, ibid., pp. 10-11 and we cannot imagine how lexical or phonetic traits are classified and integrated in his analysis. Retestable procedures should have been indicated in his paper.

2.2 Analyses of ALIFO

In the following lines, we shall have an overview of the previous studies of ALIFO. But there are only few articles on the quantitative analysis of ALIFO.

WOLF (1977) analyzed the standard forms based on 23 maps. We can see 76 research points (from 0 to 75) of ALIFO in Fig.1. In the outskirts of Paris (points 0, 1, 2, 3, 4, 5, 6, 8, 13, and 27), the standard forms are observed most frequently. Especially, at the points 2, 4, 5, 8, and 13, not only the forms but the pronunciations are almost the same as those of the standard language. On the contrary, at many points in Eure-et-Loir prefecture (points 17, 24, 25, 26, 30, 31, 37, 38, 39, 46, 47, 48, 50, 54, 55, 58, and 59), the dialect form is completely different from the standard French. WOLF pointed out that the Loire Valley in the south shows an intermediate stage between the above two areas except for Tours (point 73), where the standard forms can be observed as often as in the environs of Paris.

Figure 1: Research points of ALIFO

KAWAGUCHI 1994 analyzed 20 maps and classified them into the following three categories.

(1) maps in which dialect forms are current in half of the research points
(2) maps in which standard forms are used at 46-62 points among the total 76 points
(3) maps in which standard forms are attested at almost all points

Having examined 8 maps of category (2), KAWAGUCHI supposed that the diffusion of standard French might take two different directions starting from Paris. One goes straight towards the west through the points 6>8>16>34, while the other moves first southward through 13>15>60, and then westward, along the Loire River, through 60>53>54>62>65>73>74>75, ibid. p.269 and CARTE 2. Like WOLF, he also pointed out that many non-standard forms are used in Eure-et-Loir. He suggested that the dialect forms found in Eure-et-Loir was related to the survival of old forms, ibid. p. 270 and CARTE 3. After this pilot analysis, KAWAGUCHI added 32 maps to his database. In our analysis, 51 maps of ALIFO will be analyzed, one map being omitted for its irrelevance.

3. Characteristics of ALIFO Data

3.1 Working Hypothesis

In the field study of ALIFO, French native informants born around the year 1910 were surveyed. Dialect forms are registered in order of their frequency, but the notes are added when they are rare in use. It would be safe to think that dialect forms at a given point represent the most frequent forms. However, if we look at ALIFO maps, we can find more than two answers for a single question in many points. This means that there are more than two forms in concurrence for a given question at those points. We must therefore posit the following two working hypotheses.

Hypothesis 1: The area in question is under the progress of standardization.

The findings of WOLF and KAWAGUCHI as well as the social situation of this region lead us to presume that the progress of standardization had already advanced particularly in Paris and its surrounding area in the 1970s.

Hypothesis 2: The standardization process is both one-way and irreversible.

Even if dialect forms and standard French coexist in a given point, we can assume that dialect forms are in decline and standard French is continuously in progress. If standard French and dialect form are completely different, the dialect form will be replaced by a form similar to standard French, i.e. a phonetic variant of standard French. In case the dialect is close to standard French in form, but not in pronunciation, we should suppose that the dialect form has a tendency to shift phonetically to the standard form. In the area shown in ALIFO, where standard French and dialect forms coexist, dialect forms can never be predominant over standard French. In this sense, the standardization process in this region is not only one-way but also irreversible.

How can we then determine the most representative form among two or more answers at every point of ALIFO? We take into consideration the following two opposite cases.

Case 1: If there is an answer similar to <u>standard French</u>, it is chosen unconditionally as representative. If there is any single standard form accepted at a given point, we consider this point as the point of standard French. As a consequence, this procedure will bring out in full relief the points which do not accept standard French.

Case 2: If there is an answer similar to non-standard French, it is chosen unconditionally as representative. As diametrically opposed to case 1, this procedure will explain the survival of old forms and the direction of standardization.

In comparing the results of these two different procedures, we believe that the historical stages of standardization will be explained.

3.2 Creating the Database

Dialect forms are transcribed electronically and registered at each point of the ALIFO database created by KAWAGUCHI. Based on this database, he also created, for the present analysis, a new database in assigning a specific value (generally 1 to 3, but in some cases even 4 and 5) to each point according to the following criteria.

Value = 1: Standard form
("standard" means here "in agreement with" the pronunciation of the Dictionary of MARTINET and WALTER 1973)

Value = 2: Phonetic variant of standard French
Value = 3: The others (dialect forms which cannot be regarded neither as standard nor as its variant)

In some cases, he assigned the values 4 and 5 (see Table 1). In order to solve the problem of the evaluation of more than two variants at a given point, we divided the present database into two types of data according to the procedures of Case 1 and Case 2. The procedure consists in selecting automatically the most representative value assigned at each point. "Standard preference form data" (=Case 1) chooses the minimum value, and "non-standard form preference data" (=Case 2) the maximum value. In the next section, we name the former "SP-data" and the latter "NP-data".

4. Analysis

4.1 Simple Statistics

We will first investigate the situation of standard French in 51 maps with the simple statistical method. By calculating the values of NP-data and SP-data, we can depict the results on two maps, Fig. 2 for NP-data and Fig. 3 for SP-data respectively.

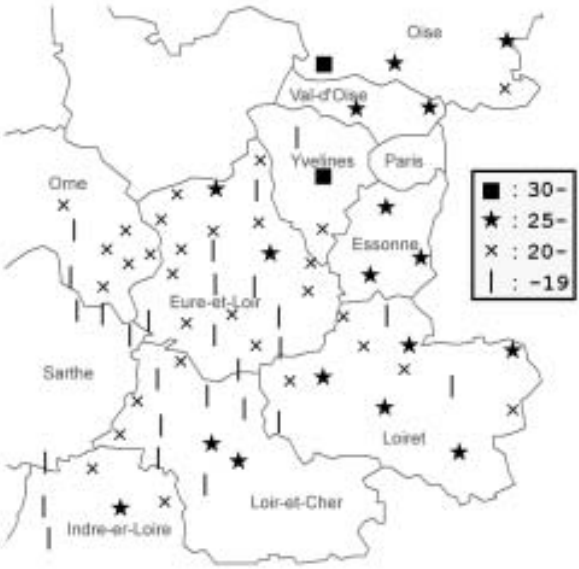

Figure 2: Simple statistics of NP-data

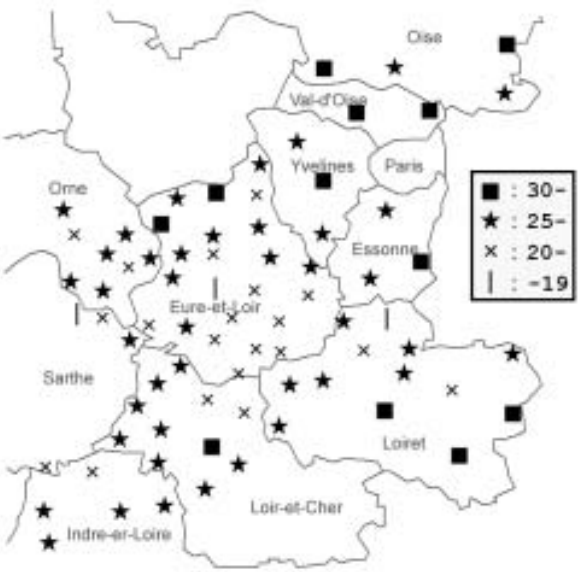

Figure 3: Simple statistics of SP-data

In Figs. 2 and 3, it is clear that Fig. 3 of SP-data demonstrates the standardization process and Fig. 2 of NP-data reflects the dialect situation before the standardization. We can confirm that the standardization goes slowly in Eure-et-Loir prefecture (see the legends | and × respectively). In Fig. 2, the standardization spreads from the northern area to the southeastern area, whereas in Fig. 3, the expansion of standard French circumscribes Eure-et-Loir. This tendency is roughly demonstrated in KAWAGUCHI 1994. He also assumes two directions of standardization. However, KAWAGUCHI's assumption is solely based on the words for which the standardization has already progressed to some extent. It seems difficult to discern such two courses of standardization in Fig. 2 and 3, because both data are shown in averaged figures.

Simple statistics can show us no more than the different degrees of standardization. The comparison between SP-data and NP-data is not sufficient, if one wants to clarify the lexical variation in the standardization process, e.g., the fact that some words are likely to be standardized and some words are not. Even in the areas where the standardization has relatively advanced, it is not always the case that the same words have been standardized at a given point. In other words, in the quantitative analysis of standardization, it is important to calculate at the same time, not only the words which have already been standardized, but the words which have not been standardized. Multivariate analysis is convenient for that purpose.

4.2 Cluster Analysis

4.2.1 Selection of Methods

Our attention will be focused on the fact that the patterns of word usage are common to some points of ALIFO. The cluster analysis is a popular method for the further classification of SP-data and NP-data into some different groups. The application of the cluster analysis must choose a suitable distance and the clustering algorithm. Since the scores of SP-data and NP-data are in ordinal scales, we selected the Manhattan distance and the complete linkage method. The calculation for the cluster analysis was effectuated by STATISTICA 2000 (Release 5.5.).

4.2.2 Cluster Analysis 1 (non-standard form preference data)

The results of NP-data is shown in the dendrogram of Fig. 4 and the map of Fig. 5.

Two major clusters, A and B, are easily distinguished in Fig. 4. Geographically speaking, Cluster A (● , ★) is attested in the outskirts of Paris, while Cluster B (□ , ▽ , | , −) in the southern area, see Fig. 5. Cluster A, considered as the core area of standardization, is found in the northern part of

Eure-et-Loir and also in the northern part of Loiret. The geographical distribution of Cluster A seems to reconstruct the early stage of standardization which originated from Paris.

Cluster A is further divided into two subclusters. Cluster A1 (●) includes, on the one hand, Oise and Val-d'Oise prefectures where the influence of Picardie dialect can not be excluded (points 0, 1, 2, 3, and 5), and on the other hand, some points in Essonne and Loiret prefectures. Cluster A2 (★) seems to circumscribe the outskirts of Paris, but is concentrated in the region more or less influenced by Normandie dialects.

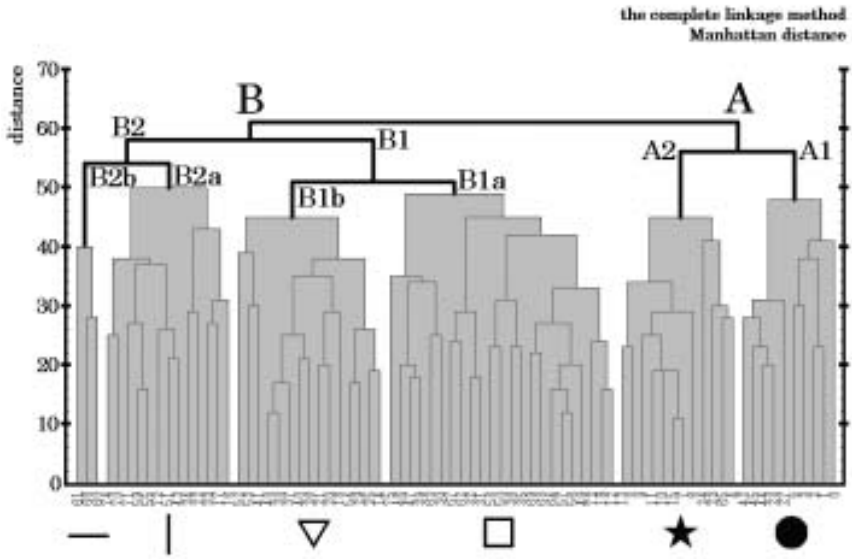

Figure 4: Dendrogram of NP-data
(non-standard form preference data)

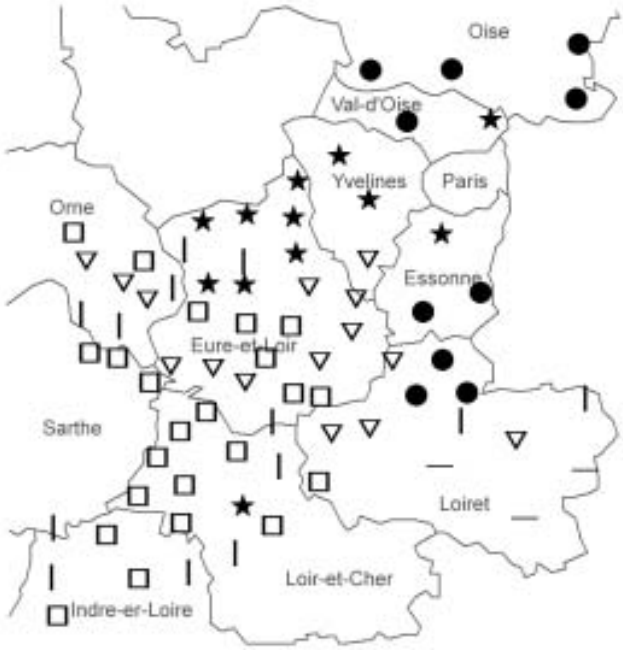

Figure 5: Map of clusters in Fig. 4

Cluster B is attested in the southwestern part of our region, but shows a slightly complicated distribution. The subcluster B1 (□ , ▽) is found mainly in the southern part of Eure-et-Loir and Loir-et-Cher. The subcluster B2 (| , −) is distributed in both the north and the south, sandwiching therefore Cluster B1.

Cluster B2 is separated from Cluster B1 according to the progress of standardization. It means that Cluster B1 represents the area where the standardization began late, while Cluster B2 shows the tendency towards the standardization. In addition, it can be said that the north part of Cluster B2 seems to follow Cluster A2 and the south part Cluster A1. As a consequence, we can here discern two different courses of standardization in both the north and the south of Eure-et-Loir. This confirms the assumption of two directions of standardization in KAWAGUCHI 1994.

4.2.3 Cluster Analysis 2 (standard form preference data)

Now we will examine the results of the cluster analysis of the SP-data. On the right side of the dendrogram of Fig. 6, Cluster A (●,▲,▼,◎,回) seems to represent standard French. The points belonging to Cluster A have been more numerous than those in NP-data (see the legends ● , ★ in Fig. 4). This means clearly that the standardization process has constantly advanced in ALIFO.

In Fig. 7, unlike the results of simple statistics (see ■ , ★ in Fig. 2), the area of standard French does not seem to expand. Cluster A occupies the whole western part of the outskirts of Paris, and also the points located around the edge of the western part of ALIFO (see especially ◎ in Fig. 7). It can be said that as Cluster A spreads over the western edge, the standardization expands throughout ALIFO.

The distribution of Cluster B (| , − , × , ∧) appears to be sandwiched by Cluster A. Although Cluster B2 (∧) occupies the northern part of Eure-et-Loir, it belongs to the standard French area. Cluster B1 (| ,−,×) is attested in the south of Cluster B2. Cluster B1 will be further subdivided into Clusters B1a and B1b, and Cluster B1a having two subclusters of B1a1 and B1a2 (see the dendrogram of Fig. 6). Cluster B1a1 (−) located in the south of Eure-et-Loir is circumscribed by Cluster B1b (×) and Cluster B1a2 (|). The relative distance among these subcategories of Cluster B will be interpreted in the following increasing order: B1a1 and B1a2<B1b<B2. This distance corresponds to the geographical distance from the area of Cluster A1a (●) near Paris, i.e. the area of standard French. Thus, if we leave the area of A1a for the southwest, we can find first B2 (∧), then B1b (×), and finally B1a2 (|) and B1a1 (−). This relative distance is correlated to the progress of standardization.

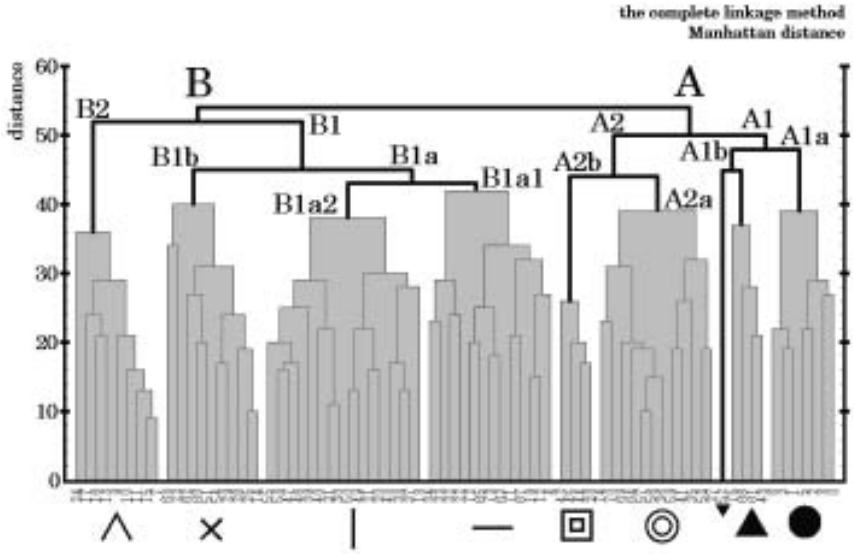

Figure 6: Dendrogram of SP-data
(standard form preference data)

Figure 7: Map of clusters in Fig. 6

This fact may confirm the presence of two different courses of standardization, i.e., one on the north side of Eure-et-Loir where the speed of standardization is slow, and the other on its south side. The correlation between the classification of the clusters and their geographical distribution is clearer in Fig.7 than in Fig. 5. However, for a better understanding of the course of standardization, it is necessary to verify the data of the eastern Normandie and of the southern Orléanais and Touraine, though these regions are located outside of ALIFO.

4.3 Multi Dimensional Scaling

4.3.1 Application of Multi Dimensional Scaling

In the previous section, we tried to classify SP-data and NP-data into some groups by means of cluster analysis. The progress of standardization can be seen in the SP-data and the decline of dialects in the NP-data. We could show not only the diachronic distinction between the SP-data and the NP-data, but also the process of standardization and the survival of the dialect forms. However, the problem still lies in the fact that the same values from 3 to 5 are assigned indiscriminately to different variants (see Table 1 in Appendix). Therefore, if the quantitative analysis is focused on the dialect forms, the results can be quite different. In order to examine more objectively the geographical continuity and the mutual similarity of the research points, we will apply to our data the technique of "Multi Dimensional Scaling (MDS)".

MDS is the method of arranging cases in n-dimensional space so that the relation between the cases would reappear as clearly as possible based on the matrix of the distance or similarity between the cases. Since we usually set two dimensions in MDS analysis, the relation can be visually displayed on the plane of two dimensions. In our MDS analysis of dialects, the points are regarded as cases and the similarity between the points is calculated. The relative locational relationship between the points obtained by MDS must reflect to some extent the actual geographical relationship, since, generally speaking, linguistic changes advance gradually from one point to its neighboring point. Our objective is to examine the geographical expansion of standard French through the distribution of research points arranged by MDS.

4.3.2 The Calculation Method of Dialect Similarity

Fig. 8 explains briefly the calculation method of dialect similarity between research points. The matrix of dialect similarity must be elaborated before the application of MDS. For each question, the answers of each point are compared one by one, and the numbers of the cases of congruity are counted and registered in each cell. Then the similarity matrix was processed

by MDS. All the questions of 51 maps of ALIFO were taken into consideration. Different from the cluster analysis of SP-data and NP-data, MDS investigates all at once the congruity between the points of ALIFO, so that all the answers in 51 maps can be examined efficiently.

database

	Word Forms		
	Q1	Q2	Q3
Point 1	*a*	*j*	*x*
	b		
Point 2	*a*	*j*	*y*
Point 3	*b*	*k*	*x*
	c		*y*

matching

Q1	Point 1	*a*	○	*a*	Point 2
	Point 1	*a*	×	*b*	Point 3
	Point 1	*a*	×	*c*	Point 3
	Point 1	*b*	×	*a*	Point 2
	Point 1	*b*	○	*b*	Point 3
	Point 1	*b*	×	*c*	Point 3
	Point 2	*a*	×	*b*	Point 3
	Point 2	*a*	×	*c*	Point 3
Q2	Point 1	*j*	○	*j*	Point 2
	Point 1	*j*	×	*k*	Point 3
	Point 2	*j*	×	*k*	Point 3
Q3	Point 1	*x*	×	*y*	Point 2
	Point 1	*x*	○	*x*	Point 3
	Point 1	*x*	×	*y*	Point 3
	Point 2	*y*	×	*x*	Point 3
	Point 2	*y*	○	*y*	Point 3

similarity matrix

	Point 1	Point 2	Point 3
Point 1		2	2
Point 2	**2**		1
Point 3	**2**	**1**	

Figure 8: Calculation of the dialect similarity

4.3.3 The Result of MDS

In our MDS analysis, we plotted the dialect similarity on the two dimensional plane. The results of MDS show only the relative relationship of research points and they are susceptible to free rotation (orthogonal rotation or oblique rotation), to enlargement, or to reduction. The procedure of rotation is not based on the objective judgment, but on the investigator's arbitrariness. In order to determine the most suitable rotation, the scatter diagram with new axes was plotted on the original geographical location of the points (see Fig. 9). The legends on the scatter diagram represent the different prefectures.

As in the factor analysis, the variables distant from the origin of coordinates should be treated as peculiar behavior of data in the MDS analysis. On the reversed axis of the first dimension (DIM. 1), the area adjacent to Picardie is placed on the right (= north) and Indre-et-Loire on the left (=

south). On the axis of the second dimension (DIM. 2), Orne is located on the top (= west) and Loiret on the bottom (= east). This constellation of dialect similarity reflects more or less the real geographical distribution of the points (see Fig. 10).

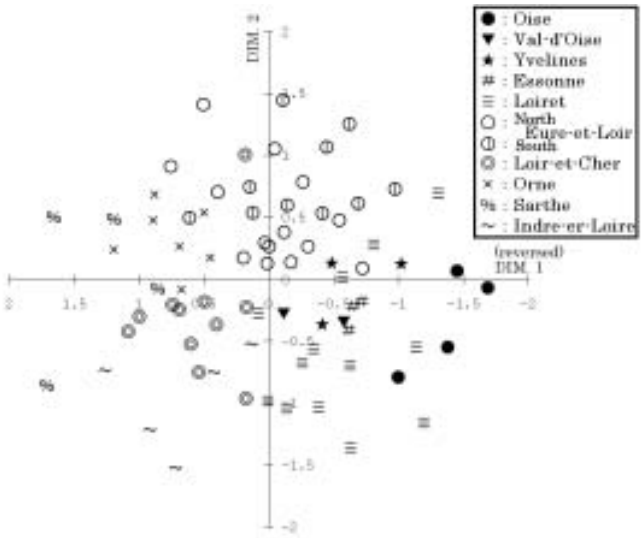

Figure 9: Scatter diagram of MDS (with the legends of prefectures)

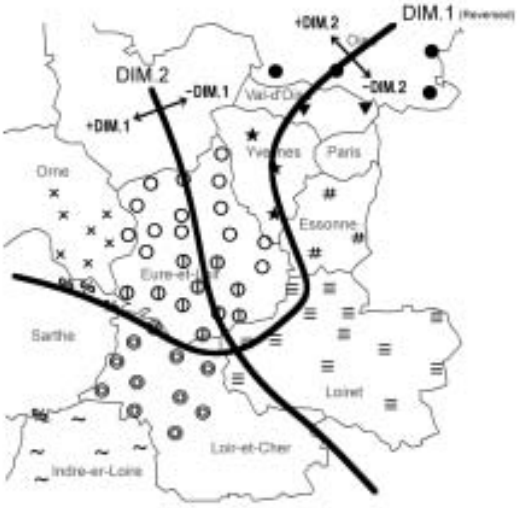

Figure 10: MDS output with two axes plotted on real ALIFO map

In our cluster analysis, several clusters could reflect the actual geographical relationship. But the correlation between the subclusters was not very clear. This correlation will be clarified to some extent in the MDS analysis.

The case adjacent to the origin indicates the case weakly influenced by each factor of MDS. On the contrary, we may assert that the points with the highest score of standard French are situated on the right (= north) near the origin of coordinates (see the legends ★, ▼, ● in Fig. 9). We can assume that Paris is also located near the origin, but is not the origin itself. Generally speaking, dialectal characteristics do not necessarily disappear under the use of standard forms. And the language of Paris is also influenced by Ile-de-France dialect which has a close historical relation with Picardie dialect. In this sense, it is not surprising to find Paris in the northern area rather than at the origin of the two axes.

On the other hand, looking at the position of Eure-et-Loir, the northern part and the southern part are reversed in Fig. 9. Some points of the southern part (⦶ in Fig.10; Cluster B1a1 in Fig.7), where the standardization is going on slowly, are located far from the origin, while the northern part (○ in Fig. 10), where the standardization is accelerating, is close to the origin. We can also see that the southern course of standardization is situated near the origin of coordinates.

The MDS analysis enables us to recognize that the similarity among the points of ALIFO reflects the geographical continuity quite faithfully. Such continuity is about to disappear from ALIFO with the progress of standardization.

5. Conclusion

The present paper is an example of dialectometrical analysis of l'Ile-de-France and l’Orléanais by means of two multivariate methods, i.e., the cluster analysis and the multi dimensional scaling. With the multivariate analyses, we could give a visual presentation of the courses of standardization, which had already been pointed out in previous studies. We could clarify the relationship between the geographical distribution and the dialect distribution. Multivariate analyses allowed us to test in a more objective way the validity of hypotheses postulated in the simple statistical analyses.

Our statistical procedures are highly reliable for their retestability. Our particularly important point of view is to examine not only the standardization process, but also the diachronic aspects of standardization. The diachronic description of standardization was realized through the utilization of two distinct types of synchronic data, i.e. standard language preference data versus non-standard language preference data.

Finally, some problems still remain. In this paper, we analyzed only less

than ten percent of the enormous bulk of dialect data of ALIFO. We had also no time to examine the lexical aspects of standardization. The correspondence analysis seems to be a more suitable multivariate analysis for lexical diffusion. But it will be a future subject.

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Appendix: Table 1 Database of 51 maps

Atlas No	Title	Value 1 to 5 (1 = standard form)	Meaning
Sowing and Cultivation			
28	semoir	1. sėmwàr; 2. sèmwèr, sėmwa^{e}, sėmwa^{ê}r, sėmwe, sėmwèr, sėmwé, sémwèr, semoé, semwé, smwàr, smwèr, sœ^{u}mwàr, sumwèr	seeder
29	semer	1. smé, sėmé; 2. (¢-sum), õ-soèmè, õ-sumè, sèmé, soèmé, sumé; 3. (õn-é-ãkuvréy), ãblèvé, ãblévé, àfyé, ãsmãsé, ¢té, fèr-là-kuvráy, fèr-lá-smáy, fèr-lé-kuvrày	to scatter (seeds)
35	herser	1. ersé; 2. àrsé, a^{ê}rsé	to harrow
37*	seigle	1. sègl; 2. sè$_{ê}$gl, sèg, ség; 3. sèy	rye
41*	avoine	1. àvwàn, à$_{v}$wàn, à$_{v}$wa^{e}n; 2. àvwèn; 3. àwàn, àwèn	oats
47	coquelicot	3. kòk, kó; 4. pàvó, pã, pã^{e}, pã¢é, pãsó, pãsyó, pẽsyó, põsó, põsyó, põsyó$_{u}$, pwẽsyó	poppy
Plough and Ploughing			
62	charrue	1. ¢àrü; 2. ¢árü, ¢á^{e}rü; 3. ¢árü-vèrswèr, ¢árü$_{y}$, ¢èrü$_{y}$, ¢a^{o}rü, ¢èrü; 4. bràkér, kutèt, vèrswè, vèrswa^{ê}r	plough
Hay			

117	regain	1. rgẽ, rėgẽ; 2. rgẽ, rgẽ^{i}; 3. pé:si, rpu:s, rpusi, rpusõ, rpusrõ, ryẽ, ptit-èrb, ?, x	second crop of hay
119	râteau à foin	1. rátó, ràtó; 2. rätó, rä:tó, ra^{o}tó, rátyó, ra^{o}t'yó; 3. rátèl, rátó-à-jàvlòté, rátó-à-mẽ, ra^{o}tó-à-fó¢é, ròtó; 4. fò¢è, fó:¢è, fó:¢é, fó¢è, fó¢é, +	rake
121-1*	faux	1. là-fó; 2. là-fà$_{o}$	scythe
136*	andain	1. ãdẽ; 2. õdẽ; 3. àkòlé, ròt, ròtẽ, rõdẽ, sãgl, urdõ	swath
138*	faner	1. fáné, fàné; 2. fànè, fènè, fėnè, fėné, fné; 3. dèròtè, ègáyè, èga^{o}yé, fèr-lá-fánás, fèr-lé-fwẽ, jté, rfoèné, +	to make hay
Harvest			
163*	faucher	1. fó¢é; 3. àdósè, èroèlvé, jėté, ku¢é, mwàsòné, pik'é, plàsé, ràmé, vèrsé, +, x	to reap
167*	javelle	1. jàvèl; 2. jàvè$_{ė}$l, jàvél; 3. jàvlè, jàvló; 4. bràs, bràsé, bràse^{i}, bràsi, jèrbé, mãdé, pàjó, wázõ, wa^{o}:zõ, wa^{o}zõ	swath
184*	glaner	1. glàné, gláné; 2. glàné$_{y}$, gla^{e}né; 3. !glèné, g'ėné, glèné, gléné	to glean
Threshing			
188*	fléau	1. fléó; 2. flàyó, flèyó, fléo^{oè}, fléyó, fliyó; 3. fló, flu, fyà$_{ó}$, fyó, fyó$_{u}$, fèlyó, fèyó, fėlyó, fėyó, fiyó; 4. bàt, ba^{o}t	flail
192	cribles	1. kribl; 2. krib; 3. !krüb, !krübl, kroèbl, krüb, krübl, krübl (v), grib, gribl; 4. gèrl, goèrl, pàswèr, pá:swèr, páswár, páswèr, páswer, pas'swèr, tàmi	riddle
193-1	van	1. ẽ vã; 2. ün-va^{o}n; 3. ẽ -pti-vã, ẽ -vã-à-mẽ, le-pėti-vã, +	winnowing basket
Orchard and Vegetable Garden			
248	plantoir	1. plãtwàr, plãtwár; 2. plãtwa^{è}r, plãtwà$_{è}$r, plãtwè, plãtwèr, plãtwé; 3. platoé$_{y}$, plãt¢u, plãtoé, plãtoé$_{y}$; 4. ¢e&viy, ¢éviy; ¢ẅiy, fi¢ó, fi¢wèr, pi$_{ė}$kè, pik¢u, pi$_{ė}$kè, pikè, piké, pikèt, piké-à-plãté, pikó, pikwé, rpikwé	dibble
256	cerise	1. sėriz; 2. sèriz; 3. g'iñ, gin, giñ	
259*	noisetier	1. nwàztyé, nwàzėtyé, nwáztyé; 2. nòzėt'é, nòzė$_{ė}$tyé, nòzėtyé, nwàzètyé, nwàzėtyèr, nwàzt'yé, nwázėté, nwázt'é, nwázt'yé, nwa^{è}zėt'é, nwa^{è}zètyé, nwa^{è}ztyé, nwaztyé, nwètyé, nwèzèt¢é, nwèzėt'é, nwèzėtyé, nwèzt'yé; 3. nòziyé, nuzéyé, nuziyé, nwàziyé, nwéziyé; 4. ku:dr, kudr, kudriyèr, kudriyé, X	hazel tree
269	pomme de terre	1. pòm-dė-tèr; 2. pó$_{t}$-tèr, pò$_{t}$-tèr; 4. kàrtu¢, pátát, pàta$_{ė}$rn, pàtàt, patat, patat (v), pétàt, pėnifl, pėnüf, pėtòk, pėtoèy, pètò¢ (r), pètòk (r), toè¢, trü¢, trüf, trüf (r), trüf (v), !trüf!, !trüf (r), !trüf (r.v.), !trüf (v), trüfl, trüfl (v), trüfyė, truf (r), trufl	potato
276	haricots verts	1. àrikó-vèr; 2. àrikó-vè, àrikó-vér, àrikò-và$_{è}$r, àrikò-vèr, àrikò-vér; 3. àrikó, àrikó-à-filè, àrikó-ã-làt, àrikó-ãn-ègẅiyèt, àrikò-ãn-ègẅiy; 4. fàyó, filè, foèv, klàkèt, kòtis, làt, pèzàr, pwá, vàrdyó	french bean

Wild Flowers and Cultivated Flowers			
277	herbe	1. èrb; 2. àrb	grass
279	chiendent	1. ¢yẽdã; 2. ¢yẽdã-à-filè, ¢yẽdã-à-kòrd, ¢yẽdã-à-trèn, ¢yẽdã-brã¢, ¢yẽdã-d-grèv, ¢yẽdã-fisèl, ¢yẽdã-ki-fil, ¢yẽdã-ki-kur, ¢yẽdã–ki-trén, ¢yẽdã –kurã, ¢yẽdã-lis, ¢yẽdã-lõ, ¢yẽdã-ràsin, ¢yẽdã-trènã, ¢yẽdã-tré:nã, ¢yẽdã-trénã, ¢yẽdã-trẽné; 3. grã-¢yẽdã, gro-¢yẽdã, gru-¢yẽdã; 4. ¢yãdã, ¢yãdã-mõtã; ¢edã ; 5. èrb-à-¢yẽ, fisèl, kudròl	couch grass
281	chardon	1. ¢àrdõ; 2. ¢àrdrõ, ¢èndrõ, ¢èrdrõ, é¢àrdõ	thistle
289-1	colchique	1. la-kòl¢ik, lė-kòl¢ik; 2. lå-¢nàrd, du-¢nàr, dü-¢nár, là-¢nàrd, lė-pòryõ, lė-sàfrã, lė-sàfrã-sóvàj, lė-tu¢yẽ, lé-vèyoéz, ?	autumn crocus
296	aubépine	1. óbépin ; 2. óbèpin; 3. óbépẽ ; 4. èpin-blã¢, ėpin-blã¢, épin-blã¢; 5. épin-dė-là-sẽt-vyèrj, fó-óbépin, mè, mé, putẽ-blã¢, ?, x	hawthorn
297	chèvre-feuille	1. ¢èvrėfoèy; 2. ¢èvrėfoėy, ¢èvrėfoéy; 3. ¢ėvrėdfèy; 4. dü-bru-d-bik, dü-brubiké, là-bàrb-ó-bõ-dyoé, lå-bårb-ó-bõ-dyoé, lė-bru-d-biké, sèrfoèy	honey-suckle
Forest Woods			
312*	peuplier	1. poépliyé, poèpliyé; 2. pèpliyé, pépèyé, pépiyé, pépliyé, poèpliyè$_{y}$, poèpoèlyé, poépèyé; 3. poép, po^{oe}pl, poèp, poèpl; 4. bòyár, boyar, buyàr, buyá, buyar, lyár, ?	poplar
Wood and Wood Works			
319*	bois	1. bwá, bwà; 2. bwa, bwá$_{ò}$, bwá$_{u}$, bwá$_{y}$, bwa^{e}, bwa^{o}, bwa$^{o}_{ò}$; 3. bwè, bwè (v), bwé (v) ; 4. bwò ; 5. plãti	wood
321	arbre	1. àrbr, árbr; 2. àbr, àrb, á:br, á$_{ò}$br, á$_{o}$br, ábr, árb, a^{è}rb, a^{o}br, èrb; 3. òrbr, áb, a^{o}:b, a$^{o}_{o}$b, a^{o}b	tree
324	aubier	1. óbyé ; 2. obyé; 3.bu, frã-bwa$^{o}_{o}$, frã-bwó, mwèl, ó:bèl, ó:boé, ó:bur, ób, óbèl, óbòr, óbór, óbo^{u}r, óboèr, óboé, óbu, óbur, !óbur, ?, X	sapwood
346*	scier	1. syé, siyé; 2. sèyé, séyé; 3. vyòrné	to saw
Springs and Rivers			
367	ruisseau	1. rẅisó; 2. rẅisò, rẅiso:; 3. rü:syó, rüsò, rüsyó, rusyá, rusyí$_{u}$, rusyò, rusyó, rusyó$_{u}$, rẅisyó, rẅisyó$_{u}$; 4. rü, ryó; 5. byà, èvyèr, fósé, fósé-d-ègu, kòrd*ó, kulà-d-ó, kulã-d-yó, kulyó, kur-d-yó, láj, la^{o}j, lè, ròjér, vàlé, +, ?	brook
375*	chemin	1. ¢mẽ, ¢ėmẽ; 2. ¢mẽ-d-dèbàrd, ¢mẽ-t-plèn, ¢mẽ-t-tèr, ¢mẽ-t-tràvàrs, ¢mẽ-t-tràvèrs, ¢mẽ-d-èksplwátá:syõ, ¢mẽ-d-èksplwátásyõ, ¢mẽ-d-èsplwátásyõ, ¢mẽ-dü, ¢mẽ-pàrdü, ¢mẽ-pèrdü, ¢ėmẽ, ¢ėmẽ-dü-¢ã, ¢ėmẽ$_{ñ}$, ¢ėmẽ$_{y}$; 3. kmẽ; 4. bré, ¢àryèr, ¢áryèr, ¢èryé, dé-èzãs, kü-t-sàk, sòrti-t-¢ã; tiré-d-ẽ-¢ã, tiré-dü-¢ã	way
376*	sentier	1. sãtyé; 2. sat'é, sãt'é, sãt'yé, sãt*é, sãt'yé, sãtk’^{t}é, sãt'yé-d-mültyé, sãk*yé, pti-sãtyé; 3. sãt, sãt-d-pyé, sãt-dė-pyé, ptit-sãt; 4. !ròt, ¢mẽ-t-pyé, ¢mẽ-vèr, èzãs, páspyé, pti-¢mẽ-t-pyé, ròt, ròtẽ, rẅèl, trèt, vèrdè	path

378	ornières	1. òrnyèr; 2. àrnyèr, òrñèr, òrñér, òrñyèr, òrnyér, ónyèr, ónyér, órñér, órnyèr; 3. bàrs, brè, brè-d-vwàtür, brèy, bré, bréy, ¢àryèr, frèyé, fréyé, rwàs, rwáj, rwẽ	rut
Sky and Atmospheric Phenomena			
409*	soleil	1. sòlèy; 2. sólèy; 3. su:lè, su:lé, sulè, sulèy, sulé, !sulè, !sulé, sòlè	sun
426	il pleut	1. i ploé; 2. i-pyoèt, i-pyoé; 3. i-¢è-d-ló, i-¢è-d-lyó, i-¢è-d-yó, i-¢è-dė-lyó, i-¢é, i-¢é-d-ló, i-¢é-d-lyó, i-¢é-d-yó, i-¢é-dó, i-¢wè-dė-lyó, i-tõb, i-tõb-dė-ló, i-tõb-dė-lyó, i-tõb-dė-yó, sà-¢é, sà-¢oé, sà-¢wè, sà-¢wè-dė-lyó, sà-pyoé, sà-tõb	it rains
428	averse	1. àvèrs, ávèrs; 2. àvèrsé, àvàrs; 3. àrèl, àré, àrñ, ákwàné, ¢áblé, ¢ódré, ¢ódròné, èrnòpé, fòrtė-nẅé, fòrtė-pẅi, gàlàrmyó, gàlàrñó, gàlàrné, gàlàrnyó, géyé, géyéy, géyõ, gi:lé, gilé, giné, gribõd, jibulé, jilé, kàrñó, kàsté', màkėryó, muté, nẅé, nẅéy, õdé, pi:sé, pisé, ràgàsé, ràgàté, ràgàtéy, rnápé, siklé, sẅé, usé, usté, vãvòl	shower (of rain)
430-1	arc-en-ciel	1. ẽn-àrk-ã-syèl; 2. ẽn-àrg-ã-syèl, ẽn-àrk-ã-syó, ẽn-àrt-ã-syèl, aerk-ã-syèl, un-àrg-ã-syèl, un-àrk-ã-syèl; 3. l-àr¢-d-nòé, l-àrk-ó-syèl	rainbow
432*	pluie	1. plẅi; 2. pẅi; 3. plé, pli, ploé, ploé$_y$, pyé, pyoé, x	rain
437	tourbil-lon	1. turbiyõ; 2. étèrbiyõ, éturbiyõ; 3. àtèrbu, àtrbó, àturbu; 4. bòn-ó-küré, ¢ãdèl, dėmwàzèl, fòl, fòli:, fudr, sàrvãt-dė-küré, siklón, vãtuz, vãvòl, vyèl, vyèy, vyoèy, vyoéy, x	whirlwind
440	vent du nord	1. vã-dü-nòr; 2. vã-dü-nór, vã-dü-nó; 3. vã-dü-ó, vã-ó, vã-óu$_o$, vã-d-aó, vã-ó, vã-d-àó, vã-d-ã-ó, vã-dè-ó; 4. vã-d-àmõ, ó, l-vã-è-biz, vã-d-biz, vã-d-gàlèrn, vã-d-là-gàlèrn, vã-d$_o$-biz, vã-dyó, vã-mègr, x	north wind
443*	eau	1. ó; 2. ò, yó	water
Wild Animals			
463	geai	1. jè; 2. jé; 3. jèg, jàkó, ják, jáká, jákó, jakó, jëtà; 4. kòlà, kòlá, pyáró, pyèró, rákó, rákó$_u$, ra^{o}kó, ri¢àr, rikár, rikar, +	jay
464	merle	1. mèrl; 2. màrl; 3.màrló, mèrló; 4. mèrlüzyó, mé$_i$l, mél	blackbird
483	serpent	1. sèrpã; 2. sàrpã, sėrpã; 3. vàrmėñé, vèrmèñé, vnẽ, +, ?	snake
Breeding			
518*	pis	1. pi; 2. !pè, pè, pé; 3. trèyõ	udder
646-1*	chat	1.ẽ-¢á; 2. ẽ -kàt	cat

! represents that the form in question is suggested by the informant.

Corpora of Spoken Spanish Language – The Representativeness Issue[1] –

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1. INTRODUCTION

By the end of XXth century, characteristic feature of Linguistics was the use of linguistic corpora, as it was also for the Spanish language. As we know the use of collections of materials for research is not new, nevertheless the development of computers has allowed us to store and access huge bodies of materials, at speeds of astonishment, allowing better possibilities for their study and application.

Corpus linguistics is based on the idea that the description of the language cannot be made just from the intuition of the linguist, but that it requires the handling of a set of real language samples. In fact, a corpus is just a sample set of language materials, which can be written texts (*textual corpora*) or transcriptions of the oral language (*oral corpora*).

According to the definition by John Sinclair "a spoken language corpus is a corpus consisting of recordings of speech which are accessible in computer readable form, and which are transcribed orthographically, or into a recognised phonetic or phonemic notation" (1996: 28). The main goal is to acquire large amounts of data that reflect the natural use of language, therefore emphasis is usually put on the naturalness and spontaneity of the recording, as well as on registering speakers from real speech communities. Those speakers are in fact representatives of a community or group.

In the last twenty years, technological development has allowed for storage of extensive collections of spoken language from different regions, in this case belonging to the Hispanic world. For that reason, next to the geolinguistcs of the sounds, morphems and words, we also have a geolinguistics of the speech, of the oral language, constructed through *corpora*.

The purpose of these pages is multiple. On the one hand, we will present the most important oral *corpora* of Spanish language created or published since 1990. Attention will be paid to the *corpora* built for application in speech technology, as well as those having as their goal the study of the lan-

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guage itself. And special attention will be paid to those *corpora* offering geolinguistic varieties of the Spanish language. Secondly, we propose to reflect on one of the most important and difficult aspects in the elaboration of oral *corpora*: representativeness of the gathered materials. The aim is to analyze whether the criteria used for the speakers' selection are suitable and appropriate for the study of the spoken language and for its applications.

2. ORAL *CORPORA* OF SPANISH LANGUAGE: TYPOLOGY

Oral or spoken language *corpora* can be grouped in two main categories. The first category is that of *corpora* created for the study and development of speech technologies; their purpose is to develop applications for training and evaluation of recognition systems. The *corpora* of the second category are those created for the linguistic study of the spoken language. Both categories can be divided in two groups: *corpora* with a specific object, within the oral language (specialized *corpora*), and *corpora* which gather spoken language in general, not focusing on a specific level or aspect (general *corpora*).

In the category of corpora with application to speech technologies, it is worth highlighting the following general *corpora*:

- **AHUMADA.** Universidad Politécnica de Madrid. Speaker Recognition. <http://www.atvs.diac.upm.es/publicaciones/docs/Ort98b.pdf >. 1998.

- **ALBAYZÍN.** Universidad Politécnica Cataluña. Universidad Autónoma Barcelona.
Universidad Politécnica Madrid. Universidad Politécnica Valencia. Speech Recognition.
<http://gps-tsc.upc.es/veu/ >. 1991

- **EUROM 1.** Universidad Politécnica Cataluña. Universidad Autónoma de Barcelona. Automatic Speech Recognition. <http://gps-tsc.upc.es/veu/LR/LR_EuromI.php3>. 1993.

- **ROARS** (Robust Analytical. Recognition System). Universidad Politécnica de Valencia. <http://www.newcastle.research.ec.org/esp-syn/text/5516.html>
<http://www.iti.upv.es/~prhlt/index.html >1990.

- **SALA I.** Universidad Politécnica de Cataluña. Latin American Speech Recognition. <http://www.sala2.org/ >. 1996.

1.- General Oral *Corpora* with application in speech technologies.

In the same group, several specialized *corpora* are available, most of them mainly developed since 1990. Among the most important are the fol-

lowing:

- **ACCOR.** Universidad Politécnica de Valencia. Co-articulation Processes. <http://www.newcastle.research.ec.org/esp-syn/text/7098.html>. 1990.

- **CEUDEX.** Telefónica I+D. Microsoft. Development Systems. <http://www.telefonica.es/index/comun/tid.html> < http://www.telecom.tuc.gr/paperdb/icassp97/pdf/author/ic970823.pdf >. 1995.

- **MATE.** Prosodic labelling for machine-person interaction. Universidad Autónoma de Barcelona. Telefónica I+D. < http://mate.nis.sdu.dk/>. 1998.

- **MULTEXT.** Universidad de Barcelona. Prosodic labelling. <http://www.lpl.univ-aix.fr/projects/multext/index.html >. 1994.

- **SPATIS** Telefónica I+D. Information about flights. <http://www.telefonica.es/index/comun/tid.html >

- **SPEECHDAT.** European Commission. Development Systems. < http://speechdat.org>. 1994

- **TANGORA.** IBM España. Universidad Politécnica de Madrid. Continuous Speech Recognition. Automatic Dictate. <http://www-4.ibm.com/software/speech/es/> . 1992

- **VESTEL.** Telefónica I+D. Digits, Numbers, and Orders Recognition. <http://www.telefonica.es/index/comun/tid.html >, <http://www.tid.es/presencia/publicaciones/comsid/esp/articulos/vol52/artic3/3.html >. 1992.

2.- Specialized Oral *Corpora* with application in speech technologies.

All these references can be extended and further analyzed from the information provided on-line by Joaquim Llisterri <http://liceu.uab.es/~joaquim/teaching/Language_resources/spoken_res/Corp_oral_esp.HTML> or in the website of the "Office of the Spanish in the Society of Information", a branch of the Cervantes Institute <http://www.cervantes.es/seg_nivel/lect_ens/oesi/index.htm>.

To date, the most important oral *corpora* for the general study of the Spanish language (always in our opinion) are the following ones.

ALCORE. "Alicante corpus oral del español". Universidad de Alicante. 2002
(Azorín 2002).

ARTHUS. "Archivo de Textos Hispánicos de la Universidad de Santiago". http://www.bds.usc.es/. 1987.

CE. "Corpus del Español". Illinois State University. National Edowment for Humanities. Brigham Young University. <http://www.corpusdelespanol.org/>. Mark Davies. 2002

CECBNA. "Corpus del español conversacional de Barcelona y su área metropolitana". Universidad de Barcelona.
(Vila 2001)

CIEA. "Corpus Integral del Español Actual". Coordinated by: El Colegio de México. <http://www.iued.uni-heidelberg.de/institut/abteilung/ spanisch/index.html>. 2000.

CLUVI. "Corpus Lingüístico de la Universidad de Vigo". Spontaneous Speech.
Bilingual Castilian - Galician. <http://www.uvigo.es/webs/sli/index.html>. 2002.

CREA. "Corpus de Referencia del Español Actual". Real Academia Española.
<http://www.rae.es>. 1998.

CORLEC. "Corpus Oral de Referencia de la Lengua Española Contemporánea".
Universidad Autónoma de Madrid. <http://ftp.lllf.uam.es/pub/corpus/oral>. 1991

C-ORAL-Rom. "Corpus Oral de las Lenguas Romances".
Universidad Autónoma de Madrid. <http://lablita.dit.unifi.it/coralrom>. 2001.

CUMBRE. Corpus CUMBRE del español contemporáneo de España y de Hispanoamérica. Editorial SGEL. Available 2 million words sample. 2001 (Sánchez and Cantos).

DIES-RTVP. "Difusión Internacional del Español - Radio, Televisión, Prensa".
Coordinated by: El Colegio de México. 1992.

PILEI. "Proyecto para el Estudio de la Norma Culta de la Principales Ciudades de la Península Ibérica y de Iberoamérica". Lope Blanch (1986); Samper, Hernández, Troya (1998). Since 1964 (Pusch 2003).

PRESEEA. "Proyecto para el Estudio Sociolingüístico del Español de España y de América". Coordinated by: Universidad de Alcalá. <http://www.linguas.net/preseea>. Since 1996. (Moreno Fernández 1997; Gómez Molina 2001).

SOC-AND. "Sociolingüística andaluza". Universidad de Sevilla. 1983-1992. Extension from PILEI Project (Pineda; Ropero; Ollero)

VALESCO. Valencia. Colloquial Spanish. Universidad de Valencia. <http://www.uv.es/~valesco>. 1995 (Briz 1995).

3.- General Oral *Corpora* for linguistic study.

Naturally, other *corpora* exist, gathered for the study of specific Hispanic areas: the corpus for Conversation Analysis gathered in Alcala de Henares (ACUAH) (Moreno 2001), "Vernáculo Urbano de Malaga" (VUM) (Alvar and Villena 1994), ALMECOR (Almería, Spain), "Corpus Sociolingüístico de Caracas (Venezuela)" (Bentivoglio and Sedano 1993) or the spoken language corpus from the "Linguistic (and ethnographic) Atlas of Castile-La Mancha". (http://www.uah.es/otrosweb/alecman), but they have been handled internally by research groups or they have not been published or freely distributed, although in some cases they have been incorporated in the "Corpus de Referencia del Español Actual" of the Real Academia Española, like Santiago de Compostela or Caracas *corpora*.

On the other hand, in the category of *corpora* for linguistic study, there are several specialized corpora, among which we emphasize these:

- **ADPA** - "Análisis del Discurso Público Actual". Universidad de La Coruña.
Discourse Analysis. <http://www.udc.es/dep/lx/proj/adpa/>. 1994

- Acquisition, Development, and Representation of Semantic Categories in school age children. UNED.

- Individual Differences in Language Acquisition. Universidad de Barcelona.

- Children's Speech Corpus. CSIC-UNED.

- **COVJUA.** "Corpus oral de la variedad juvenil universitaria del español hablado en Alicante" Universidad de Alicante (Azorín 1996).

- Léxico de la Norma Culta. Lope Blanch. (1986). Since 1964.

- DISPOLEX. "Disponibilidad léxica".<http://www3.usal.es/dispolex/>. Since 1991.

- VARILEX. "Variación léxica". University of Tokio.
<http://www.lenguaje.com/herramientas/Varilex/Varilex.asp>
<http://gamp.c.u-tokyo.ac.jp/~ueda/varilex/art/vx7d.pdf>
<http://gamp.c.u-tokyo.ac.jp/~ueda/varilex/index.html>. Since 1995

4.- Specialized Oral *Corpora* for linguistic study.

Only the enumeration of those projects allows us to perceive that the field of the linguistic *corpora* has had remarkable activity during the past few years, although it is true that its development level is not parallel to the linguistic *corpora* for the English language (see http://devoted.to/corpora, by Mark Davies). In general the projects have been created and executed from the Spanish-speaking countries, mainly Spain, with a more than satisfactory level of collaboration within multilingual and European projects.

3. REPRESENTATIVENESS IN ORAL *CORPORA* OF SPANISH LANGUAGE

Representativeness of the language samples is one of the fundamental aspects in the *corpora* formation. Generally, when speaking of representativeness, one thinks of the capacity of (written) texts as samples that represent textual types. Those types are usually defined by their themes or the communicative contexts in which they take place.

In regards to the spoken language, the situation is somewhat different. One of the criteria most used to distinguish types or varieties of language is the "register", as explained by M.A.K. Halliday, that is to say, a variety of language according to its use. From this point of view, according to the objectives of each corpus, advertisements instructions, debates, interviews, or lectures can be distinguished, among others (field of speech), as well as familiar, formal or professional texts (tenor). It is important that the samples adjust strictly to an established typology of previous form, avoiding the cases in which the limits between different types are not very clear. The samples of spoken language must represent linguistic uses that take place independently of the process of *corpora*'s elaboration.

On the other hand, *corpora* developed for their application in the speech technologies gather very specific samples of speech: they are generally formed by brief words or sequences; and therefore, it is not possible to talk of proper discourse. In this case, representativeness of the texts is not a problem because they are samples produced specifically for *corpora* elaboration

and they do not have meaning outside it.

But, for oral *corpora* one must consider another type of representativeness. In addition to the variety according to use, it is necessary to pay attention to the varieties of language according to the user, which Halliday calls "dialect". The type of producer of the language samples cannot be indifferent in order to construct oral *corpora*.

Studies on linguistic variation explain that competence and performance are tie to four dimensions: the time and the geography, on the one hand, and society and the situation, on the other hand. It is not appropriate, therefore, to elaborate or to work with *corpora* whose speaker type (user) is not well known, or whose origin is not well designed according to the four parameters determining linguistic variation: time, space, society and situation. If suitable care is not taken with these criteria, the oral *corpora* achieved would not be truly representative of the spoken language, though it would be valid for application to certain scopes.

3.1 REPRESENTATIVENESS IN APPLIED CORPORA OF SPANISH LANGUAGE

As mentioned, the *corpora* created for application in fields like the speech technologies usually do not present problems of representativeness. In general, these *corpora* are built with *ad hoc* linguistic uses.

In terms of the speakers' representativeness, the handling of two types of criteria is common: sociological and geographic. The sociological factors handled for the creation of *corpora* with technical applications are gender and age, although these factors are really used more as individual factors than as social parameters. In the age factor, for example, several age groups are distinguished at regular intervals, usually of 10 or 15 years. This allows that the selection of speakers for this type of *corpora* does not look for linguistic users' representativeness but a diversity of qualities of voice.

Identifying areas in which different varieties of Spanish are supposedly used and selecting speakers from these areas apply the geographic criteria. The way in which it has been put into practice, nevertheless, only partially has to do with the Spanish Dialectology. In technical *corpora* like SPEECHDAT or SPEECHDAT CAR, for the Spanish of Spain, 5 and 4 "regions" have been distinguished respectively. For the first, the areas are Northwest (Galicia and Asturias), North (Basque Country and Navarre), East (Catalonia, Valencia and Baleares), South (Andalusia, Murcia, Badajoz, the Canary Islands and cities of North Africa) and Center (rest of Spain); the second one reduces the zones to four, uniting the North and the Center areas.

In our opinion, these "dialect" divisions seem to be due more to the coexistence of different languages than to the varieties within the Spanish lan-

guage: territories are distinguished in which bilingual speakers exist and the monolingual Spanish zone is divided into just two areas: Center and Andalusia. It is true that in the bilingual zones Spanish language has some particular characteristics, due to contact with another language, but these characteristics are no more differentiating than those in non-bilingual areas. Therefore, for the purpose of a correct geographic division, several objections can be found: for example, it does not make sense to distinguish the Northern zone (Basque Country and Navarre) from the Center area; for that reason we tend to agree more with the criteria followed in project SPEECHDAT CAR; nor is it correct to include Andalusia and the Canary Islands in the same area, "South", because of their important phonetic differences. If the real aim was to gather samples of speakers from well-differentiated zones, from a linguistic point of view, the criteria followed in these *corpora* have not been applied correctly.

The geographic division practiced in the project SPEECHDAT across Latin America (SALA) also presents some problems. In this project, 8 zones are distinguished in order to make the recordings. Curiously, one of them is Brazil and Spanish-speaking countries make up the other seven: it seems that all the territories are linked to the same language, or that Brazilian Portuguese does not have very important dialect variations. The division of the Hispanic area is suitable because it is based on information from Alvar (1996) and Lipski (1996), specialists in Hispanic Dialectology.

Finally, another significant aspect is the lack of rigor, not to say lack of knowledge, used to handle terms and concepts from Dialectology and spoken language. It is not suitable, for example, to affirm in the project EUROM1 that their 60 speakers have been selected among a total of 100 to assure a wide dialectal variation, because the variation is not a number question, but a speakers' profile question. In the project SALA, the concepts of "zone", "region" and "dialect" are distinguished. The limits of each zone match with the political borders; each zone is divided in regions about which is said that they are "homogeneous" from a phonetic point of view and dialects are defined by morphologic, syntactic and lexical criteria. Many theoretical and practical problems arise when subordinating a grammar based division to a lexical and phonetic division, most simply because geographic limits of the features of each linguistic level do not have any reason to coincide. The point is a dialect must be defined as a whole, by its phonetic, grammar and lexical characteristic. From this point of view, it is common to talk about dialects and sub-dialects. So far, it is possible to reach these general conclusions:

1) Spanish from Spain dialect and sociolinguistic criteria have not been followed in a completely suitable way.

2) Terms and concepts related to the linguistic variation are used with a remarkable lack of rigor.

3) It seems that representativeness according to the type of speaker does not matter. The language collection seems more a quantitative than qualitative question.

If the aim is to create *corpora* for technological applications, perhaps it would be more appropriate to identify the cities with more demographic weight and with a more relevant linguistic personality and to select the speakers from these locations instead of trying to identify dialect borders with dubious success. Among the urban speakers, the most closed to certain sociolinguistic profiles, a sort of identikit picture, could be selected.

3.2 REPRESENTATIVENESS IN ORAL CORPORA FOR THE LINGUISTIC STUDY

Currently most of oral *corpora* prepared for the general study of the language are young, having been elaborated during the last ten years. This is the reason why variation in time cannot receive attention. So excluding the time factor, these oral *corpora* can be classified according to the way in which geolinguistic, sociolinguistic and stylistic parameters are attended to. This way, *corpora* could include geographic samples of one or of several modalities, one or several sociolinguistic varieties or one or several stylistic modalities. *Corpora* with broadest goals are those trying to gather samples from different dialect varieties of spoken Spanish. In the following table, a relation of some of these *corpora* is offered, explaining whether they include materials from one or more geolinguistic, sociolinguistic, and stylistic varieties.

Project	***Dialects***	***Sociolinguistic Varieties***	***Style Varieties***
ACUAH	Madrid	Several	Conversational
ALCORE	Alicante	Several	Interview
CE	Several	-------	Several
CECBNA	Barcelona	-------	Interview/Conversational
CIEA	Several	-------	--------
CLUVI	Galicia	-------	Spontaneous-bilingual
C-ORAL-Rom	Madrid	Several	Formal/Informal
COVJUA	Alicante	Youth	Conversational
CREA	Several	-------	Interview

CREC	Madrid	--------	Several
CUMBRE	Several	--------	TV-radio programs
DIES-RTVP	Several	Mass Media	Several
ILSE	Almería	Several	Interview
PILEI	Several	High Education	Interview
PRESEEA	Several	Several	Interview
SOC-AND	Sevilla	Several	Interview
VALESCO	Valencia	Several	Coloquial
VUA	Granada	Several	Interview
VUM	Málaga	Several	Interview

5.- Spanish *Corpora* with information about geolinguistic, sociolinguistic and stylistic variation.

Among *corpora* designed for the general study of language, we must pay attention to two different types, being those of most interest in relation to the subject of representativeness: the Spanish varieties *corpora* and the reference *corpora*.

3.2.1 Spanish varieties Corpora

Corpora including samples from different geographic origins deserve special attention, due to their complexity. Very often these *corpora* try to reflect geolinguistic Spanish variation and among them we highlight the following ones: "Proyecto de estudio coordinado de la norma lingüística culta de las principales ciudades de Hispanoamérica" (PILEI), directed by Lope Blanch, project "Difusión internacional del español por radio, televisión y prensa", coordinated by Raul Ávila from El Colegio de México, project " Corpus Integral del Español Actual ", coordinated by another Mexican expert, Luis Fernando Lara, and "Proyecto para el Estudio Sociolingüístico de España y América", coordinated by Francisco Moreno-Fernández from the University of Alcalá (Spain). It could be said that these corpora are the ones that have most rigourously applied the criteria of Dialectology and Sociolinguistics for the speakers' selection. In these corpora, therefore, we can find a suitable representativeness according to language users.

PILEI

PILEI project was officially born in 1964 as a collaboration by experts from the Hispanic world. The project was founded in order to determine the main linguistic features of each geographic norm, using speakers with a high level of education to identify what characterizes each norm and what differ-

entiates them. A plan of recordings of samples of spoken language, gathered from speakers of different genders and generations, was elaborated.

Since 1998, a compilation of the speech samples has been presented in CD-Rom, prepared in the University of Las Palmas and coordinated by Samper, Hernandez and Troya. This "Macrocorpus de la norma lingüística culta" offers the transliteration of 84 recording hours, proceeding from parallel samples of twelve Hispanic cities: Mexico, Caracas, Santiago de Chile, Santafé de Bogota, Buenos Aires, Lima, San Juan de Puerto Rico, La Paz, San José de Costa Rica, Madrid, Sevilla and Las Palmas de Gran Canaria. In 1999, the Real Academia Española integrated these materials with other samples - Alcalá de Henares (Madrid), Santiago de Compostela or Alicante - to create the oral sample of "Corpus de Referencia de la Lengua Española" (CREA). Davies also include those materials in his "Corpus del español".

DIES

Directed by Raul Avila, of El Colegio de México, a linguistic research project was begun in 1988 in order to study the Spanish language used in mass media of Hispanic-America and Spain. The project is called DIES-RTVP and it is structured such that each Spanish-speaking region has a coordinator who, following some general guidelines, is in charge of gathering samples of media. One of the guidelines is to gather 1.200 words text units, representative of different types of programs that can be found in each studied area: news, sports, or soap operas, for example. At the moment, research groups from Mexico, Spain, Argentina, Bolivia, Colombia, Costa Rica, Chile, Puerto Rico, Dominican Republic and the United States are working parallelly in this project.

CIEA

Fernando Lara, its coordinator, initiated the "Corpus Integral del Español Actual", a collaboration between several teams in Spain, Hispanic-America, and in the United States. Its main aim is the elaboration of electronic *corpora* of the spoken and written Spanish, using materials from the time period 1975 to 1995. This spoken language corpus tries to reach 1.000.000 words per country in order to guarantee qualitative and quantitative representativeness. According to the information provided by Nelson Cartagena, there are presently researchers from Argentina, Bolivia, Chile, Spain, Mexico, Uruguay, Venezuela, and US Southwest are working on this project. The *Corpora* from Spain, Mexico, Chile and North American are already complete, according to Cartagena, and they are in the process of gathering material from the remaining Hispanic-American regions.

PRESEEA

The target of the "Proyecto para el Estudio Sociolingüístico del Español de España y de América" (PRESEEA) is to construct *corpora* of spoken language from a variety of Hispanic cities, which collect the linguistic uses from speakers of different backgrounds. The idea is to coordinate the work by different research groups who volunteered for the project, yet for all involved to commit using a common methodology that will allow for later comparison of language samples. Among the cities envolved in the study so far, are: Alcalá de Henares (Madrid), Barranquilla, Bogotá, Cádiz, Guatemala, Las Palmas de Gran Canaria, Madrid, Málaga, México DF, San Juan de Puerto Rico and Valencia. PRESEEA relys a website (www.linguas.net/preseea) where information on all the teams and their members is offered, as well as documents, updates on project development and links with complementary or instrumental websites. Most importantly one can also find samples of spoken language, transcribed according to the "Text Encoding Initiative" international guidelines. These samples can be consulted freely. PRESEEA was born in 1996 within the "Asociación de Lingüística y Filología de la América Latina" (ALFAL).

3.2.2 Reference Corpora

In the last ten years, several *corpora* have been elaborated under the denomination of "Reference *corpora*" or with that aim: the "Corpus Oral de Referencia de la Lengua Española Contemporánea" (Reference Oral Corpus of Contemporary Spanish Language) (CORLEC), elaborated in the Universidad Autónoma de Madrid at the beginning of the nineties (Marcos Marín 1991), the "Oral Corpus of the Romance Languages" (C-Oral-Rom), a European project in which the Spanish contribution is also made by the Universidad Autónoma de Madrid (A. Moreno 2001), the "Corpus del español" by Mark Davies, and the "Corpus de Referencia del Español Actual" (CREA) by the Real Academia Española.

The oral *corpora* of the Universidad Autónoma de Madrid have focused on the representation of different types of texts depending on their linguistic use. The typology of spoken language samples with which it works is wide and diverse. Nevertheless, from our point of view, due attention has not been paid to the representativeness of the samples according to the user, and it is not sure that the speakers' selection has become fit to the regular patterns in Dialectology and Sociolinguistics.

It is true that these projects deal with speakers of differing characteristics, but the sociolinguistic diversity seems be a consequence of data collection and not the departure point of an organized sampling procedure. In the CORLEC project, the speakers' characteristics are specified, but those char-

acteristics have not determined their selection. C-Oral-Rom project considers speakers according to sex, age, education and profession, but the authors don't offer details about how they are proceeding to divide generations, educational levels or types of professions. Not one mention is made about what speech community or communities are being studied. Geolinguistic and sociolinguistic representativeness of these materials is more than doubtful, although according to the research reports it appears as one of the determining criteria for the collection of oral texts.

On the other hand, in the reports by the Universidad Autónoma de Madrid, they affirm that they are looking for spontaneous speech, yet they neglect to explain how spontaneity level is determined. In C-Oral-Rom, attention is given to formal and informal speech, but they never explain how different styles samples are obtained, or what methodological resources are used for it.

One of the innovations of C-Oral-Rom with respect to CORLEC is "legality" of the texts in the new corpus, since permission by the speakers is now obtained. The interesting point is that this is explained as a necessity due to the corpus' commercialization and the previous lack of attention is justified explaining that scientists are not concerned with legal issues. It is enough to review the bibliography on sociolinguistic methodology (in English and Spanish) to verify that the sociolinguistic field has been considering these questions and offering diverse solutions for quite some time (Milroy 1987).

In terms of CREA project, of the Real Academia Española, it suffices to comment that, so far, it is the most important and most accessible corpus in the Spanish-speaking world and for the researchers of that language. Representativeness problems that CREA raises, as far as the spoken language is concerned, are not its own, to a large extent, because it has assumed those of the different *corpora* that it is composed of (Pino and Sánchez 1999). Something similar could be said about the "Corpus del español" by Mark Davies of Brigham Young University. This is a 100 million words corpus with 6,800,000 words of spoken language. Texts from PILEI and CORLEC are included as well as parliamentary reports and journal interviews. Mark Davies' corpus of Spanish has been funded by the NEH and it can be used in the Internet (www.corpusdelespanol.org) and include a search engine that allows a wider range of searches than almost any other large corpus in existence.

4. CONCLUSION

Spanish language has diverse oral *corpora* elaborated by public initiatives (Polytechnical Universities of Barcelona, Madrid and Valencia, Univer-

sidad Autónoma de Madrid, Real Academia Española) and private iniciatives (Telephone I+D, IBM). Many of these projects have been developed within European programs (see list of European projects provided by the Cervantes Institute) or have received the support of the government of Spain. Most of the specialized projects have been developed in Spain, specially the *corpora* applied to speech technologies, whereas their development in Hispanic-America has been little or null. It is possible to point out just a few *corpora* gathered in some Hispanic-America areas for the general study of the language, for instance the project on Spanish from Caracas.

One of the main problems for oral *corpora* elaboration is representativeness. And it is possible to distinguish types of representativeness according to variation in language use and variation in the user. One can appreciate that, while the first has been handled correctly way in most circumstances, the second has received unequal attention. Three frequent research lines in corpus linguistic of Spanish language may be distinguished:

a) Research groups elaborating *corpora* for application in speech technologies. Normally they work with samples of language from diverse regions and diverse types of speakers. Speakers are selected more because of their individual characteristics (gender and age) than because of sociolinguistic profile. Geolinguistic areas are identified with criteria of doubtful methodological value. In general, speakers' representativeness is not suitable, although materials can be valid.

b) Research groups elaborating *corpora* for the general study of the language and for its application in different scopes. Generally, they announce a preoccupation with geolinguistic, sociolinguistic, and stylistic questions, but the methodological procedures followed are unclear; fundamental information used to determine representativeness of the samples is missing. Sociolinguistic factors receive attention only after the materials are collected. Speaker representativeness does not seem to be the most suitable.

c) Research groups elaborating *corpora* for the study of spoken language from a certain community or diverse regions of the Hispanic world. They generally obtain a suitable representativeness, but very frequently do not store the materials in an appropriate and accessible way.

In general, one can observe a disconnection between the groups oriented toward application of the materials in the new technologies and those that worry about the study of the language. These latter obtain representative materials, but it is not always valid for their technological application. Never-

theless, those that do not always obtain *corpora* useful for application normally do obtain a suitable representativeness.

Finally, it is important to comment that future challenges for Spanish oral *corpora* are similar to those in other languages. Lori Lamel and Ronald Cole explained some of these challenges in 1996 and they are still pertinent today: how to design a compact corpora that can be used in a variety of applications; how to design comparable corpora in a variety of languages; how to select statistically representative test data for system evaluation; how to select (or sample) speakers so as to have a representative population with regard to many factors including accent, dialect, and speaking style.

The Spanish language does not yet have the number of *corpora* that the English language has, but already has on a long and varied series of linguistic collections. The future for the elaboration of *corpora* with spoken Spanish language is linked to a larger collaboration between experts in computer science and automatic treatment of linguistic samples and experts in the study of the spoken language, mainly dialectologists and sociolinguists.

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Methods of "Hand-made" Corpus Linguistics - A Bilingual Database and the Programming of Analyzers -

Hiroto UEDA *(University of Tokyo)*

1. Introduction[1]

On this occasion I wish to present my personal methods of linguistic corpus analysis, in which I use a relatively small database and hand-made analyzers, and to compare my procedures with other methods used in the mainstream of English corpus linguistics, which depend on large-scale data and pre-fabricated, and sometimes commercialized, applications. With respect to English studies, which are considered to be in the vanguard of corpus linguistics, we have a great deal of information, and this extensive literature is available in multiple forms of communication media[2] . As for Japanese linguistic studies, many researchers have been publishing their works, both qualitative and quantitative, in specialized periodical journals[3] . Unfortunately, in the area of Spanish corpus linguistics, with its short history and relatively small number of researchers, many questions remain without answers, and much is expected from future developments[4] .

Those who intend to begin the corpus study of a language in which scientific approaches are not well developed, as is the case in Spanish, need to pay full attention to the recent achievements in advanced studies of more extensively studied languages such as English or Japanese. Naturally, such

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[1] I am most grateful to Professor Paul Rossiter of the University of Tokyo and to Professor Philip King of the University of Birmingham for their kindness in reading and commenting this paper.

[2] For a general view of English Corpus Linguistics we can consult for example: Sinclair (1991), Aijmer and Alternberg (1991), Saito (ed., 1992), McEnery and Wilson (1996), Stubbs (1996), Thomas and Short (1996), Saito, Nakamura and Akano (1998), Takaie and Suga (1998), Meyer (2002).

[3] Various articles in *Nihongogaku* (Meizishoin) vol. 22, April 2003, a special issue on "Corpus Linguistics", are useful for obtaining the most recent information. Also, Keiryo Nihongogaku Kenkyukai (2003) has published in CD-ROM form a voluminous collection of previous quantitative studies of the Japanese language.

[4] See, for example, Alvar Ezquerra y Villena Posada (1994), Etxeberría Murgiondo et al. (1995), Marcos Marín (1996), Irizarry (1997), Blecua et al. (eds., 1999), Kock (ed., 2001). Moreno (to be published) describes the current state of Spanish corpus linguistics.

researchers will not be able to follow all the numerous studies published in specialized periodicals, but it is necessary to keep pace with their general tendencies so as not to commit the error of reinventing the wheel.

Certainly, the huge number of researchers and the long history of study of these languages may guarantee a high level of investigation. Perhaps for this reason, my personal impression is that in Japan students of Spanish feel obliged to study English and French, whereas students of French may study English, but not Spanish; and very few students of English learn French and even fewer learn Spanish. Students of Portuguese generally appear to learn Spanish as well, while only a very small number of students of Spanish study Portuguese. There seems to be a rough correlation between the number of foreign language learners and that of published studies of the same language. A similar tendency can be observed in our academic environment. We observe a certain unidirectional standard with concentration on a sole polarity which is global English. The same may be true in the case of corpus linguistics. Once I attended a presentation on "the method of corpus linguistics", which actually treated only English matters. According to that speaker, "corpus linguistics" was synonymous with "English corpus linguistics". Although in practical terms the synonymy might be established, corpus studies of other languages are also being carried out and, in that sense, due attention to these others should be paid.

2. Pre-fabricated applications or hand-made programs?

The choice between using pre-fabricated applications or programming our own corpus analyzers may be compared to the issue of economy of new terms for new concepts, as treated in general linguistics. André Martinet (1960: 168) explained the distinction between syntagmatic and paradigmatic economy observed in the creation of new terms:

> To remedy the lack of specificity of a term we can use a device other than its simple replacement: we can specify a term of general meaning by conjoining it with another term likewise of general purport. *Machine* and *laver* are both terms with a wide range of application; but a *machine à laver* is a well-defined object. To secure satisfaction of their communicative needs men will thus have a choice between increasing the number of the units of the system (the housewife may, for instance, speak of her *Bendix*) or increasing the number of units used in the spoken chain of utterance (in that case the housewife will say 'ma machine a laver').

The same is true in our case of using software for corpus analyses and statistical calculations. That is, when a researcher adapts software which he or she is fully accustomed to, this is comparable to the situation in which we

refer to a new reality with a new term by means of combination of old terms. On the other hand, to install a new commercialized or freely distributed package of software for use in corpus analysis onto our hard disk seems like adding a completely new term, such as *Bendix*, to our vocabulary. My impression is that the general tendency in the practice of corpus linguistics is the latter, but the two ways each have their own pros and cons; and therefore we should discern well their characteristics. The following passage of Martinet offers us a suggestive idea:

> In the first case [*Bendix*] there will be syntagmatic economy, one moneme instead of three, the two syllables and six phonemes of /bẽndiks/ instead of the five syllables and ten phonemes of /maʃinalave/. In the second place [*machine à laver*] there will be paradigmatic economy, since we shall avoid a new item in the list of substances [sic. substantives] which the speaker must memorize and among which he must choose when he speaks.

The burden which adopting a new application places on researchers is considerable. Useful facilities and operations which seem simple to well-versed users of an application may sometimes construct an insurmountable barrier for beginners. What impression would it make on us if someone referred to a machine with the new unknown term *Bendix*, when we would have understood without any problem if they had used such a simple term as *machine à laver*? Considerable expertise is necessary in order to have sufficient skill to use a corpus and statistical software packages, and this constitutes a difficulty for students who use only general purpose applications such as *Microsoft Word* or *Excel*. The ability to understand and interpret results through repeated trial and error and to account for meaningful conclusions is required. Unfortunately very few people are able to do this. Let me quote Martinet's analogy:

> In principle what will determine the final choice between one solution or the other will be frequency of use. If the object in question is mentioned frequently, it will be more economical to adopt a brief designation even if this involves an increased burden on the memory. If, on the other hand, there are only a few occasions when the object has to be named, it will be more economical not to burden the memory but to preserve the longer syntagm.

I agree with this opinion. For specialists in corpus analysis and skilled researchers using very specialized software, it is important to develop their skill and to understand the theoretical background of each operation. Linguists, however, engage in many other activities: field work, documentary research, and the heterogeneous daily tasks related to language education. Returning to the analogy with housework, the housewives never

attend only to washing machines. The same will be true in our research environment. I think we share the impression of being surrounded by a lot of information of names and manners of operation of many instruments. According to Martinet (ibid.: 168-179), the solution depends on the frequency of use. But he offers another factor:

> Other factors, of course, come into play. In our example *Bendix* is handicapped with respect to its competitor by the fact that it designates only machines of a certain make, so that another housewife will speak of her *Hoover* or her *Servis*.

Compared with proper names, the composite term *machine à laver* has the advantage of being understood by everyone. Anyone who knows the meaning of the three words *machine*, *à* and *laver*, and who knows syntactical rules, can easily understand the meaning of the syntagm. The situation when using a specialized application is the same. The general reader will not be able to know how to treat the results of operations or calculations, if the algorithm of the software is a black box for him. Sometimes I hear the observations of students who discuss the conclusions of other investigators without knowing how to use the software in question. They say: "I don't understand well, but I guess he would like to say such and such a thing". It is risky to continue with this attitude in this age full of an overwhelming quantity of information. In fact, apart from the two solutions, we have a third which Martinet explains as follows:

> In many cases, the brief designation consisting of a single moneme is an abbreviated form without regard for the etymology of the underlying long form: *ciné* for *cinéma* for *cinématograph*, *métro* for *chemin de fer métropolitain* and this fact suffices to inhibit its general adoption by a conservative community.

Martinet treated the case of the abbreviations *ciné*, *cinéma*, *métro* in the same manner as proper names[5]. The case of acronyms like *OS* for "operation system", *PC* for "personal computer" or *FEP* for "front end processor" is of the same nature. This situation is similar to the case in which researchers use widely distributed software and add some partial improvements. This places less of a burden on users, because they feel secure in their knowledge of applications to which they are well accustomed. The communicative function is assured, because the new term relates to another well-known term in some way. The conclusion of André Martinet (ibid.: 169) is:

> What one may call the economy of a language is this permanent search for equilibrium between the contradictory needs which it must

[5] The inhibition of "its general adoption by a conservative community" is applicable also to the case of proper names.

satisfy: communicative needs on the one hand and articulatory and mental inertia on the other, the two latter in permanent conflict.

As is well known, this is the structural view of language proper to Martinet, who introduced the concept of "economy" to interpret linguistic reality. If we are faced with a choice of two methods, then we can either treat it simply as a binary choice, or we can consider modifying either or both of the choices, or we can also reject both and go for the do-nothing option[6] . Nevertheless, I wish to repeat that it is not necessarily a question of a binary choice, because an eclectic method of corpus analysis with some amendments to familiar software is also possible.

3. The development of a linguistic corpus

It is hard for those who begin to study Spanish in a university course to reach the level of native speaker-like performance. It is hard also for those who learn Spanish after the critical period of language acquisition to acquire the native speaker's linguistic intuition. To compensate for this lack, one of the most usual practices is to collect examples extracted from documents and write them on index cards. If it is difficult to give any judgment on the usage of the expressions in question, we can depend on the cards as evidence.

Such examples may demonstrate the existence of certain linguistic forms, but they cannot demonstrate the non-existence of forms. A card may demonstrate that such-and-such a phenomenon occurred in a specific text, but theoretically no card can prove conclusively that a phenomenon did not occur. And even with positive evidence, we can not be confident in affirming the existence of a phenomenon, because only one piece of evidence, which is known as "hapax", may not be sufficiently significant for us to be able to give a conclusive judgment.

It may be expected that these problems of the demonstration of the non-existence of a phenomenon and of hapaxes may be resolved by the use of a sufficiently large-scale corpus. Certainly, a corpus consisting of 100 million words is likely to have more potential than that of 10 million words. And even without a corpus on hand, we can consult the whole Internet almost instantaneously. We can obtain a certain amount of data for our immediate purposes, even though we do not know the real characteristics of this huge pool of data provided by the Internet.

These available data, however, do not necessarily provide us with

---

[6] In the pre-computer age, scholars worked with printed documents. And even now the collection of linguistic examples on cards is one of the most useful ways of gathering linguistic data.

precisely the information we need. For instance, as far as I know, no Spanish-Japanese bilingual database has so far been published either in the Internet or CD-ROM. The following is the image of the first page of the bilingual corpus from which Carlos Rubio and I elaborated a Spanish-Japanese dictionary (Rubio and Ueda, eds., 1993):

	A	B	C
1	No.	ESPAÑOL	日本語
2	1	A Alberto le gusta comer el tiempo.	アルベルトは海好きだ
3	2	A Ana le falta fuerza para empujar.	アナは押しきれなかった
4	3	A Ana le va muy bien el peinado en ondas.	アナはパーマがよく似合う
5	4	A Andrés se le ha caído a la cabeza el sombrero y está muy asustado.	アンドレスは帽子を頭からとってとても怖がっている
6	5	A Angelito le encanta chapotear en el agua.	アンヘリートは水の中ではしゃぎまわるのが好きだ
7	6	A Armando le tocaron 300 800 (tres mil pesetas) en el juego.	アルマンドは賭で18万ペセタ当てあてしまった
8	7	A Arturo le gusta el pan empapado en el café con leche.	アルトゥーロはパンをコーヒーに浸して食べるのが好きだ
9	8	¡A beber!	飲むぞ！
10	9	A base de muchos esfuerzos llegó a tener éxito.	彼は多くの努力によって成功を勝ち得た
11	10	A Bernardo le concepto de honrado en alto grado.	ベルナルドは非常に誠実だと思います
12	11	A bordo del avión viajan 15 (quince) tripulantes y 320 (trescientos veinte) pasajeros.	飛行機には15人の乗務員と320人の乗客が乗っている
13	12	A brazas mojitas no se pescan truchas.	にことわざ 軽いたゴボンでは魚ますは釣れない（音楽や情報を楽しんでいては目的は果たせない）
14	13	EA buen entendedor, pocas palabras.	にことわざ 理解のよい人には言葉数少な（ほ一を聞いて十を知る）にあたる
15	14	EA buen hambre no hay pan duro.	にことわざ 空腹に固いパンはない（「すき腹にまずい物なし」にあたる）
16	15	EA buena hora mangas verdes.	にことわざ 後の祭り
17	16	¡A buena hora me dicen que Antonio ha cambiado de dirección! ¡Ya eché la carta al buzón!	アントニオの住所が変わったことをやっと言ってきた遅いよ！もう手紙を投函してしまったんだから！
18	17	EA buey muerto, cebada al cabo.	にことわざ 死んだロバの尾に大麦（麦の大麦を飼料）。ロバは死んでしまった。後になって大麦をやっても無駄である。
19	18	EA caballo regalado no le fuerce el diente.	にことわざ もらった馬の歯を見てはいけない（貰い物にけちをつけてはいけない）
20	19	EA cada cerdo le llega su San Martín.	にことわざ どんなブタにもサンマルティンの祝祭日の時期がやって来る（誰も罰を逃れることはできない）
21	20	EA cada cerdo le llega su San Martín.	にことわざ どの動にもそのサン・マルティンの日がやってくる（どれにも自分の過ちの結果を思い知らされる時がやってくる）
22	21	A cada cual lo suyo.	人それぞれである（いろいろ紹介はしない使うがよい）
23	22	A cada instante suena el teléfono.	電話がひっきりなしに鳴る

Fig. 3a: Spanish-Japanese bilingual corpus

This corpus consists of 24,300 lines, divided into three columns of *Excel* sheet: Column A corresponds to a sentence-ID; column B, to the Spanish sentence; and column C, to the corresponding Japanese sentence. The digitized data, separated by commas or tab marks, are easily transferred to *Excel*. The matrix structure composed of columns and lines allows us to use multiple *Excel* functions like reordering, searching, replacing, statistical calculation, elaboration of graphs and so on, in the presentation format we desire. The linking with other applications such as *Word* or *PowerPoint* is smooth in a GUI (graphical user interface) environment.

The scale of such a hand-made corpus based on individual work is necessarily small. The treatment of the data, on the other hand, causes no problem because we know the characteristics of the data well. In the process of editing the dictionary we carried out proof reading at various times in order to remove ungrammatical or unnatural sentences.

It may be convenient to describe the quantitative characteristics of this corpus. By transferring the Spanish and Japanese parts separately into a *Word* document[7] , and using the function of *Counting Letters* in the *Tools*

menu, we obtain the following output:

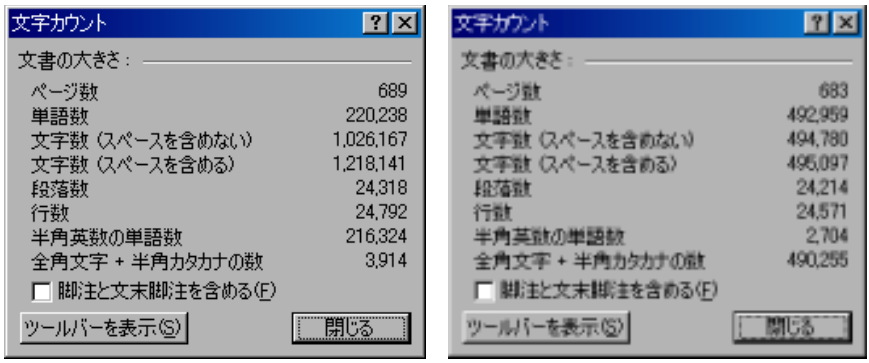

Fig. 3b: Character count of Spanish and Japanese parts

In the left figure, corresponding to the Spanish part, the word-count is 220,238; and the number of letters is 1,026,167. From this we can estimate that the average number of letters per word in Spanish is 4.66. Comparing the number of letters of Spanish and that of Japanese, we obtain the ratio of 2.07 (1,027,167 to 494,780); that is, Spanish has twice as many letters per sentence as Japanese.

A corpus of this type may be something like refined oil, while data extracted from the Internet remains crude oil. For a descriptive study, the natural state of data is valued because it reflects linguistic reality. For lexicography, and especially for bilingual lexicography aimed at developing learner's dictionaries, the crude material is not described in the dictionary as it actually is, but rather is presented in refined forms which have been filtered in accordance with some criteria and rewritten so as to be autonomous - that is, interpretable by themselves without depending on their contexts[8] .

---

[7] It is convenient not to copy and paste, but to paste with selection of form (*Paste Special*), which is "unformatted text", in this case.

[8] The situation of dictionaries for special purpose is different. For instance, Martín Alonso (1986), who focuses on the study of Medieval Spanish, attaches much importance to real examples extracted from original works. Also examples together with their sources are annotated in regional descriptive dictionaries such as those of Rodríguez Herrera (1958), Santamaría (1959), Malaret (1967), Morales Pettorino et al. (1984), Tejera (1993) and Quesada Pacheco (1995), as well as in pan-Hispanic dictionaries such as those of Richard (1997) and Steel (1990, 1999). The eight volumes of Cuervo (1994), in the same stream, constitute a monument of Spanish lexicography.

The question is not to decide which is better or which is correct, but to select the optimal material for the particular purpose of the study. For instance, in order to study the vocabulary of a specific author, we need to concentrate our attention exclusively on his or her published works. If, on the other hand, we need a demonstration of the existence of a specific word collocation in an almost unlimited linguistic context, we can depend on the Internet, which to some extent can be considered to be unlimited, in the sense that the whole huge pool of Internet information is being augmented and renewed day by day, hour by hour, continuously. The ideal might be to take into consideration not only one type but both. If not, the results of a study based solely on small-scale data would lack general application, while a study depending exclusively on Internet data would not give us the desired certainty because of its heterogeneity. It is something like the situation in which, by proving that the distilled water in a test tube and the water in the Pacific Ocean can both be reduced to the same chemical formula, we would be able to make a generalization about water's universal qualities.

4. Using the functions of applications

In widely distributed and familiar software packages there are many functions useful for processing linguistic corpora. A good example is the *Counting Letters* function, presented in the previous section. The *Find*[9] and *Replace*[10] functions are frequently used for data retrieval and text processing. The next figure shows the retrieval of the character **思** (*omou*, 'to think'):

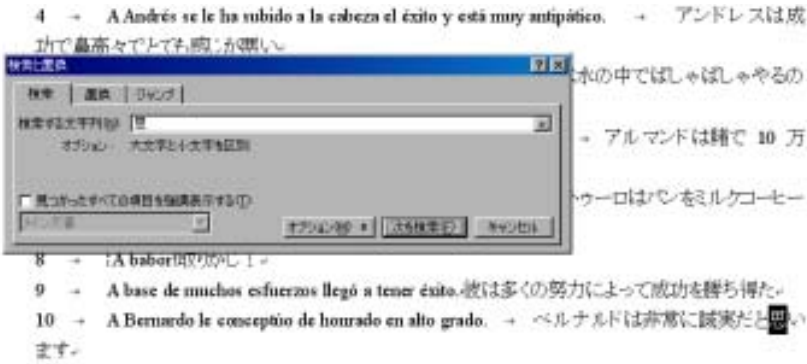

Fig. 4a: Find

[9] The operation is: *Menu > Edit > Replace> Find*.

[10] *Menu > Edit > Replace > Replace.*

The *Find* function indicates each occurrence of the expression, but once it has finished finding all the corresponding forms, there remains nothing in the document. By using the *Replace* function, the corresponding part can be highlighted in a manner the user would like (see Figure 4b). In this way, the *Replace* function is useful not only for data processing but also for data retrieval.

Fig. 4b: Replace

Sometimes we feel overwhelmed by a huge amount of data, even in the form of highlighted retrieved expressions. It would be convenient to extract only the paragraphs which contain the desired forms. Regrettably *Word* and *Excel*, even in their newest version, do not allow us to realize such an operation. However, the user who needs this operation has the possibility of using macro programming in *Visual Basic Editor*[11] . Below is an example of such programming, in this case designed to extract the desired forms:

Macro-coding (1): EXTRACT

```
Sub Extract() ' ver. 2003/07/16 by H. Ueda
  Dim i, n, m As Long: Dim intW, intB, intStart As Integer
  Dim strK, strId(60000), strSp(60000), strJp(60000) As String
```

[11] *Tools > Macro > Visual Basic Editor.*

```
strK = InputBox("INPUT Key Word" & vbCr & vbCr & _
  "Ex-1. no cre" & vbCr & "Ex-2. 思 ", "EXTRACT")
If strK = "" Or strK = vbCancel Then End

If Left(strK, 1) <= "~" Then intW = 2
If Left(strK, 1) > "~" Then intW = 3
n = ActiveSheet.UsedRange.Rows.Count

For i = 2 To n
  If InStr(Cells(i, intW), strK) > 0 Then
    m = m + 1
    strId(m) = Cells(i, 1): strSp(m) = Cells(i, 2): strJp(m) = Cells(i,
     3)
  End If
Next

Worksheets.Add after:=ActiveSheet

Cells(1, 1) = "ID.": Columns(2).ColumnWidth = 8
Cells(1, 2) = "Spanish": Columns(2).ColumnWidth = 50
Cells(1, 3) = "Japanese": Columns(3).ColumnWidth = 50

For i = 1 To m
  Cells(i + 1, 1) = strId(i): Cells(i + 1, 2) = strSp(i): Cells(i + 1, 3) =
    strJp(i)
Next
End Sub
```

Figure 4c shows a screen-shot of the macro program coding:

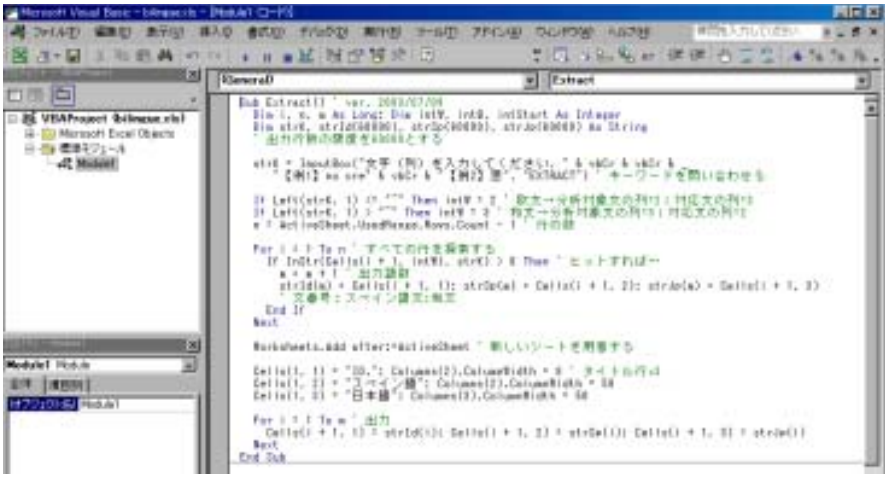

Fig. 4c: Macro codes: EXTRACT

When we run this program, the following Input Box will appear, where we put the character **思** ('to think'):

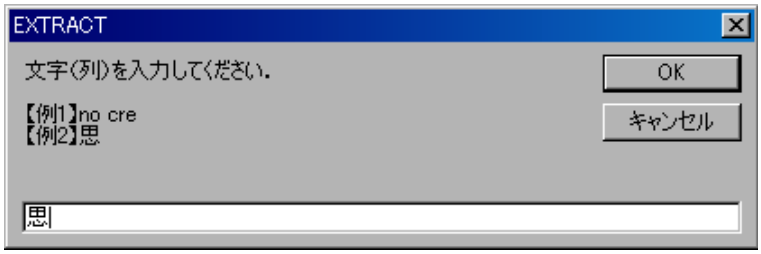

Fig. 4d: Input Box: EXTRACT

The result is Figure 4e:

	A	B	C	D
1	ID	スペイン語	日本語	
2	11	A Bernardo le conceptúa de honrado en alto grado.	ベルナルドは非常に誠実だと[思]います	
3	21	$A sala sentó le llega su San Martín $	ことわざ[]どの頃にもそのサン・マルティンの日がやってくる[]だれにも自分の	
4	79	A Domingo le gusta estar siempre en primera fila.	ドミンゴはいつも第一線にいたい[と思]っている	
5	108	A esta hora hay mucha gente aquí. Temo que $tengamos que	この時間までここは混んでいます。私たちはしばらく待たされるのではないかと[	
6	156	A Isabel se le ha metido entre ceja y ceja que quiere aprender	今年の冬はスキーを習いたいという[思]いがイサベルの頭を離れなかった	
7	280	A los muchachos les gusta tener bíceps de atleta.	少年たちは運動選手の筋肉こぶを持ちたい[と思]っている	
8	303	A Martín todos le tenemos en muy buen concepto.	私たち皆はマルティンをとてもよく[思]っている	
9	341	A mi juicio deberías esperar un poco más.	私の考えでは君はもうすこし待つべきだと[思]う	
10	368	A mi tío Ángel le dio la ventolera de irse al Polo.	アンヘルおじさんは北極へ行くなどということを[思]いついた	
11	467	A Ramón le importa un carajo lo que piensan de él.	ラモンは自分がどう[思]われているかなど少しも意に介さない	
12	471	A riesgo de que alguna persona se $escandalice de mis palabra	私の話に憤慨する人がいるかもしれませんが真実を話すべきだと[思]います	
13	490	A terror de las pruebas presentadas por el fiscal, parece que se	検事の提出した証拠によれば彼は有罪を宣告されると[思]われる	
14	520	A veces la forma puede parecernos más importante que el con	ときとして形式が内容より重要だと[思]うこともある	
15	711	Adriana siente soledad en el nuevo ambiente.	アドリアナは新しい環境で寂しい[思]いをしている	
16	750	Agustín dejó el trabajo porque le parecía rutinario y monótono.	アグスティンは仕事がマンネリで単調[思]えたのでやめてしまった	
17	804	Ahora que lo dices, tienes toda la razón. No me detuve a pensa	今君の話を聞くともっともだと[思]う。私はよく考えていなかった	
18	955	Al final Elvira se salió con la suya y su padre la llevó al cine.	結局エルビーラは自分の[思]い通りにして父親に映画に連れて行ってもらった	
19	1078	Al pobre joven el futuro se le presentaba bastante oscuro.	かわいそうにその若者の未来は暗澹[あんたんた]るものに[思]われた	
20	1085	Al principio creí que debía llamar al médico, pero vi que el enfer	最初は医者を呼ばなくてはと[思]ったが、病人はそれほど重態ではなかった	
21	1201	Alberto es un petulante y se cree alguien.	アルベルトは傲慢にうぬぼれた男で自分をひとかどの人物だと[思]っている	
22	1252	Alfonso es tímido y le cuesta conectar con la gente.	アルフォンソはひっこみ[思]案で人とつきあうのが苦手だ	
23	1286	Alfredo se declaró a su novia en la fiesta de cumpleaños.	アルフレードは誕生パーティーで恋人に[思]いを打ち明けた	
24	1297	Algún día, espero poder retribuirle toda su ayuda. Muchas gracia	いつか、あなたに助けていただいたことにお返しができたらと[思]います。どうも	
25	1342	Alvaro es muy terco y no hay quien $convenza apearse de sus ideas	アルバロはとても頑固で彼の考えを[思]いとどまらせることのできる人は誰もい	

Fig. 4e: Result: EXTRACT

This operation is fundamentally the same as collecting examples on cards. Unlike a card collection, however, the digitized texts allow us later to make use of them in multiple manners. The form of the presentation has its own advantage in offering us an opportunity for reflection. For instance, on this occasion we notice that in the Japanese affirmative form ('I think'), we have the conjunction **と** (*to*) as in **と思う** (*to omou*), and, on the other hand, in the negative form ('I don't think') the combination of two particles **とは** (*to wa*) is more frequent: **とは思わない** (*to wa omowanai*).

5. The Spanish subjunctive mood

For focusing on specific forms, such as **と** (*to*) and **とは** (*to wa*), which we observed in the previous section, the KWIC format is more convenient than simple extraction. KWIC is the acronym for "key word in context" and is used frequently in corpus analysis. The following is the macro coding for KWIC output:

Macro-coding (2): KWIC

```
Sub Kwic() ' ver. 2003/07/16 by H. Ueda
  Dim i, n, m As Long: Dim intW, intB, intStart As Integer
    Dim strK, strW, strId(60000), strMae(60000), strAto(60000),
strBi(60000) As String


  strK = InputBox("INPUT Key Word" & vbCr & vbCr & _
    "Ex-1. no cre" & vbCr & "Ex-2. 思 ", "EXTRACT")
  If strK = "" Or strK = vbCancel Then End
```

```
  If Left(strK, 1) <= "~" Then intW = 2: intB = 3
  If Left(strK, 1) > "~" Then intW = 3: intB = 2

  n = ActiveSheet.UsedRange.Rows.Count

  For i = 2 To n
    strW = Cells(i, intW)
    intStart = InStr(strW, strK)
    Do While intStart > 0
      m = m + 1
      strId(m) = Cells(i, 1): strBi(m) = Cells(i, intB)
      strMae(m) = Left(strW, InStr(intStart, strW, strK) - 1)
      strAto(m) = Mid(strW, InStr(intStart, strW, strK))
      intStart = InStr(intStart + 1, strW, strK)
    Loop
  Next

  Worksheets.Add after:=ActiveSheet

  Cells(1, 1) = "ID.": Columns(2).ColumnWidth = 8
    Cells(1, 2) = "Left context": Columns(2).ColumnWidth = 50:
      Columns(2).HorizontalAlignment = xlRight
  Cells(1, 3) = "Right context": Columns(3).ColumnWidth = 50
  Cells(1, 4) = "Correspondence": Columns(4).ColumnWidth = 50

  For i = 1 To m
    Cells(i + 1, 1) = strId(i): Cells(i + 1, 2) = strMae(i)
    Cells(i + 1, 3) = strAto(i): Cells(i + 1, 4) = strBi(i)
  Next
End Sub
```

By collocating the key word in the centre of the line and sorting by either of the contexts (the preceding or the following context), we obtain a presentation like Figure 5a:

Fig. 5a: KWIC: **思** (omou)

Scrolling down the monitor, we find the lines where effectively **思いません** (*omoimasen:* 'I don't think' in polite form) occurs with preceding **とは** (*to wa*), and we find no examples of **と思いません** (*to omoimasen*).

Fig. 5b: KWIC: **思いません** (omoimasen)

The following are excerpts from the corresponding Spanish part :

(1a) **それは移植に敏感な木なので根づくとは【思】いません** / Es un árbol muy sensible al trasplante y no creo que $arraigue. ['It's a tree very sensitive to transplanting and I don't think it will take root.']

(1b) **私たちはフランシスコにこの仕事ができるとは【思】いません** / No consideramos que Francisco $sea capaz de realizar este trabajo. ['We don't consider that Francisco is able to do this work.']

(1c) **私はこのスーツケースが** 20 **キロ以上あるとは【思】いません** / No creo que esta maleta $tenga más de veinte kilos. ['I don't think this suitcase weighs more than twenty kilos.']

(1d) **ホセのように誠実な人にそんな嘘が言えるなんて【思】いもよらなかった** / Es inconcebible que una persona tan honrada como José $pueda decir una mentira así. ['It's incredible that a person as honorable as José can tell such a lie.']

So far, we have confirmed that clauses governed by *no creer* ('I don't think') or *no considerar* ('I don't consider') use the subjunctive mood, something which is shown in elementary grammars of Spanish. Those examples correspond to **とは思いません** (*to wa omoimasen*) in Japanese. I would especially like to draw attention to the fourth example, where instead of **とは** (*to wa*), a peculiar form **なんて** (*nante*) is used. These forms **とは** and **なんて** are extremely similar in the sense that both are used to express an object of a kind of evaluation. Proceeding further, we reach the lines presented in the next Figure:

Fig. 5c: **思わない** (omowanai)

Here again, we have a combination of **とは** and **思わない** : *to wa omowanai* ('I don't think' in its plain form):

(2a) **私はそんな給料で生活必需品がまかなえるとは【思】わない** / Con ese salario no creo poder cubrir mis necesidades. ['With that salary I don't think I can cover my basic needs.']

(2b) **私はイサベルが今度の恋人と長続きするとは【思】わない** / No creo que Isabel $dure mucho con su nuevo novio. ['I don't think Isabel will last long with her new boyfriend.']

(2c) **映画が夜の** 12 **時にまで終わるとは【思】わない** / No creo que la película se $termine antes de medianoche. ['I don't think the movie will end before midnight.']

Curiously enough, however, in interrogative form **思わない？** (*omowanai*?: 'Don't you think?'), we find only **と** (*to*), excluding **とは** (*to wa*):

(3a) **授業に出ないで単位をとろうなんてずうずうしいと【思】わない？** / Eso de querer aprobar sin venir a clase pasa de castaño oscuro, ¿no crees? ['Wanting to pass the exam without coming to class is shameless,

don't you think?']

(3b) **ホルヘは両親に冷たいと【思】わない？** / ¿No te parece que Jorge es muy despegado de sus padres? ['Don't you think Jorge is very cold to his parents?']

(3c) **ほらその馬を見てごらん．すばらしいと【思】わないかい？** / Mira ese caballo ... ¡Magnífico ejemplar!, ¿verdad? ['Look at that horse, a wonderful specimen, isn't it?']

Grammatically, all four cases of combination - **と思う** (*to omou*), **とは思う** (*to wa omou*), **と思わない** (*to omowanai*) and **とは思わない** (*to wa omowanai*) - are possible and each one has its own meaning. In spite of this fact, in our bilingual database we find only examples of **と思う** (*to omou*), on the one hand, and **とは思わない** (*to wa omowanai*), on the other; and in the particular interrogative context we have **と思わない？**(*to omowanai?*). It is difficult to believe these facts are mere products of chance. Now let us retrieve the Spanish sequence of "*no creer*" ('I don't think') :

(4a) Es un árbol muy sensible al trasplante y no creo que $arraigue. / **それは移植に敏感な木なので根づくとは思いません** ['It is a tree very sensitive to transplanting and I don't think it will take root.']

(4b) Rogelio es un hombre inmaduro y no creo que $pueda llevar el negocio. / **ロヘリオは未熟で商売をやっていけるとは思えません** ['Rogelio is a very immature man and I don't think he can manage the business.']

(4c) El médico no cree que el enfermo $vaya a salvarse. / **医者は病人が回復するとは思っていない** ['The doctor doesn't think the patient is going to make a recovery.']

(4d) Cayetana tiene tales encantos que no creo que $haya hombre que se le $resista. / **カイェタナはすごく魅力的で彼女に抵抗できる男はいないと思う** ['Cayetana has such charms that I don't think there is any man who can resist her.']

Previous studies of Spanish syntax have remarked that in negative constructions, the subjunctive mood is used in the subordinate clause - although they have ignored the actual reason why the subjunctive appears in this negative context. On the other hand, we remain without an explanation as to why the subjunctive mood is also used in the affirmative context, as in *es mejor que* 'It is better that', or *es probable que* 'It is probable that'. As I will point out in a moment, the key to solving this problem can be found in this Spanish-Japanese bilingual data.

In the literature of Japanese linguistics, **は** (*wa*) appearing before a

negative predicate has been explained as a case of contrastive function. Effectively in **私は海には行きません** (*watashi wa umi ni wa ikimasen*, 'I don't go to sea'), we detect a certain contrastive meaning to, for example, **私は山に行きます** (*watashi wa yama ni ikimasu*, 'I go to mountain'). In the case of **私はペドロにこの仕事ができるとは思いません** (*watashi wa Pedro ni kono shigoto ga dekiru to wa omoimasen*, 'I don't think Pedro can do this work'), we do not detect such contrastiveness, but its meaning is rather "evaluative": 'It will be impossible for Pedro to do this work'[12] . I propose to make a distinction between a "factive" sentence such as **私は海には行きません** (*watashi wa umi ni wa ikimasen*, 'I don't go to sea'), where **は** (*wa*) possesses a contrastive meaning; and on the other hand, an "evaluative" sentence such as **私はペドロにこの仕事ができるとは思いません** (*watashi wa Pedro ni kono shigoto ga dekiru to wa omoimasen*, 'I don't think Pedro can do this work'), where **は** (*wa*) does not have a contrastive but an evaluative meaning.

Returning to the Spanish data, we find almost the same "evaluative" meaning in the cases where the subjunctive mood is used (4a-d). The negative form of *creer* ('to think') is not only a simple negation of the act of thinking, but contains a certain meaning of evaluation similar to "impossible to think", "unthinkable" or "incredible". This evaluative meaning is shared with the affirmative cases such as *es mejor que* ('it is better that') or *es probable que* ('it is probable that').

The reason why **は** (*wa*) does not appear in the interrogative context (3a-c) can be explained by the fact that these are factive sentences. The focus of interrogation is the fact of the hearer's real belief. The same is true in the case of Spanish where the indicative mood can be used, as in *¿No crees que es (ind.) maravilloso?* ('Don't you think it's marvelous?'), despite *creer* being in a negative context, where the subjunctive would normally be expected. When the meaning of evaluation is introduced, the subjunctive mood naturally appears in Spanish, and **は** (*wa*) appears in Japanese: *¿No crees que sea (subj.) maravilloso?* **それがすばらしいとは（だなんて）思わないでしょう？** (*sore ga subarashii to wa (danante) omowanaideshoo?*) 'Don't you think it's marvellous?'[13]

6. Conclusion

In this age of the twenty-first century, when large-scale collections of linguistic data and sophisticated corpus analyzers are available to linguists,

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[12] For details, see Ueda (2002).

[13] When translated into English, both the evaluative and factive sentences result in the same expression.

the use of a small hand-made corpus based on data cards collected manually, and of analyzers programmed by oneself, may seem to be scientifically backward. It would be something like a traveller who is driving a car or travelling by train looking at a jet plane crossing the sky above him. A jet plane takes us to our destination comfortably and in less time, but it does not permit us to observe the landscape or human life closely. Likewise, if we analyze a huge database which we have never read previously, and use prefabricated software whose workings are a sort of black box for us, it may be difficult for us to observe the data in detail and we may sometimes remain perplexed in the face of the huge amount of output.

Naturally, in Spanish corpus linguistics we also depend on a great deal of linguistic data published in books, on CD-ROM, or on the Internet[14] . We value such data-sources and it is not necessary to restrict our activity only to a hand-made form of corpus linguistics. But if we adhered all the time to the global standard, characterized by large corpora and prefabricated analyzers, we could not help feeling some apprehension.

Firstly, if all researchers attempted to analyze common, shared data using the same software, all of them might reach the same or similar conclusions, which would lead to a loss of originality and creativity. Unlike the natural sciences, where objectivity and confirmation of having reproduced previously obtained results are highly valued, in language studies the subjective intuition and the inspiration of the researcher, as a human being, are important. If this were not so, all research would turn out to be a simple mechanical repetition of previously proved solutions.

If we ignore our own creative capacity, we might become hunters for data and for the software of others, never being satisfied with actual possession. We would be easily influenced by others' data processing and information technology, and thus miss the chance to develop our own identity. What is important for corpus linguistics is not information about corpus linguistics, but its meaningful practice. We know well that actually in English, Japanese, Chinese or other well-studied languages a great deal of linguistic data has been accumulated, and that the research level of corpus linguistics is amazing. Although it is difficult for a beginner to develop his or her own linguistic database, and to prepare his or her own analyzer, if everyone converged into the same main stream, the study of corpora might

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[14] For Spanish linguistic corpora, see, for example, the printed indices of Kock (1991, 1992), "Macro-corpus" in the CD-Rom version of Samper Padilla (1997), the Internet site of Real Academia Española (http://www.rae.es) and that of Mark Davies (Brigham Young University, http://davies-linguistics.byu.edu). The search engines *Google*, *Goo* or *Alta Vista* are also useful.

be converted into a monocultural discipline. This monocultural state of affairs would be dangerous not only for the entire community of corpus linguistics, but also for its dominant branch (English corpus linguistics).

A personal corpus, even a small one, may serve as material to be compared with a huge database, which will ensure the real comprehension of both. For a beginner to develop his or her own analyzer, however simple it is, may enable him or her acquire the capacity to trust his or her own discretion when facing a new situation. The advantages of a "hand-made" culture are that it strengthens the ability to account for one's own work with confidence, and that it facilitates the sharing of skills and knowledge with other participants. In the history of one's study, the achievements of each stage will have proceeded logically to the next stage. The same will be true in case of corpus making and the programming of analyzers.

On this occasion I have explained methods of programming by means of macro-coding supplied in well-known software packages[15] . The learning of a programming language may be compared to foreign language learning in the sense that, while the grammatical and lexical rules are so strict as not to allow any arbitrary use, once the fundamentals are acquired, they permit us to output an unlimited number of expressions. The rules do not impose restrictions on us but rather provide us with freedom. The discussion of whether or not programming skills are necessary for students of linguistics seems at present to be inconclusive. I would like, however, to recommend students to acquire a basic knowledge of programming language as well as of foreign languages.

[15] For programming for corpus analysis in the command line of MS-DOS, see, for example, Okada (1995), Nakano (1996), Ueda (1998) and Nakao, Akasegawa and Miyagawa (2002); for the analyses in a GUI environment, see Ito (2002) and Sano (2003).

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Multilateral Interpretation of Corpus-based Semantic Analysis – The Case of the German Verb of Movement *fahren* –

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0. Introduction

The aim of this article is to demonstrate that a corpus-based semantic analysis must be undertaken under consideration of various aspects and it can offer only thereby useful suggestions for theoretical investigations. The object of the analysis is the German verb of movement *fahren*, of which collocations in corpora are analyzed. This verb appears in various usages according to context, the range of its meaning stretching from the central usage (change of place by means of a surface vehicle) to various peripheral usages such as perception of a sensation, economic condition etc.[1] To avoid unnecessarily complicated discussion I restrict myself to several phenomena in the domain of central usage. To be treated is the question of how the existence (or absence) and type of certain collocated components affect the sentence meaning. I will discuss three phenomena in detail: 1) the relation between the moving entity and directional phrase, together with its cognitive implications, 2) the ingressive reading of *fahren* considering the relation between temporal adverbials and verbal aspect (*germ. Aktionsart*) at the pragmatic level and 3) the suppression of the feature "by means of a surface vehicle". For an adequate interpretation of these phenomena one must consider various factors that are of a different character. They cannot be accounted for in any homogeneous manner. This point of view leads to the hypothesis that the meaning of a lexical item is not a homogeneous unity, but the result of diverse semantic and pragmatic processes that can be accounted for only by a multilateral approach.

1. Corpus

The collocation analysis below is based upon the following text corpora

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[1] Examples of these are: Der Schreck fuhr ihm in die Hände. (LSO 1291; as for this code cf. footnote 5.) 'The fright got into his hands (i.e. made his hands shake).' [...] wir sind gut damit gefahren. (BZ3 203) 'We've got a good deal with that.'

of the *Institut für deutsche Sprache* (IdS, Mannheim, Germany): *Mannheimer Korpus 1* and *2*, the *Bonner Zeitungskorpus*, the *Handbuchkopora 1985-87*, partially *'88*, and the *LIMAS-Korpus*.[2] The material for the analysis consists of all the instances with the verb *fahren* in these corpora.[3] The corpora include 2481 instances with the simplex *fahren* in various verbal forms. (The attributively used participle forms such as *im fahrenden Zug* ['in the running train': present participle] and *die gefahrenen Kilometer* ['the kilometers driven': past participle] are not included.) Among them are 2214 (89.2%) instances that describe a change of place with an external vehicle. They constitute the object of this article.

2. Syntactic and Semantic Structure of *fahren*

Change of place with a surface vehicle is the most frequent and thereby central usage of *fahren*. By the modus of movement it is generally distinguished from *gehen* (on foot) and from *fliegen* (by means of a flying vehicle).[4] Automobiles, trains, ships, wagons pulled by animals, bicycles etc. belong to the prototypical surface vehicle. Furthermore elevators, wheelchairs, wagons without motive power, such as wheelbarrows and baby carriages, and sledges also collocate with *fahren*.

Fahren in this usage can be used as an intransitive or transitive verb. Used intransitively the subject refers to the vehicle (THEME), as in (1), or the person who is moving with the vehicle (THEME), as in (2). In cases in which a person is the subject the vehicle can appear in the INSTRUMENTAL prepositional phrase, as in (3). There are only a few examples in which the subject refers to the transported thing, as in (4). In the transitive usage the subject is the driver (AGENT) or the vehicle (POWER SOURCE) and the accusative object is either the vehicle (THEME) or the transported thing

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[2] *Mannheimer Korpus 1* is a representative corpus of written German with ca. 2.2 million word forms and *Mannheimer Korpus 2* is conceived as a supplement to the former corpus (ca. 300 thousand word forms). The *Bonner Zeitungskorpus* consists of the articles of two newspapers of which one is published in West Germany and the other in East Germany (ca. 3.1 million word forms). The complete *Handbuchkorpora* composed of newspaper and magazine articles include ca. 11 million word forms. The *LIMAS-Korpus* consists of 500 text extracts each with 2000 word forms. Detailed information about the corpora can be found on the website of the institute: http://www.ids-mannheim.de/kt/projekte/korpora/archiv.html

[3] I obtained the material using the retrieval system at the institute on 3rd April 1990. I thank the institute and its staff for their friendly support at that time.

[4] This distinction does not hold in a few cases. According to certain contexts it is neutralized. As a result *gehen* can refer to a change of place with a surface vehicle or with a flying vehicle by implication and *fahren* to a change of place with a flying vehicle. See below 3.4.2. This phenomenon is a topic of discussion in 4.2.

(THEME, including passengers), as in (5)-(7). A PATH expression appears often as the accusative object, as in (8).

(1) Vorn fahren fünf Schmuckwagen der Genossenschaft. (LSO 10639)[5]
'Five decorated cars of the cooperative are going in front.'
(2) Ich fahr mit dir nach Kirchbach. (LBC 50)
'I will go to Kirchbach with you.'
(3) Ich wollte mit einem Auto durch die Straße fahren. (ZB7 3536)
'I wanted to go through the street by car.'
(4) Anfang Juni fuhren die ersten Transporte in Richtung Westen. (LGB 6313)
'At the beginning of June the first transport went toward the west.'
(5) Die wollen gerne schwerere Züge fahren. (D49 925)
'They want to operate heavier trains.'
(6) Am nächsten Morgen fährt mich der Hotelbesitzer in die Ladenstraße des Städtchens (WGS 2233)
'On the next morning the owner of the hotel takes me to the shopping street of the small city by car.'
(7) U-Bahnen fahren fremde Menschen durch fremde Welten. (KZ3 10169)
'Subways take foreign people through foreign worlds.'
(8) Doch ich fahre den kleinen Umweg, [...] (FZ4 3959)
'However I take the small detour.'

Semantically observed, the common feature of this usage is the movement of the vehicle. In a similar manner to the Lexical Conceptual Structure it can be noted as follows:[6]

(9) GO (LOC (x, a), LOC (x, b)) [x = surface vehicle, a, b = reference points of place]

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[5] The code after the example is the bibliographical indication of the IdS-corpora. The first three figures stand for the document and the succeeding numbers indicate the sentence number in the document. Decipherment of the document codes is listed in BRÜCKNER (1989, Anhang E).

[6] Cf. JACKENDOFF (1990), RAPP (1997). The notations below are simplified and modified in terms of RAPP (1997: 65ff.). I have no intention to claim any completeness of this notation. For example the “&” operator in the first argument of LOC could seem to be too ad hoc to many specialists for LCS. The notations here are intended only to show the semantic relations between the constituents.

This structure corresponds to (1). Other structures can be seen as synecdochial expansions from this structure:

(10) **GO (LOC (x** & y, **a), LOC (x** & y, **b))**
[x = surface vehicle, y = person or thing] (2), (3), (4)

(11) CAUSE (OPERATE (y, x), (**GO (LOC (x, a), LOC (x, b)))**)
[x = surface vehicle, y = person] (5)

(12) CAUSE (OPERATE (y, x), (**GO (LOC (x** & z, **a), LOC (x** & z, **b)))**)
[x = surface vehicle, y = person, z = person or thing] (6)

(13) CAUSE (**GO (LOC (x, a), LOC (x, b)**, GO (LOC (y, a), LOC (y, b))))
[x = surface vehicle, y = person] (7)

Various relations between the LOCs in the structure are expressed by directional phrases in the utterance. Each of the three types of directional phrase, i.e. SOURCE, PATH and GOAL, can collocate with *fahren* alone or in all of their possible combinations. In German linguistics it is generally assumed that the directional phrase concerned is a constituent whose appearance is controlled directly by the verb.[7] (Cf. HELBIG/SCHENKEL 1980: 279, ENGEL/SCHUMACHER 1978: 175) This assumption of so-called valency theory is based upon a close relationship between the verb and the directional phrase confirmed by the collocational constraint and a selectional characteristic.[8]

3. Conspicuous Phenomena from the Collocation Analysis

Some selected conspicuous phenomena from the collocation analysis are described here.[9] The selection was made according to the aspect introduced in 0. That is to say, the phenomena are to be interpreted by a multilateral approach.

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[7] Such a constituent is called *Ergänzung* (complement).

[8] At the first stage of valency theory an *Ergänzung* was an obligatory constituent which must collocate with the verb concerned. This definition was later abandoned because it came to be realized that many constituents that seem to have an essential relationship to the verb are not obligatory, but can be eventually omitted. Instead the hypothesis was proposed that an *Ergänzung* is selected by the semantic characteristics of the verb and specific to it (*verbspezifisch*) or its class (*subklassenspezifisch*). Cf. ENGEL (1977: 97ff.).

[9] A part of the analysis is treated in MUROI (1992).

3.1. Asymmetrical Distribution of Arguments with Regard to their Syntactic Realization

Each of the three types of moving entities –vehicle, person and transported thing – can appear in the subject NP, in the accusative object NP and in a prepositional phrase (PP). The structural layout in Section 2 corresponds to this possibility. However the possible constellations are not exploited equally. There is a strong asymmetry in the occupation of these syntactic positions in the practical usage. Personal participants have an extremely strong tendency to be realized as the subject. In case of vehicles their quantitative distribution is for the subject and PP nearly the same. Transported things occupy the subject position about so often as they do the position of the accusative object, but the occupation of subject position by a transported thing occurs in the majority of cases in a passive sentence (Table 1).

Table 1: Realization Forms of the Moving Entities

	Subject	Accusative Object	Prepositional Phrase
Person (1789 instances)	1776 (99.3%) (24 of them in passive)	82 (4.6%)	121 (6.8%)
Vehicle (780 instances)	311 (39.9%) (21 of them in passive)	146 (18.7%)	327 (41.9%)
Transported thing (56 instances)	24 (42.9%) (16 of them in passive)	23 (41.1%)	9 (16.1%)

In consideration of this distribution one can say that events denoted by the verb *fahren* are mostly seen from the aspect of human behavior. This phenomenon is an expression of linguistic anthropocentrism. This point of view is strengthened by the fact that nearly all the sentences that include a person in the accusative object or in a PP have a human subject. Only 12 exceptional instances have the subject referring to a vehicle when a person occurs in the same sentence, as in (7). The transported thing appears as the subject only if no person is mentioned.

3.2. Directional Phrase

As mentioned in Section 2 it is assumed that directional phrases are linked deeply with the meaning of *fahren*. Out of 2214 instances in the corpora, 1527 (69.0%) are instanced with one or more directional phrases. The frequency varies greatly among three types, however. The most frequent one is GOAL, followed by PATH, with SOURCE being the least frequent directional phrase (Table 2).

Table 2: Frequency of Directional Phrase

(Number of instances = 1527)	Instances with ...
SOURCE	118 (7.7%)
PATH	517 (33.9%)
GOAL	1119 (73.3%)

This disposition is connected with the occurrence of the moving entity. The strongly asymmetrical proportion looks even more so if we consider the instances with a moving person (Table 3). Which syntactic form the expression of the person occurs in does not essentially affect this proportion.

Table 3: Frequency of Directional Phrase in Instances with a Moving Person

(Number of instances = 1283)	Instances with ...
SOURCE	95 (7.4%)
PATH	367 (28.6%)
GOAL	1011 (78.8%)

In the case of the vehicle the asymmetry is not so strong (Table 4). But we must take that circumstance found in 3.1 into consideration. When the vehicle appears as an accusative object or in the INSTRUMENTAL prepositional phrase, the subject position is mostly occupied by a person. A sentence with a moving person has a strong tendency to have a GOAL phrase. Observing only the cases of vehicle as subject, the proportion looks different. It indicates that the moving vehicle has an affinity to a PATH phrase if no person is mentioned, namely the scene is not viewed from the aspect of human behavior.

Table 4: Frequency of Directional Phrase in Instances with a Moving Vehicle

	All instances (438)	Vehicle as subj. (196 instances)	Vehicle as acc. obj. (20 instances)	Vehicle in PP (225 instances)
	Instances with ...			
SOURCE	40 (9.1%)	23 (11.7%)	1 (5.0%)	17 (7.6%)
PATH	236 (53.9%)	126 (64.3%)	11 (55.0%)	100 (44.4%)
GOAL	233 (53.2%)	80 (40.8%)	11 (55.0%)	144 (64.0%)

3.3. Sentences without Directional Phrase

There are 687 (31.0%) instances with *fahren* denoting a movement with a surface vehicle that do not have any directional phrase.[10] The ratio is so

large (nearly a third) that talk of exceptions cannot be allowed. Naturally one must not overestimate this fact and assume for example that directional phrases are to be treated like other facultative adverbials. Although they are not obligatory, they do play an important role. Absence thereof eventually initiates an extra process for interpretation. Table 5 shows the distribution of some representative phrases in the instances without directional determination.

Table 5: Constituents in Instances without Directional Phrase

(Number of instances = 687)	Instances with ...		only with subject and ...	
Accusative object	239	(34.8%)	95	(13.8%)
Manner phrase	344	(50.1%)	131	(19.1%)
Temporal phrase	169	(24.6%)	38	(5.5%)
Instrumental phrase	76	(11.1%)	34	(4.9%)
Other adverbial phrase	162	(23.6%)	62	(9.0%)
No phrase other than subject	49	(7.1%)		

(Subject is out of count.)

The accusative object refers mostly to the vehicle in instances in which no directional phrase appears. More than half of the accusative objects have a vehicle as referent. In opposition a person appears seldom as the accusative object without combination with a directional determination (Table 6). This distribution is absolutely opposite to that of instances with directional phrases in which the vehicle appears seldom as the accusative object. (See Table 4.)

Table 6: Referent of Accusative Object (239 instances)

Vehicle	126	(52.7%)
Person	19	(7.9%)
Thing	12	(5.0%)
PATH	47	(19.7%)
Manner (speed etc.)	31	(13.0%)

3.4. Non-quantitative Analysis of Collocation

There are usage types of *fahren* that are to be found not so often, but quite remarkable to those who attempt to study the process for the interpretation of the sentence. To analyze such usages quantitatively is not so produc-

[10] If we include the expression of PATH in the form of the accusative object as in (8) in the directional phrase, the number amounts to 640 (28.9%).

tive, for the number of examples is not large and the effect of collocates (or the absence of collocates) can be pointed out only if the complex relation between the factors is revealed.

3.4.1. Ingressive Reading

The verb *fahren* sometimes has an ingressive reading describing the start of a process. An example from the corpora is:

(14) Der Zug nach Italien fährt in drei Stunden. (UZ4 6071)
'The train for Italy departs in three hours.'

The semantic structure looks as follows:[11]

(15) GO (LOC (x, a), ⌐LOC (x, a)) [x = surface vehicle]
(16) GO (LOC (x & y, a), ⌐LOC (x & y, a))[x = surface vehicle, y = person or thing]

This reading does not have any specific collocations regarding directional determinations. One might expect a directional phrase for the start point that corresponds to "a" in the semantic structure. But this expectation is not fulfilled. A restriction on the SOURCE phrase is not observed and it is rather the case in many instances that no directional phrase appears. The ingressive aspect is the linguistic correspondent to a punctual change. Therefore a temporal phrase for a point in time occurs often in instances of ingressive *fahren*. But this is not a specific collocation too, because a punctual temporal phrase occurs equally frequently in instances of other aspects e.g. in durative readings. It is however remarkable that many sentences of ingressive reading are short and have only a few constituents.

3.4.2. Suppression of the Feature "by means of a surface vehicle"

The verb *fahren* denoting a movement by means of a surface vehicle is distinguished from *fliegen* denoting a movement by means of a flying vehicle. This distinction does not hold in a few cases. In such cases *fahren* refers to a change of place not with a surface vehicle, but with a flying vehicle, for example:

(17) Mitte Januar fährt der neue Außenminister Schewardnadse nach Tokio. (QZ3 3640)

[11] Simplified and modified notations in terms of RAPP (1997: 67)

'In the middle of January the new Foreign Minister Shevardnadze goes to Tokyo.'

The semantic structure looks as follows:

(18) GO (LOC (x & y, a), LOC (x & y, b)) [x = vehicle, y = person] (“surface” suppressed)

The vehicle is not mentioned in such a sentence. The reading “by means of a flying vehicle” is motivated by the context.

4. Discussion

4.1. Cognitive Implications of the Relation between the Moving Entity and Directional Phrases

It is concluded from the collocational distributions analyzed in 3.1-3.3 that:

1) in a large number of instances with *fahren* in its central usage the event is seen from the aspect of human behavior,
2) if a moving person is mentioned, focus on the GOAL is overwhelmingly preferred,
3) if the moving vehicle is highlighted and no moving person is mentioned, focus on PATH is preferred, and
4) if the moving vehicle appears in the accusative, a directional phrase seldom occurs in the same sentence.

As to the first point, it is assumed that the preference for the aspect of human behavior comes from linguistic anthropocentrism, as stated in 3.1. The close relationship between human and GOAL on the one hand and between vehicle and PATH on the other hand is not derived from the semantic structure. All possible combinations are provided structurally. The cause of asymmetry is to be found in the point of view from which each scene is observed. This is the problem of the choice the speaker makes from all the alternative possibilities available on any given occasion. The choice is not made freely, but is controlled, or at least affected, by a certain schema.

In the case of a human, attention is paid mostly to the endpoint of the movement. There is a general interest to know which place the person reaches or will reach. When a person moves from one place to another, it is regarded as an event the person can achieve the effect on by intention. If an action or behavior is intentional, it is asked what purpose is to be found behind it.[12] In case of a movement the purpose is usually the destination. Two

examples of evidence for this assumption can be given. The first one is the existence of the metaphor “purposes are destinations” (LAKOFF 1987: 277f.). It comes from the notion that the purpose of the movement is normally to reach the destination. The second example comes from a collocation analysis. A final adverbial is much more often collocated with a GOAL phrase (in 89 of 1119 instances = 8.0%) than with a PATH phrase (in 19 of 517 instances = 3.7%). It is assumed that the GOAL is combined with the need for mentioning the purpose.

The case of the vehicle looks absolutely the opposite. If the scene is not observed from the aspect of a person, the destination is not so important. Here the route of the moving is dominant. It indicates that the purpose of moving is less focused on. The mind of expression is rather on a description of the current scene than to present it in broader contextual relations. This assumption is also confirmed by a collocation analysis. A final adverbial seldom occurs in this case. Among 196 instances in which a vehicle appears in the subject position and collocates with a directional determination, only 4 include a final adverbial.

The fourth point, i.e. the almost complementary distribution of the moving vehicle in the accusative and the directional phrase, can be also interpreted under consideration of the aspect from which the scene is observed. In most cases where the moving vehicle is in the accusative no directional phrase cooccurs in the same sentence. This fact indicates that the aspect of moving is there out of focus. Rather, the operation of the vehicle by the driver is focused on. When on the other hand a directional phrase appears, the change of place is focused on. In such a case the vehicle appears in the INSTRUMENTAL phrase (cf. Table 4). In other words, where the vehicle is in the accusative the first argument of the predicate CAUSE in the semantic structure (11) is focused on and the second argument kept in the background, while in the case of a directional phase the focus is on the second argument in (12). This collocational restraint is an issue of focusing and cannot be derived from the semantic structure alone. Instances in which both constituents appear are to be considered as an example of evidence, although they are only a few in number and are exceptional. Even here the aspect of the vehicle

[12] I do not claim that the person here is an AGENT. I assume only that the event is as whole controlled by the intention of the moving person, regardless of whether the person is a driver or a passenger. The difference of the AGENT and THEME is not established on the level of semantics. It is rather the result of the pragmatic interpretation. (Cf. also Nakau 1994: 388ff.) For example the subject of the sentence *Wir fahren jetzt nach Berlin* ('Now we are going to Berlin') refers both to the AGENT and the THEME, if the speaker is the driver and the hearer the passenger. The same is to be applied to the transitive reading: *Wir fahren unsere Gäste in die Stadt.* ('We bring our guests to the city by car.')

as a means of transportation hardly plays any role, but is regarded as an operated object, whose change of place has a certain significant effect. Accordingly the scene described in such a case is very limited, for example a traffic accident, habitual driving on a fixed route, or transport of the vehicle itself to a certain place. This limitation is not a matter of structure, but comes from the saliency assigned to aspects observed in the scene.

HARRIS (1970) assumes that the distribution of a lexical item reflects its meaning and the semantic relations between items can be defined by the difference of their distribution. But as FIRTH (1951) states, the collocation is no more than one of the factors that determine the meaning. However we have observed that the quantitative collocation or distribution is not only to be attributed to the semantics of the item. It is rather the result of the cognitive attitude with which the human beings observe, understand and express events in the world.

4.2. Reduction of Arguments and Inference

In 3.4.1 the ingressive reading of *fahren* does not prove to be distinguished by any specific collocation. It seems curious that such an idiosyncratic reading does not have any characteristic collocation and that rather a shortness of sentence is characteristic. There are indeed many instances that include only the subject and a temporal phrase for the point in time. Paradoxically, an unusual usage seems to come into existence despite the any explicit motivation.

I suppose that a revised concept of the SAUSSUREAN principle of difference and the basic idea of the relevance theory are promising for resolving this problem. According to SAUSSURE (1978) the difference is the principle constructing the structure of language. The revised concept introduced here assumes that the principle is not restricted to the structural level, namely to semantics, but is to be applied to pragmatic processes. The essential assumption of relevance theory regards the meaning of a sentence as a process that yields to maximize the relevance by comparing various assumptions about it. The relevance is directly proportional to the contextual effect and inversely proportional to the required effort (SPERBER/WILSON 1986). I suppose that the principle of difference provides the motivation to make an assumption by which an ingressive reading of *fahren* can be inferred from a sentence with little information.

Sentences such as (14) seem to be reduced at first glance. Sentences with a directional phrase, the accusative object or a manner phrase have a certain contextual effect because they add some new information to the previous context. For example information is given concerning the destination, route or start point of movement, what kind of vehicle is used, or how the move-

ment proceeds. In contrast, (14) provides no such information. If (14) should mean only that the train at certain point of time moves, one could hardy imagine a context in which it might have some significant information. However, if (14) is interpreted as the opposition to “not moving”, namely it is inferred that the sentence means a change of state from standstill to movement, then it refers to a conspicuous event and is considered as having a large contextual effect.

This approach can be applied to the case of suppression of the feature “by means of a surface vehicle”. In this case, however, the opposition “surface” vs. “flying” is neutralized. Neutralization leads not always to a reduction of contextual effect although it reduces information by the opposition concerned. The suppression occurs, because this opposition is not so important. The issue in (17) is when and to where the minister will go, not by what means. The verb *fliegen* (to fly) instead of *fahren* could even sound too detailed. Namely the opposition is here unnecessary and there is no use for it even if one paid the cost of processing it.[13] Therefore mention of an INSTRUMENTAL phrase is excluded.

5. Conclusion and Implication

Collocation analysis reveals more than the semantic structure of a lexical item. The distributions analyzed in this article seem to prove that the collocation is an effect of processes in which various factors participate. The factors are of varying character. Cognitive schemata of preference and principle of the difference are no more than some of the examples. The interaction between them is controlled by inference rules to which the maximization of the relevance belongs. This consideration tells us that the processing of sentence meaning proceeds largely at the pragmatic level. If we postulate the meaning of a lexical item as its contribution to the meaning of a sentence, we can say that a large part of it can be explored only under considerations of pragmatic processes with various factors. What factors participate in the process and what kind of inference rules are applied to it vary from case to case. As a result a lexical item can represent various aspects of events or things corresponding to the variability of interpretative processes.[14] The necessity of a multilateral approach to this is obvious.

This conclusion provides the motivation to reconsider the problem of the

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[13] The opposition of *gehen* and *fahren* is not neutralized. Here I want only to point out the effect of it. It is connected with some difference of meaning. Sentences with *gehen* as *Ich gehe nach Deutschland* ('I am going to Germany') imply that the visit continues for a long time and is combined with a certain special purpose e.g. study, job etc. On the other hand, *fahren* does not have this implication.

status of the semantic structure. Because the meaning of a sentence is processed in large part at the pragmatic level and many relevant factors are not derived from the semantic structure, its role is much restrained. There arises a series of problems: How much and what kind of sense does the semantic structure have for explorations of linguistic meaning? Is it a starting point various readings should be derived from, or inversely a common denominator in which similar examples are summarized? Each question is followed by a number of subordinate questions. The answers to those questions are to be reserved for further investigation.

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[14] HERINGER (1999) analyses some German nouns by means of his method for distributive semantics and points out the multilateral character of lexical meaning. For example from the collocation of *Liebe* ('love') it is concluded that it corresponds to various aspects under which this psychological phenomenon is observed and experienced. To name some aspects of it: "love is liquid", "love is pain", "love puts two persons into one", "love is a natural power" and so on.

Tools for Creating Online Dictionaries Judeo-Spanish[1] – A Case Study –

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1. Introduction

Judeo-Spanish is a language spoken by the descendants of the Spanish Jews who have been living for many generations[2] in Turkey, Bulgaria, Greece, Morocco and also in France, Belgium, England, the United States, Mexico, Venezuela, after the second diaspora in the 20th century. One of the largest communities of Judeo-Spanish speakers is in Istanbul (about 22,000), but there are many small communities all around the world. It can be said without any hesitation that there are no monolingual Judeo-Spanish speakers. All of the native speakers can handle two or more different languages such as Turkish, Hebrew, Arabic, Greek, French, Italian, German, English and many others including standard Spanish, depending on the place they live. There is no standard version of the language. It is mistakenly thought that Judeo-Spanish is a very homogenous language, but there is much variation[3] in pronunciation, vocabulary and even syntax or orthography[4]. Some bilingual dictionaries have been published, but they do not focus on the spoken language of today. Very little linguistic research has been done on this endangered, spoken language. Fortunately, there has been a renaissance of Judeo-Spanish literature[5], and there is a correspondence circle *Ladinokomunita*[6] in the Internet to promote the use of Judeo-Spanish. Usual topics are about Sephardic culture and language, mainly sayings and proverbs, differences in vocabulary and pronunciation, etymology, etc. One of the purposes of the *Ladinokomunita*, besides promoting the use of the language, is spreading the use of a

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[1] Also known as *judezmo, ladino, haketia,* etc. Usually, *ladino* refers to the liturgical language of the Old Testament and other religious texts used by the Spanish Jews or Sephardic Jews. Judeo-Spanish refers to the vernacular language spoken by those communities. They continue to use it today, after 500 years from their expulsion from Spain. It is similar to Spanish or Old Castilian, but it has developed into a different language.

[2] See Díaz-Mas, Paloma (1993).

[3] See Chapter 6 "Variation in Judeo-Spanish" in Penny, Ralph (2000: 174-193).

[4] In Spain, Sephardic Jews used two alphabets, the Latin and the Hebrew, but today most of them use the alphabet of the language in their country.

[5] See Romero, Elena (1992).

[6] See details on the website: http://groups.yahoo.com/group/Ladinokomunita/

standardized method of spelling with Roman characters.

One of our objectives is to compile a corpus based on the written messages of this circle, and with the help of some of its members who are native speakers, to make a dictionary of modern Judeo-Spanish which can be searched online and can be freely downloaded to be easily printed, not only when the project is finished, but at any moment of the process. As this project involves a lot of data processing, collaboration through a network between members living in different countries is absolutely essential. For this reason, all data are processed in the same database server with server-side[7] scripts developed especially for this project with free open source software or well-known commercial software packages. In this way, with an appropriate use of simple network technology, the Internet can be very helpful. In our research, we use what is recently called LAMP, that is, Linux as the Operating system, Apache as a Web server, MySQL for the handling of relational databases and PHP4 for the interface and scripts. All the programs, scripts and data are stored in a central server, so every member can easily access them. We have been using this environment for different projects such as online concordances[8], research of lexical variation in Spanish[9], a system to dynamically generate linguistic atlases[10], etc.

2. System Environment and Software Used

Our system is compatible with most operating systems such as Windows[11], Mac OS[12], UNIX[13] and Linux[14]. Only standard browsers like Internet Explorer[15] or Netscape Navigator[16] connected to the Internet are needed to access the system and obtain the results of the queries in several formats, like text files, CSV[17], Excel[18] or XML[19]. We run a Linux PC server for our tests, and all of the software needed in this environment can be obtained by GNU/

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[7] Server-side programs are executed on the web server and generate HTML code to be viewed in a browser.

[8] See Ruiz Tinoco, Antonio (2001b).

[9] See Ávila, Raúl; Samper, José Antonio; Ueda, Hiroto; Ruiz Tinoco, Antonio et al. (2003).

[10] See Ruiz Tinoco, Antonio (2001a).

[11] Cf. http://www.microsoft.com/windows/default.asp

[12] Cf. http://www.apple.com

[13] OS developed by Bell Laboratories in 1969.

[14] OS created by Linus Torvalds. Cf. http://www.linux.org/

[15] Cf. http://windowsupdate.microsoft.com/?IE

[16] Cf. http://home.netscape.com/

[17] CSV: comma separated value. A file of ASCII fields separated by values.

[18] Cf. http://www.microsoft.com/office/excel/default.asp

[19] XML: Extensible Markup Language. Cf. http://www.w3.org/XML/

GPL[20] license, which is without cost for academic use.

All the tests so far have been performed in our Linux server and partially in Windows 98/2000/XP. To take advantage of many available resources, we run a Linux PC server, operated through standard Telnet[21] and FTP[22] client programs.

As a web server, we use Apache[23] (ver. 1.3.22) because of its high reliability. At this moment, newer versions of what are in an experimental phase. Much information about what can easily be found, for example, in the very detailed manual of Charles Aulds (2000) or even free tutorials on the Net.

We have chosen PHP4[24] as the programming language; however, there are others such as PERL[25] and Ruby[26]. PHP4 is a server-side programming language, and the scripts can be easily embedded into the HTML[27] code, allowing us to prepare interactive documents very easily. This language was originally designed to be used in hypertext documents, so as to make it easy to learn and to debug it. The ever-growing users' community is always helpful. PHP4's response speed is probably the highest among script languages, especially when used with Zend Optimizer[28] and Zend Accelerator[29]. PHP4 works as an Apache module and is very well documented in introductions with CD-ROMs included, such as Meloni, Julie C. (2000), more advanced texts such as Gerken, T. and Rastchiller, T. (2000) and thick manuals like Converse T. and Park J. (2000). Another great, additional advantage is that it can be used with almost any kind of database.

We think that a standard type of database like SQL[30] is the most adequate for our aims. To manage our database, we have chosen MySQL because of its robustness and speed. There are also enough books for MySQL, such as Judith Bowman et al. (1996), Paul Dubois et al. (1999), Randy J. Yarger et al. (1999), Michael Kofler (2003) and many recent ones.

In order to easily install the mentioned software, some of the following tools can be very useful. Installers like PHPDEV[31] and PHPTRIAD[32] are util-

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[20] Cf. http://www.gnu.org/

[21] Telnet: Internet protocol to log in to a remote computer on the Internet

[22] FTP: File Transfer Protocol

[23] Cf. http://httpd.apache.org

[24] Cf. http://www.php.net

[25] Cf. http://www.perl.org

[26] Cf. http://www.ruby-lang.org

[27] HTML: HyperText Markup Language

[28] Cf. http://www.zend.com/store/products/zend-optimizer.php

[29] Cf. http://www.zend.com/store/products/zend-accelerator.php

[30] SQL: Standard Query Language

ities for automatic installation in Windows and they include Apache, PHP, MySQL and some other additional software. Also, Nusphere[33] and Abriasoft[34] have published some packages allowing the easy installation in several OSs. The server with the LAMP environment will be as shown in Fig. 1.

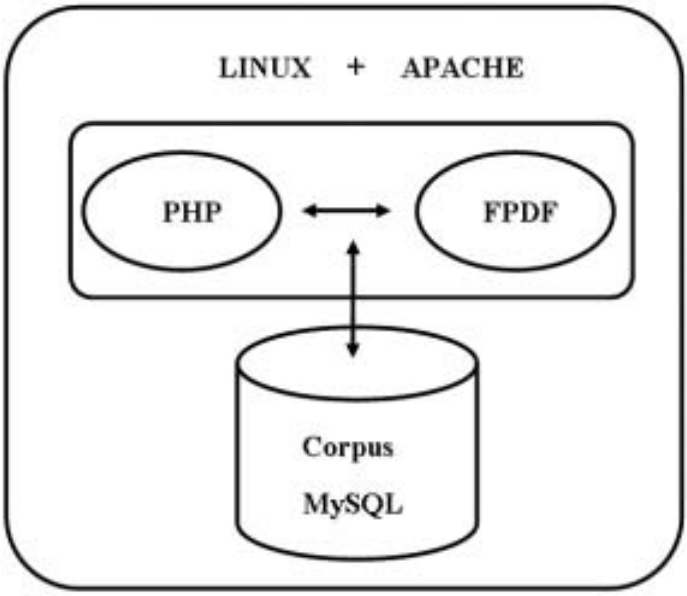

Figure 1: MySQL server

We also have to make use of a spreadsheet program to handle data in CSV format and a PHP script library to generate a printable dictionary in PDF[35] format. We use both MS Excel[36] and Openoffice[37] to handle CSV files and FPDF Library[38], but there are many other useful libraries and programs. For some statistics we have used WordSmith Tools[39], and our own scripts

[31] Cf. http://www.firepages.com.au/

[32] Cf. http://www.phpgeek.com

[33] Cf. http://www.nusphere.com

[34] Cf. http://www.abriasoft.com

[35] This is a simple way of formatting documents so that they can be viewed and printed on multiple platforms using Adobe technology. Cf. http://www.adobe.com/

[36] Cf. http://office.microsoft.com/home/default.aspx

[37] Cf. http://www.openoffice.org/

[38] FPDF is a PHP class which allows generating PDF files with PHP. Cf. http://www.fpdf.org/

[39] Cf. http://www.lexically.net/wordsmith/

were written in PHP with MySQL.

Finally, we find phpMyAdmin[40] very useful. It is a tool written in PHP to handle the administration of MySQL over the Internet and phpMyEdit[41], a code generator to display tables, add, change and delete records, sort data in ascending or descending order, etc.

3. The LK-Corpus Overview

The Corpus obtained from *Ladinokomunita* (hereafter called LK-Corpus) is presently composed of more than 6,000 messages and contains nearly 900,000 words. Messages must be preprocessed in order to clean HTML tags, etc. This process can be done easily using string manipulation functions of PHP[42] such as *strtok*, *explode* or *split*. These similar functions can be used to split the sentences of the whole corpus. In this way, split sentences or words can be put in their respective fields in a database. The following is an example of how to tokenize a text using PHP. The delimiters are "." and "?"[43].

```
//$text is the text to be tokenized
$text = "Ke es la diferensia entre la Fransia i la Espania? La Espania
exilo su djidios en 1492. La Fransia fizio la misma koza 98 aniadas antes
en 1394. La Ingletera en 1290 i tanto.";
$tokens = ".?";  //tokens
$tokenized_array = strtok($text, $tokens);
```

We can print the result to check it with a simple loop:

```
while ($tokenized_array) :
print "$tokenized_array<br>";
$tokenized_array = strtok ($tokens);
endwhile;
```

The output of the script will be as follows:

```
Ke es la diferensia entre la Fransia i la Espania
La Espania exilo su djidios en 1492
La Fransia fizio la misma koza 98 aniadas antes en 1394
```

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[40] Cf. http://www.phpmyadmin.net/home_page/

[41] Cf. http://phpmyedit.org/home.php

[42] See Wade, Matt; Adams, Paul; Cornutt Chris; Fahad Gilani, Syed; Wilton, Paul (2003).

[43] For a complete sentence splitter script, the characters included in $tokens must be changed.

La Ingletera en 1290 i tanto

There are 25,570 words (forms) which occur more than two times and 10,914 that occur more than five times. The most frequent 20 words in the corpus are as follows in Fig. 2.

Frequent Words in Judeo-Spanish			
Word	**Frequency**	**Word**	**Frequency**
de	43,720	es	9,290
ke	30,996	un	6,818
i	24,914	kon	6,622
en	24,517	por	6,353
la	23,593	me	6,118
el	18,270	una	5,890
a	16,223	lo	5,871
los	13,063	las	5,616
no	11,185	para	5,259
se	9,673	del	5,178

Figure 2: Most frequent words in Judeo-Spanish

It is basically similar to the standard Spanish frequency list[44], although we note some interesting differences such as the lower use of the definite articles.

N-grams can also be easily obtained. For example, Fig. 3 shows a list of the most frequent 3-grams.

[44] See Juilland, A. & Chang-Rodríguez, E. (1964).

Frequent 3-grams in Judeo-Spanish	
3-gram	**Frequency**
a todos los	254
os estados unidos	l228
lo ke se	189
de lo ke	184
ke no se	183
matilda de yerushalayim	182
me parese ke	181
saludos a todos	172
todo lo ke	169
ke se yama	164

Figure 3: Frequent 3-grams in Judeo-Spanish

At present, raw data must be preprocessed before being put in a database. After that the data can be distributed to the collaborators in this project in TXT or CSV format. After this step, the processed data can be accessed over the Internet by the collaborators in this project through an interface to a KWIC program and an interface for the database as we will see in the next sections.

4. PHP-KWIC

PHP-KWIC[45] is an online KWIC system written entirely in PHP with a few basic functions. Collaborators can use this system through the interface to this corpus[46] as in Fig. 4. Regular expressions can be used for complex search and the result can be saved in plain text format with the standard *save* function of Internet Explorer. Fig. 5 partially shows the result of the search for the regular expression "mansevik[oa]s*" to retrieve cases of the all forms (singular, plural, masculine and feminine) of the word *manseviko* (little or young boy).

[45] See Ruiz Tinoco, Antonio (2001b).

[46] Cf. http://133.12.37.60/diksionaryo-LK/kwic/

PHP-KWIC de tekstos en ladino (djudeo-espanyol)

1) Deskoje un korpus (agora ay 3 para eksperimentar)

- Una meskla de artikolos de Maestra Liegua
- Una parte de la Biblia ladina de Ferrara
- Letras de los miembros de LK

2) Mete una palavra kompleta o una parte.

mansevik[oa]s*

- REGEX POSIX kavzo insensitivo
- REGEX POSIX kavzo sensitivo

3) Mostrar konkordansyas

Uzo de PHP-KWIC:

"palavra" topa solamente "palavra"
(no son nesesarias las komillas)

[iy] sinyifika "i" o "y". Por egzemplo:
"gras[iy]as" topa "grasias" i "grasyas"

"." (un punto) sinyifika un karakter kualkiera. Por eshemplo:
"gras.as" topa tambyen "grasias" i "grasyas" (i tambyen "grasias", "grasuas", etc. si las ay)
"[bv].[bv]." topa "baba", "bovo", "biva", "vava", "beva", etc.
"(zo)|(zz)" topa "zo" i "zz"

El espasyo tambyen se kuenta. Por eshemplo:
" vas" topa las palavras ke empiezan por "vas", komo "vas" i "vantajes", etc.
"vas " topa as las palavras ke terminan por "vas", komo "empezavas", "esklavas", "ivas", etc.
" vas " topa solamente "vas".

Ay muchas mas otras ekspresiones regulares ke no se pueden esplikar aki por falta de espasyo.
"kavzo insensitivo" signifika ke no importa si es majiskula o minuskula.
"kavzo sensitivo" topa solamente ekzaktamente la mesma grafia.

Figure 4: Interface to PHP-KWIC for LK-Corpus

KWIC para [mansevik[oa]s*] en el korpus sigiente: "*LK*"

Korpus Linya Tekste:

LK	4154	aso, i muy maestro, i la violonista, una	**mansevika** djudiya,
LK	5750	1971 - un	**mansevik** ermozo, amable, respektuozo, atentivo, modesto i f
LK	6714	tado Papista en Italia en el kavzo de un	**mansevik** de nombre de
LK	10278	esamorada de un	**mansevik** , i malgrado esto tuvo ke despozarse kon el avuelo
LK	10488	sinava objektos arkeologicos. Un dia una	**mansevika** Judia Amerikana viso
LK	21805	una	**mansevika** estoniana. Ricardo mora en la Estonia, onde tuvo o
LK	23169	El	**mansevik** vizino de allado, David Israel (agora es hazan en
LK	25518	Estos	**mansevikos** i massevikas se riyivan ma nunka no la korektavan
LK	25518	Estos mansevikos i	**mansevikas** se riyivan ma nunka no la korektavan..
LK	33348	onde puedo a topar	**mansevika** para espozar? so de los
LK	33998	Avre el	**mansevik** i ve una ermoza donzeya.
LK	34568	nunka topi unos	**mansevikos** o massevikas de los maestros ke avlava muestra
LK	34568	nunka topi unos massevikos o	**mansevikas** de los maestros ke avlava muestra
LK	35162	sovre la	**mansevika** ke fue matada en el Shoa. Otra kantiga ke me fazin
LK	35427	Vos koro eskrivir dos byervos de unos	**mansevikos** en Estanbol ke formaron el
LK	35436	ke a los	**mansevikos** les agrada sintir "O sa sa" (una kantika humorist
LK	35445	Los felisitamos a estos	**mansevikos** ke, aunke no avlan el djudeo-espanyol
LK	35995	Yo se una	**mansevika** ke kanta en djudeo espanyol ke
LK	36002	nosotros los	**mansevikos** devemos azer por reskatar esta
LK	36028	Es un plaser grande tener una	**mansevika** komo ti es muestro kampo. Kon amista.

Figure 5: Result of the search for "mansevik[oa]s*"

Collaborators of the project can search through the LK-Corpus to check the use of the words or to look for good and natural examples that are of use to be put in the dictionary.

5. Databases and Interfaces

Our goal is to make a database with all types of information about any given entry, including categorial information, morphological or spelling variations, areas where the words or expressions are used, real examples, etc. There is a simple interface implemented with PhpMyEdit as seen in Fig. 6 and another interface with many functions implemented with phpMyAdmin as seen in Fig. 7. Databases are protected in different levels of security and a password must be used to edit or change its content. Browsing or searching the content is freely available with no restrictions at all. Fig. 8 shows the search interface for the database. The phpMyadmin database interface is not yet available to general browsing or searching. It is only for internal use at the moment. There are some commercial packages to administrate databases online, but phpMyAdmin is widely accepted as the de facto standard utility for MySQL administration.

Figure 6: A simple interface (phpMyEdit) to the dictionary database

The interface shown in Fig. 6 is a test including the entries of a limited number of words and their translations to standard Spanish, English and Turkish. In some cases, it is possible to search for examples of use and the

origin of the word. This interface has been implemented for training purposes and also to test the report scripts in PDF format as we will see later. At this stage it is a simple list of vocabulary with its translation prepared by the most active collaborator[47] of the project, but we hope that it will grow into a full multilingual Judeo-Spanish dictionary and a useful source of linguistic information about this endangered language.

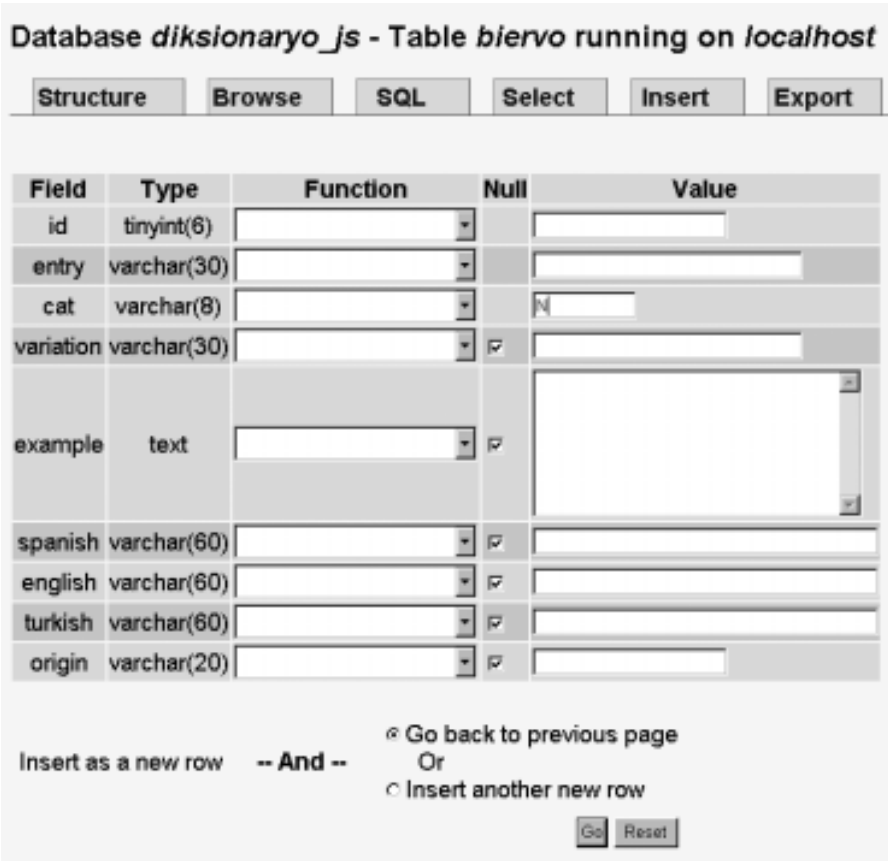

Figure 7: Interface (phpMyAdmin) to insert data manually to the database

[47] Güler Orgun, a native speaker of Judeo-Spanish, French and Turkish resident in Istambul, Turkey, and a member of *Ladinokomunita*.

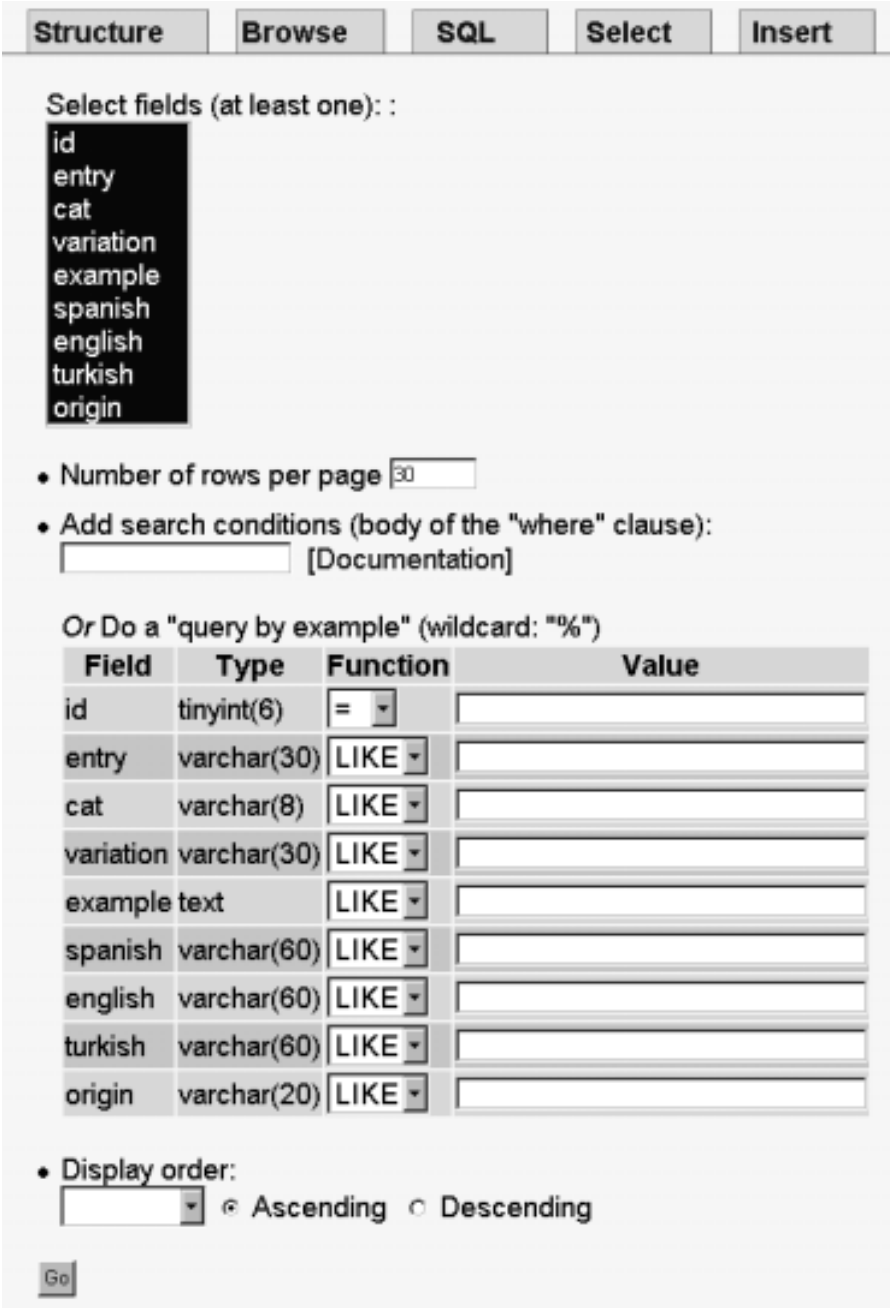

Figure 8: Search interface (phpMyAdmin) to the database

Fortunately, phpMyAdmin has many other functions which make it very attractive for research uses, but it may be a little bit sophisticated for general use by the collaborators. We are now working on a new set of interfaces of reduced functionality but easy to use.

6. Variation in modern Judeo-Spanish

Now, we will consider some variations in the morphology of modern Judeo-Spanish in LK-Corpus as an example of the use of the corpus and online tools.

Ralph Penny (2000) gives some features of Judeo-Spanish that we can easily check within the LK-Corpus, and we will see that modern Judeo-Spanish is not as standardized as can be thought. For example, he claims that the word-initial sequence *nue* is regularly modified to *mue* as in *muevo* (nuevo), *mueve*[48] (nueve) or *muestro* (nuestro), but as we will see, this is not necessarily the case in modern Judeo-Spanish used in *Ladinokomunita*. For our purpose, we search in the database using a *select* command as follows and obtain the results shown in Fig. 9.

```
SELECT word, frequency
FROM entries
WHERE word LIKE 'mue%';
```

Variation of the word-initial sequence *nue* and *mue*			
word	**frequency**	**word**	**frequency**
nuevo	102	muevo	309
nueva	77	mueva	188
nuevos	22	muevos	103
nuevas	8	muevas	83
nueve	8	mueve	23
nuestro	191	muestro	613
nuestra	143	muestra	1,350
nuestros	121	muestros	812
nuestras	39	muestras	214

Figure 9: Variation of the word-initial sequence *nue* and *mue*

From these figures it is clear that the sequence mue is more frequent than

[48] *mueve* in Judeo-Spanish is not the verb equivalent to "to move" as it is in standard Spanish, but the numeral "nine".

nue, but we can also confirm that *nue* forms are widely used and, interestingly, the proportion is quite irregular. For example, *muestro* occurs 613 times and *nuestro* 191 times, and *muestra* occurs 1,350 times while *nuestra* only occurs 143 times. Also, we can search through the PHP-KWIC interface both forms using simple regular expressions as "[mn]uev[oa]s*", "[mn]ueve" and "[mn]uestr[oa]s*".

Ralph Penny also reports that the sequence *rd* is regularly and universally modified to *dr*, as in *akodrarsi* (Sp. *acordarse*) and others, but still we find that there is much variation in our corpus for this sequence. For example, *akodro* occurs 391 times, but *akordo* occurs 112 times. The *rd/dr* variation is very frequent in many Judeo-Spanish words, but if we search for only the sequences *rd* and *rd*, we will find that there are many words that do not alternate this sequence, as is the case of proper names such as *Ricardo* or *Bardavid*; nouns such as *orden* and *saserdote* or adjectives such as *kordial* and *pardo*. Because the PHP-KWIC looks for character strings in the corpus, it is easy to search for any type of sequence.

7. Generating the dictionary

This project will probably take several years to finish. Meanwhile, the content of the database can be downloaded in PDF format which can be printed in almost any environment. It would be easier to put the static content of the dictionary in the server; however, the content of the dictionary is dynamically generated on demand using a class library called FPDF and some simple scripts written in PHP.

Currently, there are three incomplete bilingual dictionaries online that can be generated on demand: Spanish[49], English[50] and Turkish[51]. Fig. 10 shows the first page of the Spanish and Judeo-Spanish dictionary. Everyday these three dictionaries are downloaded many times from many parts of the world, as can be checked in the log file of the server.

[49] Cf. http://133.12.37.60/diksionaryo-LK/2rows-spanish.php
[50] Cf. http://133.12.37.60/diksionaryo-LK/2rows-english.php
[51] Cf. http://133.12.37.60/diksionaryo-LK/2rows-turkish.php

Diksionaryo de Djudeoespanyol a Espanyol

por Orgun, Portal / Tinoco

a (prep.): a. *Eg.:A todos los nuevos abonados, vos digo un grande "bienvenidos".*
a la fin: al final. *Eg.:Despues de munchos anyos de lavoro, al reusho a la fin a inventar esta makina.*
a la fin: al fin. *Eg.:Mete tu nombre a la fin de tu letra.*
a la letra: a rajatabla. *Eg.:Eyas observavan a la letra uno de los komandamientos.*
a las eskondidas: a escondidas (hacer algo). *Eg.:Eya se engrandeaso sin konoser djudios i se konvertio al kristianizmo, ma kontinuo, a las eskondidas, a sizer ayuno un diya al anyo porke se akodrava ke ansina uzavan en su kaza en Diya de Kipur.*
a partir de: a partir de. *Eg.:Kursos de Djudeo-espanyol a partir del 8 de noviembre.*
a pié: a pie. *Eg.:Kaminando a pie komo 10 minutos.*
abatido / abatida (adj.): estofado / estofada. *Eg.:Mos plaze komer el pishkado abatido porke es muy liviano.*
abandonár (v): abandonar. *Eg.:Nunka no mos van abandonar.*
abashada (f): cuesta abajo (ir). *Eg.:Moravamos senka de la Kale una abashadika ke se yamava El Kal De Los Frankos.*
abashada (f): resfriado, influenza, gripe. *Eg.:Estuve hazina kon una abashada muy fuerte.*
abashamentádo (adj.): con resfriado. *Eg.:Esto abashamentado.*
abashamiento (m): resfriado, influenza, gripe. *Eg.:Esto kon un abashamiento muy fuerte.*
abashár (v): bajar. *Eg.:No keriyamos abashar i suvir las eskaleras.*
abásho (adv.): abajo. *Eg.:Kale ser modesto i mirar mas abasho de si.*
abastár (v): bastar. *Eg.:De esta ropa ya abasta 4 metros: esta ancha.*
abatido / abatida (adj.): afligido / afligida. *Eg.:Desdichada de mi ! Me fui del bel din abatida i muy triste...*
abituál (adj.): habitual, normal. *Eg.:Esta sonriendo kon su dulsor abitual.*
abituár (v): habituar. *Eg.:Kale abituar a los muchachos ke aviendo el Djudeo-Espanyol.*
abolitár (v): girar, volver. *Eg.:Yo ya aboltí de mi viaje el viernes pasado.*
aborrecér (v): odiar, aborrecer. *Eg.:Todos los ke me aborrecen amasi la muerte.*
abrasár (v): abrazar. *Eg.:Los dos se abrasan i se meten a avlar.*
abráso (m): abrazo. *Eg.:Mos amasivieron kon bezos i abrasos.*
abrigárse (v): abrigarse. *Eg.:Eyos no tienen ni pan para komer ni lugar para abrigarsen.*
absolúto (adj.): absoluto. *Eg.:En la vida, todo es relativo i no ay nada de absoluto.*
abudárho (m): huevo seco salado. *Eg.:*
achákes de, por módo de (m): culpa de. *Eg.:Vino tadre achakes de la lluvia.*
achunkádo (adj.): agachado. *Eg.:Se kedo dos malas oras achunkado, no lo mirinos en la kara.*
adelantár (v): adelantar. *Eg.:Estamos perkurando para mantener i adelantar la konsentiya de esta lingua i de muesta emoza ensiyo kultural...*
adelántre (adv.): adelante. *Eg.:Siempre ke mos suyganos para adelantre i nunka para atras.*
además (adv.): además. *Eg.:Ademas vos kero resgraciar por vuestras palavras de apresiasion.*
adjektiva (m).: adjetivo. *Eg.:La verdad i la sinceridad son los dos adjektivos ke resumen mi boletin.*
adjirnsia (f): agencia. *Eg.:Me fui a la adjensia de viaje para merkar un bilyeto de avion.*
adjuntár (v): juntar, unir. *Eg.:Espero ke otros amigos de LK, se van adjuntar a este avenimiento emportante.*
adjustár (v): añadir. *Eg.:Mesitar bien, adjustar la tonet ke fue esprimida un poko para kitarle el kaldo.*
administrár (v): administrar. *Eg.:Teniya la responsabilidad de administrar el Reynado.*
admirár (v): admirar. *Eg.:Todos estos esfuerzos son de admirar.*
admirasión (f): admiración. *Eg.:Tengo grande admirasion por sus lavoros.*
adolesénsia (f): adolescencia. *Eg.:Me ritesh aboltar a mi adolesensia.*
adowár (v): reparar. *Eg.:Bushka de sizer adowar esta makina.*
adréso (m): dirección. *Eg.:Manda tu letra a su*

Figura 1/52

Figure 10: First page of the Judeo-Spanish and Spanish online dictionary

8. Some conclusions

Today, the use of server-side technologies on the Internet is very frequent with much information available, as is the case of databases and dynamic generation of printer-ready documents. On the other hand, the traditional editing of dictionaries is a time-consuming and arduous process, and its results are not always shared by all the people interested in using them. New technologies, though they need to be learned, provide us not only with a very

effective way of processing linguistic data, but also with the possibility of coordinated group research through the Net, as in the case of this database on Judeo-Spanish. We have seen that storing linguistic data in a network accessible database to generate dictionaries is not a difficult task anymore. Generating dictionaries on demand is not only a cheaper and quicker way to publish the results of research, but also a different way of producing them as we can choose the information needed by selecting only desired data with the required parameters.

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Socio-pragmatic Aspects of Workplace Talk

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Abstract

This paper describes some of the socio-pragmatic factors which influence the way people talk at work. Drawing on our extensive database of talk in New Zealand workplaces, I will discuss the importance of learning how to manage different aspects of workplace interaction. The paper focusses on two contrasting aspects of workplace interaction: firstly, on the importance of positive politeness or relational talk – namely small talk and humour at work; and secondly on a more negatively affective speech act, the speech act of refusals. In both cases, attention to socio-pragmatic factors such as the social context, social relationships, and the immediate discourse context of the talk, is crucial in expressing these aspects of language appropriately in workplace interaction.

1. Introduction

"….even advanced learners often show a marked imbalance between their grammatical and their pragmatic knowledge, or more specifically between the lexico-grammatical microlevel and the macrolevel of communicative intent and sociocultural context" (BARDOVI-HARLOG AND DORNYEI 1998:234).

When we start work or join a new workplace, we generally acquire a wide variety of new skills, including skills in communicating appropriately with our new colleagues and workmates. We learn, for instance, what kind of small talk and social talk is appropriate to establish good relations with the people we work with, and what kind of humour is acceptable in the new workplace. We learn how to get people to do things and how to disagree and refuse to do something in a way that does not cause offence. The skills we develop in learning how to do all these things with words – this range of different speech acts - are socio-pragmatic skills. Although they are widely acknowledged as very important aspects of communicative competence in the workplace (eg GUMPERZ, JUPP AND ROBERTS 1979, CLYNE 1991, ROBERTS, JUPP AND DAVIES 1992, BOXER 2002a), they have not yet received a great deal of attention in workplace communication research.

There is however a growing body of international research indicating that the ways in which different speech acts or functions of speech are appropriately expressed in different cultures is often remarkably different (eg. KASPER AND BLUM-KULKA 1993, GASS AND NEU 1996, BOXER 2002b, SPENCER-OATEY 2000, LEE WONG 2000, LITTLEWOOD 2001). So, for instance, a Japanese immigrant in New Zealand, often finds it very difficult to know how to manage socio-pragmatic aspects of workplace talk in English, because the ways in which she would make small talk, requests and complaints in Japanese is quite different. Similarly a New Zealander starting work in Japan typically finds that it is not easy to work out the appropriate ways of refusing an invitation or making a joke at work without causing offence. We acquire sociolinguistic and pragmatic skills in our native language(s) from years of immersion in a culture, mixing and working with others, but they present real challenges for second language learners. In this paper, I will describe how the material we have been collecting from New Zealand workplaces over the last seven years has helped to illuminate some of the socio-pragmatic skills which people acquire when they join a new workplace or "community of practice" (WENGER 1998, HOLMES AND MEYERHOFF 1999, HOLMES AND STUBBE 2003a).[1] And I will focus in particular on potential problems that New Zealand workers ways of expressing themselves might raise for those from a non-western culture such as Japan. First, however, let me provide a brief description of the Wellington Language in the Workplace project.

2. The Language in the Workplace Project

The Wellington Language in the Workplace (LWP) Project began in 1996 when we decided to examine ways in which people actually communicate at work.[2] The project was designed to analyze features of interpersonal communication in a variety of New Zealand workplaces, and data has been collected from government departments, commercial organizations, small businesses, factories, and even hospital wards. A great deal of earlier research on speech acts such as requests, complaints and refusals has used questionnaire data, discourse completion tasks, and role plays to collect information (see, for example, BEEBE AND CUMMINGS 1996, GASS AND

[1] ECKERT AND MCCONNELL-GINET (1992) define a community of practice as "an aggregate of people who come together around mutual engagement in an endeavor. Ways of doing things, ways of talking, beliefs, values, power relations - in short, practices - emerge in the course of this mutual endeavor " (1992:464). It is a useful term because it points to the range of shared understandings and beliefs, shared rules of interaction, and shared ways of talking which characterise workplace teams, and indicates why newcomers or those learning a second language may face challenges in using it effectively at work.

HOUCK 1999). The LWP methodology, however, has been developed to record authentic workplace interaction, using audio tapes, and more recently mini-disks, supplemented by video-recording whenever possible (HOLMES AND STUBBE 2003b). In general, we use volunteers to tape-record a range of their everyday work interactions over a period of two to three weeks. Some keep a recorder and microphone on their desks; others carry the equipment round with them.

Over the recording period, we find that people increasingly ignore the microphones and the video cameras (which are relatively small and fixed in place). The equipment simply comes to be regarded as a standard part of the furniture, and there are often comments indicating people have forgotten about the recording equipment. As a result our database includes some excellent examples of workplace interaction which are as close to "natural" as one could hope for. The complete Language in the Workplace Project Corpus currently comprises more than 2500 interactions, including small, relatively informal work-related discussions between two or three participants, ranging in time between twenty seconds and two hours, and more formal meetings varying in size from four to thirteen participants, and ranging in time from twenty minutes to four or five hours. The corpus also includes telephone calls and social talk as it occurred, for example, at the beginning of the day, at tea/coffee-breaks, and at lunchtime. The database therefore provides a wide range of examples which are valuable for examining how people in New Zealand workplaces express particular functions of speech.

3. Getting integrated at work: small talk and humour[3]

Our analyses of talk at work indicate that small talk plays a big part in constructing good workplace relationships, building up solidarity and maintaining good rapport between workers (HOLMES 2000a). Managing small talk in the workplace involves socio-pragmatic skills that most of us take for granted: knowing when small talk is appropriate or even obligatory, for instance, selecting suitable topics, and knowing when to stop and switch

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[2] See HOLMES 2000B, HOLMES, STUBBE AND VINE 1999a, 1999b, and the Language in the Workplace website: www.vuw.ac.nz/lals/lwp. The LWP Research team includes Professor Janet Holmes (Director), Ms Maria Stubbe (Research Fellow), Dr Bernadette Vine (Corpus Manager), Dr Meredith Marra (Research Officer), and a number of Research Associates. We would like to express our appreciation to all those who allowed their workplace interactions to be recorded and the Research Assistants who transcribed the data. The research was supported by a grant from the New Zealand Foundation for Research Science and Technology.

[3] This section draws particularly on HOLMES 2000a, HOLMES AND STUBBE 2003b.

back to business. These are skills we acquire with experience, but they can be challenging for those who have not had extensive workplace experience, as well as for those beginning work in a second language environment. And yet they are crucial to effective workplace integration.

These skills are rarely the focus of second language lessons, perhaps because they are taken for granted. Many textbooks which deal with “Business English” or “Business Communication” give little or no attention to the importance of managing social relationships at work. A random survey of textbooks aimed at teaching English for the workplace revealed that most devoted less than 15% of their content to social and interpersonal aspects of workplace interaction such as small talk. More advanced textbooks often ignore this area completely. Small talk is assumed to be too basic to deserve serious attention in such textbooks. Yet our research clearly demonstrates that small talk is an important component of workplace interaction, and using small talk appropriately, getting the content, placing, amount, and tone "right" can be a crucial and complex aspect of achieving workplace goals.

Using small talk and social talk appropriately in different cultures is clearly a potentially problematic area. These aspects of language use are important components of linguistic politeness, and there is an extensive literature testifying to the potential for cross-cultural problems in this area (e.g. DUFON ET AL 1994, LEE-WONG 2000, USAMI 2002). The following short excerpt, for instance, satisfies politeness requirements in a New Zealand workplace. But would it be considered adequate in Japan?

Example 1 (All names are pseudonyms.)[4]
Context: Diana, a manager of a government department, enters the office of her administrative assistant, Sally, at the beginning of the day to collect mail

1 D: good morning Sally lovely day
2 S: yes don't know what we're doing here we should be out in the sun
3 D: mm pity about the work really
4 S: how are your kids?
5 D: much better thank goodness any mail?

This is a typical example of small talk in a New Zealand workplace. It occurred as two people met for the first time that day and its main function was to oil the social wheels, to maintain good relations between Diana and

[4] Transcription conventions are provided at the end of the paper. For ease of reading, some examples have been slightly edited: eg by eliminating detail irrelevant to the point being illustrated.

Sally. Although it may seem very brief by Japanese standards, it served this purpose perfectly adequately in the New Zealand workplace where it occurred. The interaction covers standard New Zealand small talk topics – the weather, complaints about work, mention of family, health. In what follows, I will briefly discuss the topics or content of small talk in the New Zealand speech community and its distribution, or the times and places where small talk and social talk occur in New Zealand workplaces, and in doing so I will raise some questions about how far New Zealand norms differ from Japanese socio-pragmatic norms in this area.

3.1 Topics of small talk at work

Small talk in New Zealand workplaces typically focuses on non-controversial topics: the weather (eg. *cold eh, lovely day*), out-of-work social activities (eg. *wonderful concert last night*), sport (eg. *great match on Saturday, eh*), generalised complaints about the economy (eg. *stock market's crashed again I see*), positive comments on appearance (eg. *wow you're looking great*) work (eg. *how's it going?*), and so on. And whereas the ritual greeting of Japanese people focuses on location and asks *where are you going? or where have you come from?*, New Zealanders' ritual greetings tend to focus on health; they ask *how are you? or how are you going?* There is some skill involved in selecting appropriate topics for a particular workplace. In many workplaces, for example, sport is a perennial and safe topic of small talk. But you need to know about the relevant games and the latest scores to participate.

Example 2

1 A. great match on Saturday eh
2 B. yeah awesome

In others it was not so successful.

Example 3

1 A. great match on Saturday eh
2 B. what match?

In example 3, the speaker has wrongly assumed shared background knowledge. Common ground in the form of shared background knowledge, experience and/or attitudes is an important basis for successful small talk.

Clearly there is ample scope for miscommunication involving second language learners in this area. There are many well-known and well-documented cultural differences in terms of acceptable topics of small talk. Topics that are quite safe in New Zealand, such as comments on someone's physical appearance, their hair or their clothes, may cause offence to a Japanese person, for example. And topics that are perfectly normal in Japan, such

as how old you are, may surprise and offend New Zealanders.

Compliments constitute common currency of small talk in many cultures, but they are fraught with potential problems (see, for example, HOLMES AND BROWN 1987). Cultures differ in terms of what merits a compliment and who may be the appropriate recipient of a compliment. So it is worth reflecting on whether there are topics which Japanese people consider polite and acceptable in the Japanese workplace between colleagues, but which would cause offence in other cultures? These are the kinds of questions which a socio-pragmatic analysis raises.

Social factors, such as gender and how well you know someone, also influence the choice of possible topics of small talk in New Zealand, as they do in Japan. Appearance compliments between women are common small talk tokens (eg. *what a lovely suit! you're looking very smart today!*), while between New Zealand males they are very rare indeed in New Zealand (HOLMES 1988), and are almost non-existent in our workplace data. Do men compliment each other on their clothes or appearance in Japan? It is certainly inappropriate to compliment a male superior on his appearance, I understand. So this is another potential area of misunderstanding between New Zealand and Japanese businessmen.

Relative status is thus another relevant factor in both cultures, although the precise relative age of participants is not a factor in assessing this in New Zealand as it is in Japan. In our New Zealand data, as well as in WOLFSON'S American data, compliments from a woman on a man's appearance occurred only when the man was much younger than the woman. WOLFSON says: "there seems to be a rather strong if not categorical constraint against the giving of appearance-related compliments to higher-status males, especially in work-related settings" (1983:93). In New Zealand, people tend to avoid compliments "upwards", because the speaker might be thought of as flattering the higher status person, and this is not considered acceptable behaviour. Indeed it would be pejoratively described as "crawling" in New Zealand! In Japan I understand that compliments to a superior are inappropriate because they imply a lack of respect. So the same norms hold but for different reasons. This suggests that if the compliment is subtle and carefully worded in Japanese it may be acceptable. Clearly, this is another area for reflection, since getting things wrong can obviously cause offence.

Equally, there are more and less acceptable ways of *responding* to compliments. Many Asian cultures prescribe overt modesty as an appropriate response to a compliment, including denials and disagreements, whereas

western cultures tend to prescribe a gracious acceptance of some kind (HOLMES 1986, CHEN 1993). Clearly this is an area fraught with possible confusion. Especially since on other topics, English speakers are more likely to disagree in a wider range of situations than Japanese people.

Responding appropriately to workplace small talk may thus require some skill. In the workplaces we studied, responses to ritualistic or routine small talk questions would frequently elicit equally ritualistic comments about work.

Example 4

Context: Joan and Elizabeth pass in the corridor

1 E: hi Joan
2 J: hi how are you?
3 K: oh busy busy busy
4 J: mm terrible isn't it

Reference to how busy one is serves in the workplace as an ideal small talk token – a perfect topic for small talk at work. It indicates an orientation to the "proper" goals of the workplace in western terms, while also providing an acceptable account of why social relationships receive less attention than might be expected of good colleagues, and less than a Japanese person might consider polite.

What is also noticeable in such examples is the considerable skill involved in selecting the appropriate level of detail for the discussion of small talk topics. In most cases, in the New Zealand workplace, especially when they take place in passing, such interactions are very short. Just as we don't expect a blow-by-blow account of a colleague's gall bladder operation in response to *how are you*, it is considered equally inappropriate to respond in the workplace with a detailed response about your unreasonable workload to a ritual small talk query about how busy you are. More extensive social talk may occur, however, when people are beginning or ending a meeting.

3.2 When does small talk occur and how much is appropriate?

At the beginning of a meeting, small talk provides a gentle means of transition to the main business. Small talk warms people up socially, oils the interpersonal wheels, and gets talk started on a positive note. This is particularly important in Polynesian and in Japanese culture where, in some contexts, reducing preliminary social talk to a minimum, or attempting to dispense with it entirely can cause offence (METGE AND KINLOCH 1978, METGE 1995, CLYNE 1994). But it is true for all New Zealanders at work, to some extent.

So although many Japanese business people have the impression that westerners do not devote sufficient time to the social talk that Japanese consider so important, in fact, small talk is a regular feature of New Zealand interaction at work. It is typically, but not exclusively, found at the boundaries of interaction in New Zealand workplaces, as well as at the boundaries of the working day. It is almost mandatory to exchange small talk when people who work together first arrive at work, or meet for the first time in the working day, as example 1 demonstrates. An emergency or urgent task can displace it, but generally in the workplace, the first encounter of the day between work colleagues could be considered an obligatory site for small talk. The omission of small talk at such points will appear marked and is likely to be interpreted as evidence of bad manners or bad humour.

From a Japanese perspective the amount of small talk which occurs at such points may appear very truncated and short. To Japanese ears, New Zealanders seem to restrict small talk to a relatively brief exchange, and seem over-anxious to move on to business talk. Often New Zealand managers, for instance, say quite overtly, *well it's time to get down to business*, after what seems like a very inadequate amount of time devoted to oiling the social wheels from a Japanese point of view.

The ways in which New Zealanders move quickly from small talk to business was illustrated in example 1 where Diana ended the social chat by asking *any mail*? Example 5 is another example where an exchange which begins as social talk moves smoothly into business talk.

Example 5[5]
Context: Peg and her manager belong to a large commercial organisation. They are chatting at the end of a meeting of their project team. Peg is pregnant.

1. C: how is the baby
2. P: [drawls]: good: still just a baby though
3. C: right not a boy baby or a girl baby
4. P: no can't tell /it's legs crossed\
5. C: /haha you\ gonna have to wait….
6. are you feeling tired
7. P: yes but I just think it's summer too

[5] This example is discussed in more detail in a different context in HOLMES AND MARRA (2004)

8.		because I didn't you know because been in summer
9.		cos I wasn't pregnant last time or AS pregnant in the summertime
10.		so it was much easier cos I didn't know +
11.		um I had help (until) December last time (so it was easier)
12.	C:	hey you you're hoping you're gonna work [drawls]: through: /(what)\
13.	P:	/well + my\ plan is is to work full time up until the end of May
14.	C:	right
15.	P:	and then come back as we need as I'm needed after that
16.		just dependent on what happens with Daisy and Matt's group ……

Overall this conversation clearly moves from social talk to work talk but it is not clear at exactly what point the transition occurs. The discussion opens with non-work topics, Peg's baby's health and gender (lines 1-5) and Peg's own health (line 6-11), and gradually moves to the discussion of the impact of Peg's pregnancy on her contribution to the organisation (lines 12-16). In the first few lines, Peg's manager, Clara, is clearly engaging in small talk However, the discussion also addresses the implications of the information she elicits for the project team's objectives. Although the content of line 12 (*you're hoping you're gonna work through*) could be simply a further expression of interest, Peg's response (lines 13,15-16) indicates that she orients to her manager's comment as business rather than interpersonal in intent. And this is one of many such examples, where social talk moves gradually and almost imperceptibly into business talk. This kind of transition can be disconcerting for someone who expects the social talk to be more sustained and extensive, and who may therefore miss the important transition.

Clearly, the distinction between work talk and small talk is often difficult to draw in different cultures. While experienced workers move smoothly and gradually between work talk, personal talk, and social talk, discerning the boundaries, and avoiding over-stepping them are obvious potential pitfalls for someone from another culture. The cultural relativity of these patterns is illustrated by an example from MICHAEL CLYNE'S Australian data. While Japanese people tend to regard New Zealanders as engaging in minimal small talk, CLYNE (1991:21) points out that people from eastern and south-east Asia, especially Vietnamese, and northern Europe, especially Finns, do not expect small talk to occur at all within work domains. He provides an example from the factory data he collected in Melbourne where a Vietnamese woman, Giao, was puzzled because when she arrived with a request for help with a broken part, her shop steward, Liesl, responded by engaging her in small talk.

Example 6 from CLYNE (1994:148-9). (I have edited this example for ease of reading.)
Context: Vietnamese woman Giao shows her Austrian shop steward Liesl some broken parts

1 Giao: [shows broken parts to Anna]
2 Liesl: hallo Giao
3 Giao: /hallo Liesl\
4 Liesl: /I haven't seen you\ for ages ++ what's wrong? ++
5 Giao: (.....) so like this
6 Liesl: ooh they're breaking?
7 Giao: yeah see ++ I beg your pardon what you want?
8. Liesl: I haven't seen you for a long time

CLYNE comments that Giao was clearly confused by Liesl's social overture in line 4 and wondered why Liesl was talking about something other than the problem she was being asked to resolve.

So knowing *how much* small talk to use and when to extend it into more personal or social talk is a sophisticated socio-pragmatic skill which is very evident in our workplace data, but which is clearly culture-specific and thus a potential source of problems for those from cultures such as Japan where social talk at the boundaries of business encounters are is typically more sustained and extensive. The ability to move smoothly and pleasantly between social talk and work talk in different cultures is obviously a valuable skill for workers from different cultural backgrounds. Learning how to identify the relevant socio-cultural and discourse context clues is crucial in such contexts, and there is a good deal of evidence that this is an area where Japanese people are very skilled (eg FUKUSHIMA 2000, USAMI 2002). By drawing attention to the factors which are relevant in English-speaking cultures as opposed to in Japanese interaction, we can raise learners' awareness of the potential for misunderstanding.

I have suggested that even in areas involving positive politeness behaviours, such as small talk, there is potential for cross-cultural misunderstanding and learners face potential socio-pragmatic challenges in fitting into a new workplace. The potential for miscommunication is obviously greater when we consider areas involving negatively affective speech acts such as disagreements, complaints and refusals. How do New Zealanders refuse to do something they have been asked to do in the New Zealand workplace? And do their preferred strategies differ from those used by people from other cultures, such as Japan?

4. Refusals

Example 7

Context: Two young women taking a social break

1. Ang: how's your running going
2. Pam: I haven't done anything because of my toe….
3. Ang: you should do a marath- a half marathon with me
4. Pam: yeah [laughs]: probably not for a while:

(MORRISON 2003)

Example 7 illustrates that a refusal is a responsive speech act: it is the second component in a minimally two-part sequence. And, despite the token agreement expressed in the initial word *yeah* (line 4) in example 7 (a strategy commonly used to introduce disagreement and refusal in Japanese too (MORI 1999)), it is clear that a refusal is fundamentally a negative response. In line 4, Pam is saying "no" to Angela's invitation to run a half marathon with her (line 3). The core component of a refusal is a denial or an expression of unwillingness to comply with a previous request, invitation or offer. A refusal is quite clearly, then, a face-threatening and negatively affective speech act.

In many interactions, however, the refusing expression is just the heart of a highly complex speech act, which may involve extensive negotiation. In fact, a thorough description of a refusal often involves analysing lengthy negotiations between participants, involving extensive use of face saving strategies, reflecting the non-compliant nature of the speech act (HOUCK AND GASS 1996:49, GASS AND HOUCK 1999:2-3, CHEN, YE AND ZHANG 1995), although the specific utterance which comprises the refusing speech act may be quite concise.

Refusals are highly face threatening because they involve the rejection of a request which the communicator felt was legitimate to ask. In many cultures, including Japanese culture, refusals, like disagreements are avoided as much as possible (NAKAJIMA 1997, TAKAHASHI AND BEEBE 1986, BEEBE, TAKAHASHI AND ULISS-WELTS 1990). People find other ways of conveying their unwillingness to accept an offer or invitation, or refuse a request. The refuser is typically torn between their wish to resist an undesirable request and politeness constraints which require that they support the requester's face needs and self image. In western cultures, this conflict is often resolved by constructing a refusal which includes linguistic elements explicitly addressing the face needs of the requester (BESSON, ROLOFF AND PAULSON 1998). However, this is a prime area of socio-pragmatic contrast: there is a

great deal of variation in the extent to which people in different cultures take account of the face needs of their addressees when they refuse a request.

Overall, previous research on refusals has indicated that in many cultures people use rather complex linguistic and discursive strategies to refuse a request, and that they pay considerable attention to the face needs of their addressees in doing so. Our data, however, suggests that, at least in the workplace interacting with their colleagues, New Zealanders are often remarkably direct in refusing to do something, although they tend to mitigate the refusal with reasons and argumentation. Example 8 is a typical example from an interaction recorded in a white collar professional New Zealand organisation. A workplace team in a government department are discussing a range of different proposals for training courses. Turning to a specific proposal, the team leader decides a verbal presentation is required in order to deal with it fairly.

Example 8
Context: Team of eight people in government department discussing a training proposal

1. Len: um + and we would need to do a verbal for this one
2. *Bel:* *I'm not doing it*
3. [General laughter]
4. Sio: [laughs] bags not yeah
5. *Bel:* */seriously\ seriously*
6. Len: /+ that's a\ separate question [laughs] that's a separate question
7. but + as a general principle /+ last year we established\
8. *Bel:* */[laughs] I don't think (it'd) be appropriate for me to do it*
9. Len: that any existing provider that we were in danger of dropping
10. we did a verbal with + to ensure that they had had every opportunity…
11. Fem: /mm\
12. Aid: /mm\
13. Val: /I think Iris needs to do it\
14. *Bel:* */but it wouldn't be appropriate for me to do it\ would it*
15. Len: eh?
16. *Bel:* */it wouldn't be appropriate for me to\ do it would it*
17. Len: /it may\ well be appropriate for you to do it Belinda
18. [General laughter]
19. Fem: [laughs] /(oh no)\
20. *Bel:* */I don't think it is I can't\ I can't you know [voc] I'd be biased*
21. Fem: yeah
22. Len: I think we did a verbal for them last year actually

23. Fem: (no + they weren't in any contestable)
24. Bel: /(no they weren't in anything)\
25. Len: /no they weren't in\ the mix
26. *Bel: I'm definitely /biased Len [laughs]*
27. Len: /alright so they need to be they need to be\ verbalised
28. Sio: good way of getting there [laughs]
29. Len: we may be we may be quite keen on your bias
30. Val: oh no
31. *Bel: use Clive [laughs] no /I've\ had enough*
32. Len: /alright\
33. Fem: (had enough)
34. Len: /(so)\
35. Val: /I think\ that should be part of Iris's learning curve
36. [General laughter]
37. Fem: /have you been there\
38. [General laughter]
39. Val: no but /( )\
40. [General laughter]
41. Val: I said it first
42. [General laughter]
43. Len: it was- how about /um\
44. Cel: /I think\ I see a pair already set up here with it [laughs]
45. Len: how about ([name of next case])

I have provided a rather long and detailed transcription of this example because it is such a typical example of the way a refusal is negotiated in many of the white collar professional workplaces in which we recorded. Belinda is here the person who is "doing refusing" and despite the initial impression that she is refusing baldly and succinctly, *I'm not doing it* (line 2), her refusal is actually very extended, and makes use of a range of strategies (italicised lines in transcript: ie. lines 5, 8,14, 16, 20, 26, 31).

Interestingly, Belinda's initial bald-on-record statement *I'm not doing it* (line 2), actually anticipates the assignment of the verbalising task to her. In other words, she is refusing before she has been asked to take on this case. This reflects her anxiety to ensure she is not given the case, which is further apparent from the range of reasons she supplies to support her refusal. She first underscores her refusal with the repeated intensifiers *seriously, seriously* (line 5), and then provides some modification of its directness with supporting arguments. She argues that she is biased, a point she makes implicitly (and then repeats) by stating it would not be appropriate for her to take the

case (line 14, repeated in line 16), and then explicitly *I'd be biased* (line 20), and more strongly, reinforced with an intensifier (*definitely*) and a positive politeness marker in the form of her manager's name, *I'm definitely biased Len* (line 26). Finally she provides and alternative suggestion, a common and usually effective strategy when making a refusal, and then states firmly *no I've had enough* (line 31).

This is an interesting refusal which is embedded in a great deal of high energy, overlapping talk and laughter. Most of the team members are aware that this particular task is not an easy or desirable one because the people involved are known to be difficult to deal with. Thus the general laughter over Belinda's determination not to take on the case is in part sympathetic laughter and in part amusement at her wriggling efforts to evade it. Len, who is a flexible and sympathetic manager, clearly hears Belinda and it becomes apparent later accedes to her arguments that she should not have to take on the case. However in this excerpt he initially refuses to respond to her plea, arguing that who does the verbal is a *separate question* (line 6), and attempting to keep the discussion on the issue of whether a verbal is required. As she persists and others take up the issue of who will do the verbal, he refuses to explicitly reassure her that he will not assign her to the case; indeed he makes the point that she may be the most appropriate person and if so she will be required to do it. So he refutes her claim of inappropriateness quite directly: *it may well be appropriate for you to do it Belinda* (line 17) and then equally directly dismisses her arguments that her bias makes her an unsuitable person for the job: *we may be we may be quite keen on your bias* (line 29). However the tone of Len's comments and the general context of jocularity and laughter make it apparent that Len is teasing Belinda at this point, although her anxiety is so high that she cannot appreciate this,

There are three points worth highlighting in relation to this refusal. Firstly, Belinda is prepared to express her refusal very explicitly and directly, even to someone of higher status who has the power to ignore her wishes and direct her to do the task. All the research evidence suggests that this would not occur in a Japanese workplace, unless the people knew each other very well indeed (eg. BEEBE AND TAKAHASHI 1989a, 1989b).

Secondly, when her initial refusal is not accepted, Belinda complexifies it with first implicit and then explicit supporting arguments. Thirdly, the refusal is skilfully "managed" by the group, and by the manager in particular, using a range of strategies, involving first an attempt to rule the discussion of the issue as irrelevant and off-topic (*that's a separate question* (line 6)) and

then a mixture of teasing and jocularity which reduce the face threat of the refusal (BROWN AND LEVINSON 1987, HOLMES 2000b). This is a common pattern in our data. People use humour and jocular abuse to defuse tension and potentially problematic interactions (HOLMES AND MARRA 2002, HOLMES 2000b). In other cultures, avoidance, indirect strategies, and mitigation of the refusal are far more common means of managing such a situation (eg. HOUCK AND GASS 1996, GASS AND HOUCK 1999).

What these examples suggest is that between New Zealand workers who know each other well and who work together regularly, refusals may be very direct, *even when they are directed to someone of higher status and power.* However, note that in such cases, in white collar workplaces, subsequent utterances typically ameliorate and modify the bald-on-record refusal by providing reasons and arguments to support it. As mentioned above, research on refusals suggests that refusers pay a good deal of attention to the face needs of those being refused, often using avoidance or elaborate politeness strategies to mitigate the face threat of the refusal. In authentic refusals in New Zealand workplace contexts, attention to the face needs of others often appears to attract rather less attention. Other ways of ameliorating the refusal come into play, and in particular the use of arguments and suggestions which address the *transactional* imperatives of the organisation. So, apart from one use of his name (line 26), Belinda does not address Len's face needs. Rather she raises issues of appropriateness and bias, and suggests another more suitable person (in her view) to do the job, strategies that address the requirements and responsibilities of the organisation to complete the job. The personal face threat which her refusal constitutes to her manager takes a background position, and is dealt with rather by the team as a whole who defuse it with their laughter and jocular response to her refusal.

Turning to a different kind of workplace, expressions of refusal between close workmates and team members were even more direct and confrontational in the Wellington factory in which we recorded. (See DALY ET AL 2004.). The direct and confrontational language used between team members in the factory would raise eyebrows in middle class professional workplaces in New Zealand, and would very likely cause great offence to Asian people who overheard such language (cf CLYNE 1994). Their style of refusal certainly did not conform to the standard polite refusal behaviour described by KLINE AND FLOYD (1990), or the kind of behaviour reported in previous studies of refusals (BESSON, ROLOFF AND PAULSON 1998, GASS AND HOUCK 1999, SAEKI AND O'KEEFE 1994, WOOTTON 1981).

Even in this context however, we found that workers paid a good deal of attention to the socio-pragmatic context as well as the precise discourse context in selecting an appropriate way of refusing a request (DALY ET AL 2004.). Direct and face threatening refusals occurred only with close team mates. Outsiders were treated more circumspectly. Hence in refusing a request from a person who was not a team member, and whom she therefore did not know so well, the team leader who had been quite direct in refusing a team mate, adopted a much more conciliatory style, and one which would appear perfectly appropriate to an outsider or second language learner. Clearly contextual factors, and especially the relationship between participants, is very relevant in accounting for the way a refusal is encoded. In example 9 Ginette is refusing Francie, a status equal from outside the team who works across the factory in a quality assurance role.

Example 9
Context: Ginette, a Pacific Islander, team coordinator of the Power Rangers team, talking to Francie, a Maori woman of about the same age, who is not a team member.

1.	Fran:	do you have an NCR[6] for that (boxes) over there
2.	Gin:	yeah I've I'm waiting for a number + +
3.		I need to see Vicky about the NCR thing
4.		I haven't got a number for it yet
5.	Fran:	oh how would you get it
6.	Gin:	when I get to see Vicky +++
7.	Fran:	oh how's about you just give it to me now +
8.		take a copy of that + so I can compare it
9.		and I'll take the number then +++
10.	Gin:	(where are they) + do you want it right now
11.	Fran:	if it's possible [laughs]
12.	Gin:	it's just I've left a + I've got um Jennifer's working +
13.		going through it as well
14.	Fran:	oh okay is it possible tomorrow then
15.	Gin:	I'll get it to you tomorrow morning yeah

Francie's initial request is direct and clear *do you have an NCR for that box over there* (line 1). Ginette does not have the required number and so she must refuse to satisfy Francie's request. Ginette's refusal is conventionally

[6] A NCR is a Non Conformance Report, or a sheet filled out when a product is not up to standard.

polite and extended. She prefaces her refusal with a polite conventional agreement marker *yeah*, and then elaborates in the form of a full explanation *I've I'm waiting for a number I need to see Vicky about the NCR thing I haven't got a number for it yet* (lines 2-4).

Francie does not simply accept this refusal to comply with her request. She follows up with three further distinct attempts to elicit what she wants *oh how would you get it* (line 5), and then *oh how's about you just give it to me now* (lines 7-9) and finally *okay is it possible tomorrow then* (line 14). The pauses (marked +) following Francie's requests indicate Ginette's reluctance to respond. Ginette's request for clarification *do you want it right now* (line 10) buys her time before she provides another explanation (lines 12-13) for why she cannot give Francie the NCR right now. Finally they negotiate a compromise (lines 14-15) and the transaction is satisfactorily brought to completion.

The careful negotiation evident in this exchange indicates that Ginette is being conventionally respectful of Francie's face needs. While pursuing their transactional goals (Francie to see the relevant NCR and Ginette to ensure her team's paper-work is in order before it is checked by Francie), the two women skilfully avoid confrontation and direct disagreement or challenge. They have worked together for ten years, and clearly have a friendly relationship, but even so, Francie's status as a non-member of the Power Ranger's team is evident from subtle differences in the way they interact.

By contrast with the direct ways in which she refused requests from members of her team, in her interaction with Francie, Ginette uses a range of negative politeness strategies to convey her refusal in an acceptable way: avoidance strategies (lines 2-4, 10, 12-13), pauses (lines 2,9), explanations (lines 12-13), and hedges (line 12). In particular, there is no trace of any expletive in this exchange. As Ginette commented to us "I don`t treat anyone different, although the vocab used may differ." (Data from interview). In fact, Ginette was well aware that the team had a unique style of interaction, one that only core team members could be expected to use and understand: "Our team created a culture that we were all comfortable with…. when someone new joined us we obviously took the path of easing them into our culture."(Data from interview).

So our data indicates that New Zealand workers respond very sensitively to contextual factors in choosing how to encode refusals, behaviour which will be familiar to Japanese people, even if the precise range of strategies dif-

fers. Between team members people use direct, concise and apparently confrontational strategies, without elaboration or mitigation, and frequently reinforced by the use of expletives. Refusals with people outside of the team tend to be longer, more indirect, and include only mild expletives, if any. Team membership is thus a crucial factor in determining how such speech acts are constructed, negotiated and interpreted. Team members treat each other like family members or intimates in using direct and confrontational pragmatic strategies rather than conventionally polite ways of expressing complaints and refusals. Giving weight to such contextual factors will be behaviour familiar to those from Japanese culture, even if the precise ways in which refusals are appropriately encoded may be less familiar.

The generalisation which appears to hold across a number of New Zealand workplaces, then, seems to involve, most crucially, how well people know each other and how long they have worked together, a finding consistent with those of some previous ethnographic studies of western interactions (eg. D'AMICO-REISNER 1983, WOLFSON 1988), and overall with WOLFSON'S (1988) "bulge" theory, which claims that people are most polite to friends and acquaintances and least polite to intimates and strangers. In close-knit work teams, even status differences do not predict the use of indirectness or explicit politeness strategies to mitigate refusals. People tend to be direct and explicit in refusing people they have worked with over a long time, and/or with whom they have well-established relationships. Similar social factors have been identified as important in Japan. Both familiarity and status are clearly important sociolinguistic variables, though the precise ways in which these factors are assessed tends to differ from western communities (DOI 1981, USAMI 2002). And while age and gender are relevant factors in most societies, the precise weighting attributed to them tends to differ in different cultures, including Japan.

Our New Zealand research also demonstrates that in genuine face-to-face spoken interaction, a refusal may vary from a single one line direct utterance through to a highly complex and lengthy negotiation, incorporating a number of face saving strategies to accommodate the non-compliant nature of the speech act (c.f. HOUCK AND GASS 1996:49). The implications of this finding for teaching materials is also non-trivial.

5. Implications for teaching English

The examples discussed in this paper suggest that the materials needed to develop sociolinguistic and socio-pragmatic competence in the English-speaking workplace are not likely to be found in the standard textbooks used

to teach English as a second language. While some textbooks recognise that many speech acts cannot be confined to a single turn, as implied by data collection methods such as Discourse Completion Tasks (see BEEBE AND CUMMINGS 1996), few provide examples based on authentic recorded data or natural speech from genuine workplace interaction. Many textbooks suggest concise and compact formulae for learners to use in situations where they want to engage in small talk, refuse, disagree or complain. As the analysis in this paper demonstrates, such recommendations differ in many ways from the complex reality uncovered by recordings of the way people actually do these speech acts in real life situations.

In responding to the inadequacy of current materials for preparing NESB learners for workplace interaction, we have based our approach on our authentic materials. Our approach emphasises the development of socio-pragmatic awareness, and provides opportunities for developing the ability to analyse authentic data as well as produce appropriate speech acts in context.

Let's look at just one example of the kinds of exercises which our data suggests may be useful to EFL learners who want to develop socio-pragmatic competence in handling the realities of workplace talk (BROWN, DALY, HOLMES, NEWTON AND STUBBE fc).[7] The exercises we are devising involve a sequence in which learners proceed through four components:[8]

1. *Noticing / awareness raising:*
   listen and respond to questions about authentic workplace refusals
2. *Performing:*
   role-play set scenarios involving refusal based on our authentic data
3. *Analysing:*
   analyse role plays identifying the refusal strategies used and relevant contextual factors
4. *Practising:*
   using the strategies repeatedly until fluency is attained

Here is an example of a scenario based on our data which can be used in such exercises:

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[7] BOXER AND PICKERING (1995) provide examples of exercises based on authentic data which have been devised to develop competence in responding appropriately to "whinges" in informal conversational situations.

[8] This sequence is a work in progress and represents just one possible iteration.

5.1 Sample Scenario

You are working in a warehouse preparing orders for distribution. The supervisor has not organized the work roster properly and as a result you are overworked and behind on the orders. Another member of your team asks you to help him with a task he is supposed to be finishing because he wants to leave early.

Analysing video-recordings of authentic authentic workplace interactions, along with the transcripts as an additional tool, the teacher/trainer can ask learners to address such questions as:

- How and where in the interaction is the refusal communicated?
- How would you interpret the refusal?
- How does the addressee interpret the refusal?
- What can we tell about the relationship between people on the basis of the way they are talking to each other? i.e. what social messages are being communicated along with the refusal?
- How are these messages being communicated (wording, tone, non-verbal language)?
- How would you rate the politeness of this refusal?
- How would you rate the appropriateness of this refusal?

The materials being developed for EFL teachers and workplace communication trainers based on our factory data suggest specific activities, questions for discussion and reflection, and provide a range of suggestions as to how the learners may be guided in developing socio-pragmatic competence in the workplace. These materials offer a means of comparing the ways in which people from Asian cultures expect to handle a refusal with the ways in which speakers of New Zealand English express refusals, and thus provide the opportunity for improving intercultural understanding.

The goal of such training is to orient those from different linguistic and cultural communities not only to the new language that they will need to acquire in the English-speaking workplace, but also to an understanding of the way that language is used within a specific socio-cultural context, and to the messages that different ways of using the language carry.

6. Conclusion

Our research in a wide range of workplaces indicates that the sociolinguistic and socio-pragmatic demands of integrating into a new workplace are often very daunting. Learning ways of interacting which are appropriate and normal in a workplace is an important aspect of fitting in and becoming an

integrated member of the workplace as a community of practice. Socio-pragmatic competence is an often under-estimated aspect of successful intergration into a workplace team. According to a Robert Haft International 1985 survey only 15% of workers are fired because of lack of competence. The remaining 85% are let go because of their inability to get along with fellow employees. Talk is crucial in this process.

Even those born and brought up in an English-speaking speech community may find the process of learning how to do things appropriately with words at work very challenging. Fitting into the workplace involves learning the sociolinguistic and socio-pragmatic rules of expression which are particular to the specific community of practice one is joining. Managing workplace discourse, knowing how much small talk to use at the beginning of a meeting, how to make a joke, how to disagree without causing offence, and how to refuse effectively - these are examples of areas which can present pitfalls to people from cultures with different norms from those of their co-workers.

Our research strongly supports an approach to teaching and training which is firmly based in the workplaces in which people are working. Our analyses of the complexities of authentic workplace interaction suggest that second language teaching materials must move beyond short formulaic responses and artificially constructed text book dialogues, which bear little relation to genuine workplace talk. The analysis in this paper indicates that distinctive ways of doing things develop in particular communities of practice. Our experience suggests that teachers therefore need to make use of multi-media resources for work-oriented communications skills courses, based on authentic interaction in the organisations and factories in which their students will be working. In sum, our research provides evidence that the expensive and complex business of collecting and analysing authentic workplace interaction has worthwhile practical outcomes for those engaged in preparing people for the communicative demands of the workplace.

Transcription conventions

All names are pseudonyms.

YES	Capitals indicate emphatic stress
[laughs]	Paralinguistic features in square brackets, colons indicate start / finish
[drawls]	
+	Pause of up to one second
++	Two second pause

..../.....\...	Simultaneous speech
..../.....\...	
(hello)	Transcriber's best guess at an unclear utterance
?	Rising or question intonation
….	Some words omitted
[voc]	untranscribable noises

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What Do We Mean by "second" in Second Language Acquisition

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Abstract:

If we examine a variety of SLA (Second Language Acquisition) publications, we see that there are two general uses of the term 'second'. The first use of 'second' is related to the sequence of acquired linguistic knowledge, as reference to a 'second language' presupposes the existence of a first language (L1). In a large proportion of SLA research there is at work what Vivian Cook (e.g. 1996) has referred to as the 'monolingual bias'. This bias involves not only the assumption of a single and unified L1, but also the belief that this L1 remains completely intact throughout an individual's contact with other languages.

The second use of 'second' refers to the physical and geographical context of language learning. In some cases a distinction is made between naturalistic contexts, in which the learner is immersed in the language but not exposed to formal instruction, and classroom contexts, where formal instruction is the norm. Classroom contexts, in turn, are generally differentiated as either foreign (learning the target language outside the environment in which it is normally used) or second (learning the target language in the environment in which it is normally used).

In this paper I problematise these two uses of 'second' in SLA. In the first part, I examine the monolingual bias, first from a linguistic standpoint and second, from a sociolinguistic standpoint. I then go on to suggest that 'second' is in perhaps a misnomer when it is used to refer to the experiences of individuals who have had contact with three, four, five or more languages in their lifetimes. In the second part, I make the point that while it is right for SLA scholars to distinguish among classroom/ naturalistic and foreign/second contexts, they should also bear in mind that none of these contexts provides learning opportunities in anything like a predictable manner. I close with some thoughts on the continued use of 'second' in SLA,

1. Introduction

Second Language Acquisition (SLA)... is the common term used for the

name of the discipline. In general, SLA refers to the learning of another language after the native language has been learned. Sometimes the term refers to the learning of a third or fourth language. The important aspect is that SLA refers to the learning of a nonnative language *after* the learning of the native language. ... By this term, we mean both the acquisition of a second language in a classroom situation, as well as in more "natural" exposure situations. (Gass and Selinker 2001: 5)

This definition is taken from a recent comprehensive survey of Second Language Acquisition (hereafter SLA). In it, the authors take two general angles on the "second" in SLA. First there is reference to an assumed "native language", defined as "the first language a child learns" (Gass and Selinker 2001: 5). Using the term "native language" in this way, the authors are leaving out of their definition situations where two or more languages are learned from birth or this first language is later in childhood supplanted by another language which becomes the individual's primary language. This approach to SLA, I believe, is consistent with a certain monolinguistic bias that pervades the field. The monolingual bias involves not only an assumption of one L1 (native language), but also the belief that the L1 remains completely intact despite contact with a second language.

In this definition, there is also mention of a second important point, which is the possibility that a learner may be learning her/his third, fourth or even fifth language. This possibility is in fact far more common in the world than is often entertained by SLA researchers and it suggests that the term "second" itself might not even be the appropriate.

A second angle on "second" in this definition has to do with the context of learning. Gass and Selinker first make a distinction between classroom and naturalistic settings, and then go on to further break down the former category, contrasting two types of classroom language learning: foreign language learning and second language learning. The former term "refers to the learning of a nonnative language in the environment of one's native language" (Gass and Selinker, 2001: 5), for example, Japanese speakers learning English in Japan or English speakers learning Japanese in the UK. By contrast, second language learning is "the learning of a nonnative language in the environment in which that language is spoken" (Gass and Selinker, 2001: 5), for example Japanese speakers learning English in the U.K or English speakers learning Japanese in Japan. In contrasting these two different classroom situations, Gass and Selinker emphasise that "[t]he important point is that learning in a second language environment takes place with considerable

access to speakers of the language being learned, whereas learning in a foreign language environment usually does not" (Gass and Selinker, 2001: 5).

Thus, as regards context, "second" is seen to be three distinct general scenarios organised along two axes: +/- classroom and +/- language in the community. These three scenarios are represented in quadrants 1, 2 and 4 in figure 1 below. Quadrant 3, self-directed foreign language learning, has not, to my knowledge, been explored in much detail in SLA, although Tae Umino (2002) provides an interesting account of London-based learners of Japanese following televised Japanese as a foreign language course and Mark Warschauer's recent work (e.g. Warschauer, 2003) is beginning to inform us about internet based language practices.

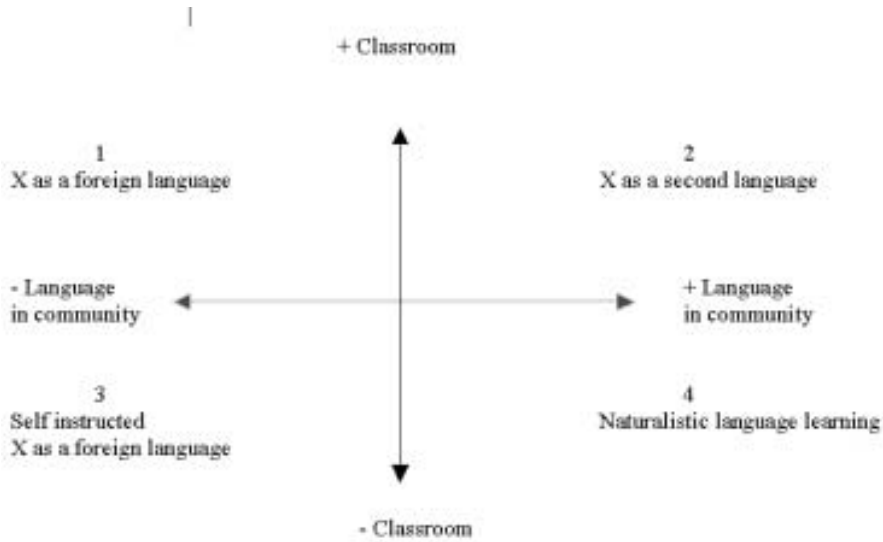

Figure 1: "Second" context scenarios

In citing Gass and Selinker, I do not wish to single them out for criticism. Rather, I wish to make the point that what they write may be taken as fairly standard response in the SLA literature to the question: "What do we mean by "second language" in SLA?" Thus their view would likely be accepted by those who take Universal Grammar perspective on SLA, those who focus exclusively on information processing, those who subscribe to input-interaction-output models and those who adopt a sociocultural approach. However, despite this seeming consensus on what we mean by "second" in SLA, it is my view that we need to unpack some of the assumptions behind these defi-

nitions, that to present "second" in the above manner is far too partial and overly streamlined. Thus, this paper is about problematising the two angles on "second" set forth above. In the Section 2, I examine the monolingual bias, first from a linguistic standpoint and second, from a sociolinguistic standpoint. I then go on to discuss how "second" is perhaps not the right term to use when considering the experiences of individuals who have had contact with three, four, five or more languages in their lifetimes. In the Section 3, I make the point that while it is right for SLA scholars to distinguish among classroom/ naturalistic and foreign/second contexts, they should also bear in mind that none of these contexts provides learning opportunities in anything like a predictable manner.

2. Monolingualism in SLA research

2.1 The monolingual bias

As presented by authors such as Gass and Selinker, the "second" in SLA implies a unitary and singular "first" as a predecessor and this is part of a certain monolingual bias in much SLA research. However, as has often been pointed out by sociolinguists such as Edwards (1994) and Romaine (1996), monolingualism is certainly not the norm in the world: Bi and multilingualism are. In addition, in countries which traditionally have assimilated immigrants, making their children into monolinguals (e.g. the US and Britain), recent changes in the nature of immigration (e.g. communities which retain a and maintain a large proportion of their home culture and language such as Latino Americans and British Sikhs) have meant that in schools in many urban areas, a high proportion of students are bi or multilingual. These students often have variable competence in two, three or four languages, when they take up the formal study of secondary school French or Japanese. Finally, even in contexts that are identified as monolingual, it is quite likely that these individuals are, in any case, multi-dialectal. By multi-dialectal, I mean that they will have a command of two or more variants of the language of which they are said to be monolingual speakers.

On the subject of language unity and integrity, so fundamental to L1-L2 discussions, a half century ago, Martinet (1953) put matters into perspective for linguistics:

There was a time then the progress of research required that each community should be considered linguistically self-contained and homogenous ... Linguists will always have to revert at times to this pragmatic assumption. But we shall now have to stress the fact that a linguistic community is *never* homogeneous and hardly self-contained ... linguistic diversity begins next

door, nay, at home, and within one and the same man. (Martinet 1953: vii)

As Romaine (1996) explains, views on bilingualism have varied over the years, from Bloomfield's (1933) view of full command of two separate languages to Haugen's (1953) more modest standard of the ability to produce complete and meaningful utterances in more than one language. Edwards (1994) begins his discussion of bilingualism by making the point that "[e]veryone is bilingual", justifying this seemingly audacious statement by explaining that there is "no one in the world (no adult, anyway) who does not know at least a few words in languages other than the maternal variety". (Edwards 1994: 55)

In SLA textbooks, we find some reference to the question of multilingualism. The following is from Ellis (1994):

> Many learners are multilingual in the sense that in addition to their first language they have acquired some competence in more than one non-primary language. Multilingualism is the norm in many African and Asian countries. Sometimes a distinction is made between a "second" and "third" or even "fourth" language. (Ellis 1994: 11)

Such contexts lead Ellis to critique the use of "second" in SLA:

> However, the term "second" is generally used to refer to any language other than the first language. In one respect this is unfortunate, as the term "second" when applied to some learning settings, such as those in South Africa involving black learners of English, may be perceived as opprobrious. In such settings, the term 'additional language' may be both more appropriate and more acceptable. (ibid)

Nevertheless, such words of caution are pushed into the background as researchers get on with their research based on the assumption that there is always a dominant L1. And yet if we look around us, we see that Martinet, writing half a decade ago, quite likely got it right, and that in SLA this means that talk about assumed L1s and L2s is on the whole problematic. I say this in particular if we examine two separate but equally interesting lines of enquiry that have called the monolingual bias into question, one linguistically based and the other sociolinguistically based.

2.2 Cook's multi-competence model

One problem with the monolingual bias in SLA is the way that it over-

simplifies linguistic competence. As Cook (1992, 1996, 2002a, 2003) points out, there is an assumed complete competence which the learner possesses in her/his L1 and an assumed complete linguistic competence in the target language which is possessed by L1 speakers of that language. Complete linguistic competence in the target language is generally considered to be out of reach for 99% of the individuals who begin learning the L2 in adulthood and the concept of interlanguage accounts for states of competence lying somewhere on a continuum with the L1 at one end and the L2 at the other. In short, L2 learners are seen to possess complete linguistic competence in their L1 and incomplete competence in their L2.

Cook does not deny that complete L2 competence is out of the reach of adult L2 learners; however, he does suggest a shift in emphasis away from this focus, suggesting that "[t]he starting point should be what L2 learners are like in their own right rather than how they fail to reach standards set by people that they are not by definition" (Cook 1996: 64). The "people that they are not by definition" refers to teachers and SLA researchers, with clear ideas about what L2 linguistic competence entails, who pass judgment on the L2 learner. Cook goes on to propose that "[m]ulticompetence is then a necessary basis for second language acquisition research" as "L2 learners are not failed monolinguals but people in their own right" (Cook 1996: 64). This multicompetence is defined simply as "the knowledge of more than one language in the same mind" (Cook 1996: 65).

For Cook, adopting this model means that the totality of an L2 learner's linguistic competence at any one time is not just the sum of her/his complete linguistic competence in the L1 and her/his incomplete competence in the L2; rather it is a system which contains both the L1 and the L2. He describes the situation as follows:

> Since the first language and the other language or languages are in the same mind, they must form a language super-system at some level rather than be completely isolated systems. (Cook, 2003a: 2)

This means that when language learners are asked to make judgments about the grammaticality of sentences in an L2, they will draw on whatever linguistic knowledge they have already acquired in their lifetimes. Thus Japanese-speaking learners of English would be expected to make judgments of grammaticality in English based on their knowledge of Japanese and any other foreign language knowledge they might have (for example, if besides English they have studied Chinese or German). This is not at all surprising to

most people; however, Cook also notes that language knowledge influences cut both ways. This means that individuals who have acquired knowledge of other languages find that their intuitions about their L1 are different from those of monolingual L1 speakers. Thus, English speakers who have acquired Japanese to a significant degree will find that when asked to make judgements about English, their intuitions are influenced by their linguistic knowledge of English *and* Japanese.

What Cook's and others' work (see Cook, 2002b; 2003b) points to is the idea that linguistic competence is not stored in the mind in neat compartments with clear boundaries; rather, a more appropriate image is that of a mass with no clear divisions among parts. Nor is linguistic competence in different languages stable over time, as there is constant bleeding between and among languages as well as additions and losses as regards repertoires. Apart from making sense as a means of accounting for seemingly mixed linguistic competence across two or more languages or dialects, multi-competence also articulates well with recent work in sociolinguistics. Let us see how.

2.3 Sociolinguistic views of multilingualism

Over the past decade, Roxy Harris, Constant Leung and Ben Rampton have both individually and collectively written about multilingualism in British education (e.g. Harris 1997; Leung 2001; Rampton 1999; Leung, Harris and Rampton 1997; Harris, Leung and Rampton 2002). One area of inquiry, developed in particular by Harris, has been the notion of "romantic bilingualism", defined as follows:

> The term 'Romantic Bilingualism' ... refer[s] to the widespread practice, in British schools and other educational contexts, based on little or no analysis or enquiry, of attributing to pupils drawn from visible ethnic minority groups an expertise in and allegiance to any community languages with which they have some acquaintance. (Harris 1997: 14)

Harris has focused on British young people from Asian backgrounds who are classified by their schools as bilingual, which generally means English + another language such as Punjabi. This classification is accurate if what we have in mind is that the person in question has been exposed to both English and Punjabi in his/her lifetime. However, if we understand the notion of bilingualism to imply full or high competence in two languages, then such cases prove to be poor examples. Harris *et al* (2002) cite the case of T, the

15-year-old son of a Muslim father and Sikh mother, both from India. T's parents separated when he was young and he has spent most of his life with his mother. Harris *et al* note that he has strong affiliations to Sikh culture and Punjabi language and that he claims to speak "slang Punjabi" with his friends and "standard casual Punjabi" with his mother. However, he says that on the two occasions when he visited India, he found that he could not really communicate very well, as people there spoke "weird Punjabi". Meanwhile T's English is influenced by Afro-Caribbean English -"rasta talk"- which fits into a general fascination with Afro-Caribbean culture which is common among London youth of all ethnic backgrounds.

It is interesting to examine T's story in the light of how he would be classified by his school in London, that is, as a bilingual Punjabi/English speaker. Such a classification accounts neither for the different varieties of Punjabi with which he has had contact during his lifetime nor for the varieties of English which form part of his repertoire. It also would say nothing about other forms of symbolic behaviour associated with these different varieties, from the dress and music associated with his participation in Anglo-Punjabi cultural events, to the "rasta talk" and music associated with his forays into Afro-Caribbean culture. Finally, T's classification as a Punjabi/English bilingual seems odd in the light of the Harris *et al*'s estimation that T's written Punjabi is actually worse than his written German, a language he has studied formally under far-from-ideal circumstances at the secondary school level.

In situations like T's, it is therefore difficult to define language competence in terms of one, two or three languages as observations of language use give the lie to the idea that a speaker speaks one language sometimes, then another at other times and that it is a question of describing linguistic competence in the different languages and of documenting code switching so as to establish rules. A more accurate appraisal of such situations is that in linguistic terms, individuals are displaying multi-competence and that in sociolinguistic and social terms, they are involved in an ongoing process of identity construction, maintenance and projection.

2.4 Multilingualism as multi-experiential

If the "second" does not appropriately apply to so-called bilinguals and multilinguals as they begin to learn another language, it is also misleading when it is used to refer to the experience of individuals who are learning not their first additional language, but their second, third, fourth or fifth additional language. On the one hand, there are many individuals who have engaged in the formal study of, or had naturalistic exposure to, three or more

languages in their lifetimes. On the other hand, there are individuals who from birth have been exposed to two or more languages and later in life have been exposed to still more either in formal or naturalistic contexts. Given the nature of SLA studies where, as Firth and Wagner (1997) argue, notions of L1 and L2 are over-simplified and there is not adequate attention to individuals having any identity other than that of learner or non-native speaker, there has not been much discussion of the problem with the use of "second" as an all purpose cover term for all language learning experiences which are understood to be beyond the L1.

A notable exception is a collection edited by Jasone Cenoz and Ulrike Jessner (2000) devoted to the learning of English as a third language in different European contexts. The contributions to this volume provide the reader with a good idea of the number of different routes Europeans follow to the learning of English, either formally or naturalistically. These different routes are outlined by the editors in the Introduction and include situations such as speakers of Frisian who are proficient in the majority language of the Netherlands, Dutch, and who study English as foreign language in school or Turkish speaking immigrants to Germany who are educated in German and study English as a foreign language in school. Given the pervasiveness of third language acquisition and more importantly, the qualitative differences between such situations and the idealised situation, dominant in SL, of the monolingual learning an L2, Cenoz and Jessner suggest that third language acquisition (TLA) "is a more complex phenomenon than second language acquisition (SLA)" (Cenoz and Jessner, 2000: ix). Because those who define SLA always include TLA in their definitions, what Cenoz and Jessner suggest amounts to a secession movement: "second" does not represent TLA (or multilingual acquisition in general); therefore, TLA is a separate field of inquiry. I return to his point later.

Given the paucity of publications like Cenoz and Jessner (2000), some of the most interesting accounts of multiple language learning experiences are to be found in autobiographical accounts, such as those presented in Belcher and Conner (2001) and the work of Aneta Pavlenko (2001). From such accounts we learn about a diverse range of factors which give the lie to the notion of compartmentalised competences in different languages. For example, multilinguals often comment on the relative ease or difficulty of learning different languages. Thus a speaker of Spanish and English might argue that learning Italian is easier than learning Japanese. In addition, multilinguals often report that they are, in effect, different people in different languages. The following excerpt is taken from an interview I conducted with a Japa-

nese woman who in her lifetime has had extensive contact with English:

> Well, when I'm speaking in Japanese, I really have to think about all the things, you know, what's the proper style and ... sort of kind of try to adjust myself to the identities, what the other person is projecting on me. ... I mean, how are they looking at me? And how am I supposed to match that idea? So if they're looking at me as a good housewife, a middle class woman, whatever, I kind of really try to adjust myself and that's why I think I feel so uncomfortable sometimes thinking like, you know, that's really not me, but if you kind of expect me that way, I will play that way. Whereas in English, I mean don't really feel that much. (AF 11/6/02)

2.5 Conclusion

The linguistic and sociolinguistic research discussed in this section suggests that the term "second" in SLA may be inappropriate for many reasons. First, its assumptions about monolingualism and separable L1 and L2 linguistic competences do not hold up when relevant linguistic and sociolinguistic studies are considered. Second, the term "second" is far too modest to account for the experiences of individuals who have learned two or more languages in their lifetimes. In short, there is evidence from a variety of sources that the assumptions that sustain use of the term "second" oversimplify matters considerably, are not warranted, and indeed might be misleading.

Of course there has been disagreement among SLA scholars about this point. For example, as Firth and Wagner (1997) point out, there has been a tendency to discuss individual learners in terms of their L1s and the target language, L2, framing both as homogeneous entities. For these authors, there is little or no problematisation of these concepts or whether or not any of the individuals involved manifest any of the variable language behaviour documented by researchers such as Harris *et al*. Multilingualism and previous language learning experiences, multiple or not, are not generally discussed, either. However, Scholars like Susan Gass disagree with Firth and Wagner. While recognising that the concept of the native speaker is not as simple as it sounds, she feels nonetheless that such considerations need to be put on hold if we are to investigate SLA as a cognitive phenomenon. This view is borne out if we examine accounts of SLA research in major international journals where researchers generally provide a minimal amount of information about learners' backgrounds- such as "L1 speaker of French"- and then do not relate this information to their research.

Of course, one possible explanation for the near total omission of other language experience is that "second" is being used in a technical sense as a synonym of sorts for whatever the language is that is being acquired and that it is more economical to reduce learners to one language background and one target language in order to investigate more important matters for theory building such as describing interlanguages and cognitive processes related to acquisition. Nevertheless, even if we accept such an argument in favour of keeping this term as regards talk of L1s and L2s, there is still a problem with "second" as a term for the different contexts in which SLA takes place.

3. Context

As we observed in the introduction to this chapter, "second" has a meaning in terms of the actual context where language learning take place and in general, SLA researchers have tended to focus on three general contexts: the foreign, the second and the naturalistic.

3.1 Foreign language context

The foreign context is the context of millions of primary school, secondary school, university and further education students around the world who rely on their time in classrooms to learn a language that is not the typical language of communication in their community. This means English for many of the world's people, but it also means widely studied languages such as Spanish, Japanese and Mandarin Chinese, to say nothing of hundreds of other languages that appear on national curricula around the world. Conditions in these contexts vary considerably as regards teacher/student ratios, teacher preparation, intensity (hours per week), accommodation, technological backup, availability of teaching materials, the relative importance of learning the foreign language and so on. This means that any talk of the foreign language classroom in generic terms is problematic. For example, if we compare one context which I know very well- Spain- with the one in which I find myself at present- Japan, we see that there area as many different, as there are similarities.

First we might consider the cross linguistic issues involved in the learning of English in these two contexts. Although English is not a romance language (as Spanish and indeed Catalan and Galician are), it does contain a large number of cognates with romance languages, due to the historical roles of Latin and French in its formation. In addition, despite having quite different syntactic and morphological characteristics, the two languages do share an alphabet, many associations between script and phonology and many common notional and functional categories (see Wilkins, 1976). By contrast,

Japanese is linguistically more distant from English than Spanish is, both in terms of script and phonology. In addition, as Lakoff (1987) argues, the Japanese conceptual system, with classifiers like *hon* (indicating long thin objects and the activities and trajectories associated with them), is notably different from European based conceptual systems.

As regards the actual contexts of EFL, we also observe differences, although at first similarities are more in evidence. In Spain, English is first and foremost seen to be a *lingua franca* to communicate with fellow Europeans who do not speak Spanish or any of the other languages recognised as "co-official" by the Spanish state (e.g. Basque, Catalan, Galician), as well as people from other parts of the world. More specifically, English is seen as a key skill in the job market as there exists the notion of English as the international language of business and trade. Both of these reasons for studying English are addressed in a school system that provides from the age of ten (and earlier in some cases) about three hours of EFL instruction a week and the huge network of language schools primarily for adults (state-run "official language schools", major international schools and numerous smaller neighbourhood academies). As regards teaching methodology, Spain is similar to other European countries in that the work of European educational institutions over the past thirty years has brought communicative language teaching (CLT) to the forefront. Although practice varies across schools and even within departments, the methodology employed tends to combine a traditional concern with formal aspects of language learning (e.g. grammar) with task-based methodologies which require students to focus more on getting a message across than on grammatical accuracy (see Ribé, 1997).

The situation in Japan is not on the surface very different from that of Spain. Children are meant to learn English in school and there are a lot of language schools for adults. In addition, there is the notion that English is the language of communication with the rest of the world, both for leisure and business. As regards teaching methodology, there has been push towards CLT over the past two decades, although the integration of this approach to Japanese educational culture has probably proven to be more problematic than in the case of Spain. Perhaps the biggest different between Japan and Spain, as regards context, is the association of English in Japan with *kokusaika*, the process and policy of internationalisation which has been prominent in Japan over the past twenty years. Spain has not the experienced the same kind of economic and social change as Japan has since World War and therefore its orientations to internationalisation, globalization and national identity are considerably different. As a consequence, the role of English in

international, global and national processes is also different. Thus, while Japan and Spain are both EFL contexts, they are not oriented to EFL in the same way.

3.2 Second language context

Like the foreign language context, the second language context implies a formal classroom setting. However, it is different in that the classroom is situated inside a community where the target language is spoken. This is the context of refugees, migrants and immigrants around the world who have enrolled for language classes focusing on the language of the community where they have taken up residence. Second language contexts vary along the same lines as the foreign language contexts, that is, as regards teacher/student ratios, teacher preparation, intensity, accommodation, technological backup, availability of teaching materials, the relative importance of learning the foreign language and so on-. However, unlike foreign language contexts, they all potentially include multiple opportunities of contact with the target language outside the classroom. Such contact is deemed to be important by SLA researchers because it exposes the learner to a wide variety of genres, target language structures and lexis, away from the confines of teacher-controlled procedures and discourse. In a sense the idea is that the ESL context potentially offers the best of two worlds: the classroom setting with guided and mentored learning and the naturalistic context with exposure to rich and varied input.

However, simply being in a second language environment is no guarantee that a learner will be exposed to richer input which will then allow him/her to learn the subtleties of the language faster and more completely. Talburt and Stewart (1999) recount the story of Misheila, an African American university student on a five-week study abroad programme in Spain, most of which took place in Madrid. The programme, which combined language and culture classes with informal socialising, was designed to develop her intercultural and communicative competence in Spanish. However, it turned into a nightmare as Misheila's exposures to naturalistic input were rendered useless by her affective state during the entire period she was in Spain. Misheila was from a middle class background and had grown up in a predominantly European American setting. While she acknowledged that she had encountered racism during her lifetime in the United States, she somehow expected her stay in Spain to be racism-free. However, after just one week in Spain, she made it clear that she was not enjoying herself and that she would not be in a hurry to return to Spain once the study abroad programme had ended. It seems that whenever she went out, men constantly made sexually explicit

comments about her physical appearance. Having observed that a high percentage of prostitutes in Madrid were dark skinned, Misheila believed that these remarks sexualised her as an "African" woman.

The extent to which this affective interference limited Misheila's linguistic development is not discussed by Talburt and Stewart. However, the race and gender issues arising from Misheila's story lead us to question the notion that a second language context is necessarily conducive to linguistic development as well as the view that language is primarily if not exclusively about such linguistic development. In other words, just as the classroom part of the second language context can vary as regards factors such as teacher preparation, materials, class size and intensity, so too can the outside the classroom part. This outside the classroom is the be-all for the third context I explore here, the naturalistic context.[1]

3.3 The naturalistic context

The naturalistic, uninstructed second language acquisition context is, in a sense, the opposite of the foreign language context because it involves no formal instruction and the learning of a language which is spoken in the surrounding community. In this case, the learner makes her/his way through a variety of interactions necessary to day-to-day life and must rely on her/his background knowledge, learning strategies and intuitions to get by. This is the situation of the millions of refugees, migrants and immigrants around the world who find themselves immersed in a new language context and simply must get on with their lives. Well-known early SLA documentations of learners in such contexts include Schumann's (1978) study of Alberto and Schmidt's (1983) study of Wes. However, in recent years, two pieces of research have shed further light on the notion that "being there" does not necessarily lead to smooth and unfettered language learning.

Bremer, Broeder Roberts, Simonot and Vasseur (1996) studied the acquisition of four different languages- English, French, German and Dutch- in

[1] Somewhere in between the foreign and second language contexts are what Ellis (1994) terms "official language contexts, that is contexts where a language which is not the L1 of all members of a community is nonetheless chosen by a government as the official language of one and all. As Ellis (1994) and Long (1998) note in general discussions of SLA, and as authors from official language contexts, such as, Sridhar and Sridhar (1986) and Sridhar (1994) have lamented in more specific reference to such contexts, there has not been much published in mainstream SLA which deals specifically with this context. Canagarajah (1999) is an exception of sorts with his compelling account of English language teaching, and learning in the postcolonial context of Sri Lanka.

the UK, France, Germany and The Netherlands by immigrants who were L1 speakers of Italian, Spanish and Arabic. Bremer *et al* were interested in how issues such as the social distance between the learner and L2 speaking interlocutors impacted on naturalistic language learning, However, more importantly, these researchers challenged three assumptions about L2 learners in naturalistic contexts.

The first assumption is that naturalist contexts provide learners with more opportunities to be exposed to input. While this was the case as regards what are known as gate keeping encounters (e.g. visits to the doctor, contacts with government agencies), it was not the case as regards less bureaucratic and social encounters. Indeed, very often it was found that the participants in the study used anything but the target language in their contacts with friends and relatives.

The second assumption challenged by Bremer *et al* is that in one-on-one gate keeping encounters negotiation for meaning takes place as both interlocutors involved in an exchange work towards mutual understanding. The researchers found that in most cases it was the immigrant who was responsible for guaranteeing understanding and the flow of the exchange and that the presumably more competence interlocutor often contributed very little.

The third assumption challenged by Bremer *et al* is that in naturalistic exchanges, learners are given plenty of opportunities to develop their speaking and listening skills. What the researchers found was that in many cases learners were being assessed by their interlocutors as regarded their competence in the L2 and were not given the space needed to develop such competence. They were in effect caught in the no-win situation of needing the language in order to communicate while needing to communicate in order to learn the language.

In Bonny Norton's view, Bremer *et al* do not go far enough in their analysis, failing to debunk yet another assumption often made about naturalistic settings, that is that "those who speak regard those who listen as worthy to listen, and that those who listen regard those who speak as worthy to speak" (Norton 2000: 8). Thus, in addition to moving away from the notion that a naturalistic setting provides abundant and useful opportunities for the learner to interact in the L2 and learn through such interaction, as Bremer *et al* do, Norton argues that there should be additional interest in how learners develop identities as what Bourdieu (1977) calls "legitimate speakers", that is how they come to be accepted and fully functioning members of *communities*

of practice (Lave and Wenger 1991) which they inhabit and engage with. In her study of five immigrant woman in Canada, she did just this. One of the woman, a Polish immigrant named Eva, exemplifies the process of moving from an attributed (by others) status of incompetent foreigner in the workplace to legitimate co-worker who is "worthy to speak and listen". Through conversations with Eva and Eva's diary entries, Norton is able to reconstruct the story of how she struggled when she took on a job at a restaurant where she was the only employee who was "not Canadian". At first, she was marginalized in workplace conversations and even exploited by her fellow workers who assigned to her the work that no one wanted to do. Observing that she tended to do the most onerous work and that she did not talk very much, the restaurant manager assumed that she could not deal with customers and so never gave her the opportunity to do anything but the most menial of jobs. However, slowly but surely her social contacts with her fellow workers outside the confines of work allowed her to take on an identity far removed from that of silent co-worker. Eva came to be seen as a European with an interesting background, someone who in addition to Polish and English also spoke Italian. Feeling more respected as a co-worker and human being, Eva began to lose her self-consciousness about speaking English and eventually engaged in far more interactions in English, both on and off the job. Eventually, she was able to carve out an identity as a fully functioning co-worker who was "worthy to speak and listen".

The lesson to be learned from the work of researchers like Bremer *et al* and Norton is that first, the actual exposure to the target language in naturalistic contexts is often far less than might be expected because there are a number of variables which together conspire to limit both the quantity and quality of input. Immigrants in different contexts find that language learning does not depend exclusively on engagement in conversational interaction with native speakers. Indeed such contacts are often very limiting as regards input and the overall affective climate is so negative, due above all to pressure to perform and conform, that they are rendered of little use. In addition, as Norton points out, even where immigrants become part of a well-defined group, such as workers in a restaurant, there is still a long battle to achieving legitimacy as a group member. This is no less true among younger immigrants as McKay and Wong (1996) and Toohey's (2000) studies of immigrant children in California and Canada, respectively, have shown.

3.4 Conclusion

In this section we have examined the three general contexts which have been explored by SLA. Foreign language contexts vary immensely, depend-

ing on factors such as the international economic projection of the country, the extent to which learners will ever really have the opportunity to put their knowledge of the target language to use and socio-historical factors related to the educational system and attitudes in general about foreignness. In addition, there are the many official language contexts around the world which have not often made it onto the SLA playing field, although they do at least merit some mention in general SLA texts. Second language contexts, often seen as ideal in that they combine formal learning with naturalistic exposure, are equally problematic because contact with the target language can vary considerably. In addition, as we observed in the case of Misheila, such contact might do more harm than good as regards the learner's identification with the target language as well as her/his attitude towards it. Finally, naturalistic contexts are equally problematic, not least because access varies considerably and very often it is how the individual learner negotiates and carves out an identity in the target language and not the context itself which ultimately shapes relative success or failure.

4. Discussion

In SLA in general, the term "second" is used in a loose fashion to refer to where the language being acquired is situated chronologically in an individual language acquisition experience (second as "after first") and to where, in physical terms, this process is taking place. In this paper I have attempted to show how these uses of second are problematic because they over simplify knowledge of languages and the contexts where such knowledge is acquired. But what do I propose as solution to the problematic nature of the term?

Rampton (1997) has suggested that terms like "other" or "additional" might be more appropriate. I see these terms as preferable to "second" if what we seek is a different umbrella term for the different language learning experiences and contexts which are currently discussed by SLA researchers. However, another way to look at this situation is to see the wide variety of lingualisms and contexts cited in this chapter as sufficiently diverse as to constitute separate areas of inquiry, as Cenoz (2000) suggests. In short, can we ever hope to elaborate a general theory of SLA which would cover all contexts?

It seems then that both SLA insiders, such as Long and Gass, and those who have been critical of mainstream SLA, such as Firth and Wagner and Rampton, would all agree that it is important to keep a diversity of individuals and contexts together under a general super-ordinate category of SLA. The problems arise when it comes time to decide how much of this diversity

should be taken on board by researchers. In context of this paper, it is a question of how ideas about multicompetence, more sociolinguistically and socially informed views of multilingualism and the social complexities of foreign, second and naturalistic language learning contexts can all be taken on board by SLA researchers. If more and more researchers do take on this expanded and more ambitious agenda, then what becomes of the "second"? What is the alternative?

Above, I cited Rampton's two suggestions, "additional" and "other". Of the two, "additional" seems the most likely substitute as it carries with it several distinct advantages. First of all, it captures the notion, essential in Cook's work, of the ongoing accumulation of linguistic knowledge. It allows us to get around the implicit reference to a unitary and singular L1, as it is agnostic on this point: "additional" could apply to any language learning experience, irrespective of the learner's previous language contact. Second, it works much better than "second" as a reference to the different contexts discussed in this paper. It would allow us to avoid confusion of using "second" to refer to the "foreign", "second", "naturalistic" and "self instructed" contexts represented in Figure 1. Not using the same term for a super-ordinate and a subordinate category, as is currently the case when we use "second", certainly allows for greater clarity. Third and finally, "additional" is an altogether less vague term than "other" which might carry the meaning of "previous" or even "instead of".

Notwithstanding this reflection on the "Second" in SLA, changing SLA to ALA (Additional Language Acquisition) would be the kind of seismic shift that academic fields seldom if ever impose on themselves. Indeed, "Applied Linguistics", a vague name for a collection of diverse interests in language in the real word, still exits after (and despite) decades of debate about what it means (e.g. issues 7/1, 8/1 and 9/1 of the *International Journal of Applied Linguistics*; Davies, 1999; McCarthy, 2001). This being the case, how can we expect SLA to change?

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Integrating Applied Linguistics Research Outcome into Japanese Language Pedagogy – A Challenge in Contrastive Pragmatics –

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1. Background

The Japan Foundation survey of 1998 revealed that there were more than 2 million learners of Japanese throughout the world. The most recent survey conducted in 2003 is expected to render more figures. In Japan, a recent increase in non-Japanese labor resource has initiated municipal efforts for Japanese language support by professional and non-professional personnel. The Japanese language pedagogy is confronted with a new phase, to match the domestic and overseas trends. Quick departure from the traditional methods of teaching a small number of devoted scholars of Japanese, and search for workable means to give assistance to meet the new needs are urgently required of the professionals. The field of applied linguistics, where basic research in the study of language structure, language use in society, learner behavior and learning process, language planning and language policy are actively discussed, should assume a major role in offering the research outcome to both teachers and learners.

In this paper I would like to introduce a case of research concerning the aspects of language use in society, particularly the types of pragmalinguistic communication gap that is created inevitably between native and non-native speakers of Japanese in native vs. non-native speaker contact. There will be various factors involved in the causes of discrepancy in native vs. non-native speaker communication. While structural aspects of interlanguage errors by non-native speakers might cause informative misunderstanding, pragmatic interlanguage failure could cause mutual malfunctioning of human relationship. I would like to focus on the latter type of communication mishaps, by way of introducing a multi-cultural/linguistic contrastive research project done by The National Institute for Japanese Language team, and relate the research outcome to possible application in the Japanese language pedagogy.

2. Research

2.1 Scope

The research I introduce here was a part of the research project the National Institute for Japanese Language conducted in 1994-1998. The major theme of the research project was to survey the state of the Japanese language in the international societies. The research topic of the research group in discussion was to survey the conflict in cross-cultural communication, with a special interest in socio/pragmalinguistic features. The objective of the group research was to induce the concept of cultural/linguistic norms in various societies, analyze the result, and find out its implication in cross cultural/linguistic communication.

2.2 Research Outline

The research scheme was a multi-cultural/linguistic survey of communication strategies. The data was collected by way of a series of structured interviews with those informants who had experienced working in the five overseas communities and had daily contact with people in their respective work places. Six everyday life situations were chosen as the topics of communication. They were as follows: (1) Passing a stranger in a hallway, (2) Serving tea in an office, (3) Dinner time in a family, (4) Asking for a special favor of an officer, (5) Ignoring a waiting line in an office, and (6) Seeking for an answer to a request. The videotape clippings from television dramas were chosen as partial stimulus to the informants in order to give visual aid for a good grasp of the situations.

The series of the structured interviews was conducted in the six countries: America, Brazil, France, Japan, Korea and Vietnam. The informants in the overseas interviews were native speakers of Japanese who were assigned to work in those countries and had daily contact with local staff members. The informants in Japan were non-native speakers of Japanese who were assigned to work in Japan.

In the actual interviews, the informants were presented with the videotape scenes, and were asked to respond to the questions concerning the possible communication in the given situations. The questions were targeted to elicit what the informants would say/do if they were participants in those situations, and what difference they realized in relation to the identical setups back in their countries. Through the series of those questions, it was expected that the informants would render their observations concerning the subconscious sense of pragmalinguistic norms of their respective native language communities.

2.3 Default Status and the Concept of Norm

The objective of the data collection was to find out the underlying sense of what the informants regarded as the most "proper" ways of communication as the native speakers of the language community they came from. The type of underlying sense in question was brought to their consciousness vis-à-vis the given environment they were in at the time, when they stayed in a community where different sets of cultural/linguistic norms prevailed. The sense of "proper" linguistic behavior had been implanted in their speech/conduct code through the stages of their upbringing. It was conceivable that such first language sense could cause various types of conflict in overseas situations. The selected videotape scenes were detected to be the possible cases where pragmatic transfer and pragmatic failures would take place in non-native speaker (interlanguage) communication strategies. It was expected that the informants would observe the situations, compare their mind notes of what would be "proper" ways of communication in the respective cultural/linguistic communities, and respond accordingly to the structured interview questions.

The native speaker's sense of "proper" communication is conceptualized as the default value in communication schema. In the cases where the purpose of the communication is to perform a particular speech act such as request, apology, invitation, compliment, refusal etc., the process requires several speaker-hearer interactive moves. The most typical ways of pursuing the goal are already installed in the participants' mind in the cases where both the speaker and the hearer are native speakers of a given language community. The speakers seek for the best strategy they foresee as the most probable script in view of the communication goals, and the hearers are expected to react accordingly. This most typical discourse pattern is defined as the sets of default value in the linguistic community. Usami (2001) proposes that the concept of what constitutes the default value should not be limited to the sentence and/or single speech act level features, but should be extended to include the discourse level interactions.

In one of the interview sections, the informants were asked to watch a videotape scene where a person comes to a city hall officer and asks her for quick issuance of his missing passport. Then they were requested to choose what they would do in such a situation. The first task was to select whether they would say the statements in question. The candidate statements were as follows:

(1) tell her that you lost your passport
(2) explain that you lost your passport out of your own carelessness
(3) apologize for losing your passport

(4) stress that you need the passport for your work
(5) state that you are asking a difficult favor of her
(6) request that your passport be reissued as soon as possible
(7) apologize for all the trouble you've caused
(8) others, if there is any

After marking the statements they would choose to say in this situation, the second task was for them to rearrange those statements according to the sequence they would adopt in their first language community. The third task then was to guess what the natural sequence would be for the native speakers in the community they resided in. The additional tasks such as to answer how the ways people behaved in the two communities were different followed.

Kumagai (2003) reports that the result obtained in the USA and Japan revealed that 37% of the Japanese informants in Japan admitted that they would select number 7, while only 17% of the Japanese in the USA selected 7 for themselves. Interestingly, 29% of the US citizens in Japan answered they would say 7 in Japan. Kumagai also found out that only 4% of the Japanese informants in the USA said they would apologize if the scene took place in the USA and that 10% of the US informants in Japan said they would apologize if they were in the USA. Both Japanese and American informants took the negative stance (23% and 26% respectively) for apologizing in the given situation if that took place in the USA.

3. Discussion

Throughout the interview process the informants were asked to reflect on their discourse strategy in both native and non-native speaker perspectives. The result shows that the decision to adopt discourse strategies is made according to the informants' assessment of the respective default status in the given linguistic community, and the default value is activated to their conscious level through observation of the videotape visual aid. The default value assessment thus obtained could then be characterized as field dependent. By field dependency I mean that the informants' judgment was based on the particular communicative variables they observed in the given situation. They did not come up with the answers according to their stereotype image of the language community as a whole.

The default value assessment was also situation dependent. The research team came to be aware that the informants' answers varied in many different ways. Since they were always presented with the videotape scenes, the answers were given as the result of their assessment of what they would do/say in the given scenes. What they saw and what they judged formed their answers. Simple dichotomy of the stereotype sense of norms according to the

linguistic ethnicity was not working as far as this interview series could tell. The answers were selected not because the scene was in Japan or in the USA, but because it was the city hall counter and there were other people waiting in line...etc.

The visual impression of the concrete figures in the videotapes contributed to the informants' answers. Their observation of the age, gender, personality, outfit etc. made difference in their answers. The third characteristics, then, of the default value would be person dependency.

The recognition of the default value was elicited when the informants were asked to compare their assessment in the contact situation, i.e., when they lived in the second language community where their daily life requirement was to establish a working relationship with native speakers of the community they faced. One of the important research findings in this research was that in many cases the informants' answers were similar or different in terms of the variables other than the linguistic ethnicity. The answers the large number of the US informants in Japan selected, i.e., that they would apologize in the office scene in Japan, exemplifies the type of adjustment they would adopt living in the community where different types of default status prevail. The adopted strategy suggests that the informants would "play the role" of the participants in a given situation according to their assessment of what proper ways of behavior are expected of them. The pragmatic failure would be avoided through the series of such observation, assessment, and adjustment.

It is also significant that the US informants in Japan reported that they were aware of the different pragmatic functions of the Japanese expression of apology. The Japanese expression "Mooshiwake arimasen (I apologize)" is often uttered when there is a judgment that the purpose of the speech act is more easily attained if the strategy of apologizing is taken initially. The US informants in Japan were aware that different communication strategy and different pragmatic function existed in the identically translatable expressions between Japanese and English.

It is the task of the researchers to identify and describe the subconscious linguistic sense of default value in the linguistic communities, pursue cross-cultural pragmatic research by observing the process of adjustment people adopt in order to cope with the contact situations, and offer the basic set of data for its possible application to language pedagogy. It is then the task of curriculum writers and teachers to plan the best possible teaching strategy to incorporate the research outcome. The learning process should not be the knowledge-based instruction, nor should it be the one-sided adaptation to the target language patterns. Another perspective of cross-cultural training where the learners are encouraged to "play the language game" by learning to

"change the gears" according to the communication needs. The research introduced here, I believe, is a sure step forward in that direction.

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Computer Assisted Language Learning (CALL) – Moving into the Networked Future –

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Over the past 30 years computer assisted language learning (CALL) has developed into a thriving interdisciplinary enterprise that now influences many aspects of language education. In recent years, advances in the application of synchronous network technologies in CALL have raised the possibility of developing important new areas of research. This paper will examine significant developments within the field, with particular regard to theory and pedagogy. In conclusion, the discussion will highlight several promising areas for future research.

Introduction

A major trend in contemporary CALL has been the application of network technologies that enable synchronous (real-time) interaction between users. Language educators have not been slow to realize the potential of synchronous tools such as LAN-based network writing tools, chat environments, MOOs (multi-user object-orientated domains) and immersive virtual reality (VR) worlds to create cooperative learning environments that support meaning-focused interaction in the target language (TL). Given the emphasis on the role of interpersonal interaction in recent SLA research (Ellis 1999) and the potential of synchronous tools to provide TL communication that is meaning-focused and speech like in nature (Pellettieri 2000), much recent CALL research has focused on the examination of the discourse produced in these environments (Sotillo 2000). These research efforts have attempted to determine from a variety of theoretical perspectives, the status of this discourse and have further sought to identify the linguistic and interactional features of this new form of communication. This research effort is ultimately concerned with identifying the features of computer-mediated communication (CMC) that may be relevant to the processes at work in SLA (Ortega 1997). This paper will provide an overview of significant developments in the above field from both a theoretical and pedagogical perspective. The discussion will also highlight areas with potential for future research.

Synchronous environments in CALL

Early attempts to utilise synchronous network technologies in CALL

centered on the application of LAN-based writing environments. A well-known example of this type of technology is the Daedalus writing environment (http://www.daedalus.com/) originally developed for the English program at the university of Texas. This writing tool was designed to provide a LAN-based collaborative learning environment. Daedalus incorporated a conferencing tool and heuristics for prewriting and topic development. An early study of native speaker (NS) interaction in Daedalus by Bump (1990) found evidence of increased written output and the emergence of collaborative learning behaviors on the part of students. Bump speculated that these behaviors were the result of reduced turn taking pressures due to the egalitarian nature of much online discourse. A study of Portuguese language students interaction by Beauvois (1992), found that the anonymity afford by synchronous discussion resulted in increased learner participation, motivation and TL output. Following on this early work, a study of non-native speaker (NNS) interaction by Chun (1994) found that participation in a Daedalus-based class discussion project fostered the development of communicative competence on the part of learners. A comparative study of NNS interaction in oral classes and networks by Kern (1995), found that students in Daedalus produced increased written output and number of turns. Moreover the learners also produced a wider variety of discourse functions than in oral discussions. These early studies highlighted the potential of synchronous technologies in CALL. Further advances in these technologies were stimulated by the emergence of the Internet in the mid 1990`s, and educators have been active in applying these new tools in the language classroom.

Chat systems

Synchronous chat tools may be found on bulletin boards, servers and web sites in many parts of the Internet. These tools enable users to communicate via text in real-time. A well-known example of a chat tool used extensively in education is Internet Relay Chat (IRC). As with most other chat tools IRC contains many thousands of "channels", virtual spaces where individuals may log in anonymously using an alias and converse with other users through the use of text-based commands. In chat tools such as IRC users written output is displayed as soon as it is sent, enabling users to monitor their linguistic output. User interaction in IRC is fast paced, multidimensional and primarily social in nature (Werry 1996). The capacity of chat tools such as IRC to promote learner-centered, cooperative learning, has made these environments a focus of investigation in contemporary CALL research.

Various chat tools have been utilized in a number of CALL research projects. A study of interaction in a chat environment by Kitade (2000),

found that learners of Japanese as a second language engaged in self-correction and various TL repair behaviors. On this basis of this research, Kitade concluded that these environments provide for collaborative and comprehensible learner interactions. A further study by Blake (2000) found that in a task-based project involving chats between native and non-native speakers of Spanish, learners engaged in the negotiation of meaning. Blake also reported that the data recording capacities of chat programs may be utilized in CALL, by providing learners with opportunities to enhance their metacognitive awareness through examination of chat files. Following on this research, a paper by Hudson and Bruckman (2002), reported that non-native speakers of French in a chat-based project engaged in meaning-focused interaction. This study also found that learners exhibited reduced levels of inhibition when communicating online. The promising findings of these studies have also been replicated to a degree in other more recent projects. These studies focus on the investigation of learner interaction in novel forms of CMC environment.

MOO environments

Of the many forms of CMC now being applied in CALL text-based virtual worlds known as MOOs, bring a new dynamic to second language pedagogy and research (Peterson 2001). MOOs share a number of the properties of other forms of synchronous CMC. MOOs consist of many user-created theme-based virtual spaces known as rooms. Users may access a MOO world by completing an alias-based login protocol. This feature of MOOs provides for the anonymity on the part of users. On completion of this procedure, users are free to navigate and communicate within the MOO via a series of text-based commands. Recent versions of MOOs are browser-based and utilise hypertext and multi-media content.

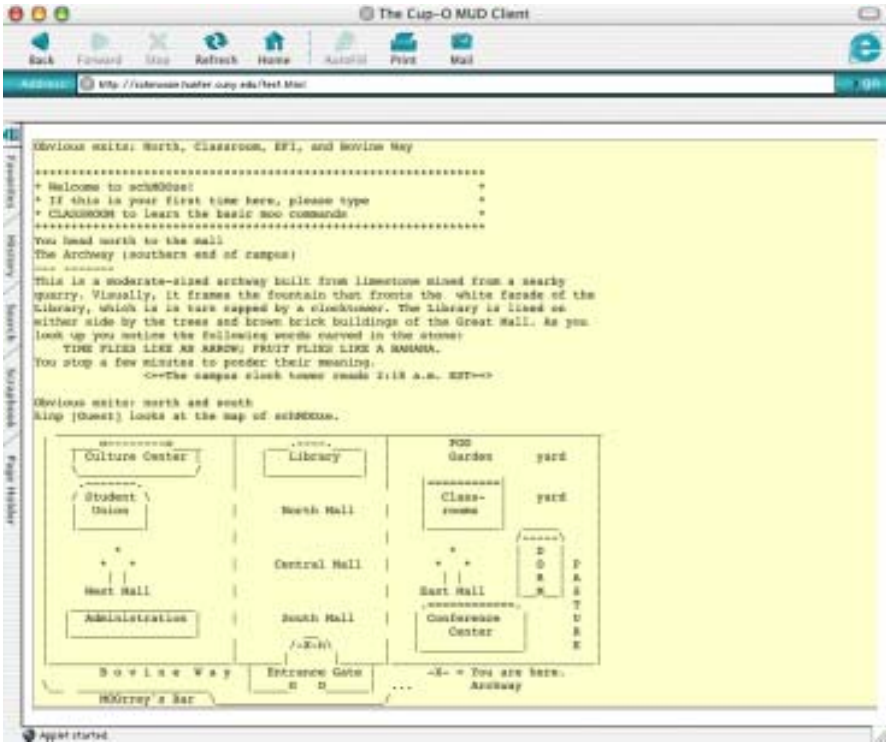
MOOs offer a higher degree of permanence and community than is found in most other forms of network-based learning environment (Kotter 2003). Several factors may account for this situation. As MOOs evolved from MUDs (multiple-user domains), text-based role-playing games, they engender a higher degree of participation and commitment on the part of users. These behaviors are reinforced by the fact that in MOOs users can create and develop an individual online persona. A hierarchy of user access coupled to the object-orientated nature of MOOs enables learners to actively participate in the creation and manipulation of objects within the MOO environment, thus creating a sense of ownership and proximity.

A further unique feature of MOOs is that they adopt many of the learning metaphors of the traditional classroom. In MOOs learners can utilize virtual

projectors, cameras, VCR`s and TV`s to create, share and manipulate virtual objects. This aspect of MOOs makes them ideally suited to task-based learning.

Moreover in MOOs, learners may participate in virtual lectures, seminars and presentations.

The networked nature of MOOs also facilitates collaborative learning projects with students in other countries. Thus providing opportunities to communicate with a wider range of interlocutors than may be found in many conventional classrooms. MOO virtual worlds therefore offer a richer more structured learning environment than most other forms of CMC. (Figure 1: Screen capture of a web-based ESL MOO Schmooze University http://schmooze.hunter.cuny.edu:8888/)



Although the application of MOOs in CALL is a relatively recent phenomenon, a small body of preliminary research studies have been conducted. In a study describing a small-scale tandem learning project involving students of German as a second language based in America and learners of English based in Germany, Donaldson and Kotter (1999) found that participation in MOO-based learning promoted collaborative learning, motivation

and the development of learner autonomy. These findings were corroborated in a study by Von der Emde et al. (2001). This study focused on the interaction of two groups of undergraduate language students based in Germany and the United States. The researchers reported that learning in MOOs appeared to support learner-focused exploratory learning and peer teaching. A further significant aspect of the interaction between the two groups was a high degree of TL use, and the development of autonomous learning behaviors that fostered a sense of community between participants.

Immersive 3D virtual reality

Recent developments in CMC technologies have focused on providing a richer, more immersive level of interaction than has been provided by conventional chat technologies. Several applications of network-based virtual reality technologies have raised the possibility of utilizing these advanced environments in CALL. One the most promising of these new tools is the Active Worlds virtual environment (http://www.activeworlds.com). This tool incorporates a 3D virtual reality authoring component, designed to facilitate the creation by users, of interactive virtual worlds. In Active Worlds users can communicate and navigate in real-time through the use of text and hypertext-based commands. This environment is object-orientated, thus enabling users to create virtual content within the world. (Figure 2: The Active Worlds interface)

Active Worlds also attempts to provide users with a sense of presence within the environment through a recent innovation in VR technologies. Namely the use of avatars; graphical embodiments of self. These virtual images are capable of replicating a number of the non-verbal cues that are absent in most chat environments. For example, in Active Worlds avatars can make gestures and show emotion on screen in real-time, thus enhancing communication. Moreover these images may also traverse virtual space and interact with other avatars, providing for a novel and motivational communication experience.

Although research into the use of advanced VR technologies in CALL remains at an early stage, a study of NS NNS interaction in an Active Worlds environment conducted by Toyoda and Harrison reported interesting results (2002). These researchers found that despite technological problems, learners of Japanese as a second language adopted various strategies to overcome communication problems with native speakers. This study also found evidence of the kind of negotiation of meaning that is held to be an important influence on the processes at work on SLA. While the findings of this preliminary study have yet to replicated, these results merit further investigation in the future.

Synchronous environments in CALL: Directions for future research

The implementation of synchronous tools in CALL raises a number of significant research issues. The results of existing research highlight new areas that require investigation. Given the emergent nature of CMC-based communication, future studies will explore the unique linguistic properties of this new form of communication. Future research efforts may also attempt to clarify the status of this new form of discourse.

At present, the status of CMC discourse remains unclear. Some view this form of discourse as distinct from speaking and writing, while others perceive a blend of these forms (Herring 2001). In the context of CALL, researchers will have to take into consideration the effects of the constraints on interaction imposed by communication in synchronous CMC, and the possible effects of these factors and other variables (such as proficiency levels, computer skills and learning context) on language learning in CMC.

A further issue that will be of importance to advances in this area of CALL is the nature and extent of learner negotiation. As negotiation is now viewed as an important element in the processes at work in SLA, future research activities will focus on investigating the degree to which negotiation takes place in synchronous environments and the ways this behaviour can be

fostered. In this context, the design and role of online learning tasks is of particular relevance. Researchers may also consider the effectiveness of participation in the various types of CMC, in order to identify the forms of synchronous communication that are best suited to the support of learners` interlanguage development. This research will also form part of the wider debate regarding future development in CALL.

Much of the current debate regarding the future of CALL is based on two conflicting views regarding the most effective way to proceed. Interactionist researchers (Chapelle 1997) take the view that the most fruitful way forward in CALL is to base research efforts on the findings of SLA research. This school of thought argues that only this approach provides a principled basis for development in CALL research.

In contrast, opponents of this view (Harrington & Levy 2001, Salaberry 1999) claim that CMC introduces a radically new dynamic in CALL, that requires the consideration of a wider set of research variables than is found in traditional interactionist research. In short a new broader approach, encompassing recent conceptions of the social nature of cognition. While a consensus on this debate has yet to emerge, future research projects in CALL will require a perspective that accounts for the many variables that influence interlanguage development in synchronous environments. Part of this future research effort will also require a focus on the impact of the various forms of CMC on language teaching pedagogy.

Many early studies on the application of CMC environments in CALL highlighted the fact that the anonymity afford by learning on-line, coupled to the egalitarian nature of electronic discourse, has the effect of radically impacting on traditional conceptions of teacher and learning (Peterson 1997).

Many proponents of the utilization of CMC in CALL argue that network-based learning empowers learners disadvantaged in conventional classrooms (Warschauer et al. 1996). Moreover from this viewpoint, participation in network-based learning engineers a major shift in classroom dynamics from the traditional teacher-led view of learning toward a learner-centered model. The role of the teacher in the on-line classroom is therefore transformed to that of facilitator. This positive view of online learning has developed in the context of the emergence of new conceptions of SLA that stress the role of social interaction in learning. What remains to be explored however, is a perspective that informs CALL research on not only why learning takes place but also how language acquisition occurs and may be supported in CMC. This major undertaking requires a range of both empirical and qualitative studies that examine the many cognitive and affective factors that

may promote learning in CMC.

Conclusion

The application of synchronous CMC environments in language education represents an important new development in the field of CALL. The increasing implementation of CMC technologies has introduced a new dynamic to CALL research and pedagogy that is rich in possibilities. In terms of theory, the existing literature indicates that these tools when combined with task-based pedagogies create the conditions in which language acquisition processes are fostered. Moreover the data recording capacities of computers provide researchers with a unique opportunity to assess the nature of learner communication in networked classrooms. The study of the discourse produced in CMC environments provides a new perspective on learners' interlanguage development (Blake 2000). In the future, researchers will continue the task of exploring this new form of communication with a view to increasing our understanding of the many complex processes at work in networked learning. There remains a need for research studies that adopt qualitative and quantitative approaches, in order to explore how the various forms of synchronous environment utilized in CALL influence learner interaction. This research effort will also focus on the development of new pedagogical approaches that capitalize on the advantages offered by CMC technologies. In a wider context, the rise of CMC-based CALL highlights the need to more fully explore how technology, teaching and learning interact in the networked classrooms of the 21st century.

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Beyond the Novelty – Providing Meaning in CALL –

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Introduction

The use of Information and Communication Technologies (ICT) in higher education experienced phenomenal attention from researchers and creative teachers during the late eighties and throughout the nineties. The interest moved from simple response (either it was right or it was wrong) marginally intelligent (ISEMONGER, 2003) CD-Rom or Net-based systems, pronunciation tutors, and slightly more interactive HTML audio-video comprehension quizzes, to cross-cultural real-time face-to-face discussions with other target language speakers through video-conferencing, and in more recent times, CD-ROMs and web-based data-base systems that are user specific, adapting to the needs of individual learners (defined by ISEMONGER as appropriately intelligent), to name only a few. However, in Japan, partly due to the fact that innovative teachers and researchers in ICT-based education received little to no support, many of the results from courses that have used ICT have been questionable or inconclusive.

ICT cannot be considered as the saviour for all of Japanese educational woes, but it also cannot be considered as an isolated phenomenon outside the realm of education. Unfortunately, many higher educational institutions in Japan have adopted the attitude that simply placing ICT in the curriculum equates to providing quality education. The introduction of ICT into the classroom will not bring about the social or learning transformation that is currently required in Japanese education. The likely scenario will be a continued tendency to utilize previously learned skills, methods and ways of learning (WARSCHAUER & MESKILL, 2000, FIELD, 1999). Furthermore, although ICT can provide opportunities for empowerment, knowledge creation, and life-long learning, the development, implementation and pedagogical practice may have the reverse affect (JUNGCK, 1987) and will do little to advance students' critical enquiry, analytical skills, understanding of learning or cross-cultural appreciation (SAKAMOTO, 1992, VAN DUSEN, 1997, WARSCHAUER, 1999, FIELD, 2003).

ICT-based education should be about synthesizing and manipulating

knowledge, skills and information to prepare life-long effective and efficient critical learners for the 21st century. Using ICT in education is about being an active member of a local and global community (FIELD, *ibid.*). ICT-based language education is more than 'grammar skills and drills': it is about developing deeper cognitive and academic skills not only in the target language, but of the target language culture/s. It is about the development of general and specific knowledge. It is about educators providing new opportunities for students to extend their language and everyday experience with other learners in real-world discourse. It is about providing opportunities for students to develop critical thinking, which will provide skills to manipulate and navigate the flood of information available in the shrinking world.

The average Japanese language student must complete compulsory language units of study, which they generally find boring; they therefore do not approach language learning with a purpose of production and communication. Many are, therefore, not motivated and rote learn material to acquire a satisfactory result on the requisite grammar tests. Recent theoretical and pedagogical trends in second language acquisition favour sociolinguistic, socio-cognitive and/or sociocultural pedagogy over the drilling, testing, and language appreciation activities that once dominated. Cultural and social learning contexts produce outcomes on bilingual proficiency that cyclically influence attitudes and cultural values. The introduction of ICT into the classroom may enable students to engage with their learning in new ways. However, the provision of opportunities to learn and produce language through ICT alone may not produce language or communicative proficiency.

Student Use and Understanding

FIELD (2002b) described in a study, in which first year university students interacted on a bulletin board (BBS) as part of their second language education classes, that "there was clear evidence of a gulf between student expectations about the CALL class and its curriculum, pedagogy and objectives" (p.7). He argued that a common theme expressed by the students was the computer nature, as opposed to the language nature, of the Computer-Assisted Language Learning (CALL) classes. Interestingly, he found that the students were confused about the purpose of the CALL class, believing it was a computer skills course, and many wanted more speaking components built into the course. The students' overall comments at the beginning of the year, however, were favourable to using ICT. This was particularly evident with students who had already experienced using and communicating through ICT prior to their CALL classes. During the CALL classes, comments posted to the BBS suggested that the general attitude to ICT and

CALL remained unchanged. The students that had a positive attitude toward ICT and CALL felt a degree of freedom through computer-mediated-communication (CMC). This was primarily attributed to a sense of release from the Japanese cultural (C1) obligations imposed on the speech act and on the interlocutors. The students that expressed negativity toward ICT and CALL continued to voice concern about the 'computer-v-language' learning dilemma. For some students there was direct evidence that a process of adaptation and/or non-acceptance of the use of ICT in the classroom was occurring.

The students' general attitudes at the end of the study period did not reveal a dominant preference to using ICT. Thirty six percent favoured CALL, fifty-five percent favoured a non-CALL curriculum, the balance were undecided or favoured both. The predominant reason for their preference continued to be influenced by attitudes to the computer. Students' comments throughout the study supported that, as a group, there was no clear pattern of adaptation to ICT. By the end of the research term, which was one semester into the students' second academic year, a small trend away from favouring the use of ICT was evident.

Although much can be concluded from this study, one interpretation should be mentioned as it is indicative of a trend that is common: the novelty value of new technology and ideas. Often, teachers and students get "seduced by the bells and whistles, the power of technology" (MURRAY, 1998), which in itself, speaks volumes about the misunderstanding of using ICT in education, as a computer (with Internet access) in every classroom is an ideal unrelated to education (BLAIR, 1996, BANNAN-RITLAND, HARVEY AND MILHEIM, 1998, SELWYN, 1999, FIELD, 2002B). For many students, participating in any class is akin to a child with a new toy: at first the child is totally absorbed in the new toy and enjoys a honeymoon period. After an indefinite time, the toy is gradually neglected: students generally drift into previously learned patterns and criticisms about the efficiency and effectiveness of the educational system and learning strategies are revived, often, sadly, including non-attendance and 'product' focused education.

FIELD (2002) went on to argue that the degree and speed of any influence from ICT on language use may be related to, what he described as, "the reliability" and "validity" the student attributes to the ICT interaction. He stated that "the degree of transference from ICT-based communication to a face-to-face event may depend on the extent the user attributes reliability and validity to the ICT event" (p.931). In other words, the more the learner perceives that

the ICT-based text, information and interaction is reliable and has some validity to their understanding and learning, the more the process will influence language use in a face-to-face event. This may, therefore, enable the students to go beyond the novelty of the new toy to utilizing it more effectively for their learning.

FIELD's study showed that students were having both negative and positive experiences when encoding the ICT-based text (50% reciprocally); however, up to two-thirds were shown to have a negative experience when decoding the text. The negative experience was considered a lack of confidence in the reliability and validity of the ICT-based text. Decoding the text is argued to be influenced by various filters including; anonymity, communicating in a non face-to-face situation, the nature of the Internet, and the inability to trust the other in the ICT communicative act. Therefore, it is hypothesized that when encoding or decoding filters are negative, reliability and validity of the text is perceived as low. Reciprocally, when encoding or decoding filters are positive, reliability and validity of the text will be perceived as high.

Any interaction may, therefore, have any one of four outcomes: positive-positive, positive-negative, negative-positive, or negative-negative. It was hypothesized that a positive-positive experience would potentially facilitate transference to a face-to-face event more than a negative-negative or negative-positive engagement in the ICT-based interaction. A false reality is created in a positive-negative engagement or negative-positive engagement. Such case is when a learner has a perception of release from first language (L1) and cultural (C1) expectations in the ICT-based interaction, but continues to communicate and interact in expected and learned C1 strategies. It is hypothesized that a false reality scenario has only a low potential to affect face-to-face L1 and C1 communication and interaction. A negative-negative coding-decoding experience was considered as a situation that would not promote language in a face-to-face event.

FIELD's argument may provide a solution to the 'novelty' dilemma often encountered in the higher education classroom in Japan; that is, students need to be provided with learning material in an environment that is applicable to their lives, to their learning situation, to the skill and academic level, and they must believe that the transactions within the ICT environment are more than 'a game'. If Field's premise is correct, providing such learning material in a supportive learning environment would enable students to interact beyond the fading of the novelty honeymoon, and furthermore, would

facilitate greater transference in higher level face-to-face communication. Moreover, evidence to support the hypothesis would provide valuable data that could be used as a catalyst to substantiate the need to modify much of the pedagogical and attitudinal approaches to learning in higher education in Japan.

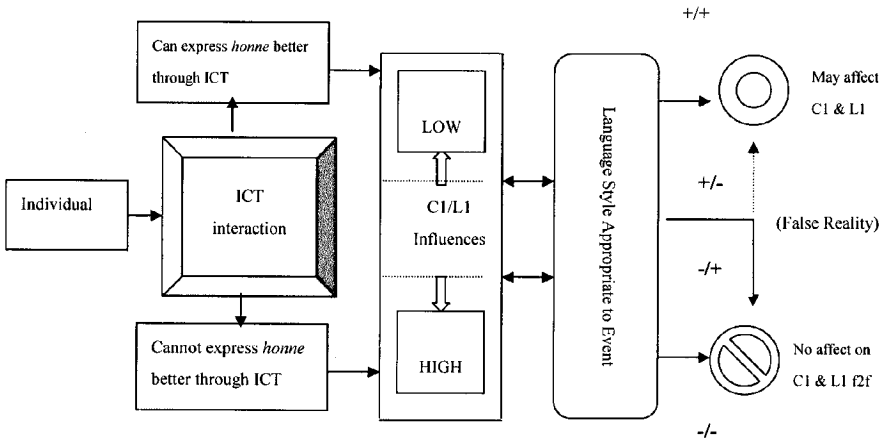

Figure 1: Affects of ICT interaction and communication on C1 and L1 (FIELD, 2002)

The Study

A case study involved ten students (6 female and 4 male) in an elective intermediate level general English course at Waseda University, considered an elite private Japanese university. All interactions were conducted as part of their regular class activities and did not involve interaction with non-members of the course. The course outline was published in the departmental handbook and students were expected to have read the outline prior to enrolling in the elective unit. The course met twice a week: once in a non-computer classroom and once in a computer lab. The students came from differing faculties within the University, including Economics, Law, Literature, Political Studies, and Psychology. One student was Korean; one student belonged to an English speaking club at the University and her major was English Literature; two students were in their first year, four were in their second year; three in the third, and one in his fourth.

Although the course was conducted over one academic year, the data for this study came from classes in the final semester of the 2002 academic year.

A Bulletin Board (BBS) was established and all class members were registered as members. Access to the BBS was available both within and outside the University server. Students were encouraged to access the BBS and post comments out-of-class as well as during in-class activities. The first three classes in the lab were designated to introducing the BBS and ensure that all members were able to log-in and post messages. This approach proved to be particularly valuable as one female student found using a computer very stressful having had only limited experience during her high school education. All other students experienced no difficulty in using the computer, though the initial all-English environment was daunting for some[1]. Accordingly, students were allowed to access and navigate the BBS in their preferred language, but were required to post to the BBS in English only.

The initial use of the BBS included self-introductions, pre or post-class homework, and as a tool to pass messages to the teacher and other students. Students also asked grammatical or social questions to the teacher using the BBS - akin to a face-to-face discussion[2].

> Hi! I'm *N*.
> How was your weekend?
> I had a test of primary system administrator yesterday.
> So I studied on last Friday and Saturday... *NA (F)*

> Hi! Malco and everyone, this is *J*. Today's lesson, we talked a lot. Finally I was thirsty, so something to drink is the necessity in this class, isn't it? Well, I have a question.
> I made a next sentence in today class. "How much do you usually drink?" So many classmates asked me. "Drink what??? Alcohol?"
> I think the word "drink" includes the mean to drink alcohol. Is it unusual?
> Give me auswer, pleaseU CU,bye. *JA (F)*

The next several classes in the computer lab were designed to get the students to consider issues that had arisen in either their non-computer class, or were topics expressed in their English language learning text book. The students were also given time in the computer lab after their posting to discuss their comments face-to-face with other students.

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[1] Several students testified during the student evaluation process to this experience.

[2] Student postings have been modified only for readability. *Italics* represent modifications in the text made by the author. Names have been coded to protect their privacy.

If you had to live abroad for 5 years:
What problems do you think you would experience?
What would you miss most about your home?
What is your definition of 'home'?
Do you agree with the expression 'home is where the heart is'? Why/why not?

Posted by teacher

If I havd to live abroad for 5 years;
1. Some problems which I'd experience
I'd miss to take a bath. I really like ohuro in Japan and it is one of the most relaxing times in a day. And maybe I'll miss Japanese halthy foods. I like flesh food especcially seafood. So I can't live in the country which doesn't have enough seafood.
2. My definition of home
The atmosphere which people feel relax and comfortable. It mey be a house, but also some community like a society.
3. I agree with the expression " Home is where the heart is".
As I mentioned above, home is one atmosphere which some people make, so I agree with this idea.
FA (F)

What is evident from the above postings is that the students did not completely utilize the opportunities that come with writing to check the text for grammatical and spelling oversights. Furthermore, there is evidence of code-switching (mixing) between two languages, though this has been kept to a minimum. The author assumes that the student used the term for 'bath' (*ofuro*) in her text to *emphasize* the thing she would particularly miss as the English word is not expected to be outside her personal lexicon. At the end of the class, the teacher went through each student's posting projecting them onto a large screen. Each posting was evaluated for grammar and spelling oversights, firstly by the student who wrote the posting, and when s/he could not improve the text, by any other member in the class, and finally as the last resort, was corrected by the teacher. Interestingly, students were able to recognise the majority of their errors.

Other ICT-based classes were used for various tasks; for example, to discuss personality tests or develop a town plan for a resort city in Japan that was faced with the dichotomy of attracting tourist money while retaining its natural environmental attractiveness.

> I have got informations about the rough breakdown of tax revenue from these cite; http://www.yomiuri.co.jp/e-japan/nagano/kikaku/013/6.htm and http://www.infocreate.co.jp/hometown/karuizawa/karuiz.html .
> About 50 percent of the tax revenue are from villas. This is a great advantage for local people. Current trend seems to be toward Karuizawa development such as outlet mall and mansions.
> This is the first time for me to know that population in Karuizawa from June to August grows 10 times larger than that of local residents, unbelievable...
> In fact, number of households is 6000, on the other hand that of 2nd houses is 13000. From the fact, we can see easily how much tax will come from 2nd houses.
> For conservation of nature, Karuizawa town office have strict reglations on building houses. From architect's view, these laws is not kept well. http://www.karuizawa.jp/kankokyokai/green/10/guest10.htm
> This is one problem related to Karuizawa development plan.
> I suggest that they separete Karuizawa into two area, develop-able-area and non-develop-able-area completely. In the non-develop-able-area, nature will be conserved. This complete separation of Karuizawa area is my final opinion.
> *KB2 (M)*

Students were also encouraged to utilise the BBS out-of-class to communicate with other members of the group. This was predominantly used to ask questions of the teacher; such usage was an indication that the students were beginning to feel comfortable using the BBS as an extension of their classes away from face-to-face communication.

> Maru, we met on the back alley near the Goken this afternoon, didn't we? When you said something like "How are you, gentlemen? it's too hot ...", then I could only replied "I'm fine"...
> I felt the reply is too stereotyped to make native speakers laugh at hearing a typical Japanese English and lacks in interest, too.
> I hope to improve my reply paterns and get away from a poor 2bit circuit (Q → A. or Q →×).
> In this case what kind of phrases can I use?
> *KB2 (M)*

The final classes for the semester were designed to encourage the students to utilize the BBS to share opinions about a specific topic. The final task (for the course) was a debate[3] that was held in the last non-lab class of

the semester. Prior to allocating students into groups (members for the groups were selected by the teacher in an attempt to make the groups balanced in terms of conversation skill, argumentative skill, confidence, and gender), a general theme that would help generate thoughts for the debate topic was posted to the BBS and students were asked to write their comments (either pro or con) in response. Students had also begun to talk about and develop thoughts and language about themes in the computer classes prior to posting[4]. The variables in the BBS thesis statement were generated from the teacher's summation of the students' discussion. The teacher also interacted on the BBS by replying to the students' postings to help stimulate further thought, but no attempt was made to correct grammar or spelling. The thesis statement posted to the BBS was: *If we can solve the economic, religious, ethnic and military differences between nations or peoples, then we will be able to live in peace*.

BBS

Most students attempted to address the thesis statement although two students did not post any comments. There was no reason forthcoming for this non-posting during the student evaluation. Several students posted only once, three students posted twice, and two students posted three times or more during the classes. As the exercise of combining ICT-based interaction with face-to-face interaction was to take the students beyond the novelty, and allow them to deepen their learning and language in a non face-to-face event prior to utilizing it in a face-to-face verbal situation (albeit a staged event), there was little attempt to edit grammar. Only four student's postings are selected as a sample to emphasize the level of language use and depth of personal thought. A cross-section has been selected, from one posting to a 'string' that developed. The teacher's interaction has also been included.

> I have a friend, he does vorunteer works.
>
> One day, we talked "How can we solve the economic differences between developed country and developing country?"He usually bought coffee beans from Brazil.And, his job is writing a paper about world economy, religious, war, and environment.
>
> He said "I want to tell many people the problem of the world" And, he hopes the people become aware of the problem.And, he said "It's impossible to solve the economy difference."Because that rich man Mr. ビルゲ

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[3] Students were taken through the basic processes of a debate in their regular class and the structure for any formal debate was posted to the BBS.

[4] The general discussion started from a current affairs event, namely the situation in Afghanistan.

イツ (*Bill Gates*) remit large money.But, he can't solve the problem.
So, many American pray to God for peace of world.But, I think if they believe God, they must not make war.Because,God say " **汝、戦うことなかれ** (*Old Japanese suggesting the concept that 'You should not fight'*)"If they are good Christianity, they naturally notice their mistake.
But, almost people (include of myself) worry about only their peace and safety.How can we mind peace and happy of any other people??*SA (F)*

Hello! I am *MA*.
Frist of all,my opinion is that the assumption is impossible. It is beyoud reality.
About economic,there are many political regimes in the corrent world.Accoding to them, it does not always happen that the more people work, the more money they earn. The religion and ethnics are deeply related to human mind and thought.I think nobody can make it equal.When it comes to culture, language and race which are rare, it is valuable to preserve.Everyone all over the world is diferent,and that differences make our life interesting.The military problem represent how strong the country is. So I think it is related to economic problem.
Second, even granted that we could make the economic,religious ethnic and military differences equal, which does not mean we will be able to live in peace. The conflict will happen because how people think is different from parson to parson.
So, I think that perfect peace world would never exsist.*MA (F)*

MA,
It may be the reality that peace will never exist because people are different all over the world, but ideally, what could we do to make us all be able to live in peace a little more than we do? *Teacher*
(No reply).

Hello, I'm FA. About this topic, I have some opinions. First, do people can solve the religious differences? I think the religion has an important part for foreign people. And a lot of people except most of the Japanese believe a kind of religion. Besides, the cause of many international conflicts occurring today is the religious differences between ethnics. From this view, the statement "People will be able to live in peace." seems to be impossible.
Second, do people can solve the economic differences? As Mr. Field said in the class, it is impossible to equal the economic situation of all the countries. If we tried to do that, the natural resources all over the world

would be limited and we would not be able to live under the same standard as today. Probably, we would have to lower our living standard. So from this view, I do not agree with the statement "People will be able to live in peace," either. *FA (F)*

FA,

Would you say, therefore, that if people did not have 'religion' then we could make a big step forward to peace? How about the other idea that it is not the religion that creates the 'war', but rather the the people who follow the religion? That is, people change the 'rules' to suit themselves.

On economics: Should the developed countries not try to help the less economically developed countries as we all cannot have the same standard of living?

teacher

My openion: Religions have an important part for people who believe them. And under statu quo, there are many international disputes all over the world which caused by religions differences. Actually, the thing that creates the war is people who follow the religion. However, who could change the religion?

For example, Christianity is parted in some communions and most of the Christians believe their own religions is the best. *FA (F)*

Pro-side

Dispute always comes from difference, so this hypothesis is right to the point.

Economic equality

Wars often break up for economical inequality. So we must realize it to realize world peace. Developed countries donate money to developing countries in order to help food and water and medicine and so on. So if this equality is in practice beforehand, developing countries don't have to suffer from famine.

Ethnic differences

Inequality of ethnic deference is as often becoming wars' cause as economic deference. It is often used as slogan of war. So, it is important for realizing peace in the world.

Military deference

If we realize economic and ethnic equality on the earth, we don't have to own weapons any more. So, military equality is no more a problem this time for us.

Contra-side

The hypothesis is very attractive, while it is not realistic. To realize it is

heard to consider in terms of realistic measure.

Economic deference
The equality of economic means effort for work is not concerning to what you earn. This means to deteriorate the industry. Look at USSR in the past, for example.
Ethnic deference
The equality of ethnic is hard to consider. To abolish cultural and religious and lingual deference will be harshly restrict human freedom , mental peace, too. These things defer countries' history and thoughts and natural features. There aren't as many Christians in Japan as some African countries in ratio. Missionaries visited these countries the same. So the deference is indispensable for countries' to stay in good condition.
Military deference
The equality of military comes after the realization of ethnic and economic deferens, because most of wars happened because of these. But these two is hard to realize and even if it came true, the bad things would occur as I guessed above.

NA (F)

A response was posted by the teacher and a string of postings between the female student and the teacher continued and culminated in the following posting[5]:

I didn't mean to get into religion discussion. I would like to discuss culturel deference so it's a wrong example to take about. Because it is very subtle. So, for example some country eat rice and other country eat bread mainly. *NA (F)*

The teacher expressed how impressed he was by both the students' thinking and the effort they had gone to express their thoughts around the difficult topics. He also encouraged the students to continue to express their opinion, especially to teachers and persons that are traditionally considered by the culture to be of 'senior' status and therefore usually not challenged, and conveyed that the students can make a valuable contribution, regardless of what 'culture' had taught them.

[5] The full interaction has not been recorded in this paper as the author believes that her opening posting on the theme highlights the strength of the female student to be able to express and defend her opinions in the BBS

Following several weeks of student interaction around the thesis statement, the students were then allocated into one of two groups and the debate discussion question was posted to the BBS: *It is possible for America to live in the world without conflict and war?* The teacher allowed time in two classes for students to develop their ideas further and to discuss possible strategies for the debate. The teacher did not, however, bring any new ideas to their discussion; he only helped to develop thoughts into more concrete concepts. The students generally did not use the BBS to discuss debate strategies preferring to arrange meetings through cell phone email. Only one student used the BBS to confirm an appointment and apologize for an absence.

Debate

The debate was conducted in a non-computer classroom and a colleague was invited to participate as judge. The inclusion of 'an outsider' into the process increased the formality of the debate, and in some cases, facilitated an obvious increase in anxiety in some of the less confident students. The debate was also video recorded (this action had previously been taken during presentation components of the course and thus the students had become accustomed to the presence of a camera). The purpose of the debate was: to allow student further public speaking practice in a second language; to allow them opportunities to logically present and defend an argument; and to solidify language, concepts, and structures that had been practiced and developed throughout the course, especially on the BBS. For the purpose of this paper, the following discussion surrounding the debate has been limited to language and concepts that had been previously developed on the BBS or in the computer lab. There is no attempt to discuss student grammar, presentation skills, logical development, or clarity.

An immediate difference that was recognized between the BBS postings, in-class discussion and preparation, and the debate, was the dearth of references to religious issues. During the BBS interaction particularly, students often made reference to religious concepts, especially (mis-)understandings about US cultural nuances that they equated as being 'Christian' belief; however, in the debate, there was almost no reference to such concepts. This may be related to the teacher's posting during the BBS interaction that particularly addressed the religious discussion, and could reflect the power of the 'teacher's voice' in directing thinking and discussion.

> Everybody is talking about religion... but what causes religion? And is there a difference between religion and faith? Religion and culture? Religion and ethnicity? Even in the time when Animism (of which Shinto is one example) was the predominant philosophy, people still went to war

> against each other and killed each other... usually over economics. Do we use religion as an excuse for what is really greed and economics?
> *teacher*

The debate followed standard US-practices: an affirmative case, a negative case, rebuttal, and cross-examination opportunities. Students were required to firstly, give their presentation and arguments, and then defend them during cross-examination. Rebuttals concluded the process. Students with only limited previous public speaking experience read their argument, and found it difficult to defend, or expound, their argument during cross-examination. Other students found that words and concepts developed in the BBS discussion could mostly be re-explained during cross-examination; but in some cases, especially for students that had not interacted extensively on the BBS, when challenged with a word or concept during cross-examination, which they had introduced previously, found it difficult to defend or explain their thoughts, argument, and words. For example, *YB (M)* attended most classes, but his presence was mostly in a passive capacity, both on the BBS and in the regular non-computer class discussions. As a result, his argument during the debate was weak utilizing shallow concepts, suggesting he had not given much thought to the language or the concepts, and furthermore, during cross-examination, he found it difficult to defend, explain or develop his argument further.

Conversely, most of the other students that had actively participated, found it easier to elaborate on the concepts they raised during the presentation of their case. Students that utilized the in-class time to develop their thinking, language and concepts, also seemed more adept in handling the oral event. It is difficult, however, to isolate whether it was the BBS interaction, regular class participation, or a combination of both, that aided the students in this process. Only one student, who was Korean, and who did not actively participate on the BBS, preferring to discuss current affairs face-to-face with the teacher, was confident to present and defend an argument. This may reflect a cultural or personality trait that may not be natural in Japanese society. Notwithstanding, the judge criticized all student arguments as lacking support and depth, and she was not persuaded by their discussion to either the affirmative or negative case. The judge was also critical of several students' poor attempts to try to make themselves understood by other interlocutors, which led to confusion during cross-examination - including, at one point, cases being reversed.

Student Evaluation

Following the debate, students completed a questionnaire that asked them to evaluate the use of the BBS in the course, as a tool for their language learning, in the development of ideas, and as a tool to help them before face-to-face discussion. Questions were posed in English, but students were encouraged to respond in their preferred language. Only three students responded in Japanese and the author's English translations have been checked by a native speaker. Comprehension of terms was clarified prior to completion.

Most students wrote that their understanding of the BBS was either a tool to convey information or communicate with others. All students responded that they used the BBS for class work (assigned by the teacher), but only four students stated that they used it for communication of non-class assigned tasks. One student complained that it was troublesome (a first year female student whose exposure to computers was limited) and one student was critical of both the teacher and the students because she felt that they did not use the BBS sufficiently. When asked to provide additional comments about the use of the BBS as a tool for communication or language development, one female student responded that she felt the topics presented on the BBS should be student led, based on student experiences (travel, events they organize) and this would further facilitate students to speak English. On the other hand, the Korean student argued that the students needed to be encouraged to think more deeply about topics and this would benefit their language development and understanding. Another female student criticized the BBS as being *ad hoc* and not planned and this was supported by another member who asked for the course schedule to be put on the BBS (it was available in the handbook). This feedback highlights the difficulties of providing a course that will sufficiently cater to all students needs, but it can be attributed to the lack of uniform pre-requisite evaluation standards of student ability before they register for any course - this includes both language proficiency and ICT competency.

The majority of students did not find it easier to express their real opinion and feelings (*honne*) through the BBS than in a face-to-face event. Two students did not respond, two found it easier, and one changed her mind during the course.

> I thought it was convenient at first, but gradually classmates began not to use it so I found face-to-face was better after all. *FA (F)*

The reasons for not finding the BBS easier to express real opinions and feelings were not consistent as one student expressed concern that s/he was known in class anyway, and therefore knew the 'face' and the other students personally, while another expressed that s/he did not know the other students personally enough to express such opinions and feelings. Another argued that s/he did not have the linguistic depth to convey real opinions and feelings on the BBS and therefore preferred face-to-face events. The students that found the BBS easier argued that the act of writing allowed them time to develop their thinking and language before expressing it and because they do not have to face another person to express their real feelings and opinions.

Seven of the ten students did not express any negative feelings when encoding (composing) text for the BBS (with reference to reliability and validity - simply defined as 'truth' in the text). Two students expressed a conditional to the validity of the text, arguing that it depended on their mood, the time available to interact, and their language ability: "I didn't have enough language ability to write down what I wanted to express". Only one student did not respond, and the same student did not respond to the reliability and validity of other student's text on the BBS. All other students responded that they did not have a negative experience when decoding (reading) the text. Generally the students argued that "we don't lie in English", "it was really what I thought", "because I can get a deeper understanding of other students" and "it gave me a chance to reconsider".

When students were asked to consider whether they were able to discuss the topics and concepts raised on the BBS in face-to-face situations prior to communicating on the BBS, three students did not respond, two answered that they were able to, and five wrote that they were not able to.

> Maybe, no. I often hesitate to say true mind in a face to face, (maybe for its Japanese culture).
> *YB (M)*
> I learnt many academic words and terms due to the entrance exam, but I do not know many words for real life. Therefore, it was hard for me to keep conversing (transl).
> *Anon.*
> Since starting to discuss some issues on the BBS, we have become to discuss the same topic outside of class and this has made a strong bond among the class (transl). *SA (F)*

Therefore, only 20% of this focus group were able to discuss the topics

in a face-to-face situation before interacting on the BBS. Furthermore, only 30% did not believe that the BBS helped their language learning and use in other events outside of the BBS and the course. Only one of those students argued that the language on the BBS was used only within the course. The other two students believed that the BBS did not help because they had not used it enough. On the other hand, 50% believed that interacting on the BBS helped them learn and use language in other situations (only two students did not respond to the question).

Yes, I could read language on the Web by my eyes and communicate face-to-face. It helped me when I speak English. *SA (F)*

Yes, I learnt different expressions in English. *Anon*.

Conclusion

The use of the BBS in conjunction with regular non-computer based second language classes was found to be a useful supportive tool to help students generate thoughts, develop concepts, practice language, pass information and communicate with other members of the class in a community-based environment, which allowed them some degree of individual autonomy over the process. Some students found that the difficulty level of the tasks was too high and that limited their ability to interact; thus highlighting the need for a positive 'valid' engagement.

My English is too poor to express my opinion... *Anon*.

Half the students expressed disappointment that the BBS was not used more often and several requested that it should have been used on a daily basis - need for positive 'reliable' engagement. One female student felt that many students were not motivated to interact and another felt that the teacher was responsible for motivating the students and encouraging them to develop their thinking to deeper levels. Although the author agrees with the student's comments, it highlights the dilemma that students and teachers face at university: students lack skills to self-educate and interact, forcing some language teachers to struggle between the roles of entertainer, teacher and academic. In this study, the teacher expressed his delight in the student interaction and completion of tasks, however, some students felt that the interaction was insufficient. The teacher responded that he had hoped all students would interact more on the BBS, but as the course was an elective unit, and as some students did not receive credit for their study, he did not want to take a 'strong-arm' approach, but tried to coerce the students to interact.

Notwithstanding Japanese educational pedagogical and learning issues,

the students' comments highlight the need to develop ICT-based educationally applicable tasks that students find meaningful for their learning. The student evaluation of the use of the BBS revealed that students had differing opinions and needs and therefore it may be impossible to provide a course that caters for the needs of all the students in the current Japanese university language education system, which does not effectively evaluate communicative or academic language ability. Moreover, cultural influences will further affect student need and understanding and this should be considered in the design of any educational course. The reality remains that CALL is not suitable for all learners.

The level of participation was hypothesized to be related to the degree of reliability and validity and only in a positive-positive event will the likelihood of transference to a face-to-face event be greater. 70% of the students reported that their encoded text was reliable and valid and 90% wrote that the other students' text was also reliable and valid. In other words, 70% had a positive encoding experience and 90% had a positive decoding experience. According to Field's hypothesis, therefore, the likelihood of transference to a face-to-face event should be higher than if the students had experienced a negative encoding-decoding encounter. During the face-to-face event (debate) it was outlined that students who had interacted on the BBS were more able to defend and clarify their argument than students who had been more passive members of the class. This observation, in part, confirms the hypothesis that a positive-positive encounter will more likely result in transference to a face-to-face event.

However, the result was not consistent for all cases as the Korean female student did not actively participate on the BBS, but was still able to present her argument, and then defend it during cross-examination. This has been attributed to a cultural and/or personality difference, reflecting different curriculum design and pedagogical implementation is required for students from different cultural groups if they are to achieve maximum benefit from the ICT tools. For the Japanese, on the other hand, ICT can be a useful tool supplementing part of the language education curriculum, especially in task-based educational processes that require the student to develop new ideas and concepts in conjunction with language acquisition.

This is also supported by the result that only two students stated that they were able to discuss the issues in the second language prior to discussing it on the BBS, and three students believed that they did not use the language or concepts in an another event outside the BBS or the course. Conversely, 50%

stated that they were not able to discuss the concepts or use the language prior to interacting on the BBS before the course, and 50% believed that the interaction helped them to use the second language and the concepts outside the course and BBS interaction. Subtracting the Korean student, and the identified passive male student (who did not comment), the result is increased to about 80% of the focus group believing that the use of the BBS enabled them to utilize concepts in the second language in other events outside the BBS.

What can be concluded from this case study is that there is evidence that FIELD's hypothesis may be a useful theory upon which to build ICT-based language education courses in Japan. Although there were indications that student interaction was waning (from student evaluation), providing students with learning material that is applicable to their lives, to their learning situation, to the skill and academic level, and when they are provided with a supportive learning environment that encourages personal development, the students will interact beyond the fading of the novelty honeymoon, and furthermore, ICT-based interaction should facilitate greater transference in higher level face-to-face communicative events.

A Final Word

The results from this case study are indeed promising and in many ways confirm Field's (2002) hypothesis that a positive encoding-decoding interaction may have positive influences on face-to-face events. Several concerns need to be raised. Firstly, the study needs to further consider C1 influences on language use and on the interaction as outlined in Field's model. Secondly, although this paper focused only on the BBS-debate interaction and process, it is acknowledged that the students also had opportunities to discuss some of the concepts in other classes, and face-to-face with the teacher, and this may have influenced their ability to re-use the language and concepts in other events. This suggests that the use of ICT alone will not bring about radical new learning, and that ICT is most optimal when combined with other pedagogical approaches. Thirdly, there was some evidence from both student comments and from BBS interaction (especially during the final stages of debate preparation) that student participation was waning. Unfortunately, this indicates that some students did not move beyond the novelty phase. Suggestions to alleviate this situation have been outlined above by both students and by the author, but an additional recommendation must be made: the need to radically modify the educational process in Japan. Finally, as a case study, the author recognizes that the results may only be case specific; however, he firmly believes that they indicate that Field's previous results, which were

conducted at another university in another area in Japan, may be applicable nationally. In other words, the trend may be indicative; thus the results provide a solution to the link between ICT-based language learning and face-to-face use in Japanese higher education, and will provide a method of taking students beyond the novelty value of learning.

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Why Do We Need to Analyze Natural Conversation Data in Developing Conversation Teaching Materials? - Some Implications for Developing TUFS Language Modules[1]-

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1. Introduction

This paper outlines the aims and scope of a series of research projects conducted by the Discourse Research Group at Tokyo University of Foreign Studies (TUFS) in 2002 and 2003, and summarizes some of the major findings obtained from them. These projects were conducted as Applied Linguistics Projects in TUFS's 21st Century COE Program called the Center of Usage-Based Linguistic Informatics ("Linguistic Informatics COE Program" hereafter), and some of the findings were presented as a series of four presentations at the 1st International Conference on Linguistic Informatics held at TUFS in December 2003[2].

A main goal of the Linguistic Informatics COE Program is to "innovate foreign language education by developing superior foreign language educational material and transmitting it through the Internet (The official website of the Linguistic Informatics COE Program).[3] For this aim, the TUFS Language Modules, new e-learning materials which cover 17 languages, are being developed from the research accomplishments of the program.

As members of the Applied Linguistics Project Group in this COE Program, the main role of the Discourse Research Group is to study actual interaction and seek implications for the development of the Language Modules,

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[1] I would like to show my gratitude to Takashi Suzuki for his cooperation from various points of view.

[2] At the First International Conference on Linguistic Informatics - State of the Art and its Future - Computer Assisted Linguistics, Corpus Linguistics and Applied Linguistics, held on December 13th - 14th , 2003, the Discourse Research Team presented the following four papers which are related to each other by an underlying theme - the development of teaching materials from natural conversation data: 1. USAMI 2004 (an earlier version of the present paper) 2. SUZUKI et al. 2004, 3. SEKIZAKI et al. 2004, and 4. XIE et al. 2004.

[3] http://www.coelang.tufs.ac.jp/english/coe.html

as well as to evaluate the modules that have been already developed and explore ways in which those materials can be improved.

In recent years, the need to analyze actual interaction and to incorporate natural conversation data into language teaching materials has been recognized increasingly. In the field of English language teaching, attempts have been made to use actual conversation for language teaching, both in the form of teaching materials (SLADE AND NORRIS 1986, STUBBE AND BROWN 2002) and suggestions on how to incorporate such data into language teaching (BARDOVI-HARLIG AND MAHAN-TAYLOR 2003, HOLMES 2004). In Japanese language teaching, however, there has been little research and discussion regarding the use of natural conversation as conversation teaching material, and thus to date such materials consisting of natural conversation data virtually do not exist in Japanese.

In this paper, I will discuss why the analysis of natural conversation data is necessary in developing conversation teaching materials, drawing on the results of empirical research conducted by the Discourse Research Group at TUFS. First, I will summarize the major findings of three research projects conducted by our research group and outline their implications for the development of conversation teaching materials. Those projects are, 1. an analysis of English teaching material that consists of natural conversation data, 2. an analysis of the process of developing a multilingual corpus of spoken language and how functions of language behaviors are realized in natural conversation, and 3. an analysis of 'requesting' in actual Japanese conversation[4]. Next, I will compare natural conversation data with the language of created skits presented in the Japanese Dialog Module ("the D-Module" hereafter) of the TUFS Language Modules, and outline the differences found between the two kinds of dialogs.

2. An analysis of English teaching material consisting of natural conversation data

As mentioned above, there are virtually no teaching materials consisting of natural conversation data in Japanese, and the situation is similar also in the field of English language teaching although there are some exceptions. One of such exceptions is "Talk That Works" (STUBBE AND BROWN 2002), a video-based communication training kit based on the findings of the Language in the Workplace Project at Victoria University of Wellington in New Zealand[5]. We took up Talk That Works ("TTW" hereafter) as a sample of teaching materials made up of natural conversation data, and analyzed it to

[4] These three projects are reported in detail in SUZUKI et al. 2004, SEKIZAKI et al. 2004, and XIE et al. 2004, respectively.

seek implications for the development of such materials in Japanese. Here I summarize some of the major findings and their implications[6].

First, through our analysis of TTW as teaching material, we found that by incorporating natural conversation data and giving learners opportunities to be exposed to such language, TTW gives learners a fresh perspective on the nature of conversation. It facilitates learners to view conversation as interaction between participants rather than merely as successive production of linguistic forms. TTW's focus as teaching material includes discourse processes (e.g. turn/floor taking, topic management, the joint negotiation of meaning, etc.), pragmatic/discourse features (e.g. fillers, feedback, hedges, discourse markers, etc.), politeness strategies, and repair strategies, most of which can be categorized as interactive linguistic behavior. Raising learners' awareness of the interactive aspect of conversation can help them communicate more smoothly and naturally in the target language, and this can be done most effectively through using natural conversation samples.

Next, we analyzed the natural conversation data included in the TTW video to investigate how the functions featured in the D-Module are realized in it[7]. Out of the 40 functions featured in the D-Module, we selected 7[8] which appear most frequently in the TTW data and coded the discourse sentences[9] carrying these functions using the following coding scheme.

Type-1: One of the seven functions is realized in the discourse sentences and is accompanied by a corresponding linguistic form[10].

Type-2: One of the seven functions is realized in the discourse sentences but is not accompanied by any of the corresponding linguistic forms for that function.

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[5] The main aims of the Language in the Workplace Project are to study spoken communication in New Zealand workplaces and to explore possible applications of the findings for those workplaces. This project is led by Dr. Janet Holmes, whose paper (HOLMES 2004) appears in this volume.

[6] For methodology and detailed results, refer to SUZUKI et al. 2004 (this volume).

[7] The same set of 40 functions is used for the D-Module of all the 17 languages in the TUFS Language Modules.

[8] The seven functions are 'Asking for information (about attributes)', 'Stating an opinion', 'Making a comparison', 'Giving a reason', 'Giving a direction', 'Giving an example', and 'Giving advice.'

[9] A discourse sentence is defined as 'a sentence in interaction'. It can be a "single word sentence" or a structurally incomplete sentence as well as a structurally complete sentence, as long as it fulfills a substantive function within the conversation. (See USAMI 1997, 2002, 2003, and SUZUKI et al. 2004 for detail.)

[10] In this study, a "corresponding linguistic form" is defined as "a linguistic form featured in the English D-Module to represent the function" or "a linguistic form which is considered to represent the function from its literal meaning or conventional usage."

In addition, we also extracted discourse sentences in which any of the corresponding linguistic forms is used, but the function itself is not realized. We call this kind of discourse sentences Type-3.

We found that in the TTW data, linguistic forms often considered to be 'typical' of a particular function (e.g. "I think -" for 'stating an opinion', or "because" for 'giving a reason') do not necessarily occur as frequently as might be expected. In the TTW data, functions are more often (57.1%) realized without typical linguistic forms, sometimes with forms not usually associated with that particular function, or sometimes without any explicit forms at all. When a function is realized in discourse without a corresponding linguistic form, context at various levels was found to contribute to the realization of the function. This suggests that when developing conversation teaching materials, we must select forms or discourse patterns to be presented for a particular function carefully so that they will reflect the ways in which that function is realized in actual conversations, either with or without a corresponding linguistic form.

In sum, we saw that the analysis of natural conversation data and the actual incorporation of it into teaching material can be beneficial in more than one way; it can raise learners' awareness of the interactive features of conversation which have been often neglected in more conventional teaching materials, and it can also enable the development of conversation teaching materials which present functions in discourse in more realistic ways.

3. The development of the "Multilingual Corpus of Spoken Language by Basic Transcription System (BTS) - Japanese 2"

In order to facilitate and promote studies on language use by preparing natural conversation data which can be shared by researchers, the Discourse Research Group at TUFS has been developing "the Multilingual Corpus of Spoken Language by Basic Transcription System (BTS)". The latest addition to this corpus is "Japanese 2 by Basic Transcription System for Japanese (BTSJ) "[11], which consists of natural conversation samples corresponding to various functions in discourse. In developing this component of the corpus, we obtained some findings regarding how functions are realized in actual interaction in Japanese. I summarize them here as they provide implications for the development of conversation teaching materials[12].

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[11] A corpus of spoken Japanese was included in the "2002 Progress Report of Applied Linguistics Projects conducted by the Discourse Research Group". This corpus is now considered the first Japanese component of the "Multilingual Corpus of Spoken Language by Basic Transcription System (BTS)" although it was not named as such at the time of the publication.

"The Multilingual Corpus of Spoken Language by BTS- Japanese 2 by BTSJ" consists of discourse samples extracted from actual Japanese conversation. In order to investigate how functions in discourse (e.g. 'stating an opinion', 'requesting', 'apologizing' etc.) are realized in natural conversation in Japanese and compare it to how they are presented in teaching materials, we collected samples in which one of the 40 functions featured in the Japanese D-Module of the TUFS Language Modules is realized. We used a coding scheme similar to the one used for the analysis of the English data in TTW (i.e.Type-1 to 3) to analyze how the 40 functions are realized. Among our findings, the most relevant to the present paper can be summarized as follows.

1. While the majority (73.9%) of the extracted examples are Type-1 (The function is realized with a corresponding linguistic form), there are several functions which have more Type-2 examples (The function is realized without a corresponding linguistic form) than Type-1. These functions include 'giving something to someone', 'asking for information (about experience) ', 'asking for information (about skills and ability) ', and 'asking about responsibilities', among others. When a particular function is realized without the presence of a corresponding linguistic form, it is realized through, or accompanied by, other means or elements which include different levels of context, the local and global structure of discourse, discourse/pragmatic features such as repetition or ellipsis, and politeness strategies.
2. The frequency of occurrence of each function and the ratio of Type 1-3 for each function both fluctuate depending on the participants' relationship with each other (i.e. strangers or friends).

The following example demonstrates both point 1 and 2 above. In our natural conversation data, it was observed that the function 'asking for information' (about existence/place) is realized frequently without a corresponding linguistic form. This is particularly true in stranger to stranger situations, where the ratio of Type-1 to Type-2 is 44% to 56% (16 examples to 20 examples). More specifically, questions asking for information about existence/place between strangers (typically questions about the other participant's hometown, place of residence etc.) are often expressed as incomplete sentences without explicit linguistic forms corresponding to this function[13]. This

[12] SEKIZAKI et al. (2004) describes in detail the aims and process of the development of this corpus, and discusses some of the main issues and findings of this project.

[13] The corresponding linguistic forms for this function include "*Doko* [where]" or "*Dochira* [where (a more formal form than *Doko*)]" which often appear near the end of a discourse sentence in Japanese, and hence are likely to be omitted if ellipsis occurs.

can be interpreted as a politeness strategy used to avoid direct expressions asking for such information about the other participant. (See SEKIZAKI et al. 2004 for detail.)

Just as we saw in the previous section with the TTW English data, these results in Japanese also suggest that in order to develop conversation teaching materials which can effectively help learners carry out various functions in discourse, it is not enough to merely present linguistic forms for each function. We must design and/or use materials that raise learners' awareness of other elements of natural conversation, such as discourse structure, discourse/pragmatic features, and politeness strategies, which are found to contribute to the realization of functions. We should also provide learners with examples that show functions are often carried out in different ways according to social factors such as the participants' relationship with each other.

4. An analysis of 'requesting' in natural conversation

In the research project described in section 3 above, we analyzed the 40 functions featured in the Japanese D-Module, focusing on form-function relationships. In order to investigate more closely how functions are realized in natural conversation, we selected 'requesting' for a detailed analysis and extracted discourse samples of this function from "the Multilingual Corpus of Spoken Language by BTS- Japanese 2 by BTSJ". We then compared these samples with those in the Japanese D-Module's unit featuring 'requesting.'

While 'requesting' is a very common function which occurs frequently in our everyday lives and is included in many conversation teaching materials, it is not an easy function for non-native speakers to master. This is because requesting is essentially a face-threatening act, which requires the requester to employ culture-specific strategies to show consideration to the requestee. Failing to do so appropriately 'can cause a malfunctioning of human relations' (NISHIHARA 2004), or result in the request being rejected by the requestee.

We analyzed the samples at the discourse level, focusing on how the requester proceeds to have his/her request granted, and what kind of 'politeness behavior' can be observed. In each 'request sequence', we coded 5 acts performed by the requester: 'getting attention', 'prefacing', 'assessing the prospect', 'a supportive move', and 'an explicit request utterance'[14]. We also coded the existence of 'thanking' and 'apologizing' as politeness behavior.

The differences between the request sequences in the natural conversation data and the Japanese D-Module can be summarized as follows.

1. While most of the natural conversation samples include 'an explicit request utterance', there are also some which do not include one. The D-Module skit for this function features a conversation with 'an explicit

request utterance'.

2. The request-sequence samples in the natural conversation data have 7 patterns (i.e. combinations) of requester's acts, although the Japanese D-Module has to select only one pattern as an example.
3. Both 'thanking' and 'apologizing' cooccur with ‘requesting’ as ‘face-redress behavior’ in the natural conversation samples. Only 'thanking' is featured in the Japanese D-Module.
4. In the natural conversation samples, when the requested action is not performed immediately, it is typically followed by an expression by the requester which functions to reinforce the request, such as "Yoroshiku (onegai itashimasu)" [literally, "(I beg you to) take care of it well"]. The skit in the D-Module does not include this kind of expressions even though the requested action is not carried out immediately in the skit.

From these findings, we obtained the following implications for the development of conversation teaching materials.

There are natural conversation samples in which no 'explicit request utterance' is used, but the request is communicated (and granted) through combination of other acts such as 'assessing the prospect' (e.g. "Print-o motte imasuka? [Do you have the handout?]"), and 'a supportive move' (e.g. "Print o nakushi mashita. [I've lost my handout.]"). If a learner of Japanese is not aware of this kind of discourse pattern, he or she may not be able to understand the intention of the speaker when another person uses this pattern to make a request. If a learner can identify this pattern, he or she can consequently respond to it in a natural and appropriately cooperative manner. Elementary level learners can also use this pattern to make a request even if they haven't learned the language needed to carry out this function. For these reasons, discourse patterns such as this should be introduced to learners of all levels including beginning level learners.

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[14] Definitions and examples of each act follow. (Refer to Xie at al.2004 for detail.)
Getting attention/An utterance which urges attention, such as "Anoo. (Umm.)"
Prefacing / An utterance which communicates the general purpose of the conversation, such as "Chotto onegai shitai koto ga aru n da kedo. (I wonder if you can do me a favor.)"
Assessing the prospect / An utterance which assesses the prospect of successful implementation of the request, and/or checks the existence of obstacles, e.g. "Getsuyobi, X sensee no jugyo deta? (Did you attend Prof. X's class on Monday?)"
A supportive move / An utterance preceding the explicit request utterance, used to lesson the burden placed on the requestee either practically or psychologically, e.g. "Sono print o nakushi chatta n de... (I've lost that handout, so...)", or "Waruikedo...(Sorry to bother you but...)"
An explicit request utterance / An utterance which includes a typical linguistic form used to make a request, such as "-shite kurenai?", "-shite moraenai?", or "-shite kudasai." (All these expressions can be roughly translated as "Will you (please) -?"

In only 9% of the natural conversation samples in which a request is granted, the request is communicated through 'an explicit request utterance' alone, or 'an explicit request utterance' and 'getting attention'. In the rest of the samples, 'assessing the prospect' or 'a supportive move', or both of them, always precede an 'explicit request utterance'. Also, since requesting imposes a burden on the requestee, it is often followed by a 'face-redress behavior' such as 'thanking' and 'apologizing'. In order to facilitate learners to be able to make requests in authentic and smooth manner, we must present more than one pattern of requesting, which include not only the 'explicit request utterance', but also the surrounding elements that make up requesting sequences in natural conversation.

Having outlined the major results of three projects undertaken by our research group, I will now proceed to demonstrate why the analysis of natural conversation data is necessary in developing conversation teaching materials.

5. A comparison of natural conversation data and created skits

The studies I summarized in section 2-4 and the present study are all related to each other by an underlying theme: the development of conversation teaching materials using natural conversation data. In this section, I compare features of natural conversation with those of the language in created skits in the D-Module.

5.1 Features of the skits in the TUFS Japanese Dialog Module

In this sub-section, I introduce a skit in the Japanese Dialog Module in order to outline the general features of the skits in the D-Module. In the following created skit, which is supposed to be a telephone conversation, A, the caller, makes a reservation for a graduation party at a hotel. Only A's utterances are shown here.

D-Module Skit -"Making a reservation"

A: Moshi moshi, shaonkai no kaijō o *yoyaku shitai n desu kedo*.
A: Sangatsu nijūgonichi no rokuji kara desu.
A: Sanjūmei de hitori ichimanen gurai de *onegai dekimasu ka*?
A: Hai, Tamura to mōshimasu.
A: Sore kara, hoteru no panfuretto o *okutte hoshi in desu ga*.
A: Hai, jūsho wa, Tokyo-to, Fuchū-shi, Asahi-chō, san no jūichi no ichi desu.
A: Hai, *dewa, yoroshiku onegai shimasu*.

D-Module Skit -"Making a reservation" (English Translation)

A: Hello, *I would like to* make a reservation for a place to hold a graduation party.
A: We plan to hold it from 6 o'clock on the 25th of March.
A: *Would it be possible to* pay about 10,000 yen per person for a group of thirty people?
A: Yes, my name is Tamura.
A: And *could you also please* send me a brochure of your hotel?
A: Yes, my address is 3-11-1 Asahi-cho, Fuchu city, Tokyo.
A: Yes, *thank you very much*.

The characteristic features of this created skit can be summarized as follows.

1. There are no unnecessary or irrelevant utterances or expressions, such as fillers, repetitions, or repairs. This makes it easier for learners to focus on the target patterns and functions, such as "I would like to..." and " Would it be possible to ...".
2. Because there are no unnecessary or irrelevant elements, the skit is rather short. Within seven utterances, the speaker has managed to make a reservation and also has asked to have a brochure sent out. The parts which are italicized, such as *'...tai'*, *'ongegai dekimasu ka?'*, *'..te hoshii'*, *'yoroshiku onegaishimasu'* are all typical linguistic forms used to make requests or express desires.

In the D-Module, there are set forms that can be easily focused upon, and the functions associated with them are relatively easy to pick up. The expressions used in the skit and how they are used there also seem to be quite natural. The D-Module also features notes for the function of each and every utterance. This information will be very helpful for learners. In sum, it can be said that for beginners, this teaching material is constructed very well.

However, this does not necessarily mean that this kind of teaching materials alone is sufficient for teaching Japanese. To demonstrate this point, in the next sub-section, I will analyze features of a telephone conversation recorded by our research group, which can then be compared to the created skit we've just presented.

5.2 Features of natural telephone conversation

The following conversation was recorded with the cooperation of a group of TUFS students, who were asked to control the contents of the conversation so that they would match as closely as possible the contents of the D-Module skit shown in the previous section. However, since we could not ask the students to make an actual reservation, we instead asked them to call

up a hotel to make an inquiry about reserving a place for a party. What follows is a transcript of an actual telephone conversation, although the other speaker's utterances are not shown here. We call this conversation a 'natural conversation' here purely for the sake of comparison. We believe that naturalness of a conversation is a relative matter, and we readily admit this conversation is structured to a certain degree. A, the caller, is a student from an undergraduate class, whose permission we secured to use this dialog for this study.

Recorded conversation -"Making a reservation" (in Japanese)

A: Hai, osoreirimasu. (hai) *Ano desu ne*, chotto shaonkai o sochira de **kangae... dekiru kanā to omoimashite, kangaete orimashite** chotto otoiawase no denwa o sasete itadaite iru n desu kedo mo.

A: Hai. *Ēto desu ne*, daitai ninzū sanjyūnin gurai de, hito ata (hai?) rino *anō* ninzū ga sanjūnin gurai de, (ee) *anō* yosan hitori ichimanen to **iu...yutta** katachi de kōsu nado wa sochira dewa kikaku sarete rasshaimasu ka?

A: *A*, sayō de gozaimasu kā.

A: *A*, wakarimashita. Moshi, *ano*, **sōitta...hokani wa** ato onedan te aru n desu ka? **Daita**... (hai?) .

A: Hoka no ryōkin settei toka mo arimasu ka?

:

(continues)

Recorded conversation -"Making a reservation" (English Translation)

A: Yes, thank you. (yes) *Well*, I'm calling to make an inquiry, please. We're **thinking ..., thinking of, wondering** if it will be possible to, hold a graduation party at your hotel.

A: Yes. *Well, let's see*, we are a group of about thirty, and per (pardon?) person, uh, we are about thirty people (yes) uh, do you have **something, something like**, course menus available for about 10,000 yen per person?

A: *Oh*, is that so?

A: *Oh*, I see. If, *uh*, do you also have **other**, course menus **like**... at different prices? **Abou**... (yes?).

A: Do you have other course menus at different prices as well?

:

(continues)

The characteristic features of the natural conversation can be summarized as follows.

1. Many types of fillers, such as 'ano desu ne' ('well'), 'eeto desu ne'('let's see') etc., which are italicized in the transcript, are present.
2. The actual conversation is overall longer than the skit in the D-Module.
3. Examples of self-initiated repair are present, which are shown in bold in the transcript. Although the speaker is a native speaker, we can see that the speaker is thinking carefully as she speaks.
4. The word '*hai*' (literally, 'yes') which is put in parentheses in the transcript, is used for more than one function; it is used to show that the participant is listening to the other person(backchannels), and also to ask the other person to repeat something (Similarly to 'Excuse me?' or 'Pardon me?').
5. The expression *A, sayō de gozaimasuka* is a slightly old-fashioned, rather polite expression often used in service industries. The use of an expression like this by a student who wouldn't normally use it is a reflection of accommodation to the other participant, who is an employee of the hotel in this dialog.
6. Apart from one expression, *Chotto otoiawase no denwa o sasete itadaite iru n desu kedo* ("I have called to make some inquiries about..."), which is a fairly direct way of expressing that one wishes to make an inquiry, there are no other direct expressions of 'request' or 'desire' to be found, but the information needed is obtained by asking a series of questions.

In the next section, I would like to make a comparison of the utterances made by the person making the reservation in the Dialog Module, and those made by the person making an inquiry in the recorded natural conversation.

5.3. Comparison of the D-Module skit and the recorded natural conversation

As we have seen, what we notice immediately when comparing these two dialogs is that in the D-Module skit, the speaker describes what she wants very concisely with precise forms. The D-Module skit is also shorter in length than the actual conversation. As we saw in section 4 of this paper, our research group found that in natural conversations in which 'requesting' is realized, such acts as 'getting attention', or 'prefacing' are observed frequently before 'explicit request utterances', whereas in the D-Module skit, 'explicit request utterances' are presented without much preceding talk. This, in addition to the presence of various fillers we saw in the recorded dialog, can account for the difference in length of the two kinds of dialogs.

Although there are not many direct expressions of 'request' or 'desire' in the actual conversation, the information the caller needed is obtained without a problem. This is in line with the findings of our projects described in section 2 and 3 of this paper which report that functions can be frequently car-

ried out without the presence of typical linguistic forms corresponding to the function in actual conversations. We saw that discourse context, such as surrounding talk and the patterns in which utterances are sequenced, as well as other kinds of context, help realize functions in natural conversations.

In natural conversation, as can be seen from the backchannel 'hai' in the recorded dialog, one form can be used for various functions, whereas in created skits, typical forms tend to be reserved for the function they supposedly correspond to. This is supported by one of our findings in the analysis of TTW that in the authentic conversations in Talk That Works, (English) backchannels are used for different functions, with global context being one factor that influences the function of them (SUZUKI et al. 2004).

Another distinctive feature of the natural conversation is that there are several instances where the speaker repeats or repairs her own utterances, which are shown in bold in the recorded dialog, whereas nothing in the D-Module skit is repeated or repaired. Even though the speaker is a native speaker of Japanese, particularly in situations where one is making inquiries, we can often see people thinking and choosing their words at the same time as they speak. Since actual conversations involve mutual interaction with another person, depending on what the other person has said, one may quickly change what he or she was planning to say. Especially in the case of younger people who are not used to using honorifics, they tend to think as they speak, and therefore repairs occur frequently.

The last difference between the created telephone conversation and the natural one is how backchannels are used. In the case of languages such as English or Chinese, there is a tendency for one person to finish the basic sentence before the other person speaks, at least at the beginning of many interactions. However, in Japanese, it is possible to insert backchannels often in the middle of utterances at certain points, as we can see in the D-Module skit.

6. Conclusion

As we have seen throughout this paper, natural conversation data provides us with insights into how language is used to carry out different functions or to realize speakers' intentions. We believe that even elementary-level learners, as well as intermediate to advanced-level learners, will benefit from being exposed to (recordings of) actual conversations. They may not be able to reproduce much of what is said, but they can notice some of the features of natural conversation, such as backchannels, self-repairs, fillers, humor, turn-taking, or discourse patterns.[15] In fact, these features are essential elements of

[15] KONTRA 2003 demonstrates how non-native speakers' awareness of pragmatic aspects of communication could be raised through classroom activities even at elementary levels.

effective communication and must be introduced to, and noticed by, learners of Japanese in early stage of their learning.

For example, the extensive use of backchannels discussed in section 5.3 is one of the important elements which form the mutually interactive style of communication in Japanese (MIZUTANI 1988, MAYNARD 1993, CLANCY et al. 1996). This style of communication has been termed *kyōwa* ('cooperative talk') and is regarded as characteristic of spoken Japanese (MIZUTANI 1988, 1993). For speakers of languages other than Japanese, the backchannels or overlapping that occurs in Japanese may sometimes be felt to be interrupting or even violating another person's speaking turn, and thus may seem rude. However, in Japanese conversation, backchannels are essential for smooth communication, and therefore must be introduced to non-native speakers. For this purpose, natural conversation data would be the natural choice over created skits to be used in teaching materials.

Also, as discussed in section 4, Japanese people sometimes 'request help' by simply explaining the problematic situation, and waiting for the hearer to offer help. If a learner of Japanese notices this pattern through watching or listening to actual conversations, he or she will at least be able to comprehend what is happening when someone uses this pattern to request help. In this way, for elementary level learners, natural conversation data can be especially useful for improving comprehension skills. This point clarifies some issues and offers a new perspective on the way in which the teaching of listening in the past has been a test of the student's ability to understand the literal meaning memory rather than being a test of their ability to understand the speaker’s intention over extended discourse. However, if we believe language education to be about facilitating smooth communication, we can see that in order to listen and understand, one has to understand the intention of the speaker who is making the utterance. Thus in actual communication it is necessary to understand the intention/functions that can arise in dialogs from a whole range of utterances. How to make teaching materials from natural conversations in order to practice such skills requires more attention from now on.

Finally, I would like to add that it is not our intention to suggest that teaching materials should only feature natural conversation data, or that there is no place for created skits in language teaching. Natural conversation data and created skits both have their own advantages and disadvantages as teaching material. We believe that the two kinds of dialogs should both be used in language teaching so that they can supplement each other.

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An Analysis of Teaching Materials Based on New Zealand English Conversation in Natural Settings – Implications for the Development of Conversation Teaching Materials –

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1. Introduction

In virtually all the Japanese language textbooks and teaching materials which focus on conversation, most of the target items or skills are presented through non-authentic dialogs that are written specifically for the purpose of teaching those items/skills. Although non-authentic data has its own advantages over authentic data, it has been pointed out that there are considerable disparities between the two kinds of data (NUNAN 1989, 1999), and that being exposed only to non-authentic data can limit or hinder the learning process (NUNAN 1999). The need for studying authentic conversation with consideration to materials development, and incorporating the results of such research into actual teaching materials, has been acknowledged in recent years.

In this context, "Talk That Works" ('TTW' hereafter), a video-based "communication training kit" developed in New Zealand in 2002, deserves attention for two reasons:

1. TTW consists entirely of authentic conversations. Such teaching materials can rarely be found also in English.[1]
2. TTW's video is accompanied by a handbook with notes which are based upon recent research on discourse and communication.

In the present paper, we will first briefly analyze TTW as teaching material and outline what strategies and features of language it focuses on. Our purpose is not so much to evaluate TTW but rather to determine what it has that conventional textbooks with non-authentic dialogs do not.

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[1] Other teaching materials of this kind include CRYSTAL AND DAVY (1975) and SLADE AND NORRIS (1986).

Next, we would like to consider the implications of the analysis of authentic data for the development of teaching materials. Since TTW is communication-training material targeted at higher-level learners, we need to examine whether, and in what ways, the analysis of authentic data can be beneficial to the development of conversation teaching materials in general, including those for lower-level learners. For this purpose, we will use the recorded conversations in TTW as data, and analyze them focusing on some of the basic functions realized there.

2. The foci of TTW as teaching material

In this section, we will outline what skills or aspects of language are focused on as study objectives in TTW. For the sake of simplicity, we refer to teaching materials consisting mostly of specially written dialogs as "conventional textbooks/materials", although we recognize that there are such materials whose dialogs are based on extensive research and resemble natural interactions closely.

In the teacher's handbook, TTW lists two groups of objectives: (1) Focus on communication and (2) Focus on discourse features, which are featured in the first half (Part I and II) and the second half (Part III) of the video respectively. The first part of the video is expected to provide learners with insights into effective communication at a macro-level and deals with such aspects of language as communication strategies and communication styles. For example, these issues are explored in Part I and II: "What is effective communication? How does the way we communicate affect workplace culture and relationships? What strategies do people use to get others to do things at work or to avoid miscommunication? How do different communication styles and processes affect the way a [factory] team work?" (STUBBE AND BROWN 2002: 3)

The second part of the video (Part III) focuses on "language and communication at the micro-level of discourse and pragmatics" (ibid.). The corresponding section in the handbook includes notes on such aspects of language as discourse processes (e.g. turn/floor taking, topic management, the joint negotiation of meaning, the joint construction of humor), pragmatic/discourse features (e.g. fillers, feedback, hedges, discourse markers), politeness strategies (e.g. indirect language, implicatures, getting people to do things), clarification and repair strategies, as well as non-verbal features and features of spoken language such as colloquial vocabulary, repetition and incomplete sentences.

Apart from non-verbal and colloquial features, we can categorize most of the study objectives in TTW as interactive linguistic behavior. Communication strategies, for example, involve more than one participant by defini-

tion, and so do such linguistic behavior as turn/floor taking, feedback, the joint negotiation of meaning, and the joint construction of humor.

While conventional textbooks tend to focus more on the production of linguistic forms on the speaker's side, TTW emphasizes the interactive nature of conversation by directing learners' attention to such linguistic behavior. To take one example, in the notes on discourse features of a recorded conversation, TTW handbook makes a reference to the minimal feedback given by one of the participants and provides the following explanation.

> Gordon has a calm, unobtrusive manner, and makes extensive use here of minimal feedback (*eg. yep, right*) both of which function to encourage Michael to keep talking. (STUBBE AND BROWN 2002: 31)

Suggestions of this kind on how to listen actively and effectively are not commonly found in the majority of conventional conversation textbooks/materials, in which "listening" tends to be regarded as passive retrieval of information.

Though it may be also possible for conventional textbooks to focus on the interactive aspect of conversation, this will require the designing of non-authentic dialogs that effectively represent the characteristic features of authentic conversations including the interactive elements. This will in turn require not only great care and effort on the part of textbook/materials developers, but also detailed and extensive research on natural interaction, especially the kind of research which takes materials development into consideration.

To conclude, by analyzing and incorporating authentic conversations, TTW offers an interactive view of conversation and also study objectives based on such a perspective, which conventional textbooks have largely failed to include.

3. Analysis of the authentic data in TTW

In this section, we will analyze the authentic data in TTW and seek implications for the development of conversation teaching materials.

3.1 Purpose

As we saw in the previous section, TTW offers a kind of objectives which are difficult to include in conventional textbooks/materials and therefore it is likely to be a valuable tool for learners who wish to improve their conversation skills in English. According to the handbook, however, TTW is intended to be used with "people who already speak English well" (STUBBE AND BROWN 2002: 2), or intermediate to advanced level ESL/EFL students.

In fact, most of the study objectives and items focused on in TTW are either those at the global level of communication or those related to meta-communication; TTW does not elaborate on more basic skills that are essential to lower-level learners, such as how to perform basic speech/discourse acts (e.g. "giving a direction", "stating an opinion" etc.). We need to examine how the analysis of authentic data can be beneficial to the development of conversation textbooks/materials in general including those for lower-level learners.[2]

As a sample of ESL/EFL conversation teaching materials targeted at lower-level learners, we will take up the English Dialog Module in the TUFS Language Modules, which is currently under development at Tokyo University of Foreign Studies as part of the 21st Century COE Project on Usage-Based Linguistic Informatics[3]. The TUFS English Dialog Module ('the D-Module' hereafter) is web-based learning materials with an emphasis on conversation, targeted at young-age, elementary-level learners. It is based on a notional functional syllabus, which is a type of syllabus widely used in conversation textbooks/materials including more recent versions adapted to incorporate more "communicative" elements. In the D-Module, a typical unit includes the target function (e.g. "Asking about time"), a non-authentic dialog, and comprehension exercises. The dialog and exercises feature the linguistic forms learners need to master in order to carry out the target function.

Using the notional functional syllabus of the D-Module as a point of reference, we will now analyze the authentic conversations featured in TTW.[4] Specifically, we will investigate how some of the functions featured in the D-Module are realized in the authentic data in TTW and seek implications for the development of conversation textbooks/materials. Although we occasionally refer to the D-Module for comparison, our intention is not to evaluate it as teaching material, but rather to find how the analysis of authentic data can contribute to the development of teaching materials in general.

3.2 Data

Our data consists of 21 conversations included in TTW video clips, total-

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[2] In the following sections, we will limit our discussions to the analysis of authentic data, rather than the actual incorporation of it in teaching materials, although the latter will likely be a logical step if the former proves to be feasible.

[3] Our discussion of the English D-Module is based on a trial version we obtained from the D-Module development team (led by Dr. Asako Yoshitomi) in June 2003. We would like to express our sincere gratitude to the development team for generously providing us with the material and giving us permission to use it for this study.

[4] In this and the following sections, the recorded conversations in TTW will be treated purely as conversational data for our analysis, rather than as part of teaching material they are meant to be.

ing 11 minutes and 25 seconds of talk. The number of participants and the topics in each conversation were not controlled. The participants are all members of factory teams in New Zealand and they include team members (factory workers) and team managers. Most but not all of the topics of the conversations are directly related to their work.[5] In the TTW video, some of the conversations are repeated in more than one section, but we only used one of the segments for analysis.

Although the TTW handbook contains scripts of all the video clips, we re-transcribed the data using the Basic Transcription System for English ('BTSE' hereafter) (USAMI 2003b). BTSE is a transcription system still in its trial stage, but has the following advantages for our analytical needs:

1. BTSE is based on "discourse sentences" rather than other units such as phrasal or intonation units. Although prosodic and pragmatic factors are also considered in the segmentation process, a "discourse sentence" is primarily a syntactically defined "sentence".[6] This facilitates comparisons between data in the BTSE format and other sentence-based data, such as dialogs in textbooks.
2. BTSE makes use of spreadsheet software (e.g. Microsoft Excel) and therefore it is suitable for quantitative analysis as well as qualitative analysis.
3. BTSE is an adapted version of the Basic Transcription System for Japanese (BTSJ) (USAMI 1997, 2002, and 2003a). Having the TTW English data in the BTSE format will enable cross-linguistic studies in the future.

Using BTSE, we re-transcribed the 21 conversations in TTW, referring to the transcripts in the handbook for unfamiliar names, unclear contexts, etc.

In transcribing authentic conversations using BTSE, one issue that needs particular attention is how to secure reliability of transcription, especially that of the segmentation of aural data into discourse sentences. We recognize that the segmentation of spoken language into sentences is a more compli-

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[5] TTW also includes interviews with team managers but we excluded them from our data since the purpose of our study is to analyze "natural interaction" and consider its implications for materials development; interviews seem to be a rather unusual form of interaction in most people's daily lives.

[6] A "discourse sentence" is defined as follows: In actual conversation, backchannels indicating attention, agreement and so forth, as well as incomplete sentences, occur frequently. Also, there are cases in which words, though grammatically only a single word, fulfill a substantive function within the conversation. Our definition of "discourse sentences" includes "single word sentences" and incomplete sentences, as well as structurally complete sentences uttered by the same speaker, even when backchannels are used, or speakers briefly alternate. However, expressions such as "let's see", which are used as fillers, are not counted as independent discourse sentences unless they are uttered in isolation, even though they are structurally complete sentences.

cated task compared with that of written language. For this reason, we checked the intercoder reliability of sentence segmentation using a portion of the data, and obtained a kappa of 0.890.[7]

3.3 Methods

This section describes the methods and procedures we used to select the functions for analysis, to identify the realizations of them in the data, and to code the ways in which they are realized.

3.3.1 Selection of functions for analysis

Our purpose here is to examine how the functions featured in the D-Module are realized in the authentic conversations in TTW. In order to include a qualitative aspect in our study, we limited our analysis to the following seven functions which appeared frequently in the TTW data, out of the 40 featured in the D-Module.[8]
<ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>, <STATING AN OPINION>, <MAKING A COMPARISON>, <GIVING A REASON>, <GIVING A DIRECTION>, <GIVING AN EXAMPLE>, and <GIVING ADVICE>.

3.3.2 Coding of functions and its reliability

The identification of functions in authentic data can become a highly subjective process if the criteria for identification are not articulated. For this reason, we first operationalized the seven functions with clear definitions and examples. Below are our definitions of the functions. (The definitions refer to "a discourse sentence in/with which the function is realized".)

<ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>

A discourse sentence in which the speaker asks about attributes of a person or an object. An attribute is defined as a quality which can be found usually, normally, or for a long period of time, and it does not include a temporary state or such appearance.

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[7] Two coders, one of whom is a native speaker of English, independently identified discourse sentences in a portion of the data, and then compared the results using Cohen's kappa as an index. When Cohen's kappa is used to evaluate intercoder reliability, a value of over 0.850 is generally considered satisfactory when the coding is of a mechanical nature, which is the case with sentence segmentation. (See BAKEMAN AND GOTTMAN 1986 and NISHIGORI 2002 for discussion.)

[8] These functions appeared in more than three discourse sentences out of the total of 291 (1.0%) in a sample data set.

<STATING AN OPINION>

A discourse sentence in which the speaker makes an assertion, judgment, projection, or evaluation. Statements which do not involve the speaker's judgment at all, such as a plain fact (i.e. the speaker believes that the truthfulness of the statement is obvious to the hearer), or a report of hearsay, and a pure expression of feelings/emotions are not included in this category. Directions from a person in a higher position are not included, but advice and suggestions are.

<MAKING A COMPARISON>

A discourse sentence in which the speaker discusses the differences/similarities or merits/demerits of two or more objects, persons, or situations etc.

<GIVING A REASON>

A discourse sentence in which the speaker states the cause of an event, emotion or situation, the motive for an action, or the basis for a decision or belief.

<GIVING A DIRECTION>

A discourse sentence in which a person in a higher position tells a person in a lower position to do something and has the expectation that this will be done. If the speaker does not have this expectation (i.e. a rejection from the hearer will not be considered non-normative), the discourse sentence will be categorized as <GIVING ADVICE>, <MAKING A SUGGESTION>, <MAKING A REQUEST>, etc., and will not be coded as having this function.

<GIVING AN EXAMPLE>

A discourse sentence in which the speaker talks about an item/items or a person/persons which belong(s) to a group or a type. The item/person is typical or representative of the group or type which the speaker is making an assertion or a judgment about, or describing.

<GIVING ADVICE>

A discourse sentence in which the speaker gives the hearer information that (the speaker believes) the hearer does not have, or recommends doing something, believing that such information or action will be for the hearer's benefit. In case a person in a higher position is forcing a person in a lower position to take a certain action, it will be coded as <GIVING A DIRECTION> rather than <GIVING ADVICE>.

After giving the functions these definitions and examples[9], two coders independently identified the discourse sentences with one or more of the

[9] Examples are not provided here for lack of space.

seven functions in the TTW data and compared the results.[10] Using 41.3% of the data as a sample, we measured the intercoder reliability and obtained a Cohen's kappa of 0.761.[11]

3.3.3 Coding of form-function relationships

After we extracted all the discourse sentences in the data in which one or more of the seven functions we selected are realized, we examined how the functions are realized there. Focusing on form-function relationships, we coded these discourse sentences as either of the following.

Type-1: One of the seven functions is realized in the discourse sentence and is accompanied by a corresponding linguistic form.
Type-2: One of the seven functions is realized in the discourse sentence but is not accompanied by any of the corresponding linguistic forms for that function.

We defined "a corresponding linguistic form" as "a linguistic form featured in the D-Module to represent the function"[12] or "a linguistic form which is considered to represent the function from its literal meaning or conventional usage". After defining corresponding linguistic forms as above, we also coded the discourse sentences which fit the following description.

Type 3: One of the corresponding linguistic forms is present in the discourse sentence but not the function itself.

3.4 Results

We show the results related to Type-1 & Type-2 first (3.4.1 & 3.4.2), followed by those related to Type-3 (3.4.3).

3.4.1 Overall distribution of Type-1 & Type-2 discourse sentences

Figure-1 shows the overall distribution of Type-1 (a discourse sentence with one of the seven functions realized with a corresponding linguistic

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[10] When more than one function is present in a discourse sentence, the sentence was coded separately for each function. For this reason, the total number of sentences with the seven functions is calculated to be larger than that of the actual number of such sentences.

[11] This is above the standard (k = 0.7) generally considered satisfactory for this kind of coding, which inevitably involves subjective judgment by the coders (NISHIGORI 2002).

[12] The D-Module features "adjectives" as one of the forms to be used to "state an opinion". However, we excluded "adjectives" from our list of "corresponding linguistic forms" since they are such a generic category that the link between the form and the function seems to be much weaker compared with the other forms.

form) and Type-2 (a discourse sentence with one of the seven functions realized without a corresponding linguistic form) among all the discourse sentences in which one of the seven functions is realized.

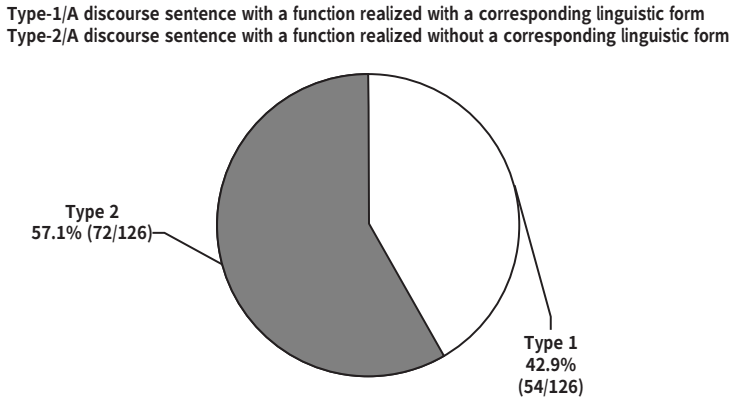

Figure 1: Distribution of Type 1 & 2 among the discourse sentences with the 7 functions

As we can see from Figure-1, more than half (57.1%) of the discourse sentences in which one of the seven functions is realized are not accompanied by any of the linguistic forms corresponding to that function.

3.4.2 Realizations of the seven functions

In this sub-section, we show how the seven functions are realized in the TTW data.

Table 1- How the seven functions are realized

The numbers after each linguistic form indicate the number of times each form appears in the TTW data. The linguistic forms in bold are those featured most prominently in the D-Module unit for that function.[13] (e.g. In the D-Module's unit for <ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>, interrogative sentences with "BE", such as "Is this/that -?", are the linguistic forms featured most prominently.

[13] Which linguistic form is featured the most prominently in the D-Module was judged based on our observation.

<ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>

TYPE-1	**Interrogative Sentences with "*BE*"**	***Is this/that -?***	4	5 (45%)
	Interjections	*eh*	1	
TYPE-2	No corresponding linguistic form		6	6 (54%)
Total			11 (100%)	

<STATING AN OPINION>

TYPE-1	Modals	*can/could, have (got) to, shall/ should, will/would, be going to*	13	20 (28%)
	think/hope etc.	***think, hope, bet***	4	
	(Dis)Agreement Backchannels	*yes, yeah, why not?*	3	
TYPE-2	No corresponding linguistic form		52 (72%)	
Total			72 (100%)	

<MAKING A COMPARISON>

TYPE-1	**Comparatives**	***better, lower***	2	2 (67%)
TYPE-2	No corresponding linguistic form		1	1 (33%)
Total			3 (100%)	

<GIVING A REASON>

TYPE-1	**Conjunctions**	***because, cos, so that***	9	11 (69%)
	Prepositions/Prep. Phrase	*thanks to,*	1	
	Infinitives	*to soften up*	1	
TYPE-2	No corresponding linguistic form		5 (31%)	
Total			16 (100%)	

< GIVING A DIRECTION>

TYPE-1	**Imperatives**	***Keep going, Make sure***	10	12 (71%)
	Modals	*have (got) to, be supposed to*	2	
TYPE-2	No corresponding linguistic form		5 (29%)	
Total			17 (100%)	

< GIVING AN EXAMPLE>

TYPE-1	**Prepositions**	***like -***	2	2 (67%)
TYPE-2	No corresponding linguistic form		1	1 (33%)
Total			3 (100%)	

< GIVING ADVICE>

TYPE-1	Imperatives	*Smile, Ask*	2	2 (50%)
TYPE-2	No corresponding linguistic form		2	2 (50%)
Total			4(100%)	

For <GIVING ADVICE>, the D-Module features "*You should/had better*", which does not appear in the TTW data.

3.4.3 Ratio of Type-3 among all discourse sentences

Figure-2 shows the ratio of Type-3 discourse sentences among the total number of discourse sentences.

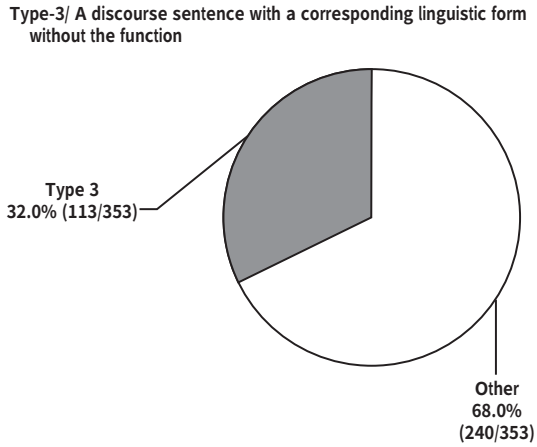

Figure 2: Ratio of Type 3 among all discourse sentences

As we can see from Figure-2, in about one third (113/353 = 32.0%) of the discourse sentences in our data, one or more of the corresponding linguistic forms are present, but not the function which the form corresponds to.

4. Analysis

In this section, we analyze each type of discourse sentences, focusing particularly on Types-2 & 3.

4.1 Functions realized with corresponding linguistic forms (Type-1)

As we saw in Figure-1, among the 126 examples where the seven functions are realized in our TTW data, 54 (42.9%) are accompanied by corresponding linguistic forms. We can see in Table-1 that most of the linguistic forms featured most prominently in the D-Module are also used in the TTW data. Although this is not clear from Table-1, on the whole, wider ranges of linguistic forms and their variants are used for each function in the TTW data than in the D-Module.

4.2 Functions realized without corresponding linguistic forms (Type-2)

As we saw in Figure-1, among the 126 cases where one of the seven functions is realized in TTW, 72 (57.1%) are not accompanied by any corresponding linguistic forms. As it is this kind of example that conventional textbooks/materials have failed to give much attention to, we would like to

take a close look at some of these cases and examine how the functions are realized without the existence of any corresponding forms.

4.2.1 <ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>

As Table-1 shows, out of the 11 examples of this function, six (54%) are realized without corresponding linguistic forms. Here we show two examples. (See Appendix for key to transcription symbols.)

<Example 1> W: Team member, G: Team manager
(W points to the paper G is holding. 'Congo' is a color name.)

1	*****	**W**	**Some of it is congo?. [↑] <ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>**
2	*	G	Congo.
3	*	W	Congo.

<Example 2> L: Team manager, C: Team member
(L has just come to C's station where she is sewing shoe parts, and starts talking to her. 'Jayne' is a style name of shoes.)

1	*	L	Yeah Jayne, you got any urgent ones here?. [↑]
2	*	C	Nah.
3	*	L	All finished?. [↑]
4	*****	**L**	**That a trial line?. [↑] <ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>**
5	*	C	Yeah a trial line.
6	*	L	Oh yeah okay well that's urgent anyhow.

Neither line 1 in example 1, nor line 4 in example 2, is equipped with grammatical or lexical clues to indicate that they are questions asking for "information about attributes". However, we can see this function is realized in those lines from the way the other parties respond in the following lines. (Line 2 and line 5, respectively.)

As Table-1 shows, there are six examples of this function realized without corresponding linguistic forms (Type-2). Not only are these examples all produced with a rising intonation, but they are all produced in a situation where the participants can actually see the object being discussed. This transparency of context is apparently contributing to the syntactic simplicity and terseness of these discourse sentences, enabling them to be produced and interpreted as "asking for information about attributes" even without formal clues.

Among the six Type-2 examples, there are also three that are produced directly after another question or after a question-answer sequence. In these cases, the participants can be considered to be already attuned to the context in which one is asking the other a question. This can also account for the absence of corresponding linguistic forms in these discourse sentences.

Thus, we can say that it is thanks to the interplay of prosodic features, physical context, and discourse context, that this function is realized in the six Type-2 examples of <ASKING FOR INFORMATION (ABOUT ATTRIBUTES)>.

4.2.2 <GIVING A REASON>

As Table-1 shows, out of the 16 examples of this function, five (31.3%) are realized without corresponding linguistic forms. Here we show two of such examples.

<Example 3> L: Team manager, S: Team member
(L notices S is wearing gloves while he is at work.)

1	*	L	And what's with the gloves?.
2	*****	**S**	**Don't want to get my hands dirty. [Smiling] <GIVING A REASON>**
3	*	L	Don't want to ruin your manicure. [Smiling] <GIVING A REASON>

<Example 4> R: Team member, J: Team manager
(J comes over to talk to R. R asks him if she can take a smoke break.)

1	*	R	<Can I>{>} , can I go for a smoko?.
2	*	J	Uh... , have you been good?.
3	*	R	Oh yes.
4	*	J	Give me a reason why.
5	*****	**R**	**Um finished the run for the day. <GIVING A REASON>**
6	*	R	# finished # tomorrow. <GIVING A REASON>[14]
7	*	J	Yeah it's pretty hard.
8	*	R	<Why?>{<}.
9	*	J	<Yeah alright>{>}.

In line 1 in example 3, L asks S why he is wearing gloves. To this, S replies in line 2 "Don't want to get my hands dirty", providing a reason with-

[14] This discourse sentence is coded as <GIVING A REASON> based on possible reconstructions of the discourse sentence, such as "Almost finished for tomorrow".

out any corresponding linguistic forms such as "because". In line 4 in example 4, J asks R (jokingly) why she thinks she deserves a smoke break now. To this, R gives a reason in line 5 "Um finished the run for the day", again without any corresponding linguistic forms. What these two discourse sentences (line 2 in example 3 and line 5 in example 4) have in common is the context in which one participant is explicitly demanding a reason as an answer. Since this context will enable almost any kind of reply by the other participant to be produced and interpreted as "a reason", the discourse sentences are freed from the necessity of explicit corresponding forms.

It is interesting to note that in both examples, additional reasons are given again without corresponding linguistic forms in the following lines (line 3 in example 3 and line 6 in example 4). As we saw with questions produced directly after another question-answer sequence (4.2.1), the participants here are already attuned to the existing context where reasons are being provided. In these discourse sentences too, discourse context is an essential factor for the functions to be realized.

4.2.3 <STATING AN OPINION>

As Table-1 shows, out of the 72 examples of this function, 52 (72.2%) are realized without corresponding linguistic forms. Here is one example.

<Example 5> L: Team manager, A: Team member
(L is leading a team meeting.)

1 * L So um that's about that's about it from me.
2 * L Does anybody uh want to bring anything up ###?.
(OMISSION THREE LINES)
6 * L Come on <anybody>{<}.
7 * A <Just let>{>} them know that we've got two styles that running out. <STATING AN OPINION>
8 * A They are all urgent. <STATING AN OPINION>
9 * A <And they running out>{<}.
10 * L <Oh yeah that's right>{>} yeah.

In example 5, the imperative form, which we designated as a corresponding linguistic form for <GIVING A DIRECTION> and <GIVING ADVICE>, is used in line 7. However, since speaker A works under L, and the content of the discourse sentence is directly related to their work, which is under L's authority, the function of A's discourse sentence cannot be <GIVING A DIRECTION>. Neither can it be "advice" since the discourse sentence is not produced for L's benefit. Since A is making an assertion based on her judgment, the function of this discourse sentence is coded as <STAT-

ING AN OPINION> according to our definitions.

Notice this discourse sentence does not have any formal clues such as "I think" or "-should" to indicate that it is "an opinion". It is, however, produced after an explicit demand from the meeting leader asking for contribution from the members. Although she is not specifically demanding an opinion rather than other kinds of comments, contributions from non-leading members at a meeting are usually limited to a fairly restricted range of comments including opinions. Therefore we can say that the situation in lines 6 to 7 is similar, in terms of discourse context, to the situation where one participant is explicitly demanding a reason from the other and where any response is likely to be treated as one (See 4.2.2).

Although what follows is a Type-1 example rather than Type-2, let us discuss it here as it concerns the absence of corresponding linguistic forms for this function.

<Example 6> L: Team manager, B: Team member
(L is leading a meeting.)

1 * L = But there's been some health and safety people monitoring the dust levels and the fumes with the um solvent and glue, you know.
2 * B Did they also ### do the noise?.
3 * B She asked, ###### uh... she, she asked about um that it's a bit noisy.
4 * L Oh I think they were doing the noise.= <STATING AN OPINION>
5 * L =I put a I put a uh question mark beside noise because I wasn't absolutely certain.

Many learners of English will probably associate the function <STATING AN OPINION> with the form "I think", which is arguably the linguistic form most frequently taught in textbooks and classrooms for this function. It is therefore interesting to note that there are only four cases of this function realized with "I think", "I hope" etc. in our TTW data (4/72 examples = 5.6%), with the exact form "I think" appearing only once.[15]

Moreover, in the single case in which "I think" is used in its literal form, this linguistic form appears to be used to add a different note to the quality of the opinion being stated, specifically "uncertainty". In example 6, when L talks about "health and safety people" checking the dust and fumes in the

[15] The other three examples are "I thought", "I hope" and "I bet".

factory, one of the workers, B, asks if they were also checking the noise level. L answers to this in line 4 with "I think", but then adds in line 5 that she wasn't "absolutely certain". In this example, L is using the linguistic form "I think" but is not stating her opinion with total confidence. On the contrary, the linguistic form "I think" appears to be used to show a certain degree of uncertainty or lack of confidence in the speaker's judgment, rather than to simply state an opinion.

In sum, in our TTW data, "opinions" are stated more frequently without a corresponding linguistic form. The linguistic form "I think" is rarely used to realize this function in our data, and when it is, it appears to be used to add an element of uncertainty to the opinion being presented.

4.3 Corresponding linguistic forms not representing the seven functions (Type3)

As we saw in Figure-2 in 3.4.3, 113 out of the 353 discourse sentences (32.0%) in our TTW data include one or more of the corresponding linguistic forms for the seven functions but not the functions themselves. This is not surprising considering many of these linguistic forms can also be used with functions other than the ones to which we assigned them. Although what function can be realized using a particular linguistic form may be dependent on various factors, let us show one case where an important factor is the global context in which the discourse sentence is situated.

We designated "yes" and other backchannels as linguistic forms corresponding to the function <STATING AN OPINION>; when used directly after another participant's opinion or judgment, backchannels function this way.[16] In the following example, however, the backchannel in line 2 is used with a different function.

<Example 7> L: Team manager, D: Team member
(L is telling D a story about her experience over the weekend.)

1	*	L	###, we we went out for dinner on Friday night [↑].
2	*	D	Yeah. ~~<STATING AN OPINION>~~
3	*	L	With Barry [↑].
4	*	L	And he was pretending to be the king of Tonga [↑]. (< laugh >)

In this example, L is starting to tell a story about her personal experience,

[16] We categorized this function as <STATING AN OPINION> because "Agreeing/Disagreeing" are not on the 40-function list of the D-Module.

which is clear from the way she lists the typical components of a narrative opening (the place, time, people involved, etc.) and how she uses a rising intonation in line 1. Since the discourse sentence in line 1 includes virtually no element of "opinion" or "judgment" of the speaker, D's discourse sentence in line 2 cannot be interpreted as having the function <STATING AN OPINION>. If we look at the way L continues her talk in lines 3-4, it is clear that "yeah" in line 2 functions as a "continuer" (SCHEGLOFF 1982), which signals L to keep the floor, rather than as a sign of agreement.

The actual function that can be realized with a backchannel may be dependent on several factors, including intonation, its location relative to the other participant's utterance, etc. An important factor in this example, however, seems to be its global context, which is storytelling performed by the other participant. Since storytellers tend to provide factual information pertaining to the story at the beginning of a narrative (LABOV 1972), rather than state their opinions, we can say that backchannels used in this context more often function as a continuer than as a sign of agreement.

In this way, corresponding linguistic forms may be used with different functions depending on the global contexts in which they are situated.

5. Conclusions: Implications for the development of conversation teaching materials

What we found through our analyses can be summarized as follows. We hope these findings can, and will, be applied to the development of conversation teaching materials in the future.

5.1 Choice of linguistic forms or patterns to be featured in conversation teaching materials

We saw in Table-1 that "I think", a linguistic form commonly taught in conversation textbooks/materials for the function <STATING AN OPINION>, is rarely used for this function in our authentic data. The use of "I think" can even suggest uncertainty of the speaker as we saw in example 6. Although we should not make sweeping generalizations based solely on our database, we can at least claim that the choice of linguistic forms to be presented in conversation teaching materials should be based on research on authentic conversations so that it will reflect what kind of linguistic forms are, or are not, used frequently to carry out the target functions in natural interactions.

5.2 The importance of contextual information

As we saw in 4.2.1 and 4.2.2, some functions are realized in different ways depending on their discourse context. For example, although many

conversation textbooks/materials present "Why-?" and "Because- " as a paired sequence, "because" is not always used to give a reason in our data, especially in a context where a reason is explicitly demanded (4.2.2). We also saw that backchannels used in the context of storytelling are more likely to function as a continuer than as a sign of agreement (4.3). This kind of fairly simple contextual knowledge can, and should, be introduced to learners from an early stage; it will help learners carry out the functions in more authentic ways and understand in what kind of context a certain linguistic form is used for a particular function.

5.3 Form-function mappings

In many conversation textbooks/materials, especially those based on a notional functional syllabus, the focus is on the mappings of forms to functions, i.e. what linguistic form(s) learners should use in order to carry out a particular function. In our data, however, we saw that functions can often be realized without a corresponding linguistic form (4.2), or with a linguistic form often used for another function (4.2.3). These results suggest that the mappings of functions to linguistic forms have to be presented with care in conversation textbooks/materials. In 4.2.3 for example, we saw how the imperative form can be used to <STATE AN OPINION>, in a certain context. Those learners who think of imperatives as a linguistic form used exclusively to "give directions/advice" may have difficulty learning how to state opinions, or how to use the imperative form for different functions. Thus, placing too much emphasis on form-function mappings in conversation textbooks/materials could hinder the learning process and must be avoided. Learners should be exposed to various ways in which functions can be realized flexibly with different linguistic forms, or without any linguistic forms, in authentic conversations.

By first analyzing TTW as teaching material, and then analyzing the authentic conversations in TTW as data, we hope to have demonstrated that the analysis of natural interactions can contribute to the development of conversation teaching materials for learners of various levels. Since functions are often realized through context without corresponding linguistic forms in authentic conversations, we believe that learners should be exposed to authentic data displaying such examples from an early stage. A possible extension of this study would be to expand the range of functions to be analyzed and to include a kind of authentic data whose context is closer to that of the dialogs in the conversation textbook/material under development. This will allow us to apply the findings of the analysis of authentic conversations more directly to materials development.

Appendix / Key to Transcription Symbols

Among the symbols used in BTSE, only those relevant for this paper are listed here.

. (period)	The end of a discourse sentence. A period is added also after a question mark.
*	An asterisk shows that a discourse sentence ends in that line.
,	Commas are used where they are conventionally placed to facilitate reading.
...	Hesitant tone.
?	A question mark is used at the end of a question. This mark is used if the discourse sentence is judged to function as a question from its intonation etc., even if it does not have the syntactic features of a question.
[↑]	Rising intonation.
< >{<}	Section of speech which is overlapped by another speaker's speech.
< >{>}	Section of speech which overlaps another speaker's speech.
( )	A short backchannel without a particular meaning is placed in brackets with the other speaker's discourse sentence.
<laugh>	Laugh.
(<laugh>)	Laugh overlapping another speaker's speech. (Placed with the other speaker's discourse sentence.)
[ ]	Paralinguistic or non-verbal features.
###	Untranscribable or incomprehensible speech. The number of # indicates the relative length of that section of speech.
=	No or shorter-than-average pause between discourse sentences.

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The Creation of the TUFS Pronunciation Module

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1. Introduction

TUFS Language Modules are being developed at the Graduate School of Tokyo University of Foreign Studies ("TUFS") as part of one of its two 21st Century Center of Excellence programs granted by the Ministry of Education, Culture, Sports, Science and Technology of Japan, with an aim to create multilingual e-learning materials covering 17 different languages. This large-scale multilingual e-learning system being developed by TUFS, which covers not only European languages but also Asian languages, is probably the first system of this kind to appear anywhere in the world. The TUFS Language Module system consists of pronunciation, dialogue, grammar and vocabulary modules. As the forerunner of the system, the first version of its pronunciation module was completed and put on the website in April 2003, in 11 out of the 17 planned languages: German, French, Spanish, Portuguese, Russian, Chinese, Korean, Mongolian, Filipino, Vietnamese and Japanese.

The Center of Usage-Based Linguistic Informatics was proposed by TUFS and selected as one of the 21st Century Center of Excellence Programs promoted by Japan's Education Ministry. The objective of the proposal is to integrate linguistics and language education by the utilization of informatics which has been dramatically developed in recent years. The new academic domain thus to be created is called Linguistic Informatics. The development of TUFS Language Modules is part of this proposal.

This paper focuses on the creation of the multilingual e-learning pronunciation materials with a particular emphasis on the process by which we formulated its design concept, and the basic structure common to all the proposed languages.

In the following sections we will discuss the existing e-learning pronunciation materials (Section 2), the design of the TUFS Pronunciation Module (Section 3), and the content of the Spanish Pronunciation Module (Section 4).

2. The existing e-learning pronunciation materials

Before embarking on the planning of the TUFS Pronunciation Module, starting with the example of French, we conducted a survey centering on domestic and overseas academic institutions which offer French learning materials on the websites. We were, however, unable to find any website containing independent pronunciation material made available to the public free of charge.

Given this situation, the following four materials were chosen for the purpose of examining different features of the existing CALL and CD-ROM materials that include a pronunciation section.

1) "CALL French Grammar" (Center for Information and Multimedia Studies, Faculty of Integrated Human Studies, Kyoto University, Japan).
   WWW page: http://sage.media.Kyoto-u.ac.jp/call/soujin/Grammaire/Grammaire.html
   A comprehensive material covering pronunciation, vocabulary, grammar and expressions, used in French CALL classes of the Faculty of Integrated Human Studies, Kyoto University.
2) Website of the Laboratory of Phonetics and Phonology, Laval University in Quebec, Canada.
   WWW page: http://www.lli.ulaval.ca/labo2256/
   A website for the study of French phonetics and phonology.
3) "Sound Reproduction - Pronunciation Exercises and Methodological Studies," Summer Seminar Program of the Institute for American Universities (Aix-en-Provence, France).
   WWW page: http://courseweb.edteched.uottawa.ca/Phonetique/Aix2000/phonetique.html
   A summer seminar program for American high school teachers of French.
4) "Learn French Now!" (Transparent Language, USA).
   A CD-ROM material for teaching pronunciation, vocabulary, grammar and expressions.

LANCIEN (1998:24-32) refers to the essential characteristics of multimedia as follows:

1. Multichannels: Various communication channels coexisting on the same base, with combined images, sounds and texts.
2. Multi-referentiality: A system closely related to hypertext and multichannels, enabling diversification and multiplication of information sources on a given topic. It diversifies the base and at the same time

expands referential fields associated with the subject.

3. Interactivity: Capability of responses to utterances, rather than one-way messages.

The comparison of the four materials in terms of utilization of multimedia is as follows:

	Kyoto Univ.	Laval Univ.	I.A.U.	T.L.
Channels	Letters Sounds			
	Animation	Photos	Sound waves, pitch curves, stress curves	
Reference	Words-sounds	Segmental sounds*, words, phrases, sentences-sounds		
Interactivity	Listening/looking/reading			Listening/looking/ reading/speaking (including recording)/writing

*Sound links are only found in the Laval University material. (NAKATA 2004)

Figure 1: Comparison of utilization of multimedia

- Channels:
All the four materials provide letters and sounds. In addition, Kyoto University uses animated illustrations and video films. Laval University uses still photos, while I.A.U. and T.L utilize sound waves, pitch and stress curves.

- Reference:
With the exception of Kyoto University, sound links are provided not only at word level but also at phrase and sentence levels.

- Interactivity:
T.L. alone utilizes the function of dictation (writing) of sounds using the keyboard and recording of learners' pronunciation.

The comparison of the four materials in terms of learning process is as follows:

	Kyoto Univ.	Laval Univ.	I.A.U.	T.L.
Input	Sounds and articulation	Sounds and articulation points	Sounds only	
Discrimination	Contrastive explanation of similar sounds	Listening exercises	Contrastive explanation French-English	N.A.

Sound-Letter association	Explanation only	Explanation only	N.A.	Dictation (words and sentences)
Sound production and its evaluation	N.A.	N.A.	N.A.	Evaluation of pitch, stress and fricatives through recording

(NAKATA 2004)

Figure 2: Comparison of learning process

- Input of the sounds of the target language:
Animated illustrations showing the movement of the tongue in diagrams of the organs of speech and video films showing the mouth when articulating vowels as provided by Kyoto University are useful for learners to understand the ways of articulating each consonant and vowel. X-ray photographs of the oral cavity used by Laval University help users to check out articulation points, but may not be so effective for learning pronunciation. Sound waves, pitch and stress curves offered by I.A.U. and T.L. visualize fricatives, intonation and stress, which can be compared by learners with recording of their own voices. While no explanation is given as to how to correct pronunciation when deviating from the model, such visualization is useful for the acquisition of prosody.

- Discrimination of phonemes:
Kyoto University and I.A.U. provide contrastive descriptions of the sounds of French and learners' mother tongue, which is very effective.

- Association of sounds and letters:
T.L. provides dictation exercises. Kyoto University and Laval University give explanation only. I.A.U. does not handle this matter.

- Sound production and its evaluation:
This is the most advanced level of e-learning which gives learners feedback and evaluation. Of the four materials, only T.L. approaches the area of evaluation, giving the three-step evaluations of: "Keep practicing"/"Good job"/ "Wow," while the criteria for appraisal is unknown.

As far as e-learning pronunciation materials are concerned, the current situation is that there are only a few materials worldwide that offer something more than model pronunciations of individual sounds. Moreover, such materials are available only in a small number of languages.

3. The design of the TUFS Pronunciation Module

3.1 What the Pronunciation Module should look like

We find on bookstore shelves a great number of conversation books, grammar books and vocabulary books for the teaching/learning of second languages, but at least in the case of Japan, with the exception of English and Japanese languages, we do not see many guides to pronunciation as published separately from general primers and textbooks. In general, only the first few pages of such textbooks are dedicated to a general guide to pronunciation, barely touching upon sounds at the segmental level, and not beyond. This is so, in spite of the fact that the importance of supra-segmental features in teaching pronunciation ought to be widely accepted as WONG (1987:21) put it: "because their major roles in communication, rhythm and intonation merit greater priority in the teaching program than attention to individual sounds."

In the classroom, again except for English and Japanese, when teaching pronunciation of a second language, the norm seems to be for teachers to explain the way of articulation in an "orthodox" fashion, using diagrams of the organs of speech, focusing on the features of the segmental sounds, using patterns of minimal pairs and phonetic symbols. However, we see not a few learners who give up learning a second language in the middle of such boring intensive pronunciation practice.

TUFS Language Modules are developed primarily by postgraduate students under the guidance of university instructors. The staff involved in the planning of the Pronunciation Module proposed to tackle the task of creating a user-friendly material with which learners can really enjoy learning pronunciation, making the most of the advantages of e-learning materials. We found it most difficult to bridge the gap between linguistics, language education and informatics, and to build up interdisciplinary dialogues. More often than not, what was common knowledge in English and Japanese language education proved not necessarily to be so elsewhere. We consequently undertook detailed discussions as to the most suitable format for the Pronunciation Module. Our discussions were at all times premised on the understanding that "pronunciation" is "a key to gaining full communicative competence" (BROWN 2001:283). With that in mind we decided to produce a pronunciation material, centered on exercises, and covering not only segmental sounds but also prosody, an indispensable factor in communication. As a result of this process, we decided to make the Pronunciation Module twofold, with both "theory" and "practice" sections. It was proposed that what could be called an "outline of phonetics and phonology" be compiled as the "theory"

section which would serve as an academic reference, and that a learning material, separate from this and with the backing of theories, be produced as the "practice" section. The "theory" section was set aside for the time being, and we concentrated on developing the "practice" section that was eventually produced as the Pronunciation Module.

3.2 The design concept

The target users of TUFS Language Modules, with the exception of the Japanese modules, are basically Japanese-speaking students who start learning the target language as beginners. We will not discuss here the English module, another exception in a different sense, which is targeted at children (although adult learners can benefit from it as well), and for this particular reason designed and developed separately.

We recognized the need to keep in mind learners' mother tongues. This is based on the premise of the Contrastive Analysis Hypothesis which FRIES (1945) summarizes by the words "the most efficient materials are those that are based upon a scientific description of the language to be learned carefully compared with a parallel description of the native language of the learner."

The planning staff of the Pronunciation Module also took into account the fact that we were producing an e-learning material. We needed a paradigm shift from materials printed on paper. In this regard, WARSCHAUER and HEALEY (1998:59) offer the following benefits of including a computer component in language instruction:

1. multimodal practice with feedback
2. individualization in a large class
3. pair and small-group work on projects, either collaboratively or competitively
4. the fun factor
5. variety in the resources available and learning styles used
6. exploratory learning with large amounts of language data
7. real-life skill-building in computer use.

TUFS Language Modules are a modular system, consisting of four modules. The modules are separate but can be combined in different ways. We decided to apply the concept of the "stand-alone module" to the various levels of each part and unit, so that learners can choose and assemble modules in any way they wish even within the Pronunciation Module.

In the process of creating materials centered on exercises, we introduced

the concept of discovery learning, which, according to BROWN (2001:29), advocates less learning "by being told" and more learning by discovering for oneself various facts and principles. RICHARDS and ROGERS (1986:99) point out within the context of The Silent Way that "learning is facilitated if the learner discovers or creates rather than remembers and repeats what is to be learned." In attempting to make the most of the e-learning material, in contrast to the normal order of explanations being followed by exercises, we devised wherever considered appropriate, to turn the table around and start off with an exercise. This was done exactly for the purpose of discovery learning.

For an adult learner, a simple process of repeated listening and mimicking does not suffice, and proper instruction in articulation is required. Indeed, some people possess a phonetic coding ability that others do not. Even if some learners find it difficult to learn pronunciation, with some effort and concentration, they can improve their competence. We decided to explain the way of articulation without recourse to diagrams of the organs of speech or phonetic symbols and jargons. In choosing examples for the purpose of practicing pronunciation, we tried to avoid low frequency words, no matter how felicitous they may be as minimal pair examples. We also paid particular attention to providing authenticity, real-world simulation, and meaningful tasks as opposed to rote learning. We also intended to see to it that, as BROWN (2001:283) advocates, "instead of teaching only the role of articulation within words, or at best, phrases, we teach its role in a whole stream of discourse."

In connection with prosody we referred to the Verbo-Tonal Method, which places its theoretical base on the verbo-tonal system related to the acceptance and production of speech sounds as theorized in the 1950s by Petar Guberina, a linguist at Zagreb University, former Yugoslavia. This method is fairly known to Japanese phonetic teachers. Its method of allowing learners to acquire a sense of rhythm and intonation by using the rhythm of nursery rhymes gives a lot of insight. This method is detailed in KRAPEZ (1971).

BROWN (2001:268) points out that "fluency and accuracy are both important goals to pursue in Communicative Language Teaching." We were determined that in designing the Pronunciation Module, we should make proper balance between "the two clearly important speaker goals of accurate (clear, articulate, grammatically and phonologically correct) language and fluent (flowing, natural) language."

The actual content of the material was left to the discretion of the writers specializing in the particular language, because of the particularity of each individual language. The planning staff of the Pronunciation Module requested the writers of the material for each language to pay attention to the following ten points:

1) Be aware that the aim is to produce self-teaching pronunciation material to develop communicative competence.
2) Keep in mind that the target users are intended to be Japanese speakers (except for the Japanese material which is intended for non-Japanese speaking learners), but are not to be limited to university students, but also including other learners such as students of secondary education. Apply contrastive analysis of the sounds of Japanese and the target language.
3) Make it user-friendly. Do not take it for granted that learners have phonetic knowledge. Try to explain the way of articulation in plain words. Avoid technical terms. In principle, do not use phonetic symbols.
4) Make clear to users what can be achieved from learning with the system, part by part and unit by unit.
5) Make the most of the fact that the modules are e-learning materials. Remember that unlike materials printed on paper, with e-learning materials, speech sounds can be listened to and imitated repeatedly simply by the click of a mouse.
6) Apply the design concept of module to the various levels of each part and each unit as well. Disregard the principle of building on previously learned information to teach new information, so that users, regardless of their learning experience, can start from anywhere they wish to.
7) Make it exercise-oriented, for the purpose of enforcing understanding of the discrimination and production of sounds. Take into account the merits of discovery learning.
8) Cover not only segmental sounds but prosody as well, which is indispensable in communication.
9) Both fluency and accuracy should be pursued.
10) Teach not only pronunciation but spelling as well. The idea is to try to apply phonics[1], an established method of teaching English sounds and spelling without using phonetic symbols, to other languages as well.

3.3 Basic structure common to all the planned languages

Pronunciation is important in language. SAPIR (1933:155) claims that

[1] WILEY 2002 and other good guidebooks of phonics are available.

"phonetic language takes precedence over all other kinds of communicative symbolism, all of which are, by comparison, either substitutive, like writing, or excessively supplementary, like the gesture accompanying speech."

Our aim in creating the Pronunciation Module was to add more value to mere brief pronunciation guides that are only incidental to a course of study, and to establish such a material that can be used continuously during the whole span of learning the target language as learners' proficiency levels increase. Our goal at beginning levels would be, as BROWN (2001:284) puts it, "focused on clear, comprehensible pronunciation," and we wanted "learners to surpass that threshold beneath which pronunciation detracts from their ability to communicate." We also considered that "fluency may in many communicative language courses be an *initial* goal in language teaching" (BROWN 2001:268). At advanced levels, however, "pronunciation goals can focus on elements that enhance communication: intonation features that go beyond basic patterns, voice quality, phonetic distinctions between registers, and other refinements" (BROWN 2001:284).

It was agreed that the following basic structure was to be conformed to in the case of all the languages.

1) Learners will first familiarize themselves with the sounds of the language they are going to learn. For this purpose, a short text with prosodic features such as a poem is provided.
2) Part 1 is entitled "For Survival." The target of this section is for learners to be able to read words, phrases and sentences of the target language well enough to make themselves understood.
3) Part 2 is entitled "For Smooth Communication." The target of this section is to enable users to learn the trick of pronunciation in terms of improving their listening comprehension. Here acquisition of fluency is pursued by means of practicing prosody.
4) Part 3 is entitled "To Master the Pronunciation of a Native-Speaker." The target of this part is to take one step further towards learning completely accurate pronunciation and acquiring the feel of the target language. Here acquisition of accuracy is pursued. Despite the eye-catching, perhaps overly ambitious sounding title, our real intention lies in simply revisiting segmental sounds for the mastery of accuracy.

The Pronunciation Module was divided into three parts so that learners can freely choose from the three parts where to start and end, and what they want to learn, depending on their purpose of learning. If their aim is to manage to make themselves understood in the target language, then only Parts 1

and 2 may suffice; if they wish to acquire the real feel of the language, then Part 3 is very important to them. The reason why we avoided using the term "level" for "part" was to allow for the current holistic, rather than atomistic, approaches to pronunciation as BROWN (2001:283) describes: "Rather than attempting only to build a learner's articulatory competence from the bottom up, and simply as the mastery of a list of phonemes and allophones, a top-down approach is taken in which the most relevant features of pronunciation–stress, rhythm, and intonation–are given high priority." The Pronunciation Module was so designed that it would be in no way at a disadvantage if learners decide to skip Part 1 and start with the prosody featured Part 2. Part 1 is not necessarily a level 1.

When we look back over the past hundred years of language education, in the Direct Method, which gained popularity at the turn of the twentieth century, "correct pronunciation" was "emphasized" (RICHARDS and ROGERS 1986:10). In a comparison of the Audiolingual Method which flourished in the 1950s and the currently recognized Communicative Language Teaching, in the former, native-speaker-like pronunciation was sought, while in the latter, comprehensible pronunciation is sought (FINOCCIARO and BRUMFIT 1983). As the Critical Period Hypothesis claims, the brain lateralization is a slow process that begins around the age of two and is completed around puberty. The crucial age ranges from five to the early teens in different research. BROWN (2001:284) points out that "generally speaking, children under the age of puberty stand an excellent chance of 'sounding like a native' if they have continued exposure in authentic contexts. Beyond the age of puberty, while adults will almost surely maintain a 'foreign accent,' there is no particular advantage attributed to age. A fifty-year-old can be as successful as an eighteen-year-old if all other factors are equal." The ultimate goal of the Pronunciation Module is in no way to achieve totally accent-free speech that is not distinguishable from that of a native speaker, but to show learners "how clarity of speech is significant in shaping their self-image and, ultimately, in reaching some of their higher goals" (BROWN 2001:285).

4. The Content of the Spanish Pronunciation Module

In this section we will show what the content of the Pronunciation Module looks like, taking the Spanish module as an example.

4.1 The Sounds of Spanish

We chose a poem that is contained in *Rimas* by Gustavo Adolfo Bécquer, a well-known Spanish poet of the 19th century.

Hoy la tierra y los cielos me sonríen;	(Today the earth and the sky smile at me;
hoy llega al fondo de mi alma el sol;	today the sun reaches the bottom of my soul;
hoy la he visto...,	today I saw her...,
la he visto y me ha mirado.	I saw her and she looked at me.
¡Hoy creo en Dios!	Today I believe in God!)

This five-line poem was chosen particularly because it contains typical prosodic features of Spanish. As far as vibrants and laterals are concerned, [ɾ] appears twice, [r] twice and [l] as many as nine times.

Figure 3: Introduction: Sound of Spanish

4.2 Part 1: For Survival

An example of the items contained in this Part is as follows:

1.1 Vowel 'u'

Exercise:

Either one of each of the following pairs you will hear is Japanese or Spanish. Click whichever you think is Spanish.

1. A B

2. A B
3. A B
4. A B
5. A B

(Only sounds are provided. The user will click A or B. The correct answer and its related explanation appear on the next page.)
1. The correct answer is A. A is "luz," the Spanish word for "light." B is the Japanese word "rusu" (meaning "absent").
2. The correct answer is B. A is the Japanese word "puro" (meaning "pro"). B is "puro," the Spanish word for "pure."
3. The correct answer is B. A is the Japanese word "uba" (meaning "nanny"). B is "uva," the Spanish word for "grape."
4. The correct answer is B. A is the Japanese word "miren" (meaning "regret, attachment"). B is "miren," the Spanish word for "look" (subjunctive (imperative), 3rd person, plural).
5. The correct answer is A. A is "casa," the Spanish word for "house." B is the Japanese word "kasa" (meaning "umbrella").

Explanation:
Notice how the Japanese and Spanish vowels differ.
Japanese has five vowels: a, i, u, e and o. Spanish also has five vowels: a, e, i, o and u. 'A', 'e', 'i' and 'o' are almost the same in both Japanese and Spanish. Be careful with 'u'. Push your lips well forward when you pronounce the Spanish 'u,' further than when you pronounce the Japanese 'u.' Try to pronounce the Spanish 'u' in the depths of your throat.

Note that no phonetic symbols are used. The idea is to put each letter or combination of letters to the sound heard on the web.

Part 1 consists of the following 22 units:

1.1 Vowel 'u'
1.2 Accentuation of a word
1.3 'l'
1.4 'r'
1.5 'rr'
1.6 'll' and 'y'
1.7 'j', 'ge' and 'gi'
1.8 'f'
1.9 'z', 'ce' and 'ci'

1.10 Silent 'h'
1.11 'b' and 'v'
1.12 'ñ'
1.13 Write /ka/ /ki/ /ku/ /ke/ and /ko/ in the Spanish spelling
1.14 Write /ga/ /gi/ /gu/ /ge/ /go/ in the Spanish spelling
1.15 Write /ha/ /hi/ /hu/ /he/ /ho/ in the Spanish spelling
1.16 Write interdental /za/ etc. in the Spanish spelling
1.17 Watch out for 'ti', 'tu', 'di' and 'du'
1.18 Read the alphabet
1.19 Memorize numbers
1.20 Memorize the names of the days of the week
1.21 Memorize the names of the months
1.22 Comprehensive exercises of Part 1

Figure 4: Part 1: For Survival

4.3 Part 2: For Smooth Communication

Part 2 consists of the following five units:

2.1 Ignoring spaces between words
2.2 Unstressed words
2.3 Syllables
2.4 Stress positioning
2.5 Intonation

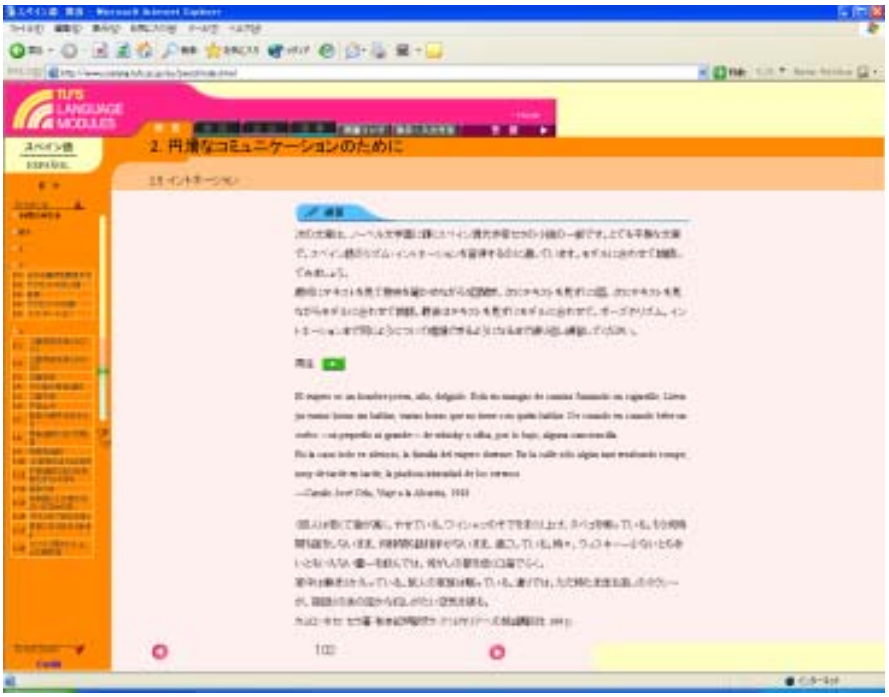

Figure 5: Part 2: For Smooth Communication

4.4 Part 3: To Master the Pronunciation of a Native-Speaker

The last part consists of the following 16 units with the aim of acquiring accuracy:

3.1 Smoothing out diphthongs (1)
3.2 Smoothing out diphthongs (2)
3.3 Triphthongs
3.4 Other vowel combinations
3.5 Two contiguous consonants
3.6 Consonant stops
3.7 Avoiding vowel devoicing
3.8 Omission in vowel combinations
3.9 Sequence of same vowels
3.10 Sound change or disappearance of 's'
3.11 Sound change or disappearance in consonant combinations
3.12 'd' at the end of a word
3.13 The 'w' sound that is used only in foreign words

3.14 The Spanish 's', as compared with the Japanese 's'
3.15 'b', 'd' and 'g' between vowels
3.16 The non-aspirated Spanish 't', 'k', 'p' and 'ch'

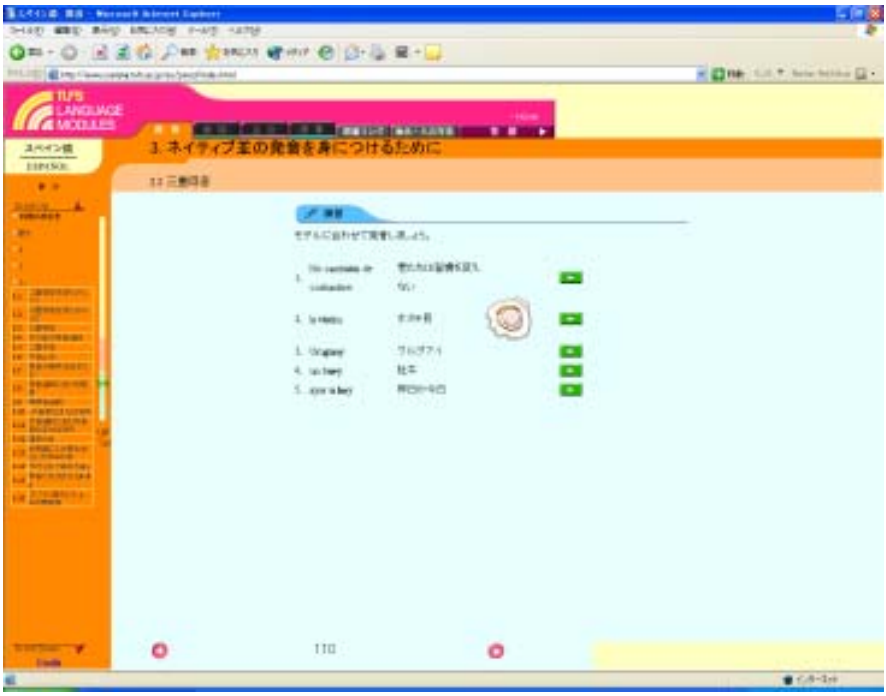

Figure 6: Part 3: To Master the Pronunciation of a Native-Speaker

The Spanish pronunciation module consists of 43 units, covering 137 pages.

Such terms as "diphthongs," "tripthongs," "devoicing," "aspiration" and so forth are used in the heading of the units only for convenience, but are never used in the explanation. Throughout the entire material from Parts 1 to 3, plain words and specific examples are used to explain phonetic phenomena and articulation. We paid particular attention to choosing as examples practical words that are useful in communication, centered on basic vocabulary. Phrases and sentences are also used.

5. Conclusion

Our aim was to develop a pronunciation teaching material to develop communicative competence and to create a state-of-the-art e-learning envi-

ronment. The eleven language versions were developed in parallel within a span of only eight months, with two months each spent on planning, the writing of each language content, preparation for and the actual recording, and building of the website.

The result may still be far from perfection. Our intention, however, is to continue to make revisions as feedback is made available to us from teachers and lecturers in class and users at home. Such revisions will be made from the viewpoint of pedagogy and human technology.

Evaluation sheets, which contain detailed questions, varying from those concerning the users' purpose of learning, needs and wants to those pertaining to the self-assessment of achievement of each unit, have been distributed to instructors and users within and outside our university. Feedback from these evaluation sheets will be useful in enabling us to gradually improve the quality of the material. Such evaluation sheets involve the concept of need analysis. Included in our future plans is to make the Pronunciation Module interactive, providing learners with appropriate feedback and correction, and facilitating task-based learning, which would lead to more dynamic teaching.

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Development and Assessment of TUFS Dialogue Module -Multilingual and Functional Syllabus –

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Key Words
e-learning, multilingual learning materials, cross-lingual syllabus, functional syllabus

0. Introduction

We are currently developing multilingual e-learning materials named TUFS[1] Language Modules under the 21st Century Center of Excellence Program "Usage-Based Linguistic Informatics." TUFS Language Modules consists of the Dialogue Module, Pronunciation Module, Grammar Module, and Vocabulary Module. For the development of the materials we adopted technologies such as Unicode UCS Transformation Format 8 and eXtensible Markup Language. We also adopted a cross-lingual functional syllabus for the Dialogue Module which is composed of 40 functions for each of 17 target languages. This paper reports the results of a survey that was addressed to the teachers of these target languages who were responsible for writing up the dialogues for the Dialogue Module. The dialogue writers were asked to reflect on the adequacy of the 40 functions that were specified as the target of the dialogues.

1. Background of this paper

1.1 A definition of e-learning and the state of the art

Advance Learning Infrastructure Consortium (2003) defines e-learning as follows:

> e-learning is proactive learning by means of information technologies used for communication and network, and its contents are edited for learners' objectives and involve interactivity between learners and content suppliers. (translated by the author)

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[1] TUFS Stands for Tokyo University of Foreign Studies.

We will follow this definition in this paper. The definition of a "cross-lingual syllabus" and "functional syllabus" will be described later in this paper.

E-Learning is now popular in various educational scenes, but in this section we limit our attention to the current situation in higher education. According to ALIC (2003:2-11), universities are making more use of e-learning systems than the past and the use extends to adult education. According to Uskov (2003), in the United States 90% of the universities are planning to provide web-based courses. Thus, e-learning is growing constantly in the higher educational systems.

1.2 Previous studies

The majority of studies about e-learning materials contain reports and assessment of the materials that were developed and are currently in use in addition to the presentation of educational models. Collis (2003) explains scenarios which constitute the mainstream of e-learning with her case study at Twente University. Uskov (2003) states the attempts of web-based education supported by National Science Foundation in the U. S. at Bradley University. Likewise, Takefuta (2002) reports the theory of three step system and the development of multimedia materials for learners of foreign languages. As for the functional syllabus, this paper refers to the studies of Wilkins (1994) and Finocchiaro (1983).

This paper reports the process of the development and assessment of TUFS Dialogue Module materials following Collis (2003) and Uskov (2003), and focuses on a cross-lingual and functional syllabus. From the viewpoint of developing multilingual and multimedia learning materials, TUFS Dialogue Module is an unprecedented attempt to use a common scheme in the development of e-learning materials for 17 languages. We tried to assess the efficiency of utilizing a notional/functional syllabus based on Wilkins (1994) and Finocchiaro (1983) in a cross-lingual syllabus and for the materials of foreign languages that belong to different language families.

1.3 Potentiality of e-learning research at Tokyo University of Foreign Studies

Tokyo University of Foreign Studies[2] has advantages in developing multilingual e-learning materials for mainly two reasons. The first is that at TUFS 52 languages are taught and researched, and teacher-training courses are offered for more than 20 languages. This makes the university one of the largest Japanese universities for training language teachers. The second is

[2] TUFS

that we have an advanced IT environment. So we have a great potential for developing multilingual e-learning materials.

1.4 The 21st Century Center of Excellence Program

The learning materials in this paper was developed under the 21st Century Center of Excellence Program "Usage-Based Linguistic Informatics" at TUFS, funded by the Ministry of Education, Culture, Sports, Science and Technology of Japan. This program aims to establish what we call linguistic informatics by integrating linguistics, language education and information technology. We are developing multilingual corpora, conducting research into each language and applying linguistic theories to practical areas as well as developing web-based training materials.

1.5 The learning materials under development

Target learners of the learning materials are freshmen except for English and Japanese materials[3]. At TUFS, all students belong to the faculty of foreign studies and take six classes a week of the languages in which they major. Almost all of the students have learned English for at least six years. They will use these e-learning materials out of class for studying by themselves.

We are currently developing multilingual materials for the following 17 languages: English, German, French, Spanish, Portuguese, Russian, Chinese, Korean, Mongolian, Indonesian, Pilipino, Laotian, Cambodian, Vietnamese, Arabic, Turkish and Japanese. The e-learning materials developed in our project consist of four modules: the dialogue, pronunciation, grammar and vocabulary modules. The next section describes how we have developed the dialogue module.

2. The process of developing the Dialogue Module

2.1 Standards adopted in our materials

For publication of the materials on the web and for their flexible use, we adopted the following standards: Unicode UTF-8[4] and XML[5] for coding the data, and JAVA Script, HTML[6], and Macromedia Flash for generating pages

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[3] The English material is for elementary school students in Japan and the Japanese material is for English speaking learners of Japanese or learners who can understand simple Japanese written in Hiragana.

[4] Unicode UCS Transformation Format 8

[5] eXtensible Markup Language

[6] HyperText Markup Language

and making them interactive and appropriate for a multimedia environment.

2.2 Unicode UTF-8

We coded the final data of our material, using Unicode UTF-8. As mentioned in Section 1.5, the learning materials have multiple target languages with different sets of characters. The characters should be able to be typed and displayed on the same web pages. However, we had to avoid giving users the unnecessary trouble of installing a new set of characters for each language material. Nikaido (2002) states the efficacy of Unicode UTF-8, while he indicates the font problem in browsing pages. Thus, we decided to code the final data of our material in Unicode UTF-8.[7]

2.3 XML

We adopted XML technologies in coding the data of the materials. Since we plan to make our e-learning materials open to the public on the World Wide Web, the data structure of our materials needed to be formatted in HTML or XML so that the data can be browsed by internet browsers. As materials for language learning, they include not only text data, but also multimedia data such as movies, sounds, and links to them. It is also necessary to generate pages dynamically to meet a variety of educational needs. Lin (2003) states the necessity of an XML database for such materials. The XML database is also effective for variable-length data that are composed of large data of language learning materials.

2.4 JAVA Script and Macromedia Flash

We decided to use JAVA Script and Macromedia Flash[8] in generating and displaying the materials. They are popular technologies on the web, and make much of interactivity and enable interactive learning with multimedia such as sounds, images and movies. As mentioned in Section 1.1, it is a prerequisite condition for e-learning materials to be interactive. The language learning materials should also use multimedia data such as sounds and movies to be effective.

2.5 Process of the development

In this section the process of developing the learning materials is

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[7] According to Unicode Consortium (2003), the combination of Unicode and XML has some problems. They aren't resolved at present but the problems are not major excluding the case of Arabic.

[8] The newest version, Macromedia Flash MX/Macromedia Flash Player 6, is capable of Unicode.

explained. We decided to use a cross-lingual syllabus and a functional syllabus in the development. The materials of each language have 40 lessons, each of which has one dialogue and one main target "function." We will explain the cross-lingual syllabus, the functional syllabus, and the process of selecting the functions in Sections 3 and 4. Each dialogue basically has two interlocutors and is constituted of five turns. Dialogue writers also provided necessary explanations of the vocabulary, grammar, a key sentence and its variations, and exercises on the dialogue and its key function.

We used the format of Microsoft Word which is based on Unicode for exchanging the data among dialogue writers, native informants and data processors. We prepared a data sheet which is understandable for the dialogue writers since they did not necessarily have sufficient knowledge about data structures. We then processed the data, designed the learning materials and recorded the dialogues in a studio. We have completed the designs for an "in-class" page to be used by teachers in the classroom. Figure 1 shows the design we developed based on the studies of Yuki et al (2003/1) for the actual procedures to develop the whole dialogue module and Yuki (2003/2: 175-186) for the development of the Spanish material.

Figure 1:

3. Cross-lingual syllabus

3.1 Its definition in this paper

As mentioned above, our learning materials aim to adopt a cross-lingual syllabus. This is one of the most significant features in TUFS Language Modules from the viewpoint of language education. One of the features of the module is that each module for the 17 languages was developed following a common framework (TUFS 2002/2). Kawaguchi (2002) states as follows:

> One of the academic purposes of TUFS Language Modules is that we develop the materials putting more weight on cross-lingual awareness than on following the previous literature. The dialogue module was designed based on 40 functions in daily life. ... We thus make use of the viewpoint of cross-linguistics in the development of modules.
>
> It is an academically meaningful study to consider evaluation of language proficiencies and examine its efficacy in multiple languages, using the cross-lingual and cross-linguistics concepts. There is no accumulation of studies of attempts to consider the language proficiency in language education and in applied linguistics of multiple languages. In this sense, it is very interesting field to investigate the cross-lingual evaluation model for language proficiency. (translated by the author)

According to the above statement, the cross-lingual syllabus here is defined as a way of evaluation and content of learning materials which are adoptable to multiple languages. The ground of the possibility of its establishment is the similarity between languages that linguists are often aware of, and which is expressed as "the cross-linguistic viewpoint." In this paper we adopt this as the definition of the cross-lingual syllabus.

3.2 The Importance of a pan-lingual syllabus

The statements above illustrate that TUFS Language Modules aim to develop learning materials adopting a pan-lingual syllabus and a model of an evaluation system for language proficiency that is applicable to multiple languages. This attempt is similar to *the Common European Framework of Reference for Languages: learning, teaching, assessment* released by the Council of Europe (COE 2003). The framework aims to establish a common basis for syllabuses, curriculums, examinations and textbooks for language education

to break the communication barrier in Europe where many languages are used. The development and evaluation of our learning materials are an experimental attempt to recognize the possibility of a similar system which contains not only European languages but also other languages.

3.3 The Importance from the viewpoint of information technology

From the viewpoint of information technology, the pan-lingual syllabus is efficient as well. As mentioned in Section 2, the e-learning materials are generated from XML databases. In the construction of databases and the development of generation systems, it was necessary to establish standards for contents and a coding method of the contents in each language. Adopting the pan-lingual syllabus made it possible to fulfill the necessities, although this is a subsidiary reason for the use of the syllabus.

3.4 Research on the learner needs and material assessment

The Common European Framework of Reference for Languages: learning, teaching, assessment stated in Section 3.2 is based on research on needs. In the course of developing our materials, however, we only analyzed existent learning materials, and did not conduct sufficient research on learner needs. Thus, we are aware of the need to conduct a needs analysis and improve the materials based on the results. In section 5 we report a preliminary attempt of surveying the needs of the language teachers through a questionnaire distributed to the developers of the dialogue materials.

4. Functional syllabus

4.1 Its definition in this paper

The dialogue module of TUFS Language Modules adopts a functional syllabus. In Wilkins (1994:27-28), "function" is defined as a communicative function and the social purposes of utterances. Wilkins adds concepts of semantic and grammatical categories such as frequency and quantity to the syllabus based on the "function," and calls the syllabus "notional/functional syllabus." Johnson (1999:305) defines the notional/functional syllabus as a syllabus in which items are arranged according to notions and/or functions. In this paper we follow this definition. In the development of the materials, however, we simply call the syllabus simply "functional syllabus."

4.2 Grounds of adopting a functional syllabus

As mentioned in Section 3, our learning materials aim to adopt a pan-lingual syllabus and to develop contents and evaluation standards which can be adopted to multiple languages. *The Common European Framework of Reference for Languages: learning, teaching, assessment* has the same goal and

puts weight on functions and notions for communication in the framework. This also justifies our attempt to use the functional syllabus. We are also motivated by the fact that the 21st Century Center of Excellence Program "Usage-Based Linguistic Informatics," to which the materials belong, emphasizes the usage of languages.

We adopted the functional syllabus to the dialogue module. In Japan the majority of language learning materials are developed following either a grammatical or functional syllabus or a combination of both. It is difficult to use a grammatical or structural syllabus in the above-mentioned cross-lingual syllabus for the following two reasons. One is that there are significant grammatical differences among the languages because the target languages in the materials often have different genealogy. It is difficult to establish a common content across the materials of all the target languages. The other reason is that the research into grammatical structures in one language may differ in degree from that in another. Some teachers and authors of textbooks make use of the descriptions of morphology and syntax of the target language for educational purposes. The difference in the amount of linguistic research accumulated among different languages, therefore, would be reflected upon the grammatical content of the materials. The adoption of a functional syllabus can also be justified because the dialogue module is designed separately from the grammar module, which deals specifically with the grammar of each language.

In terms of functions in languages, there are common situations such as greeting, inviting, advising, and so on. They should not be radically different although each language has a diverse cultural background. Developing materials based on language functions therefore, is hardly influenced the amount of grammatical research accumulated in each language. Hence, we believed it possible to find common functional factors among different languages. A functional syllabus is appropriate to our multilingual language learning materials which aim to adopt a cross-lingual syllabus. Finocchiaro (1983:19) states that the functional syllabus is adaptable to non-European languages to research and analyze the needs for investigating notions and functions to be taught to learners. The same applies to *the Common European Framework of Reference for Languages: learning, teaching, assessment*, which is one of the models of our project.

TUFS (2002/2) indicates language learning in a short time and language learning for special purposes as social meanings of the 21st Century Center of Excellence Program "Usage-Based Linguistic Informatics," to which the

development of the materials belong. Wilkins (1994:86-87) mentions that the functional syllabus is appropriate to language learning in a short time, and Johnson (1999:306) states that a notional/functional syllabus is appropriate for English for special purposes. Based on these grounds, we adopted a functional syllabus to the dialogue materials.

One characteristic of our e-learning materials is that it makes use of multimedia. It is difficult to separate functions and notions from real communication and situations in language learning materials. Images and movies can appropriately present the situations and the relationships of characters. We recorded the movies of the conversations in our studio and added backgrounds to them to provide as much situational context as possible.

4.3 Problems of the functional syllabus

The functional syllabus has some problems, although we have so far seen the motivation for using the syllabus. From the viewpoint of language generation, the notional/functional syllabus is not so effective as the structural/grammatical syllabus. Johnson (1999:307) mentions problems that arise when teachers use the functional syllabus to beginners who understand little structural information of the target language. For the problem of language generation, on one hand, a full grammar module is a solution. For the beginners' lack of grammatical information, on the other, a solution is the use of the functional syllabus in supplementary studies added to the classes in which they learn the grammar of the target language from teachers.

4.4 The process of selecting 40 functions

We decided that 40 is the appropriate number of functions for our materials, on the basis of the time period allotted for developing the teaching materials as well as the amount of time and energy that dialogue writers of each language could afford to spend on this project. We chose common functions in the following way. First, we extracted functions from the existing learning materials for the Japanese language based on the explanation of notions and functions by Wilkins (1994). We then examined learning materials for Korean and Chinese in order to analyze whether the extracted functions from the Japanese ones were valid for studying other languages and selected 50 functions (Matsumoto 2002). This procedure was also extended to Spanish and German materials created in Japan and the respective countries. We collected 71 functions in total this way. Table 1 shows this process.

Next, we reduced the number of functions to 40 by comparing the 71 functions with those listed by Brundell et al (1982). In this process, we gave

priority to the functions common to both lists, while reorganizing the different functions in terms of their similarities. Functions that appeared in both lists were retained (Process 1), and those that didn't appear in either were combined (Process 2) if they had similar characteristics, or deleted if they could not be combined (Process 3). Finally, we selected the 40 functions. Table 2 shows this process.

5. Survey and analysis

5.1 Goals of the survey

After the development of the materials mentioned in Section 2, we surveyed the dialogue writers of 17 languages. The survey was conducted following the three objectives:(1) to collect data on how each language material was developed, (2) to evaluate the 40 functions in each language and the possibility of expanding them, and (3) to measure computer literacy of each developer. The questionnaire we used is shown at the end of this paper.

We selected one dialogue writer per language and collected answers from the dialogue writers of ten languages so far: English, Vietnamese, Korean, Russian, Arabic, Portuguese, Spanish, German, Japanese and Indonesian. The number is a little less than two thirds of the 17 languages, but this collection will be completed in the near future. These ten languages represent a balanced geographic distribution: America, Europe, Western Asia, Southeast Asia and Eastern Asia.

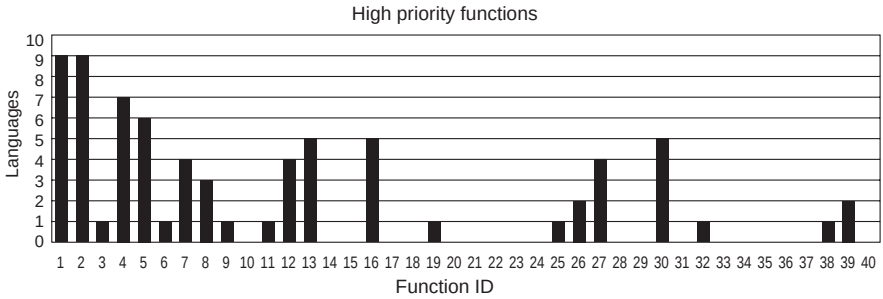
5.2 Results of the survey

The main aim of the survey was to assess the validity of the cross-lingual syllabus and the functional syllabus. Here, we limit our analysis to the results obtained for questions regarding objective (2) above, Part 3 and Question 3 of Part 5 in the questionnaire.

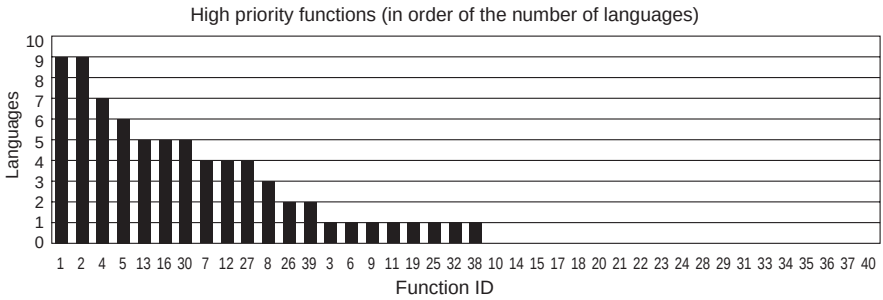
5.2.1 High priority functions for learners

The first analysis was conducted to extract the about high priority functions for learners. Many of the dialogue writers teach the target language in the classroom, and some of them write textbooks. From the viewpoint of language teachers they selected a maximum of eight functions that they want learners to learn preceding the other functions. Graphs 1 and 2 show the results. The function IDs are shown in Table 2.

Graph 1



Graph 2



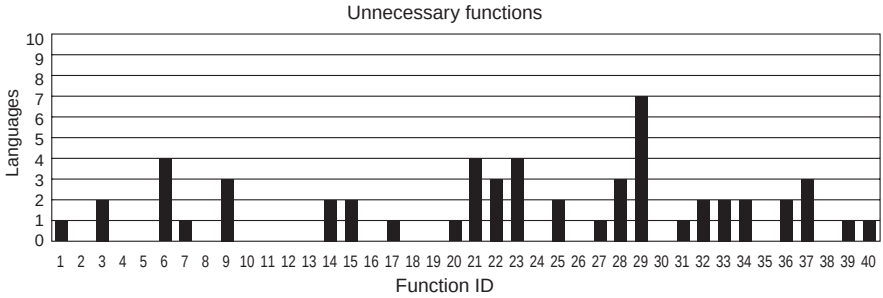
The result shows that on each item, functions (1), (2), (4), (5), (13), (16), and (30) are considered to have priority in 5 or more languages.

As a whole, 13 out of 40 functions, 33%, are considered to have priority in multiple languages. As to the majority of the languages the developers regarded seven common functions, 18%, as having priority. If the cross-lingual syllabus had no validity, and therefore languages have few high priority functions in common, the graph would show that they may be distributed equally. The results of the survey on the 40 functions are unbalanced indicating that the languages have some common functions that are given priority when taught and learned. This means that the cross-lingual syllabus is appropriate in multiple languages.

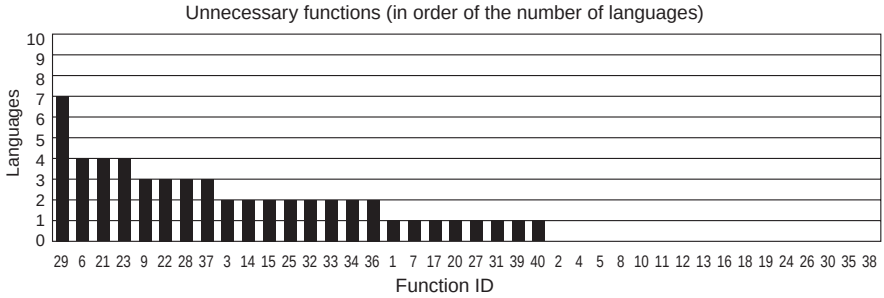
5.2.2 Unnecessary functions

Secondly, we analyze functions that are regarded as unnecessary for elementary language learners. Each dialogue writer selected a maximum of eight functions that beginners do not need to learn. Graphs 3 and 4 show the results.

Graph 3



Graph 4



The results show that function (29) is prominent. This function is "giving alternative plan/compromising," that was extracted from the Japanese language learning material in the starting point of selecting functions. The function was neither deleted nor combined, and was finally added to the list. The result thus, may be indicating that there is a difference in the degree of how necessary a certain function is between Japanese and other languages. Another possibility is that the function is merely considered to be too difficult for beginners.

As a whole, 16 functions, 40%, are considered to be unnecessary for multiple languages. The majority of the languages regarded only one function as unnecessary. We cannot make a clear statement because most of the languages give less than eight functions as unnecessary. The results, however, indicate that the languages do not have one highly common unnecessary function for beginners except for function (29).

5.2.3 Functions to be added

In Question (3) of Part 3, the dialogue writers selected a maximum of eight functions to be added to the list of 40 functions. The List 1 shows the

functions, and the numbers in the parentheses represent the number of languages.

List 1

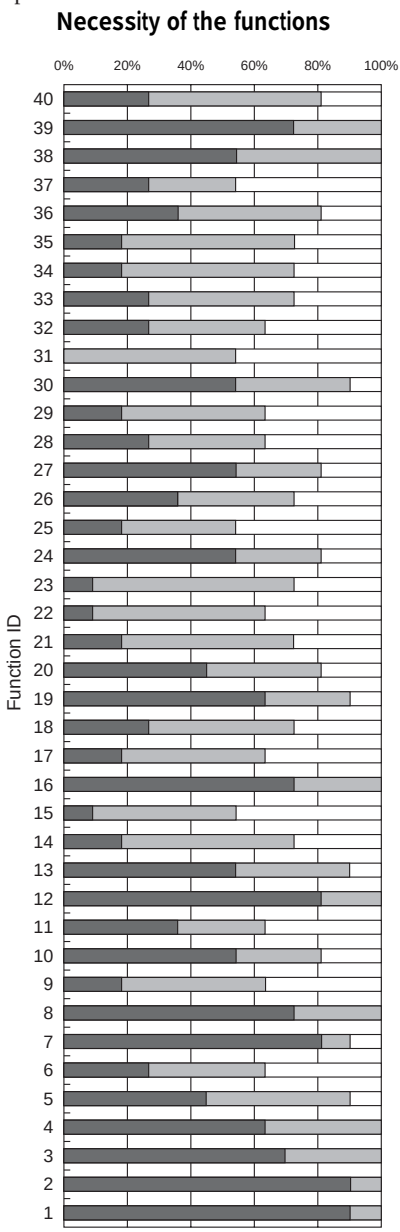
Confirming one's action in the past (6)
Asking/answering about possessions (6)
Inviting/refusing (6)
Introducing family members (3)
Greeting (first time) (2)
Greeting (in the street) (2)
Greeting (clerk-customer) (2)
Asking/answering the purpose of movement (2)
Not permitting/prohibiting (2)
Asking/answering one's plan (2)
Asking one's habits or plan (2)
Explaining processes of operating (2)
Asking again (2)
Confirming that one has done what to do (1)
Asking/explaining objects of exchange (1)
Asking/answering indefinite factors (1)
Asking/answering changes of situation (1)
Explaining with supplement or limitation (1)
Explaining situations or time with supplement or limitation (1)
Shopping (1)
Confirming/answering one's plan in supposed situations (1)
Praising (1)
Expressing one's impression (1)
Expressing being in trouble (1)
Expressing having no idea (1)
Agreeing (1)
Refusing (1)
Asking names of person or things (1)
Ordering (1)
Welcoming (1)
Asking/stating reasons (1)
Demanding/consenting (1)
Asking/stating processes of orders (1)
Asking/stating one's opinion (1)
Expressing one's emotion: surprising, regretting (1)
Various ways of answering: hesitating (1)
Hedging/agreeing (1)

Although these results do have relatively little significance in this paper, we must consider these functions to be high priority candidates that will be added to the materials in future revisions. The functions "confirming the action in the past," "asking/answering about possessions" and "inviting/refusing" are popular among material developers of different languages.

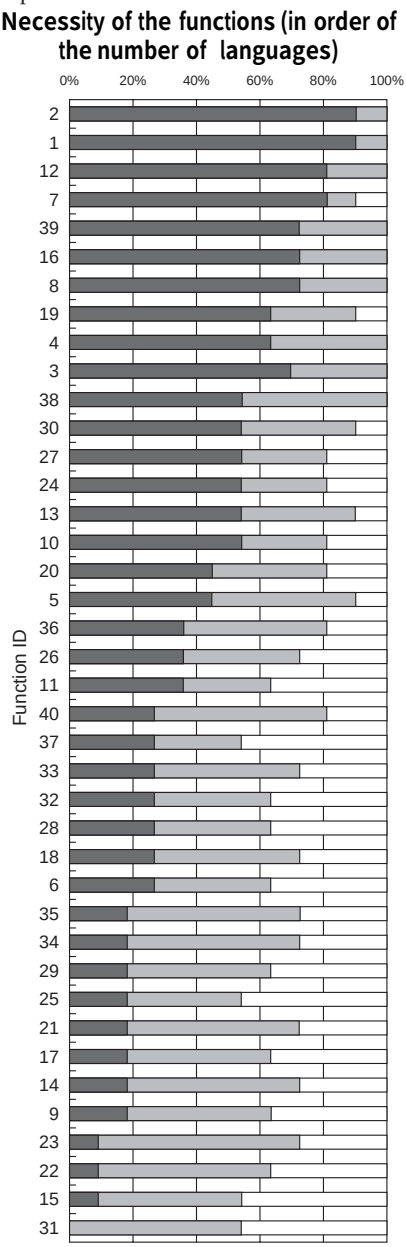
5.2.4 Necessity of each function

The final analysis is on the necessity of each function in each language situation. This necessity is graded based on the frequency of a situation where the functions are needed if a beginner of the language visits the area where the target language is spoken. The dialogue writers selected the frequency from "the situation occurs frequently," "the situation occurs," "the situation could occur," "the situation rarely occurs" and "the situation does not occur." Graphs 5 and 6 show the results. In these graphs "the situation occurs frequently" and "the situation occurs," and "the situation rarely occurs" and "the situation does not occur" are unified into "the situation occurs" and "the situation does not occur," respectively.

Graph 5



Graph 6



Black: the situation occurs.
Gray: the situation could occur.
White: the situation does not occur

The result shows that functions (1), (2), (7), and (12) are considered to occur by 80% of the developers. No dialogue writer selected "the situation occurs frequently" or "the situation occurs" for function (31), and the other functions are thought to be necessary in more than one language. Therefore, it can be said that the selection of the 40 functions are appropriate except for function (31) and function (29) as mentioned in Section 5.2.2.

The majority of the dialogue writers selected "the situation occurs frequently" or "the situation occurs" in 16 functions, 20%, in common. So the necessities of the languages on our 40 functions are judged to be highly common. Furthermore, the answers "the situation rarely occurs" or "the situation does not occur" do not hold the majority in any of the functions. As a whole, we can say therefore that the adoption of a common functional syllabus is appropriate for developing language learning materials of multiple languages.

6. Conclusion

In this paper, we have explained firstly TUFS Language Modules, their background, and the development of the dialogue module which is an e-learning material of multiple languages. We then have explained the cross-lingual syllabus and the functional syllabus used in our materials, and shown the process of selecting 40 functions for our list. Finally, we have analyzed the validity of the syllabuses. Based on the research into the dialogue writers of each language, we have suggested the adequacy of adopting these syllabuses for our dialogue materials and the validity of the 40 functions we selected.

As future plans we will revise the materials. We will complete the planned designs of the materials, and evaluate them. As for the developed materials, we will conduct surveys on the users and make use of the results to improve the materials. We are currently researching learner needs, and intend to analyze the results and incorporate feedback into the materials. The analysis of the results of the questionnaire to the developers has not been completed at present, so this, too, will be continued. In the investigation, we will attempt to clarify the common factors as well as the differences in how essential various functions are considered to be and how dialogue materials were developed among different languages.

Acknowledgements

We would like to extend our sincerest thanks and appreciation to our project leader Yuji Kawaguchi and to Professor Koji Shibano for their valu-

able advice and guidance throughout this study.

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8. Tables

Table 1. From 50 functions to 71 functions

50 functions extracted from Japanese language materials	71 functions after analysis of the other languages
Confirming/answering about things	Confirming/answering about things
Confirming existences of things	Confirming existences of things
-	Greeting
-	Greeting (first time)
-	Greeting (clerk-customer)
-	Greeting (in the streets)
-	Apologizing
Asking/answering degrees of actions	Asking/answering degrees of actions
-	Confirming/answering one's action under certain circumstance

Asking/stating one's opinion	Asking/stating one's opinion
Asking/answering the purpose of movement	Asking/answering the purpose of movement
Shopping	-
Confirming the action in the past	Confirming the action in the past
-	Asking/answering situations in the past
-	Explaining one's family
-	Thanking
Asking/answering skill and ability	Asking/answering skill and ability
Stating things that one want	Stating things that one want
Stating one's hope	Stating one's hope
Confirming duty /affirming	Confirming/negating one's duty
Confirming/negating one's duty	Confirming/stating duty
Asking permission/not permitting	Not permitting/prohibiting
Asking for permission/permitting	Asking for permission/permitting
Confirming prices/paying	Confirming prices
-	Stating prices
Prohibiting	Prohibiting
Asking/answering one's plan	-
Asking one's experience	Asking one's experience
Asking/answering present time	Asking/answering present time
Explaining with supplement or limitation	Explaining with supplement or limitation
-	Asking/answering one's taste of behaviors
Asking/answering exchanges and companions	Asking/answering exchanges and companions
Instructing/requesting actions	Instructing/requesting actions
Inviting/accepting	Inviting/Suggesting
Inviting/refusing	-
-	Saying good-bye
Asking/answering ranges of time	Asking/answering ranges of time
Asking/answering one's taste of things	Asking/answering one's taste of things
-	Introducing oneself
Instructing	Instructing/requesting
Asking/answering ways and means	Asking/answering ways and means
-	Asking/explaining one's hometown or address
Explaining situation or time with supplement or limitation	Explaining situation or time with supplement or limitation
Asking/answering situation	Asking/answering situations
Asking/answering changes of situation	Asking/answering changes of situation
Asking/answering about possessions	Asking/answering about possessions
Asking/answering situations in procession	Asking/answering situations in procession
Confirming that one has done what to do	Confirming that one has done what to do
Asking one's habits or plan	-
Asking/answering one's habits or plan	-
Explaining processes of operating	-

Asking /explaining processes of operating	-
Confirming /answering situations in sup-posed situations	-
Giving alternative plan/compromising	Giving alternative plan/compromising
-	Attracting attention
Stating things that one want	Stating things that one want
Asking/answering a point of time	Asking/answering a point of time
-	Asking/explaining procedure and order
Giving one's message	Giving one's message
-	Asking telephone numbers

Table 2. From 71 functions to 40 functions

functions71	Process	F.ID	functions40
Greeting	Process1 (retain)	1	Greeting
Thanking	Process1 (retain)	2	Thanking
Attracting attention	Process1 (retain)	3	Attracting attention
Introducing oneself	Process1 (retain)	4	Introducing oneself
Greeting (first time)	Process3 (delete)	-	-
Apologizing	Process1 (retain)	5	Apologizing
Greeting (in the streets)	Process3 (delete)	-	-
Giving	Process1 (retain)	6	Giving
Greeting (clerk-customer)	Process3 (delete)	-	-
Saying good-bye	Process1 (retain)	7	Saying good-bye
Confirming prices	Process2 (combine)	8	Asking information (price)
Stating prices	Process2 (combine)		
Asking one's experience	Process1 (retain)	9	Asking information (experience)
Confirming that one has done what to do	Process3 (delete)	-	-
Confirming/answering one's plan	Process1 (retain)	10	Telling one's plan
Confirming the action in the past	Process3 (delete)	-	-
Asking/answering degrees of actions	Process1 (retain)	11	Asking information (degree)
Asking/answering exchanges and companions	Process3 (delete)	-	-
Asking/explaining objects of exchange	Process3 (delete)	-	-

Asking date or day of the week	Process2 (combine)	12	Asking information (time)
Asking/answering ranges of time	Process2 (combine)		
Asking/answering present time	Process2 (combine)		
Asking/answering a point of time	Process2 (combine)		
Asking telephone numbers	Process1 (retain)	13	Asking information (number)
Asking/answering ways and means	Process1 (retain)	14	Saying how and why
Asking/answering about possessions	Process3 (delete)	-	-
Asking/answering skill and ability	Process1 (retain)	15	Asking skill and ability
Asking/answering about existence and place	Process1 (retain)	16	Asking information (existence and place)
Asking/answering indefinite factors	Process3 (delete)	-	-
Asking/answering change of situations	Process3 (delete)	-	-
Asking/answering situations	Process2 (combine)	17	Asking information (attribute)
Asking/answering situations in the past	Process2 (combine)		
Asking/saying one's opinion	Process1 (retain)	18	Saying one's opinion
Giving one's message	Process3 (delete)	-	-
Asking/answering one's taste of things	Process1 (retain)	19	Saying one's taste (thing)
Asking/answering one's taste of behaviors	Process1 (retain)	20	Saying one's taste (behavior)
Asking/explaining one's hometown or address	Process2 (combine)	16	Asking information (existence and place)
Inviting one's family	Process3 (delete)	-	-
Asking/explaining procedure and order	Process1 (retain)	21	Stating procedure and order
Enumerating	Process3 (delete)	-	-
Asking/answering situations in procession	Process1 (retain)	22	Asking what one is
Confirming/answering one's action under certain circumstances	Process1 (retain)	23	Saying how one acts under certain circumstance

Confirming/answering things	Process2 (combine)	16	Asking information (existence and place)
Confirming existence of things	Process2 (combine)		
Asking/answering locations	Process2 (combine)		
Asking/answering the purpose of movement	Process3 (delete)	-	-
Explaining/comparing two things	Process2 (combine)	24	Comparing (comparative and superlative degree)
Explaining/comparing more than two things	Process2 (combine)		
Suggesting/giving information about the subject	Process1 (retain)	25	Suggesting
Explaining with supplement or limitation	Process3 (delete)	-	-
Explaining situations or time with supplement or limitation	Process3 (delete)	-	-
Asking/answering reasons	Process2 (combine)	26	Explaining why
Stating reasons	Process2 (combine)		
Asking explaining reasons or one's hope	Process1 (retain)	27	Asking
Exemplifying	Process1 (retain)	28	Exemplifying
Giving alternative plan/ compromising	Process1 (retain)	29	Compromising
Asking for permission/permitting	Process1 (retain)	30	Asking for permission
Confirming/negating one's duty	Process1 (retain)	31	Confirming duty/negating
Prohibiting	Process1 (retain)	32	Prohibiting
Instructing/requesting	Process2 (combine)	33	Instructing
Instructing/requesting actions	Process2 (combine)		
Not permitting/prohibiting	Process3 (delete)		-
Asking for unacceptable thing	Process1 (retain)	34	Asking for unacceptable thing
Confirming/stating duty	Process1 (retain)	35	Confirming duty/affirming
Inviting someone to a location	Process1 (retain)	36	Inviting
Inviting/suggesting	Process1 (retain)	37	Advising
Asking one's order	Process2 (combine)	38	Demanding
Stating things that one want	Process2 (combine)		
Ordering things	Process2 (combine)		
Stating one's hope	Process1 (retain)	39	Stating one's hope
Introducing someone	Process1 (retain)	40	Introducing someone

10. Questionnaire

Questionnaire about the Development of Dialogue Module

Thank you very much for your cooperation in the development of Dialogue Module in TUFS Language Modules. This research aims to find a concrete way to develop language materials and examine the adequacy of the 40 functions in each language. The result will be used for the improvement of the module. This is for developing better materials. We are grateful for your assistance. Please answer all of the questions and hand in the answer sheet by Friday, August 8.

Chunchen LIN

Please use the answer sheet and refer to the attached function list.

Part 1: Answerers
(1)Please fill in your name

(2)Your situation
1. Teacher of native Japanese speaker
2. Teacher of native target language speaker
3. Student of native Japanese speaker
4. Student of native target language speaker

Part 2: Way of the Development
(1)Dialogue writer's information: if more than two writers take part in the development, select the answer for each writer.
1. Native speaker with experience in teaching the target language
2. Non native speaker with experience in teaching the target language
3. Native speaker with no experience in teaching the target language
4. Non native speaker with no experience in teaching the target language

(2)Translator's information: if more than two translators take part in the development, select the answer for each translator.
1. Native speaker with experience in teaching the target language
2. Non native speaker with experience in teaching the target language
3. Native speaker with no experience in teaching the target language
4. Non native speaker with no experience in teaching the target language

(3)Your criterions for the situations and settings of the dialogues
1. We had no criterion.

2. We limited the situations to particular situations. (For example: In the school)
3. We limited the settings to particular settings of characters. (For example: Story style)
4. We limited the situations and settings to particular situations and settings of characters.
5. We followed existing criterions. > Please give a concrete example.
6. We made our own criterions. > Please give a concrete example.

(4)Your criterion for selection of lexical items to be explained
1. We did not provide supplemental explanation of lexical items.
2. We had no criterion and provide supplemental explanation of all lexical items.
3. We selected lexical items following our experience of teaching and learning.
4. We followed existing criterions. > Please give a concrete example.
5. We made our own criterions. > Please give a concrete example.

(5)Your criterions for selection of grammatical items
1. We did not explain grammatical items.
2. We had no criterion and described all grammatical items.
3. We selected grammatical items following our experience of teaching and learning.
4. We followed existing criterions. > Please give a concrete example.
5. We made our own criterions. > Please give a concrete example.

(6) Your criterions of translations
1. We had no criterion
2. We made word-for-word translations so that learners can easily understand the construction of sentences and meanings of each word.
3. We made translation that learners can understand the sentences easily as the Japanese language.
4. Other > Please give a concrete example.

Part 3: 40 functions
(1) Please select a maximum of eight high priority functions from the list of 40 functions for beginners to learn the material as supplemental materials or for teachers to use for beginners. Please give us the number(s) of the function ID.

(2) Please select a maximum of eight unnecessary functions from the list of 40 functions for beginners to learn the material as supplemental materials or for teachers to use for beginners. Please give us the number(s) of the function ID.

(3) Please select other necessary functions or notions a maximum of eight for beginners of the target language to learn. If the attached list contains the function/notion, give us the number(s) in the list.

Part 4: Skills and Ways of Data Processing
(1) Developer's skill: if more than two developers, in order of the amount of dialogues.
1. I had no experience in computers.
2. I had experience in computers at the time of development, but no experience in word processor applications.
3. I had experience in word processor applications at the time of development, but no experience in XML/HTML.
4. I had experience in word processor applications and XML/HTML.

Part 5: Ways of Development of Each Dialogue and Adequacy of Functions
Please answer these questions about each function in the materials. If you are not the dialogue writer, get information from the dialogue writers. If it is difficult to do this, please give us the answer from the viewpoint of the dialogue writer.

(1) Difficulty of the dialogue in your language
1. Easy 2. Somewhat easy 3. Medium 4. Somewhat difficult 5. Difficult

(2) Adequacy of the amount of the dialogue, 5 turns and 10 lines, in your language
1. Much 2. Somewhat much 3. Medium 4. Somewhat little 5. Little

(3) Frequency of situations where the function is needed if beginners visit the area where the language is used
1. The situation occurs frequently.
2. The situation occurs.
3. The situation could occur.
4. The situation rarely occurs.
5. The situation does not occur.

(4) Native speakers' check on the target language. (Check: evaluation of the dialogue and accuracy of related information as language learning materials)
1. We didn't check.
2. We checked and corrected spelling of words and sentence from the viewpoint of grammar based on the check.
3. We checked and corrected situations and characters based on the check.
4. We checked and rewrote the entire dialogue again based on the check.

(5) Difficulty in setting the situation of the dialogue in your language
1. Easy 2. Somewhat easy 3. Medium 4. Somewhat difficult 5. Difficult

(6) Easiness of making the key sentence of the dialogue in your language
1. Easy 2. Somewhat easy 3. Medium 4. Somewhat difficult 5. Difficult

Thank you for your cooperation.

Concluding Remarks

Yuji KAWAGUCHI *(COE Program Leader)*

Two days have passed more quickly than I imagined. I hope we could benefit from this encounter of Theoretical Linguistics and Applied Linguistics, which represents the vast field of Linguistic Informatics.

In my opinion, our program of Linguistic Informatics can be compared to the construction of gothic cathedrals in the Middle Ages of Europe. At the end of the twelvth century and especially at the thirteenth century, people wanted to build higher and higher cathedrals. It seemed to be a pursuit of ultimate height to approach the devine summit.

This architect's dream was realized through the most advanced technology at that time, for example, the invention of flying buttress and ribbed vault. But human dream always needs a moralistic background. In the construction of medieval gothic cathedrals, it was the scholasticism that constituted a mental support for the efforts of building huge monuments[1]. In this way, gothic cathedrals in Europe are considered as the results of a happy marriage between medieval architecture and scholasticism.

As far as our Linguistic Informatics is concerned, it will be Theoretical and Applied Linguistics that fournish a humanistic backbone to our project. And with the assistance of computer sciences, we try to realize our ideal, but this time, not in the physical world, but in, what we call, the virtual world of the Internet. And there is no goal for our scientific pursuit. Just like a search for Holy Grail in medieval romance.

Finally, I'd like to express my deepest gratitude towards our guest speakers, colleagues, graduate students, and many collaborators of this COE program. I appreciate your two day's attendance. And I hope we will see each other again at the next conference.

Now I regret to announce the closing of this International Conference. Thank you very much for your kind attention.

Tokyo, December 14. 2003

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[1] Ervin Panofsky, *Gothic Architecture and Scholasticism*, Latrobe, Pennsylvania, 1951.

Index of Proper Nouns

21st Century COE Project on Usage-Based Linguistic Informatics 298
Active Worlds 252
Applied Linguistics Projects in TUFS's 21st Century COE Program 279
AWK 49
Basic Transcription System for English (BTSE) 299, 313
Basic Transcription System for Japanese (BTSJ) 299
Daedalus 249
Discourse Research Group 279, 280, 282
D-Module 280, 281, 283-290, 298, 300, 302-304
DÍAZ-MAS, PALOMA 180, 194
Estoria do Muy Nobre Vespesiano 64
Japan 197, 199-201, 205, 213
Japanese 197, 200-206, 209, 212, 213
Japanese 2 by Basic Transcription System for Japanese (BTSJ) 282
L'Atlas Linguistique et Ethnographique de l'Ile-de-France et de l'Orléanais (ALIFO) 99
Language in the Workplace (LWP) Project 197, 198
Leal Conselheiro 69
Leite de Vasconcelos 64
Multilingual Corpus of Spoken Language by Basic Transcription System (BTS) (~-Japanese 2, ~-Japanese 2 by BTSJ) 282-284, 292, 294
New Zealand 196-203, 205, 207, 210, 212, 213, 215
New Zealander 197, 200-205, 207
Perl 49
Real Academia Española 123, 131
Talk That Works (TTW) 280-284, 290, 295, 297-300, 302-305, 309, 310, 312
TUFS (English) Dialog module 294, 298
Wellington Language in the Workplace project 197

Names

BOONS, J. 32
BREMER, K. 234
BROEDER, P. 234
CHANG-RODRÍGUEZ, E. 185, 194
COOK, V. 221
DAVIES, M. 125
FIRTH, J. R. 177
GASS, S. 230
GOEBL, H. 100
GROSS, M. 29, 77
GUILLET, A. 32
HARRIS, Z. S. 177, 95
HOOK, D. 64
JUILLAND, A. 185, 194
KAWAGUCHI, Y. 102
LARA, F. 130
LECLÈRE, C. 32, 77
LLISTERRI, J. 122
M.-T. VASSEUR 234
MARTINET, A. 146
MORENO-FERNÁNDEZ, F. 129
NORTON, B. 235
PAUMIER, S. 89
PENNY, R. 180, 191, 192, 194

RAMPTON, B. 227
RAMÍREZ, F. 48
ROBERTS, C. 234
ROMERO, E. 180, 194
SALVÁ, V. 47
SAUSSURE, F. de 177
SILBERTZTEIN, M. 89
SIMONI-AUREMBOU, M.-R. 99
SIMONOT, M. 234
SPERBER, D. 177
WILLIAMS, E. B. 69
WILSON, D. 177

Index of Subjects

3D 252
40 functions 338
Additional Language Acquisition (ALA) 238
asymmetrical distribution 171
authentic conversation 295, 297-300, 311, 312
avatar 253
awareness 281, 282, 284, 292
backchannels 290-292
BBS 259, 263, 265, 266, 270-276
bi and multilingualism 224
bilingual corpus 150
Cervantes Institute 133
classifying adjective 47, 57
cluster analysis 106
communicative competence 320, 323
communities of practice 235
competence 214
compound verb 39, 42
Computer Assisted Language Learning (CALL) 248, 259, 260, 275
computer-mediated communication (CMC) 248
context 196, 211, 214, 215, 307-312
contextual 211-213, 311
Contrastive Analysis Hypothesis 321
conventional teaching material 282
conversation teaching material 279, 280, 282-286, 292, 294
Corpus de Referencia del Español Actual 123
corresponding linguistic form 281-283, 302-311
Counting Letters 151
cross-lingual (functional) syllabus 333, 334, 337, 338, 340, 342, 343, 347
decoding 261, 275
defining property 29, 32, 36
dialectology 131
dialogue module 333-335, 337, 339, 340, 347, 354
discourse sentence 281, 299-303, 305-308, 310, 313
discovery learning 322, 323
e-learning (~material) 333-336, 339, 341, 347
electronic dictionary 29, 41, 42, 89, 92
encoding 275
European projects 125
eXtensible Markup Language (XML) 333, 335, 336, 339
face-redress behavior 285, 286
face-to-face 260-263, 266, 271, 277
filler 281, 287, 289, 290
finite state automata 42
fluency and accuracy 322, 323
foreign language context 231
frozen sequence 40
function 333, 337-347, 349, 351
functional syllabus 298, 334, 337, 339, 340, 342, 347
geolinguistcs 120
hiatus 72
humour 196, 210
ICT-based 258-261, 264, 275-277
inference 177, 178
ingressive 174, 177

interactive linguistic behavior 296
intercoder reliability 300, 302
Internet Relay Chat (IRC) 249
Japanese conversation 280
Judeo-Spanish 180, 181, 185, 186, 189, 191-195
KWIC (~format) 48, 49, 156
Ladinokomunita 180, 181, 184, 189, 191
LAN 249
learning material 334
lexicon-grammar 29, 42, 77, 88
linguistic anthropocentrism 171, 175
linguistic variation 128
LK-Corpus 184, 187, 188, 191
local grammar 42, 88-90
macro programming 153
materials development 295, 297, 312
monolingualism 224
Multi Dimensional Scaling (MDS) 111
multi-competence model 225
multilingual material 333-335
multi-user object-orientated domains (MOOs) 248
multivariate analysis 100, 106
MySQL 181-184, 188, 194, 195
natural class 30
natural conversation (~data) 279-291
naturalistic context 234
notional functional syllabus 298, 312
number of syllables of an adjective 48-51, 53, 54, 58
object-orientated 252
PHP-KWIC 186, 187, 192, 194
politeness behavior 284
Portuguese (Modern~, Old~) 65
predicate of variation 79
preference 175, 178
PRESEEA 131
principle of difference 177, 178
prosody 319, 320, 323, 325
qualitative adjective 47
reference corpora 131
refusal 196, 197, 205-213, 215
refuse 205-209, 211, 212, 214
refuser 206, 210
refusing 197, 206-208, 211, 213
relevance 177, 178
replace 153
representativeness 125
requesting 280, 283, 284, 286, 289
retrieval script 49
romantic bilingualism 227
Second Language Acquisition (SLA) 221, 248
second language context 233
Second Person Plural 65
sentences describing a variation in value 77
small talk 196-205, 214, 216
sociolinguistics 131
socio-pragmatic competence 213-216
Spanish language 121, 125
Spanish subjunctive mood 156
spoken language 120
standardization 99
support verb 39, 42, 79, 80
synchronous 248
table 29, 30, 39
task-based learning 331
transcription 299
transference 260-262, 275, 276
TUFS Language Module 333, 338, 339, 347, 354
type frequency 57, 58
Type-1 281, 283, 302-305, 309
Type-2 281, 283, 302-307, 309
Type-3 282, 283, 302, 305, 310

Unicode UCS Transformation Format 8
(Unicode UTF-8) 333, 335, 336
UNITEX 89
virtual reality (VR) 248